CHAPTER

79

ENGINE OIL

(LEAP-1B ENGINES)

BOEING

737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

CHAPTER 79
ENGINE OIL

Subject/Page	Date	COC
79-EFFECTIVE PAGES		
1 thru 4	Sep 15/2026	
79-CONTENTS		
1	Sep 15/2025	
2	Jan 15/2025	
3	Jan 15/2026	
4	Jan 15/2025	
79-00-00		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
610	May 15/2024	
611	May 15/2024	
612	Sep 15/2024	
613	May 15/2024	
614	May 15/2024	
615	May 15/2024	
616	May 15/2024	
617	May 15/2024	
618	May 15/2024	
619	May 15/2024	
620	May 15/2024	
621	May 15/2024	
622	May 15/2024	
623	May 15/2024	
624	May 15/2024	
625	May 15/2024	
626	May 15/2024	
627	May 15/2024	
628	May 15/2024	
629	May 15/2024	
79-00-00 (cont)		
630	May 15/2024	
631	May 15/2024	
632	May 15/2024	
633	May 15/2024	
634	May 15/2024	
635	May 15/2024	
636	Sep 15/2025	
637	Sep 15/2025	
638	Sep 15/2025	
639	Sep 15/2025	
640	Sep 15/2025	
641	Sep 15/2025	
642	Sep 15/2025	
643	Sep 15/2025	
644	Sep 15/2025	
645	Sep 15/2025	
646	Sep 15/2025	
647	Sep 15/2025	
648	Sep 15/2025	
649	Sep 15/2025	
650	Sep 15/2025	
651	Sep 15/2025	
652	Sep 15/2025	
653	Sep 15/2025	
654	Sep 15/2025	
655	Sep 15/2025	
656	Sep 15/2025	
657	Sep 15/2025	
658	Sep 15/2025	
659	Sep 15/2025	
660	Sep 15/2025	
661	Sep 15/2025	
662	Sep 15/2025	
663	Sep 15/2025	
664	Sep 15/2025	
665	Sep 15/2025	
79-00-00 (cont)		
666	Sep 15/2025	
667	Sep 15/2025	
668	Sep 15/2025	
669	Sep 15/2025	
670	Sep 15/2025	
671	Sep 15/2025	
672	Sep 15/2025	
673	Sep 15/2025	
674	Sep 15/2025	
675	Sep 15/2025	
676	Sep 15/2025	
677	Sep 15/2025	
678	Sep 15/2025	
R 679	Sep 15/2026	
R 680	Sep 15/2026	
681	Sep 15/2025	
682	Sep 15/2025	
683	Sep 15/2025	
684	Sep 15/2025	
685	Sep 15/2025	
686	Sep 15/2025	
687	Sep 15/2025	
688	Sep 15/2025	
689	Sep 15/2025	
690	Sep 15/2025	
691	Sep 15/2025	
692	Sep 15/2025	
693	Sep 15/2025	
694	Sep 15/2025	
R 695	Sep 15/2026	
O 696	Sep 15/2026	
O 697	Sep 15/2026	
698	Sep 15/2025	
698.1	Sep 15/2025	
698.2	Sep 15/2025	
698.3	Sep 15/2025	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

79-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

Subject/Page	Date	COC
79-00-00 (cont)		
698.4	Sep 15/2025	
698.5	Sep 15/2025	
698.6	BLANK	
79-00-00		
901	May 15/2024	
902	May 15/2024	
79-11-01		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	Jan 15/2026	
412	Jan 15/2026	
413	Jan 15/2026	
414	Jan 15/2026	
79-11-01		
R 601	Sep 15/2026	
602	Jan 15/2025	
R 603	Sep 15/2026	
R 604	Sep 15/2026	
R 605	Sep 15/2026	
R 606	Sep 15/2026	
R 607	Sep 15/2026	
R 608	Sep 15/2026	
R 609	Sep 15/2026	
R 610	Sep 15/2026	
R 611	Sep 15/2026	
R 612	Sep 15/2026	
R 613	Sep 15/2026	
R 614	Sep 15/2026	
A 615	Sep 15/2026	
A 616	Sep 15/2026	
A 617	Sep 15/2026	
A 618	Sep 15/2026	
A 619	Sep 15/2026	
A 620	BLANK	
79-11-03		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	BLANK	
79-21-01		
401	May 15/2026	
402	May 15/2024	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	Jan 15/2026	
409	Jan 15/2026	
410	Jan 15/2026	
411	Jan 15/2026	
412	BLANK	
79-21-02		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	
413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
424	May 15/2024	
425	May 15/2024	
426	May 15/2024	
427	May 15/2025	
428	May 15/2025	
429	May 15/2025	
430	May 15/2025	
79-21-02		
601	May 15/2026	
602	May 15/2024	
603	May 15/2024	
604	Jan 15/2025	
605	May 15/2026	
606	May 15/2026	
607	May 15/2026	
608	May 15/2026	
609	May 15/2026	
610	May 15/2026	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

79-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

Subject/Page	Date	COC
79-21-03		
401	May 15/2024	
402	May 15/2024	
403	May 15/2025	
404	May 15/2024	
405	Jan 15/2026	
406	May 15/2025	
407	Jan 15/2026	
408	Jan 15/2026	
409	Jan 15/2026	
410	Jan 15/2026	
411	Jan 15/2026	
412	Jan 15/2026	
413	Jan 15/2026	
414	Jan 15/2026	
415	Jan 15/2026	
416	Jan 15/2026	
417	Jan 15/2026	
418	Jan 15/2026	
419	Jan 15/2026	
420	Jan 15/2026	
79-21-04		
201	May 15/2024	
R 202	Sep 15/2026	
R 203	Sep 15/2026	
R 204	Sep 15/2026	
205	May 15/2024	
206	May 15/2024	
207	May 15/2024	
208	May 15/2024	
209	May 15/2024	
210	May 15/2024	
211	May 15/2024	
212	May 15/2024	
213	May 15/2024	
214	May 15/2024	
215	May 15/2024	

Subject/Page	Date	COC
79-21-04 (cont)		
216	May 15/2024	
R 217	Sep 15/2026	
218	May 15/2024	
219	May 15/2024	
220	May 15/2025	
221	May 15/2024	
222	May 15/2024	
223	May 15/2024	
224	May 15/2024	
225	May 15/2024	
226	May 15/2024	
227	May 15/2024	
228	BLANK	
79-21-07		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2026	
408	May 15/2024	
79-21-07		
601	May 15/2024	
602	May 15/2024	
79-21-12		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
79-21-14		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	

Subject/Page	Date	COC
79-21-14 (cont)		
405	May 15/2024	
406	May 15/2024	
79-21-17		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
79-21-21		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	Sep 15/2024	
408	Sep 15/2024	
R 409	Sep 15/2026	
410	Jan 15/2026	
411	Jan 15/2026	
412	BLANK	
79-21-22		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	BLANK	
79-30-00		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

79-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

Subject/Page	Date	COC	Subject/Page	Date	COC	Subject/Page	Date	COC
79-30-00 (cont)								
604	May 15/2024							
605	May 15/2024							
606	May 15/2024							
607	May 15/2024							
608	May 15/2024							
609	May 15/2024							
610	May 15/2024							
611	May 15/2024							
612	May 15/2024							
79-31-01								
401	May 15/2024							
402	May 15/2024							
403	May 15/2024							
404	May 15/2024							
405	May 15/2024							
406	May 15/2024							
407	May 15/2024							
408	May 15/2024							
79-32-01								
401	Jan 15/2026							
402	May 15/2024							
403	Jan 15/2026							
404	May 15/2024							
405	May 15/2024							
406	May 15/2024							
407	May 15/2024							
408	May 15/2024							
79-35-01								
401	Jan 15/2026							
402	May 15/2024							
403	Jan 15/2026							
404	May 15/2024							
405	May 15/2024							
406	May 15/2024							
407	May 15/2024							
408	May 15/2024							

A = Added, R = Revised, D = Deleted, O = Overflow, C = Customer Originated Change

79-EFFECTIVE PAGES

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ENGINE OIL - INSPECTION/CHECK	79-00-00		601	PTF ALL
Engine Oil System (External Surfaces Walk Around) Inspection TASK 79-00-00-210-801-G00			601	PTF ALL
Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection TASK 79-00-00-210-802-G00			610	PTF ALL
Engine Oil System (Internal) Inspection TASK 79-00-00-210-803-G01			621	PTF ALL
Oil Debris Monitoring System Sensor Inspection TASK 79-00-00-210-804-G00			629	PTF ALL
Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check TASK 79-00-00-220-801-G00			636	PTF ALL
Engine Oil System (Fluids (Fuel or Skydrol) and Solids Contamination) Detailed Inspection TASK 79-00-00-211-801-G00			684	PTF ALL
Engine Oil System (Wear Metal Contamination) Inspection TASK 79-00-00-200-802-G00			689	PTF ALL
ODMS Fault Check TASK 79-00-00-200-803-G00			695	PTF ALL
Drain Tubes Inspection and Cleaning TASK 79-00-00-200-804-G00			696	PTF ALL
ENGINE OIL - DDG MAINTENANCE PROCEDURES	79-00-00		901	PTF ALL
DDG 79-31-01 Oil Quantity Indicating Systems Inoperative - Preparation TASK 79-00-00-040-802-G00			901	PTF ALL
DDG 79-21-01 Engine Oil Debris Monitoring System Inoperative - Preparation TASK 79-00-00-040-804-G00			902	PTF ALL
OIL TANK - REMOVAL/INSTALLATION	79-11-01		401	PTF ALL
Oil Tank Removal TASK 79-11-01-000-801-G00			401	PTF ALL

79-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Oil Tank Installation TASK 79-11-01-400-801-G00			411	PTF ALL
OIL TANK - INSPECTION/CHECK	79-11-01		601	PTF ALL
Oil Tank Inspection TASK 79-11-01-200-801-G00			601	PTF ALL
OIL TANK FILLER CAP - REMOVAL/INSTALLATION	79-11-03		401	PTF ALL
Oil Tank Filler Cap Removal TASK 79-11-03-000-801-G00			401	PTF ALL
Oil Tank Filler Cap Installation TASK 79-11-03-400-801-G00			409	PTF ALL
MAIN FUEL/OIL HEAT EXCHANGER - REMOVAL/INSTALLATION	79-21-01		401	PTF ALL
Main Fuel/Oil Heat Exchanger Removal TASK 79-21-01-000-801-G00			401	PTF ALL
Main Fuel/Oil Heat Exchanger Installation TASK 79-21-01-400-801-G00			408	PTF ALL
ENGINE SURFACE AIR COOLING OIL COOLER - REMOVAL/INSTALLATION	79-21-02		401	PTF ALL
Engine Surface Air Cooling Oil Cooler 1 Removal TASK 79-21-02-000-801-G00			401	PTF ALL
Engine Surface Air Cooling Oil Cooler 1 Installation TASK 79-21-02-400-801-G00			412	PTF ALL
Engine Surface Air Cooling Oil Cooler 2 Removal TASK 79-21-02-000-802-G00			415	PTF ALL
Engine Surface Air Cooling Oil Cooler 2 Installation TASK 79-21-02-400-802-G00			427	PTF ALL
ENGINE SURFACE AIR COOLING OIL COOLER - INSPECTION/CHECK	79-21-02		601	PTF ALL
Engine Surface Air Cooling Oil Cooler Inspection TASK 79-21-02-211-801-G00			601	PTF ALL
LUBRICATION UNIT - REMOVAL/INSTALLATION	79-21-03		401	PTF ALL
Lubrication Unit Removal TASK 79-21-03-000-801-G00			401	PTF ALL

79-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Lubrication Unit Installation TASK 79-21-03-400-801-G00			414	PTF ALL
SCAVENGE SCREEN PLUGS - MAINTENANCE PRACTICES	79-21-04		201	PTF ALL
Scavenge Screen Plugs Removal TASK 79-21-04-000-801-G00			201	PTF ALL
Scavenge Screen Plugs Installation TASK 79-21-04-400-801-G00			211	PTF ALL
Scavenge Screen Plug Inspection TASK 79-21-04-211-801-G00			214	PTF ALL
OIL FILTER CARTRIDGE ELEMENT - REMOVAL/INSTALLATION	79-21-07		401	PTF ALL
Oil Filter Cartridge Element Removal TASK 79-21-07-000-801-G00			401	PTF ALL
Oil Filter Cartridge Element Installation TASK 79-21-07-400-801-G00			406	PTF ALL
OIL FILTER CARTRIDGE ELEMENT- INSPECTION/CHECK	79-21-07		601	PTF ALL
Oil Filter Cartridge Element Inspection TASK 79-21-07-200-801-G00			601	PTF ALL
OIL DEBRIS MONITORING SYSTEM SENSOR - REMOVAL/INSTALLATION	79-21-12		401	PTF ALL
Oil Debris Monitoring System Sensor Removal TASK 79-21-12-000-801-G00			401	PTF ALL
Oil Debris Monitoring System Sensor Installation TASK 79-21-12-400-801-G00			405	PTF ALL
OIL DEBRIS MONITORING SYSTEM UNIT - REMOVAL/INSTALLATION	79-21-14		401	PTF ALL
Oil Debris Monitoring System Unit Removal TASK 79-21-14-000-801-G00			401	PTF ALL
Oil Debris Monitoring System Unit Installation TASK 79-21-14-400-801-G00			405	PTF ALL
OIL EDUCTOR VALVE - REMOVAL/INSTALLATION	79-21-17		401	PTF ALL
Oil Eductor Valve Removal TASK 79-21-17-000-801-G00			401	PTF ALL

79-CONTENTS

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CHAPTER 79
ENGINE OIL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Oil Eductor Valve Installation TASK 79-21-17-400-801-G00			406	PTF ALL
AIR OIL SEPARATOR - REMOVAL/INSTALLATION	79-21-21		401	PTF ALL
Air Oil Separator Removal TASK 79-21-21-000-801-G00			401	PTF ALL
Air Oil Separator Installation TASK 79-21-21-400-801-G00			409	PTF ALL
ANTI LEAK VALVE - REMOVAL/INSTALLATION	79-21-22		401	PTF ALL
Anti Leak Valve Removal TASK 79-21-22-000-801-G00			401	PTF ALL
Anti Leak Valve Installation TASK 79-21-22-400-801-G00			405	PTF ALL
OIL INDICATING SYSTEM - INSPECTION/CHECK	79-30-00		601	PTF ALL
Oil Indicating System Inspection TASK 79-30-00-210-801-G00			601	PTF ALL
OIL LEVEL SENSOR - REMOVAL/INSTALLATION	79-31-01		401	PTF ALL
Oil Level Sensor Removal TASK 79-31-01-000-801-G00			401	PTF ALL
Oil Level Sensor Installation TASK 79-31-01-400-801-G00			406	PTF ALL
OIL PRESSURE AND TEMPERATURE SENSOR - REMOVAL/INSTALLATION	79-32-01		401	PTF ALL
Oil Pressure and Temperature Sensor Removal TASK 79-32-01-000-801-G00			401	PTF ALL
Oil Pressure and Temperature Sensor Installation TASK 79-32-01-400-801-G00			406	PTF ALL
OIL FILTER DIFFERENTIAL PRESSURE SENSOR - REMOVAL/INSTALLATION	79-35-01		401	PTF ALL
Oil Filter Differential Pressure Sensor Removal TASK 79-35-01-000-801-G00			401	PTF ALL
Oil Filter Differential Pressure Sensor Installation TASK 79-35-01-400-801-G00			406	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE OIL - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) An inspection of the engine oil system (external surfaces walk around)

(2) An inspection of the engine oil system (fan cowl panels and thrust reversers open)

(3) An inspection of the engine oil system (internal)

(4) An inspection of the oil debris monitoring system sensor

(5) A detailed dimensional check of the engine oil system (oil debris monitoring system sensor, scavenge screen plugs, starter magnetic plug, and oil filter)

(6) A detailed inspection of the engine oil system (fluids and solids contamination)

(7) An inspection of the engine oil system (wear metal contamination)

(8) A check of the ODMS fault.

(9) An inspection and cleaning of the drain tubes.

TASK 79-00-00-210-801-G00

2. Engine Oil System (External Surfaces Walk Around) Inspection

(Figure 601)

A. General

(1) This task gives the instructions on how to examine the engine and nacelle for evidence of oil leakage.

B. References

Reference	Title
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-71-00-200-801-G00	Engine Vents and Drains Inspection (P/B 601)
72-22-01-200-801-G00	Front Spinner Cone Inspection (P/B 601)
72-60-00-200-801-G00	Accessory Drives Inspection (P/B 601)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Inspection

SUBTASK 79-00-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

F. Engine Oil System (External Surfaces Walk Around) Inspection

SUBTASK 79-00-00-210-001-G00

(1) Do the visual inspection of the fan blades [1], stage 1 booster vanes [2], and front spinner cone [3] as follows (Figure 601 (Sheet 1)):

(a) Examine the fan blades [1] and the inner side of the fan case for oil leakage.

(b) Do a visual inspection at the 6:00 o'clock position of the fan blades [1] and stage 1 booster vanes [2] behind the fan blades.

1) If they are wet with oil, replace the engine (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

(c) Do the visual inspection of the front spinner cone [3].

1) If it is wet with oil, look for indications of the source of the oil leakage (TASK 72-22-01-200-801-G00).

SUBTASK 79-00-00-210-002-G00

(2) Do a visual inspection of the drain masts as follows (Figure 601 (Sheet 2)):

(a) Do a visual inspection of the fan drain mast [4].

(b) Do a visual inspection of the core drain mast [5].

(c) If there is oil leakage from the drain tubes on the drain masts, do this task: Engine Vents and Drains Inspection, TASK 71-71-00-200-801-G00.

SUBTASK 79-00-00-210-003-G00

(3) Do a visual inspection of the bottom of left fan cowl panel [6] and the right fan cowl panel [7] (Figure 601 (Sheet 2), Figure 601 (Sheet 6), Figure 601 (Sheet 5)).

(a) If there is oil leakage, do this task: Accessory Drives Inspection, TASK 72-60-00-200-801-G00.

SUBTASK 79-00-00-210-005-G00

(4) Do a visual inspection of these engine areas (Figure 601 (Sheet 6), Figure 601 (Sheet 5)):

NOTE: Oil puddling on the aft sump area cover, lower area of the forward plug, aft plug, and core primary nozzle is permitted if the oil consumption rate and leakage rate from the aft sump flooding drain are within the serviceable limits (TASK 71-00-00-800-802-G00).

(a) The aft sump area of the primary nozzle [10].

(b) The 6:00 o'clock areas of the forward plug [11] and aft plug [12].

SUBTASK 79-00-00-960-001-G00

(5) If the oil consumption rate and oil leakage rate are more than the serviceable limits, replace the engine (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-860-002-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

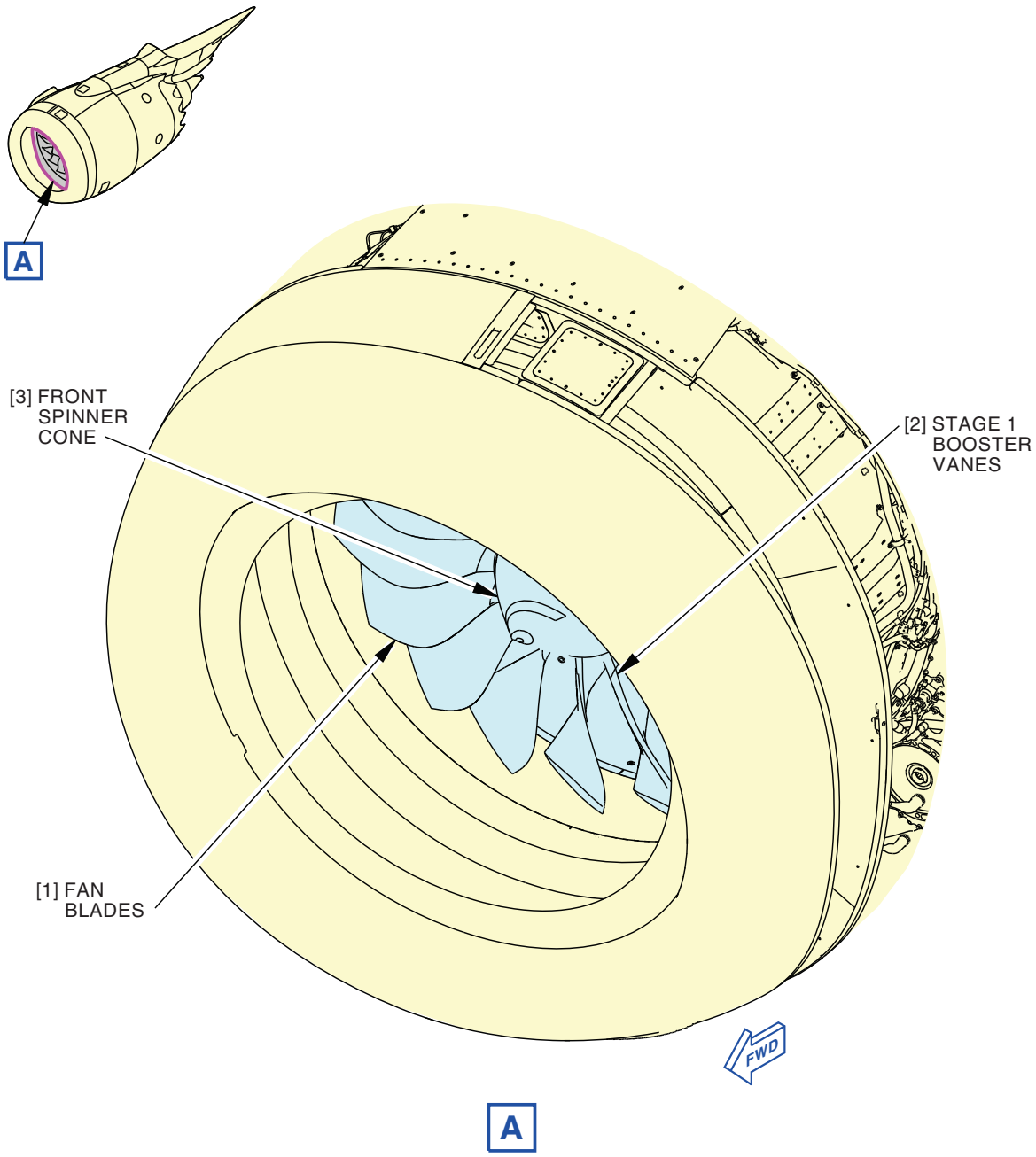
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



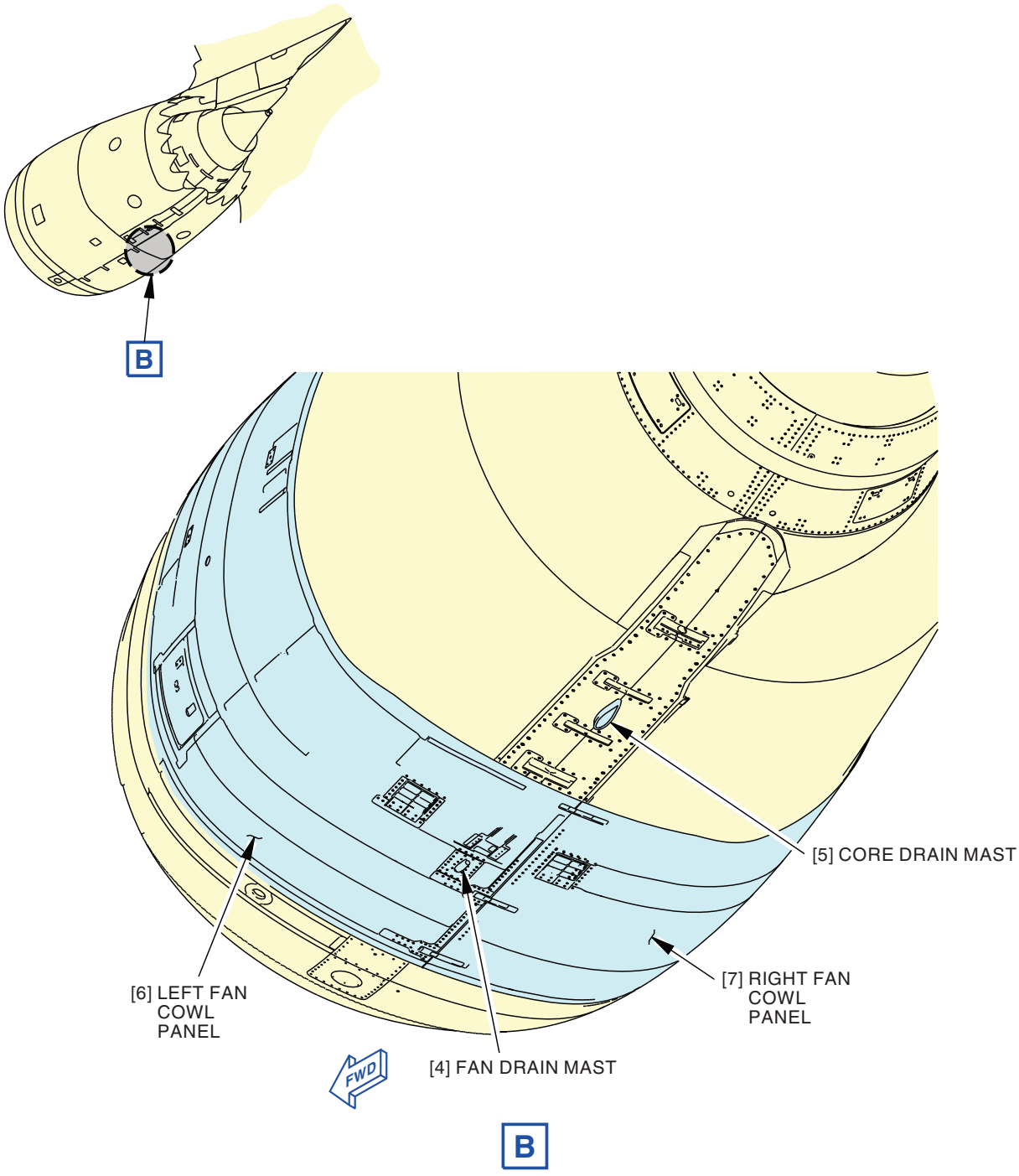
NOTE:

STAGE 1 BOOSTER VANES ARE NOT VISIBLE IN THIS VIEW.

2465655 S0000574679_V1

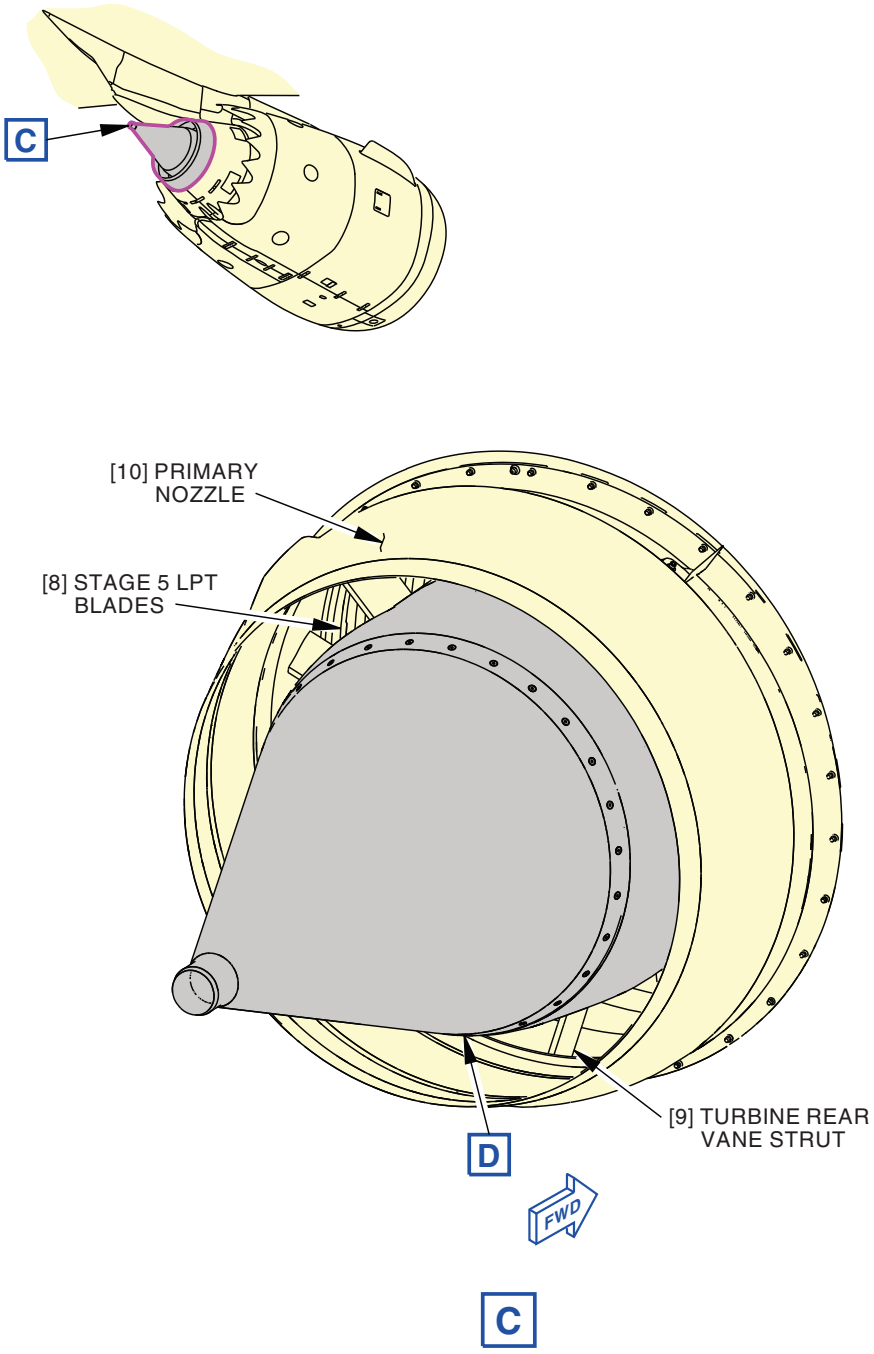
Engine Oil System (External Surfaces Walk Around) Inspection
Figure 601/79-00-00-990-809-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Oil System (External Surfaces Walk Around) Inspection
Figure 601/79-00-00-990-809-G00 (Sheet 2 of 6)

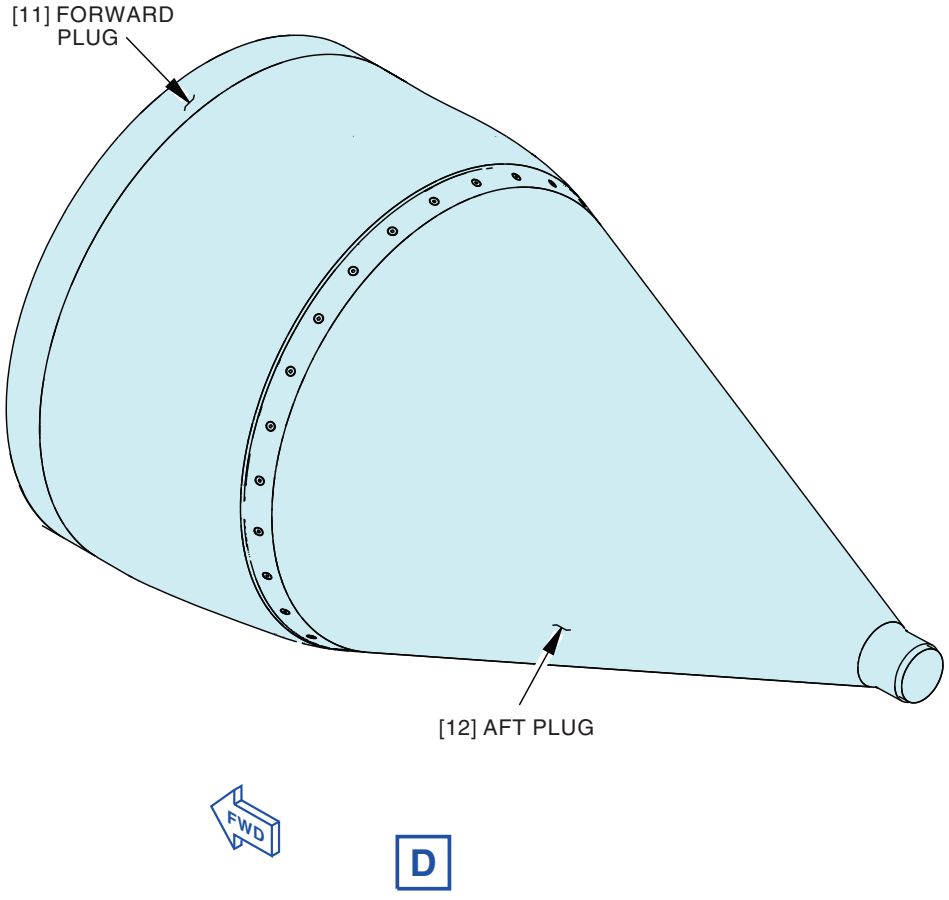
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2465684 S0000574681_V2

Engine Oil System (External Surfaces Walk Around) Inspection
Figure 601/79-00-00-990-809-G00 (Sheet 3 of 6)

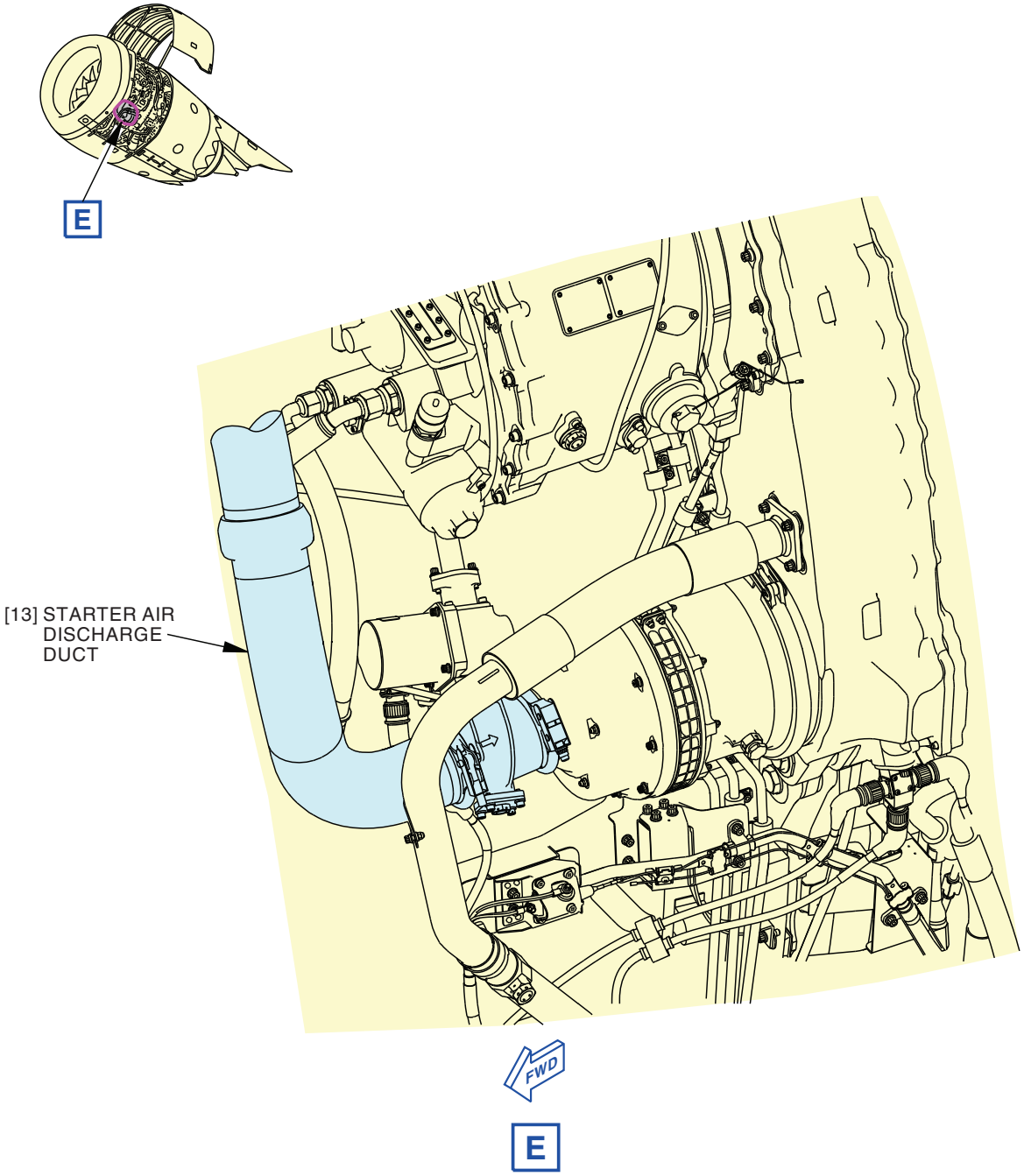
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2465651 S0000574682_V1

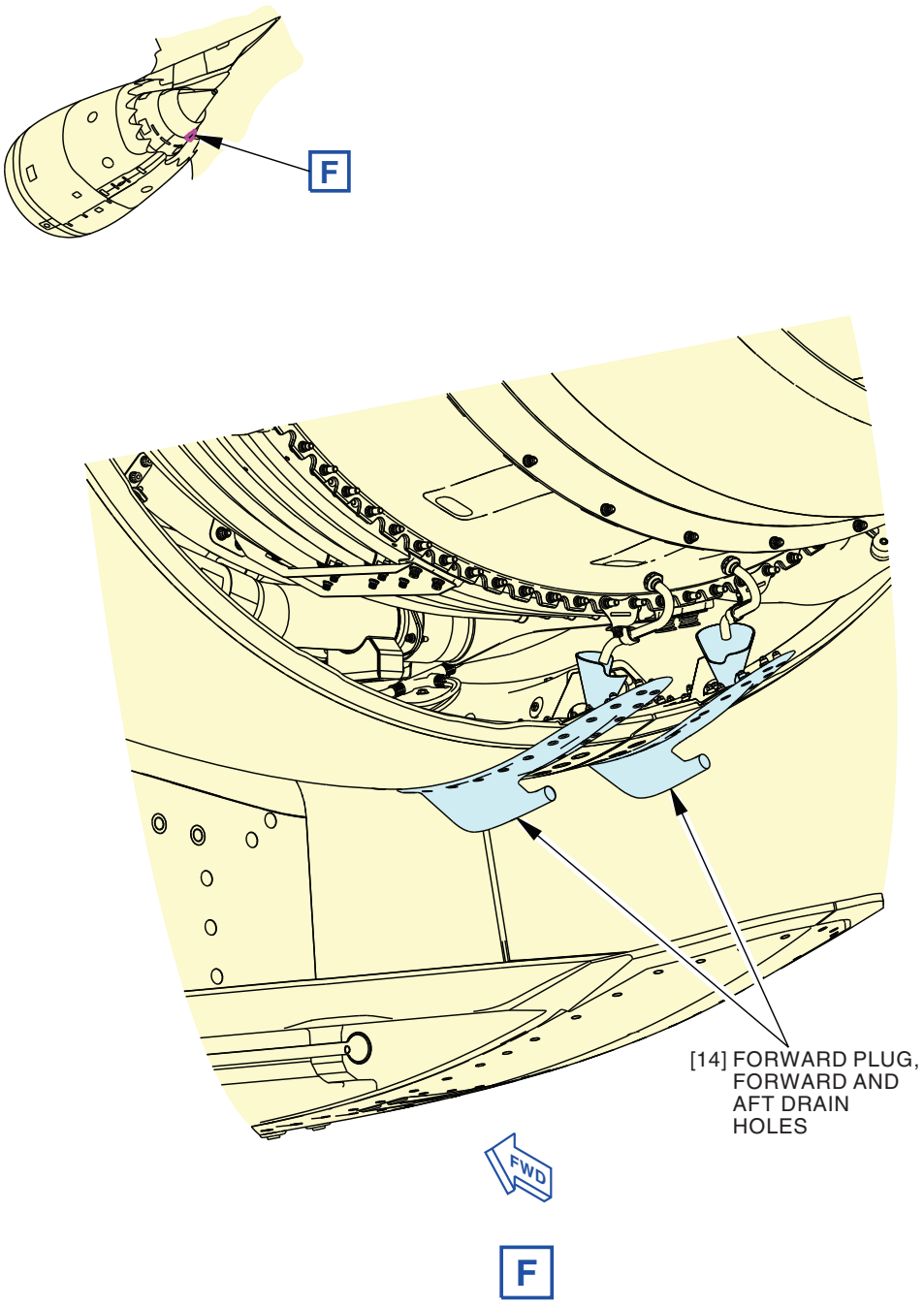
Engine Oil System (External Surfaces Walk Around) Inspection
Figure 601/79-00-00-990-809-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Oil System (External Surfaces Walk Around) Inspection
Figure 601/79-00-00-990-809-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2523943 S0000595137_V1

Engine Oil System (External Surfaces Walk Around) Inspection
Figure 601/79-00-00-990-809-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-210-802-G00

3. Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection

(Figure 602)

A. General

(1) This task gives the instructions to do a visual inspection of the of the engine for evidence of oil leakage.

B. References

Reference	Title
70-00-05-200-801-G00	Tube and Hose Inspection Criteria and Damage Limits (P/B 201)
71-00-00-910-803-G00	Start the Engine (Normal Start) (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50291 [CP2560]	Solvent - cleaning - BONDERITE C-SO IND 79 AERO (TURCO IND79)	
B50292 [CP2566]	Solvent - Degreasing - MAG-CHEM DIESTONE MTK	
B50293 [CP2584]	Solvent - Cleaning - Degreasol 99R	
B50294 [CP2656]	Solvent - Cleaning - MAGCHEM DIESTONE M-SK	
B50295 [CP2010]	Solvent - cleaning, White Spirit	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 79-00-00-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-00-00-860-004-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

H. Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection

SUBTASK 79-00-00-212-001-G00

(1) Do a visual inspection of these components for oil leakage:

(a) Internal surface of the left fan cowl panel [1]

(b) Internal surface of the right fan cowl panel [2]

(c) Internal surface of the left thrust reverser [3]

(d) Internal surface of the right thrust reverser [4]

(e) Engine surface

(f) Oil tank [5]

(g) Non-return valve [6]

(h) Lubrication unit lube and scavenge pump [7]

(i) Transfer gearbox assembly [8] and Fan frame [9] interfaces.

NOTE: Staining or wetting is permitted (e.g., Lubrication (petrolatum) used during seals, gaskets, and joints assembly).

SUBTASK 79-00-00-211-001-G00

(2) Do a visual inspection of tubes:

NOTE: Look for oil strips and drops to do this check.

(a) Worn and cracked hoses [10] are not permitted.

(b) Cracked tubes and tube coupling nuts [11] are not permitted.

(c) Do this task: Tube and Hose Inspection Criteria and Damage Limits, TASK 70-00-05-200-801-G00.

1) Repair if necessary.

(d) Leakage is not permitted.

NOTE: Staining or wetting is permitted (e.g., Lubrication (engine oil) used during tubes assembly to ensure proper seating and to minimize the risk of damage).

SUBTASK 79-00-00-212-002-G00

(3) If leakage is suspected, do these steps:

(a) Clean the suspected area with the solvent, B00682 [CP2011] or the acetone solvent, B01058 [CP1039] or the Ardrox 5503, B50290 [CP2559] or the BONDERITE C-SO IND 79 AERO solvent, B50291 [CP2560] or the MAG-CHEM DIESTONE MTK solvent, B50292 [CP2566] or the Degreasol 99R solvent, B50293 [CP2584] or the MAGCHEM DIESTONE M-SK solvent, B50294 [CP2656] or the White Spirit, B50295 [CP2010].

(b) Operate the engine at 70% of N1 or above to pressurize the oil system and to get accurate reading during leak test (TASK 71-00-00-910-803-G00).

(c) Do a visual inspection of the suspected area.

(d) If leakage is identified, tighten the nuts to the appropriate torque values and/or replace the defective components.

(e) Repeat the engine operation and visual inspection.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-00-00-212-004-G00

(4) Do a visual inspection of the attachment parts of the oil tubes.

(a) Replace the broken or cracked brackets [12].

(b) Replace the worn clamps [13].

(c) Install the missing nuts and bolts [14].

(d) Tighten the loose nuts and bolts [14] to the appropriate torque values.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-860-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-00-00-860-006-G00

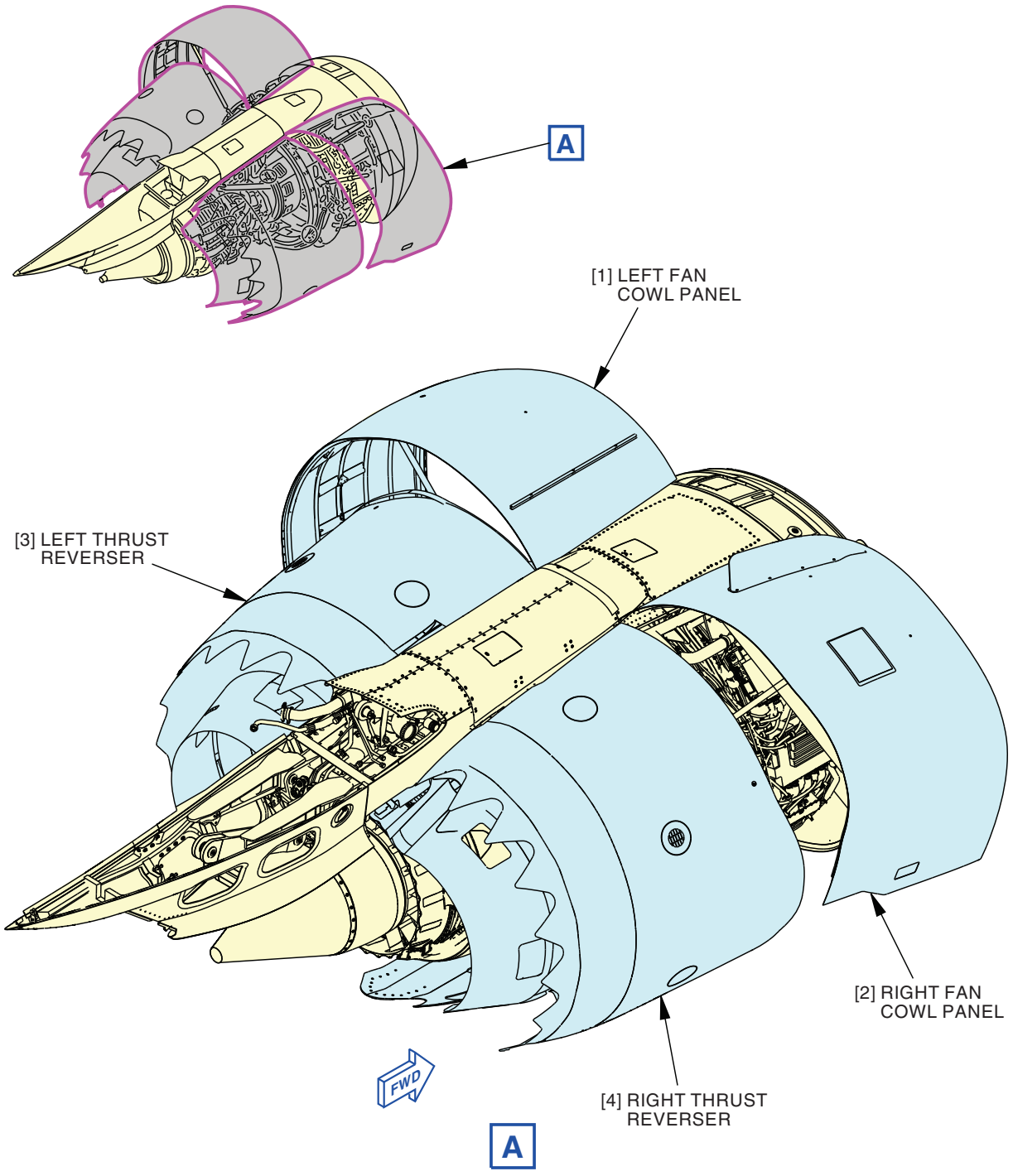
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

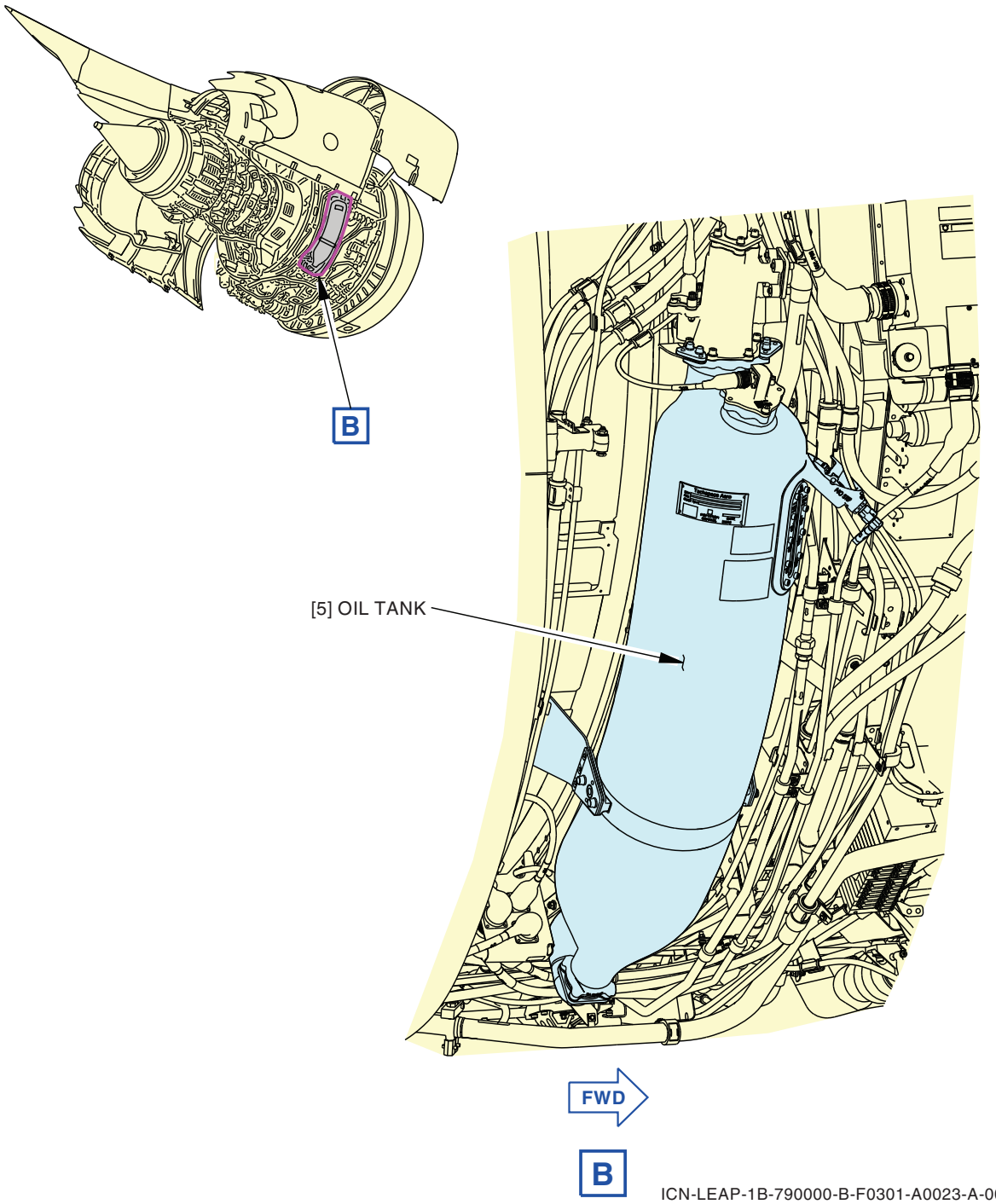
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



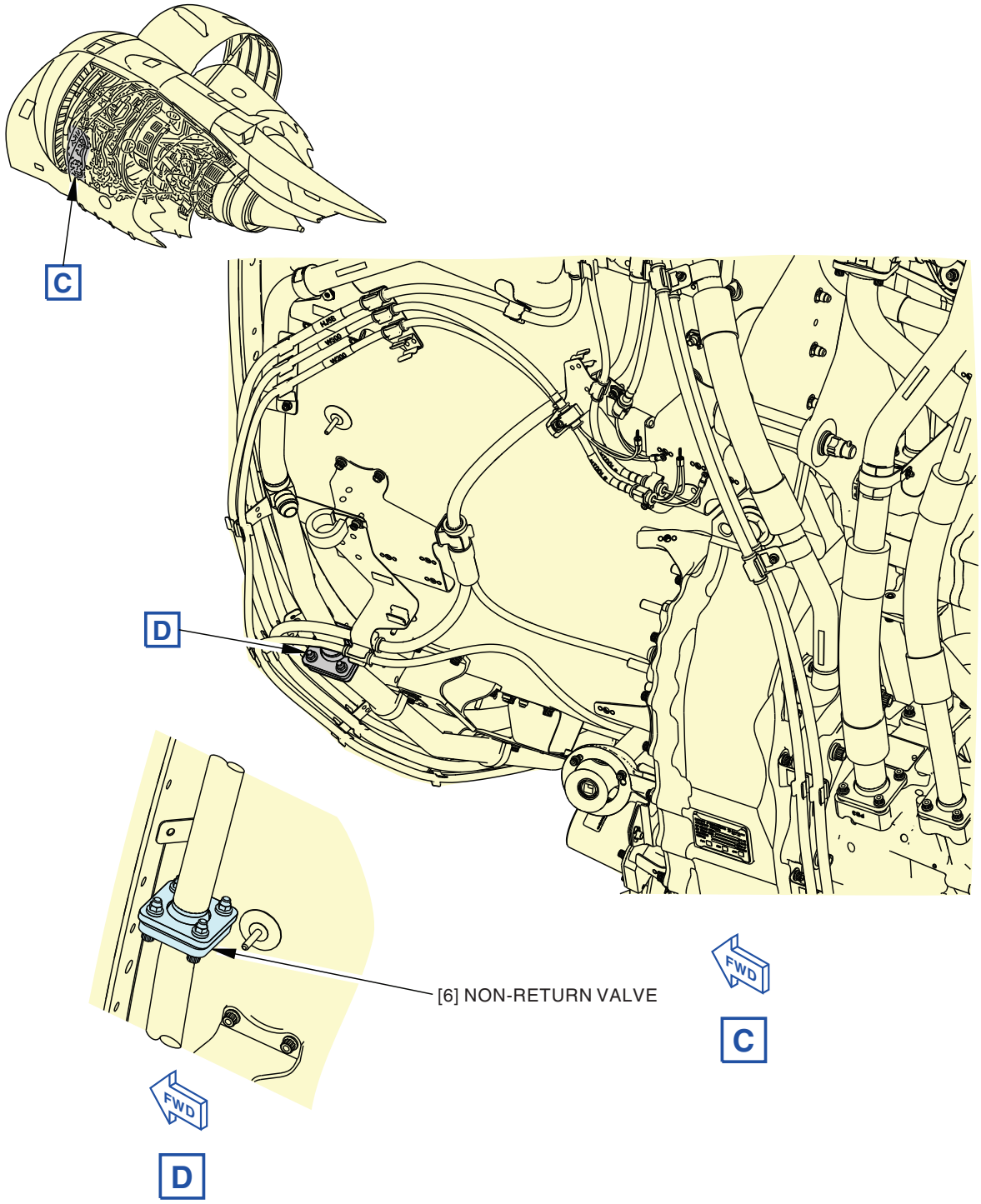
Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



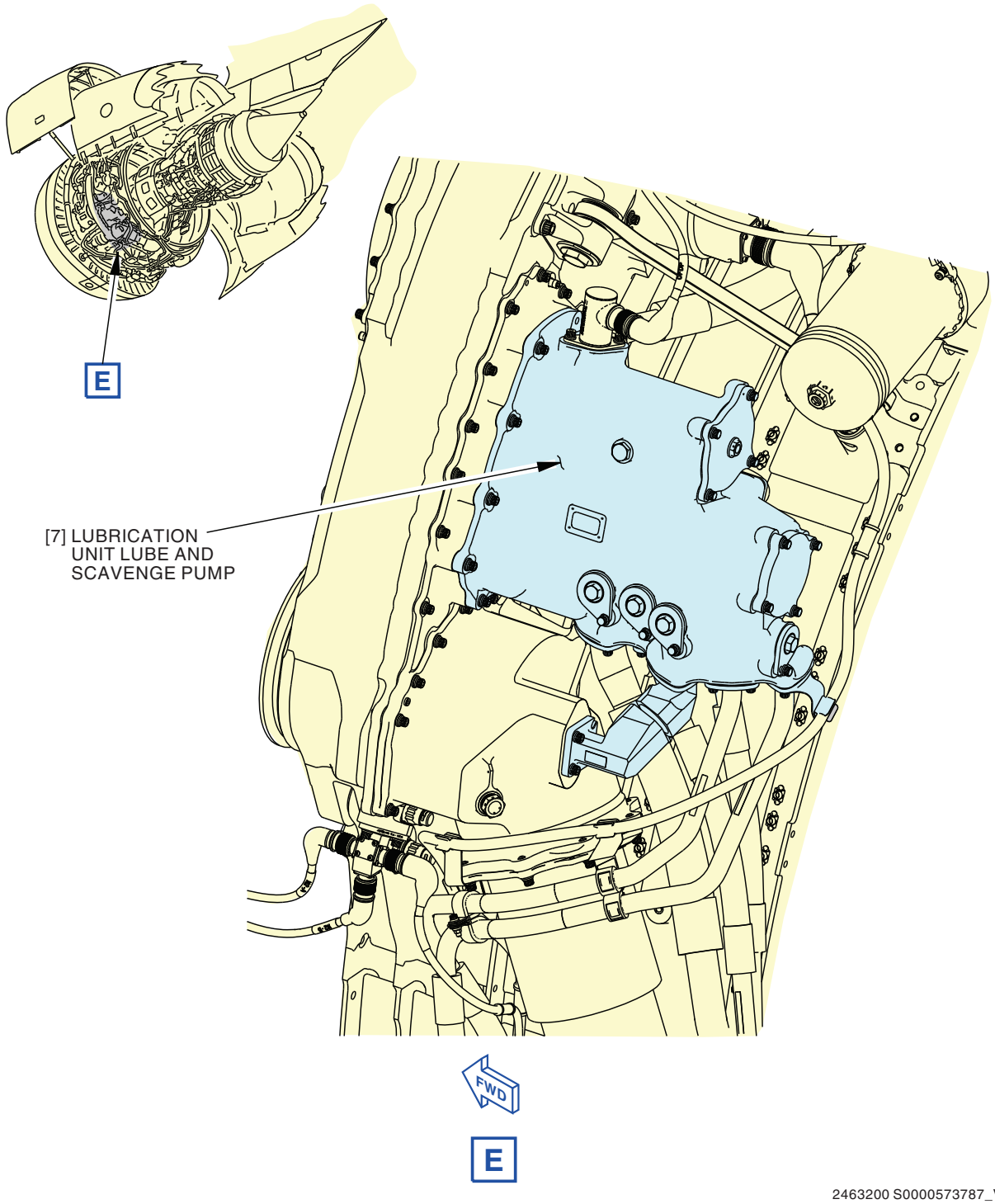
Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



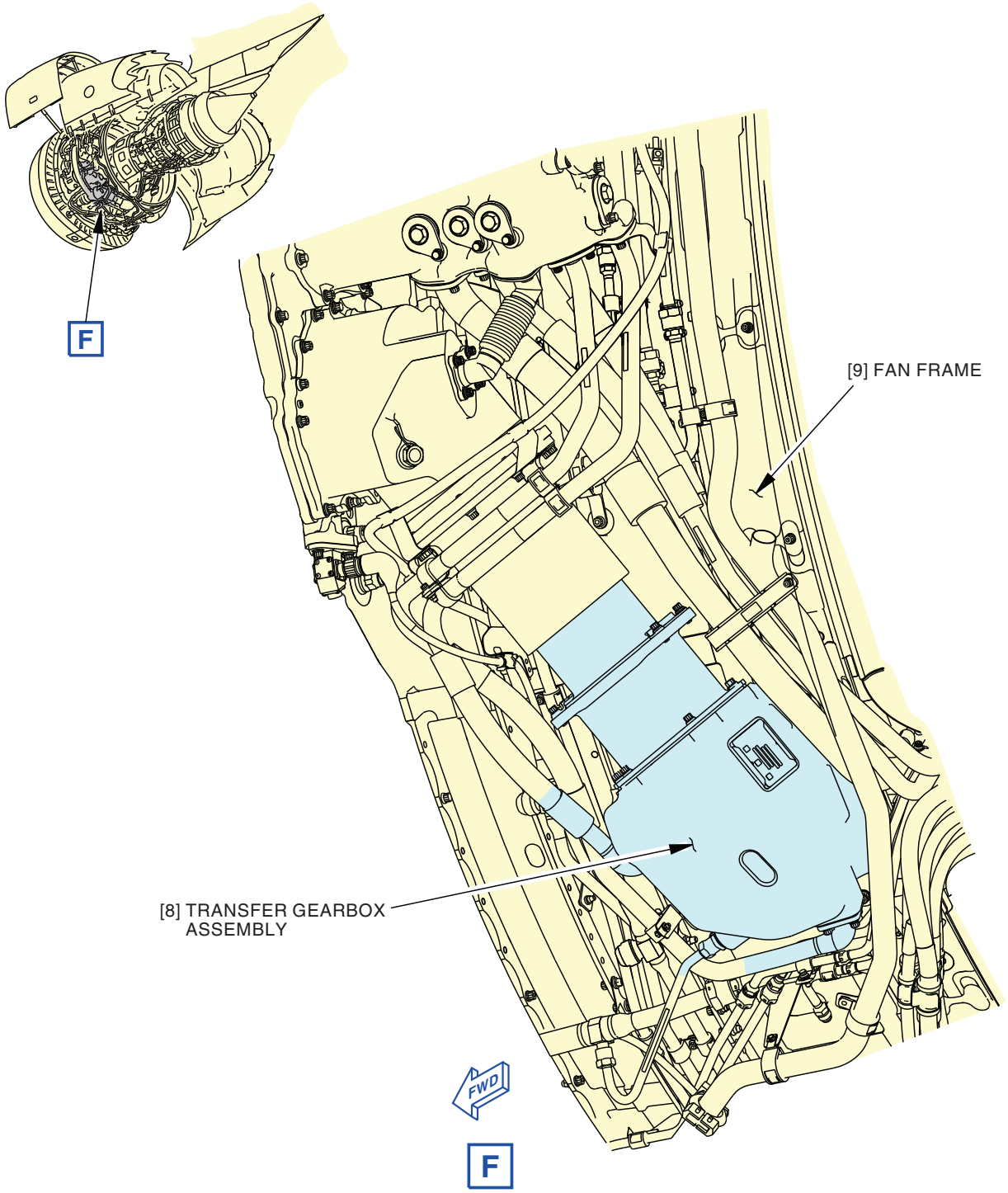
Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



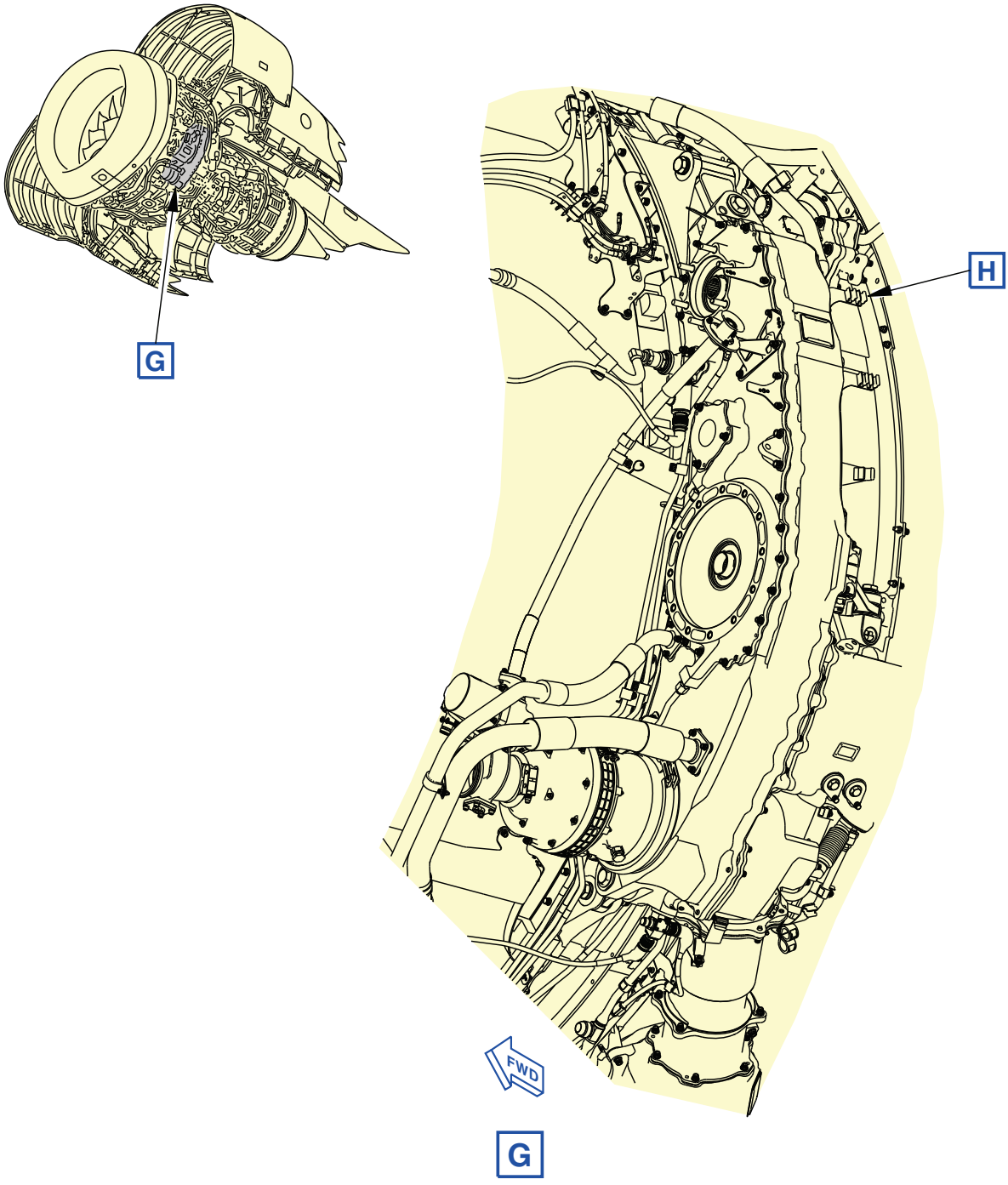
Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 5 of 7)

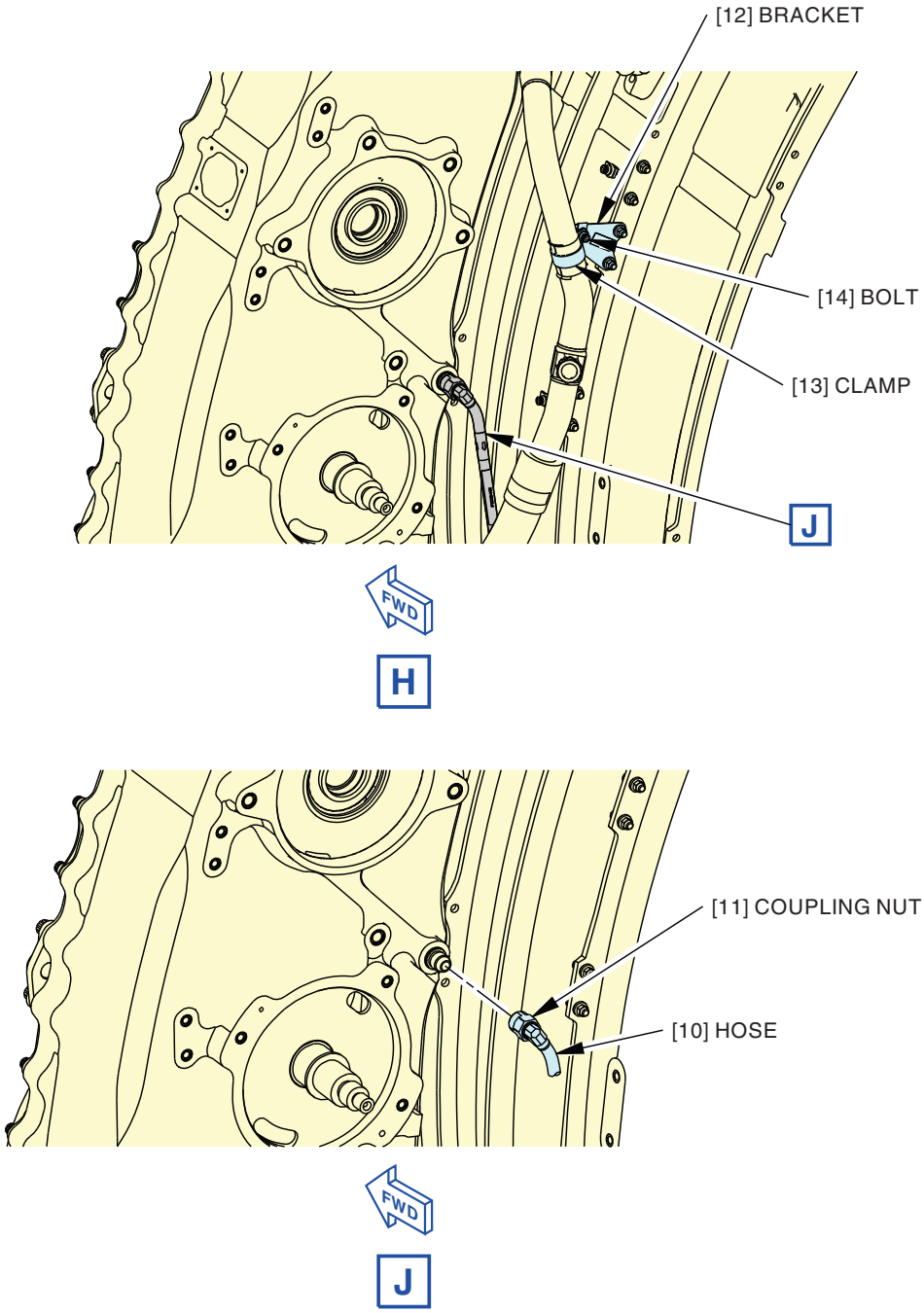
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2463211 S0000573789_V1

Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-790000-B-F0301-A0028-A-001-01
2578075 S0000613642_V1

Engine Oil System (Fan Cowl Panels and Thrust Reversers Open) Inspection
Figure 602/79-00-00-990-808-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-210-803-G01

4. Engine Oil System (Internal) Inspection

(Figure 603)

A. General

(1) This task provides the instructions on how to examine the internal areas of the engine for oil system leakage.

(2) This task also provides the steps to do if you find wet oil in these areas:

(a) Inlet guide vanes and the first stage blades of the fan booster

(b) Exhaust area.

B. References

Reference	Title
71-00-00-790-803-G00	Test No. 3 - Wet Motor Leak Test (P/B 501)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-803-G00	Booster Blades and Vanes Borescope Inspection (P/B 601)
72-00-00-290-804-G00	High Pressure Compressor Section Borescope Detailed Inspection (P/B 601)
72-00-00-290-811-G00	Low Pressure Turbine (LPT) Section Borescope Inspection (P/B 601)
72-50-03-200-801-G00	Turbine Rear Vane Module Inspection (P/B 601)
78-11-03-210-801-G00	Center Vent Tube Extension (CVTE) Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 79-00-00-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-00-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Engine Oil System (Internal) Inspection

SUBTASK 79-00-00-210-006-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET OIL STAY ON YOUR SKIN. THE SYNTHETIC OIL CONTAINS ADDITIVES THAT CAN BE POISONOUS IF THEY ARE ABSORBED THROUGH THE SKIN. CLEAN AWAY ALL OIL THAT GETS ON THE SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Do the visual examination of the engine oil system (Internal) as follows:

(a) Examine the fan blades [1] or inner side of the fan stator case for oil leakage.

(b) If the fan blades [1] or inner side of the fan stator case are wet with oil, do these steps:

1) Examine the inlet guide vanes [2] and first stage blades of the fan booster for oil leakage.

2) If the inlet guide vanes [2] and first-stage blades of the fan booster are wet with oil, do this task: Test No. 3 - Wet Motor Leak Test, TASK 71-00-00-790-803-G00.

3) If the leakage is not repairable, replace the engine (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

(c) Examine the exhaust area for oil leakage.

(d) If the exhaust area is wet with oil, do these steps:

1) Examine the primary nozzle [3] and exhaust plug [4] interfaces.

2) If there is oil leakage, examine the internal area of the exhaust plug [4] for leakage at the oil tubes (TASK 72-00-00-290-811-G00).

3) Examine the turbine rear frame struts [6] surfaces in the gaspath and the tubing interfaces on the outer side of the turbine rear frame [5].

4) If the oil leakage is found, examine the internal area of the exhaust plug [4] for leakage at the oil tubes (TASK 72-00-00-290-811-G00).

5) If the leakage is not repairable, replace the engine (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

(e) Examine the booster rotor blades stage 4 and high pressure compressor blades and vanes stages 1 and 2 for oil traces (TASK 72-00-00-290-803-G00 and TASK 72-00-00-290-804-G00).

1) If any oil traces are found, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Continue in service unless the smell in cabin is experienced or oil consumption is above the serviceable limit (TASK 71-00-00-800-802-G00).

<1> If the smell in the cabin is experienced, refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

b) Examine the Center Vent Tube Extension (CVTE) and mixer for coking (TASK 78-11-03-210-801-G00).

c) Examine the exhaust tube support for coking (TASK 72-50-03-200-801-G00).

d) Examine the injection nozzle for coking (TASK 72-50-03-200-801-G00).

e) Examine the air exhaust tubes for coking (TASK 72-50-03-200-801-G00).

f) Examine the rear vent tube for coking (TASK 72-50-03-200-801-G00).

(f) Examine the B-sump, C-sump, and Auxiliary drains for coking (TASK 79-00-00-200-804-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-080-001-G00

(1) Remove all tools used to do the inspection and clean the work area.

SUBTASK 79-00-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-00-00-860-008-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

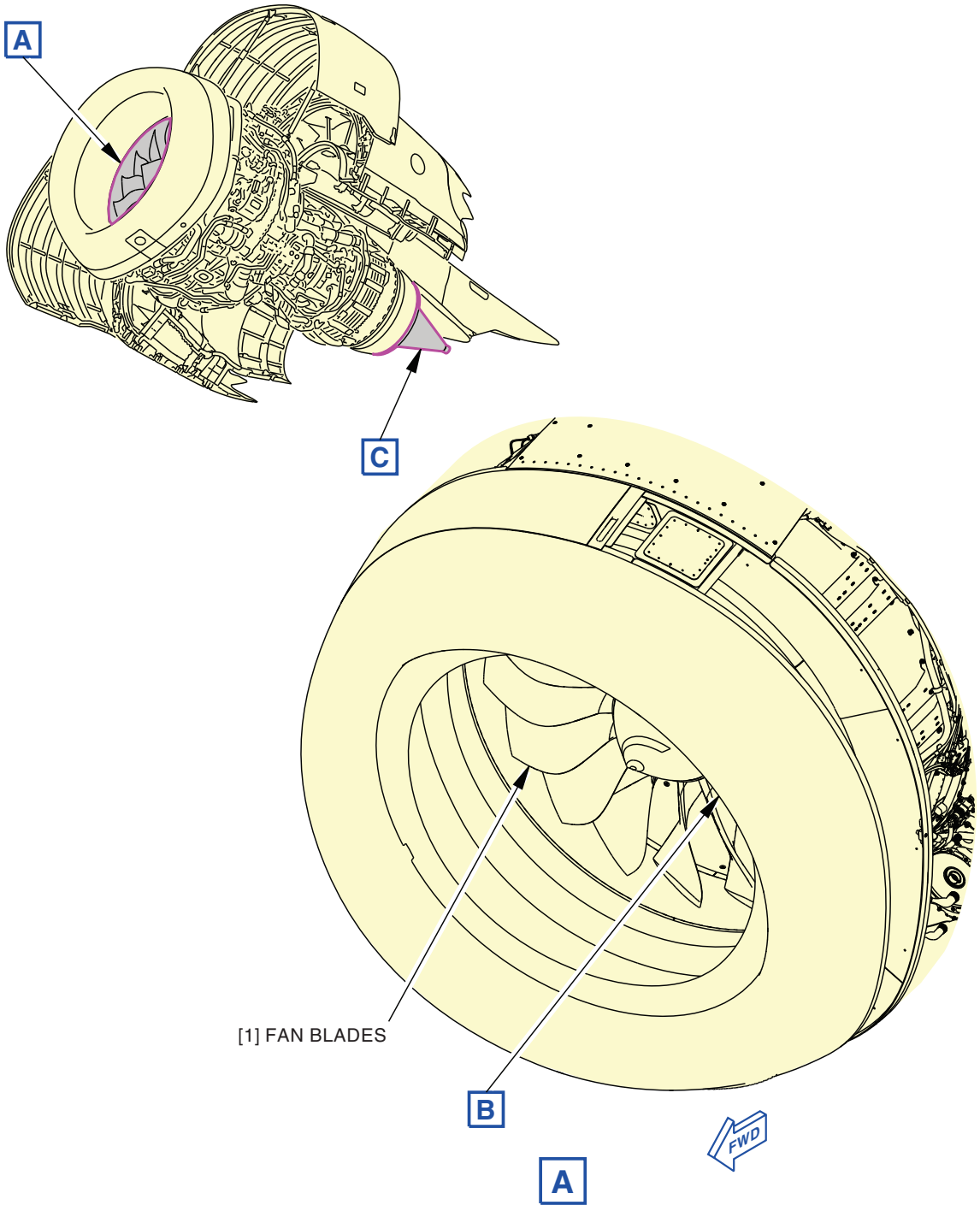
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

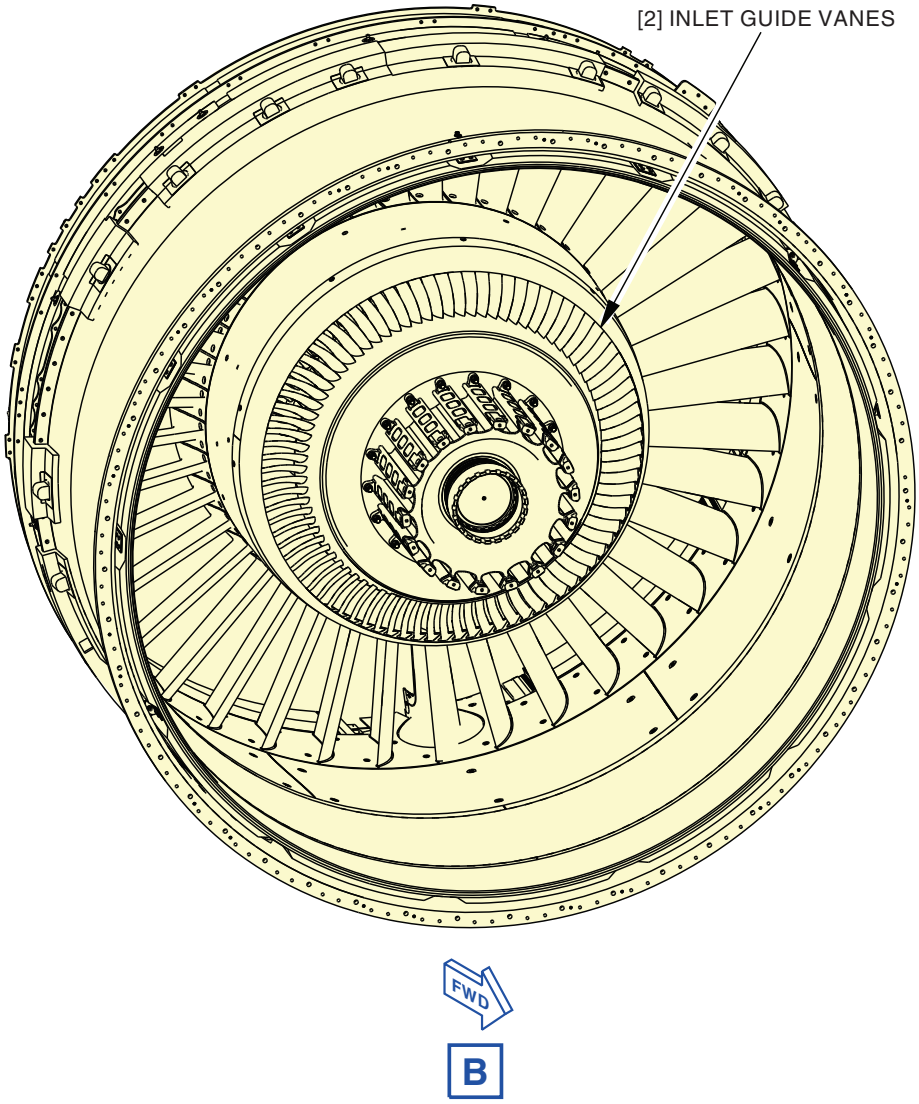
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Oil System (Internal) Inspection
Figure 603/79-00-00-990-806-G00 (Sheet 1 of 3)

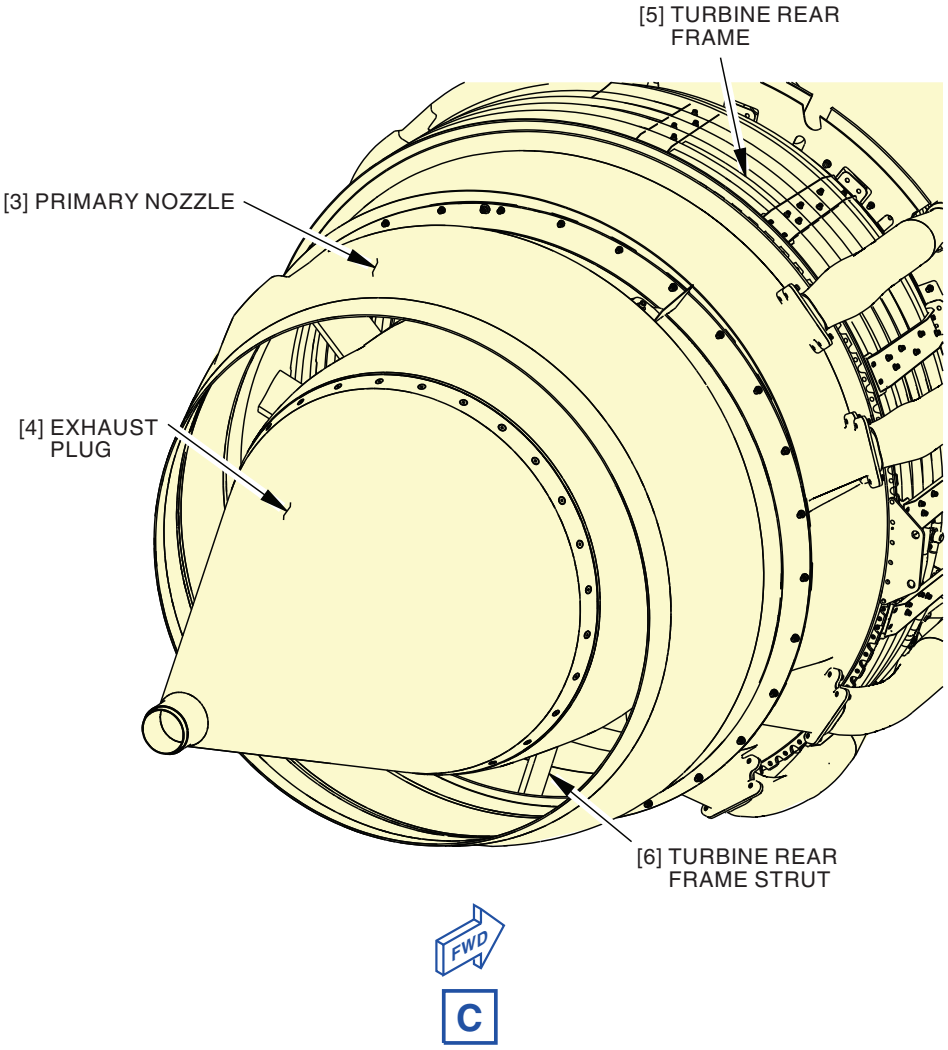
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2456152 S0000571370_V1

Engine Oil System (Internal) Inspection
Figure 603/79-00-00-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2456155 S0000571371_V1

Engine Oil System (Internal) Inspection
Figure 603/79-00-00-990-806-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-210-804-G00

5. Oil Debris Monitoring System Sensor Inspection

(Figure 604)

A. General

(1) This task gives the instructions to examine the Oil Debris Monitoring System (ODMS) sensor.

B. References

Reference	Title
71-00-00-700-802-G00	Test No. 26 - Debris Monitoring System Channel B Special Functions Test (P/B 501)
79-21-04-211-801-G00	Scavenge Screen Plug Inspection (P/B 201)
79-21-12-000-801-G00	Oil Debris Monitoring System Sensor Removal (P/B 401)
79-21-12-400-801-G00	Oil Debris Monitoring System Sensor Installation (P/B 401)
80-11-09-200-801-G00	Engine Starter Magnetic Plug Inspection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1070	Lens - Magnifying, 10X, Hand Held
STD-1280	Source - Air, Regulated, Dry Filtered, 0-30 PSIG

D. Consumable Materials

Reference	Description	Specification
B00682 [CP2011]	Solvent - Stoddard	MIL-PRF-680 Type I, II or III
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Inspection

SUBTASK 79-00-00-020-001-G00

(1) Do this task: Oil Debris Monitoring System Sensor Removal, TASK 79-21-12-000-801-G00.

NOTE: It is recommended to perform this task only when requested by another Aircraft Maintenance Manual (AMM)/ Interactive Fault Isolation Manual (IFIM)/scheduled maintenance task or CFM International (CFMI) instructions.

NOTE: If the ODMS sensor is removed for inspection only, it is not necessary to disconnect the ODMS sensor connector from the ODMS Signal Conditioner Unit (SCU).

(a) When a Customer Notification Request (CNR) is received, make sure to perform this task within CNR urgency requirements.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Oil Debris Monitoring System Sensor Inspection

SUBTASK 79-00-00-200-004-G00

CAUTION

TAG THE REMOVED PARTICLES WITH THE ODMS SENSOR NAME TO PREVENT AN ERROR IN IDENTIFICATION OF THE SOURCE OF THE MATERIAL DURING VISUAL CHARACTERIZATION AND LABORATORY ANALYSIS. AN ERROR IN IDENTIFICATION OF THE SOURCE OF THE MATERIAL COULD LEAD TO AN INACCURATE INTERPRETATION OF SERVICEABLE CRITERIA AND DAMAGE CAN OCCUR.

CAUTION

CFM DOES NOT RECOMMEND THE USE OF ADHESIVE TAPE. HOWEVER, IF A PIECE OF ADHESIVE TAPE IS USED TO COLLECT THE PARTICLES, MAKE SURE THE ADHESIVE IS NOT TOO STRONG. IF THE ADHESIVE ON THE TAPE IS TOO STRONG, THE PARTICLES MAY BE DAMAGED WHEN THEY ARE REMOVED AND/OR IT WILL AFFECT THE LABORATORY ANALYSIS OF THE MATERIAL.

(1) Inspect the ODMS sensor [1] for material presence on the magnetic tip [2] with a 10x hand held magnifying lens, STD-1070.

(a) If there are no findings on the magnetic tip [2] of the ODMS sensor [1], do these steps:

1) Check the chip count value on the Debris Monitoring System (DMS) chip counter.

NOTE: Engine Propulsion Control System (EPCS) page 2 shows the engine DMS chip count for both Engine Electronic Controller (EEC) channel A and EEC channel B.

a) If chip count value = 0

<1> No further actions are required.

b) If chip count value > 0

<1> Refer to the IFIM and do the applicable procedure(s) (79-42852 or 79-42854).

<2> Reset chip count value, do this task: Test No. 26 - Debris Monitoring System Channel B Special Functions Test, TASK 71-00-00-700-802-G00.

2) Do this task: Oil Debris Monitoring System Sensor Installation, TASK 79-21-12-400-801-G00.

(b) If you find material on the magnetic tip [2] of the ODMS sensor [1], do these steps:

1) Take pictures of the particles on the magnetic tip [2] of the ODMS sensor [1], before you remove them.

2) Carefully remove and keep the particles from the magnetic tip [2] of the ODMS sensor [1], with cotton wiper, G00034, on a thin sheet of paper.

3) After you remove all the particles, clean the magnetic tip [2] of the ODMS sensor [1] with solvent, B00682 [CP2011].

4) Dry the magnetic tip [2] of the ODMS sensor [1] with a 0-30 psig dry filtered regulated air source, STD-1280.

5) Make sure that the chip detector is free of all particles.

NOTE: If you do not remove all the particles from the ODMS sensor, a difference between the collected particles and chips count value can occur at the next ODMS sensor inspection.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

6) Do this task: Oil Debris Monitoring System Sensor Installation, TASK 79-21-12-400-801-G00.

SUBTASK 79-00-00-200-003-G00

(2) Examine particle size with a 10x hand held magnifying lens, STD-1070, and use scale paper to take pictures (View D and View F, Figure 604).

NOTE: Normal fuzz ONLY (less than 0.004 in. (0.10 mm)) is not expected with ODMS fault raised or chip count shift.

(a) If you find ONLY very fine particles like powder, less than 0.004 in. (0.10 mm) (Normal Fuzz) in any dimension on the magnetic tip [2] of the ODMS sensor [1], do these steps:

1) If chip count value = 0

a) No further actions are required.

2) If chip count value > 0

a) Refer to the IFIM and do the applicable procedure(s) (79-42852 or 79-42854).

b) Reset chip count value, do this task: Test No. 26 - Debris Monitoring System Channel B Special Functions Test, TASK 71-00-00-700-802-G00.

3) Continue the engine in service.

(b) If you find particles that are larger than 0.004 in. (0.10 mm) (Abnormal Fuzz) in any dimension on the magnetic tip [2] of the ODMS sensor [1], do these steps:

CAUTION

DO NOT APPLY TEST NO. 26 - DEBRIS MONITORING SYSTEM RESET CHANNEL B SPECIAL FUNCTIONS TEST, TASK 71-00-00-700-802-G00. IF YOU DO NOT OBEY, DAMAGE CAN OCCUR TO THE ENGINE.

1) Check the A, B, C sumps, Accessory Gear Box (AGB), and Transfer Gearbox (TGB) sumps scavenge screen plugs for further troubleshooting (TASK 79-21-04-211-801-G00).

2) Do this task: Engine Starter Magnetic Plug Inspection, TASK 80-11-09-200-801-G00.

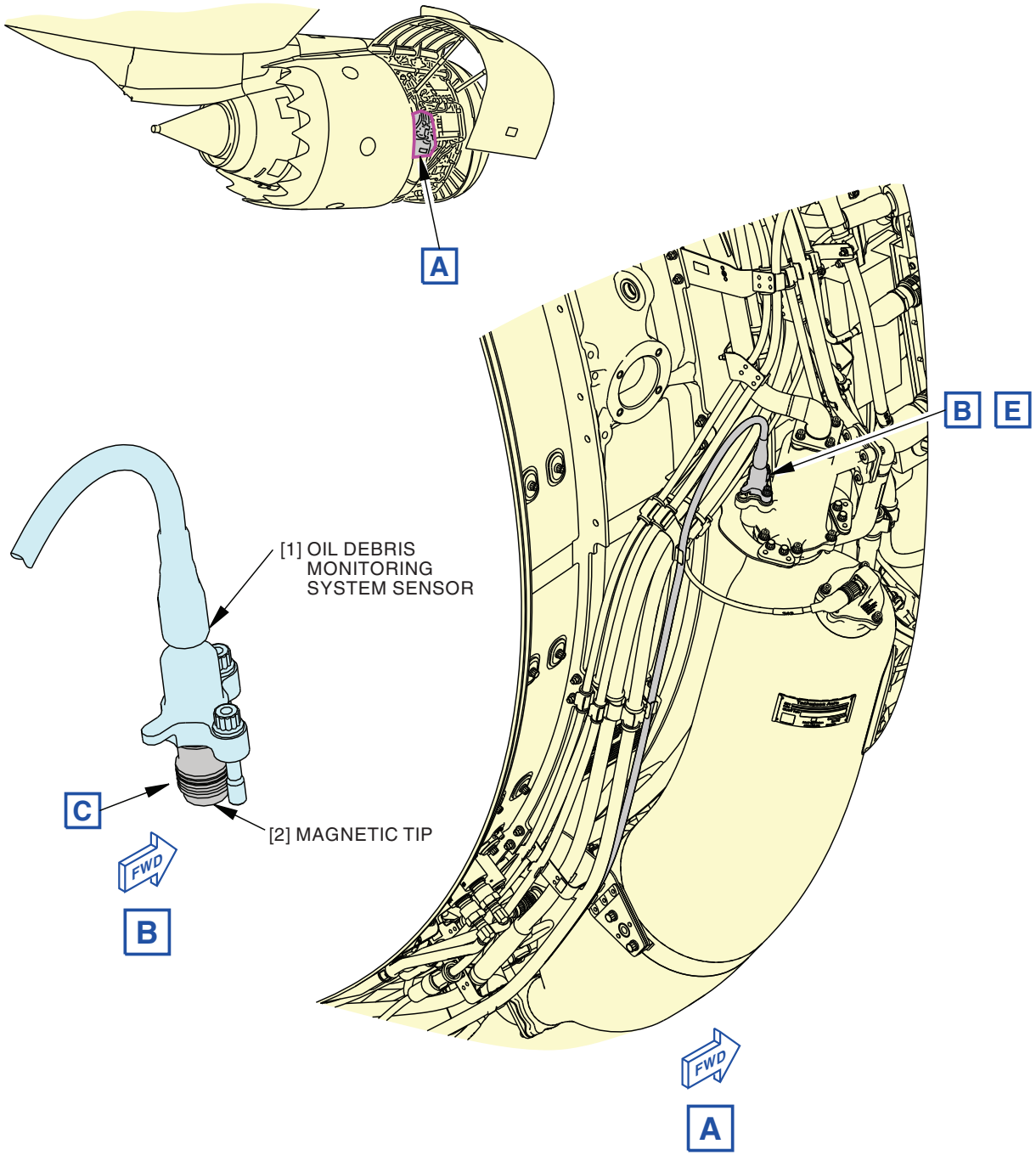
3) Analyze particles, do this task: Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check, TASK 79-00-00-220-801-G00.

4) If chip count value = 0

a) Refer to the IFIM and do the applicable procedure(s) (79-42852 or 79-42854).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

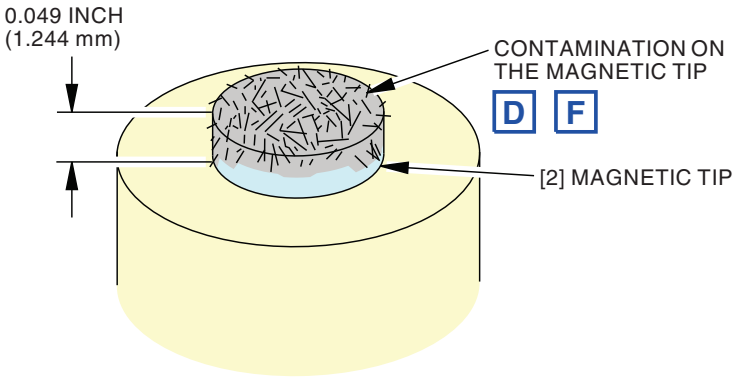


ICNLEAP1B790000D5882800017A00101

2967651 S0000747146_V1

Oil Debris Monitoring System (ODMS) Sensor Inspection
Figure 604/79-00-00-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAGNETIC TIP

C



CONTAMINATION ON THE MAGNETIC TIP

D

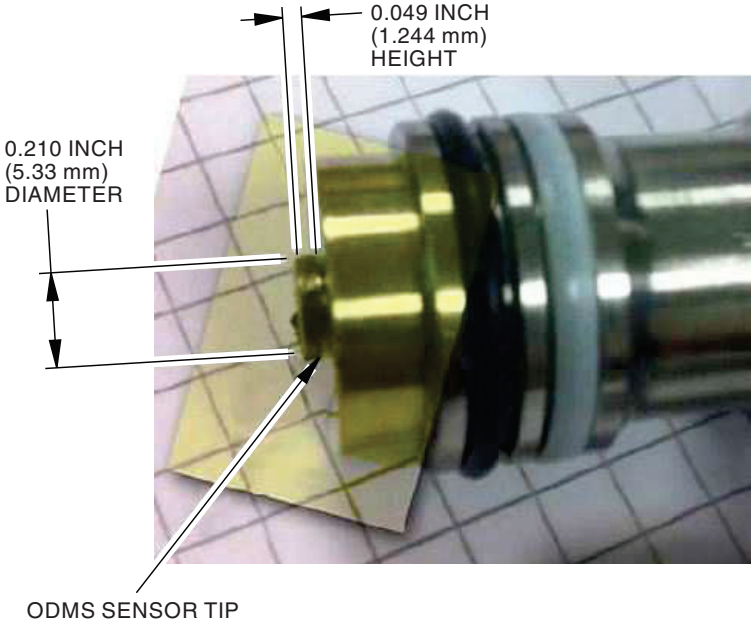
ICNLEAP1B790000D5882800018A00101
2355316 S00061516985_V5

Oil Debris Monitoring System (ODMS) Sensor Inspection
Figure 604/79-00-00-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



E



(ODMS SENSOR TIP DIMENSION)

E

ICNLEAP1B790000D5882800019A00101

2967680 S0000747143_V1

Oil Debris Monitoring System (ODMS) Sensor Inspection
Figure 604/79-00-00-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



0.0008 TO 0.004 INCH (0.02 TO 0.10 mm)
(NORMAL FUZZ SATISFACTORY CONDITION)

F



0.016 TO 0.035 INCH (0.4 TO 0.9 mm)
(ABNORMAL FUZZ UNSATISFACTORY CONDITION)

F



0.008 TO 0.003 INCH (0.2 TO 0.8 mm)
(ABNORMAL FUZZ UNSATISFACTORY CONDITION)

F

ICNLEAP1B790000D5882800020A00101

2967682 S0000747144_V1

Oil Debris Monitoring System (ODMS) Sensor Inspection
Figure 604/79-00-00-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-220-801-G00

6. Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check

(Figure 605, Figure 606, Figure 607, Figure 608, Figure 609, Figure 610, Figure 611)

A. General

(1) This task gives the instructions to do an analysis of the unwanted material found on the Oil Debris Monitoring System (ODMS) sensor tip, scavenge screens plugs, starter magnetic plug and oil filter cartridge.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
79-21-04-211-801-G00	Scavenge Screen Plug Inspection (P/B 201)
79-21-04-400-801-G00	Scavenge Screen Plugs Installation (P/B 201)
79-21-07-400-801-G00	Oil Filter Cartridge Element Installation (P/B 401)
79-21-12-400-801-G00	Oil Debris Monitoring System Sensor Installation (P/B 401)
80-11-09-200-801-G00	Engine Starter Magnetic Plug Inspection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-562	Lens - Magnifying, 10x
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Check

SUBTASK 79-00-00-860-011-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-00-00-860-012-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 79-00-00-211-014-G00

(3) Do a visual inspection of the ODMS sensor, scavenge screens and starter magnetic plug:

(a) Do a detailed inspection of the ODMS sensor for debris (TASK 79-00-00-210-804-G00).

(b) Do a visual inspection of the scavenge screen plugs for debris (TASK 79-21-04-211-801-G00).

(c) Do a visual inspection of the starter magnetic plug for debris (TASK 80-11-09-200-801-G00).

G. Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug, and Oil Filter) Detailed Dimensional Check

SUBTASK 79-00-00-211-008-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. USE THE OIL IN AN AREA WITH GOOD AIRFLOW. THE OIL IS POISONOUS AND YOUR SKIN CAN ABSORB THE OIL. THE OIL FUMES CAN CAUSE INJURY.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION YOU MUST PUT A MARK OR LABEL ON THE PARTICLES WITH THE SOURCE OF THE MATERIAL YOU REMOVE. THE MARK PREVENTS AN INCORRECT VISUAL IDENTIFICATION AND LABORATORY ANALYSIS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do the visual inspection of the particles collected from the ODMS sensor, A, B, C, AGB, TGB1, TGB2, scavenge screen plugs and starter magnetic plug to identify the physical properties of the material (shape, size, quantity, magnetic properties, color), do the following:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Examine the particles with 10x magnifying lens, STD-562.

(b) Any quantity of material that is less than or equal to 0.004 in. (0.10 mm) in all dimension (normal fuzz) is satisfactory with no need to send particles for analysis.

NOTE: Normal fuzz can be noticed on the new engines or engines after overhaul.

NOTE: Material that is larger than 0.004 in. (0.10 mm) in any dimension is not normal fuzz.

(c) Refer to NDT manual Part 9 "LEAP-00-70-30-05-02A-371B-D" and Part 10 "LEAP-00-70-30-04-02A-370B-D" to identify possible source or refer to the Field Guide for Lubrication System Debris Tool Number GEK 131722 for examples of shapes that may be seen on the ODMS sensor, scavenge screen plugs/magnetic plug and oil filter cartridge.

SUBTASK 79-00-00-211-009-G00

(2) Examine the particles to identify their potential source from within the engine and the recommendations (Figure 606).

CAUTION

MAKE SURE THAT THERE ARE NO MAGNETIC PARTICLES ATTACHED TO THE IDENTIFIED PARTICLES, IF THE IDENTIFIED PARTICLES ARE NON-MAGNETIC. INCORRECT PARTICLE IDENTIFICATION WILL CAUSE THE ERROR IN THE SERVICEABLE LIMIT ANALYSIS. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE CAN OCCUR.

(a) To identify the particles, take into consideration the following (Figure 608 and Figure 609):

- size
- shape
- quantity
- color
- nature (magnetic, non magnetic, non metallic)
- origin (sump, filter, ODMS sensor).

(b) If you found some magnetic material on the Accessory Gear Box (AGB) scavenge screen plug or on the ODMS sensor, make sure that you examine the starter magnetic plug for contamination (TASK 80-11-09-200-801-G00).

SUBTASK 79-00-00-211-010-G00

CAUTION

IF THERE IS AN INDICATION OF THE BEARING DAMAGE, DO NOT OPERATE THE ENGINE UNTIL YOU IDENTIFY THE MAGNETIC PARTICLES. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND THE ENGINE CAN OCCUR.

(3) Identify the nature of particles greater than normal fuzz from all sources (ODMS and/or scavenge screen, starter magnetic plug, oil filter).

(a) If material cannot be visually identified, send the particles to a laboratory for chemical analysis according to NDTM Part 10.

(b) Once you have identified the debris, examine the particles (Figure 606).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-00-00-211-015-G00

CAUTION DO NOT OPERATE THE ENGINE UNTIL OIL FILTER ANALYSIS IS COMPLETED. IF THE OIL FILTER IS BYPASSED BY METAL PARTICLES, DAMAGE TO THE ENGINE CAN OCCUR.

(4) In case of oil filter actual bypass, send the oil filter and the retrieved particles from the cavity of the lubrication unit for analysis.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-860-028-G00

(1) If not already done, install the oil filter cartridge element (TASK 79-21-07-400-801-G00).

SUBTASK 79-00-00-860-030-G00

(2) If not already done, install the single stage scavenge screen plugs and double stage scavenge screen plugs (TASK 79-21-04-400-801-G00).

SUBTASK 79-00-00-860-031-G00

(3) If not already done, install the ODMS sensor (TASK 79-21-12-400-801-G00).

SUBTASK 79-00-00-860-013-G00

(4) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 79-00-00-860-014-G00

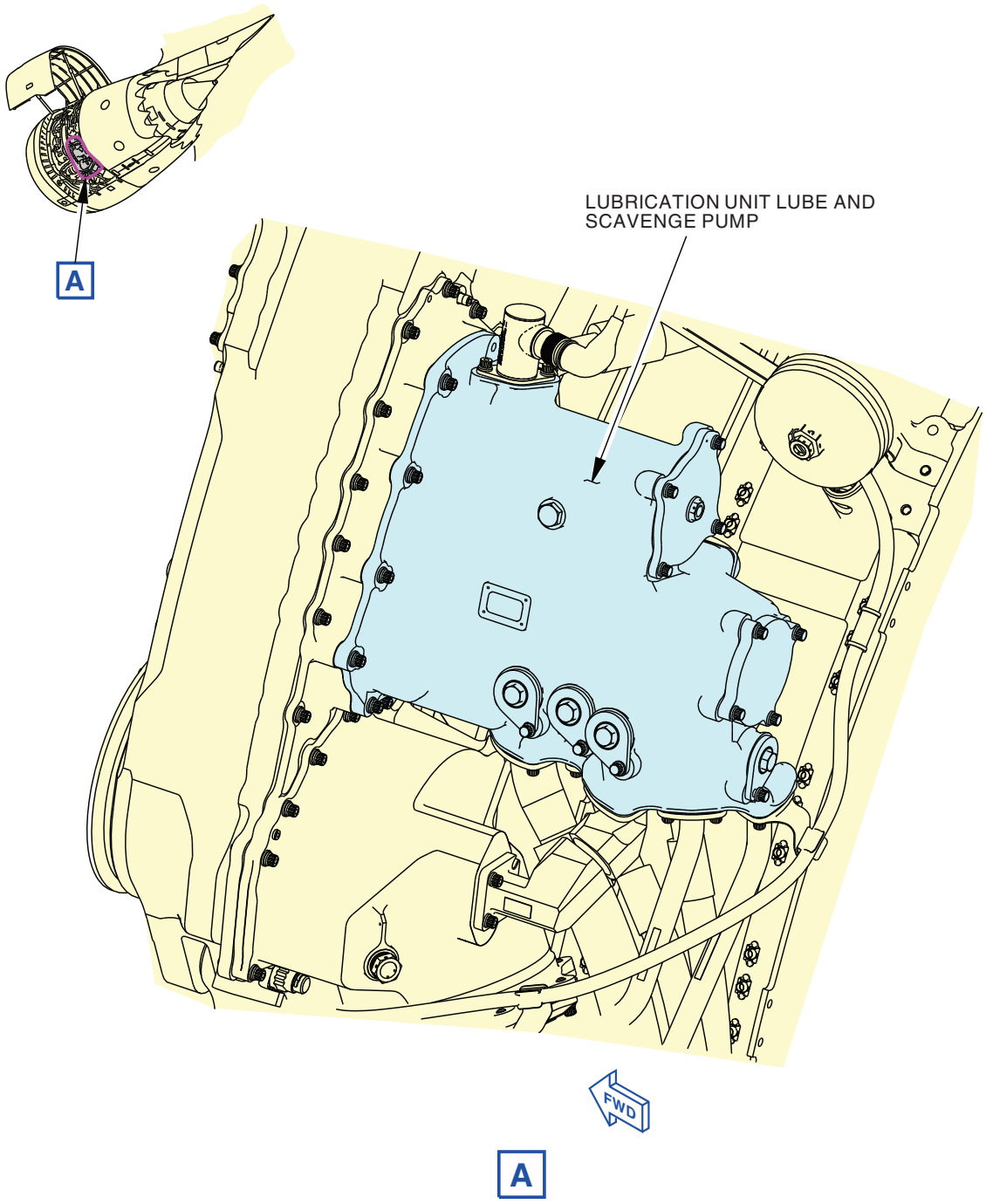
(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

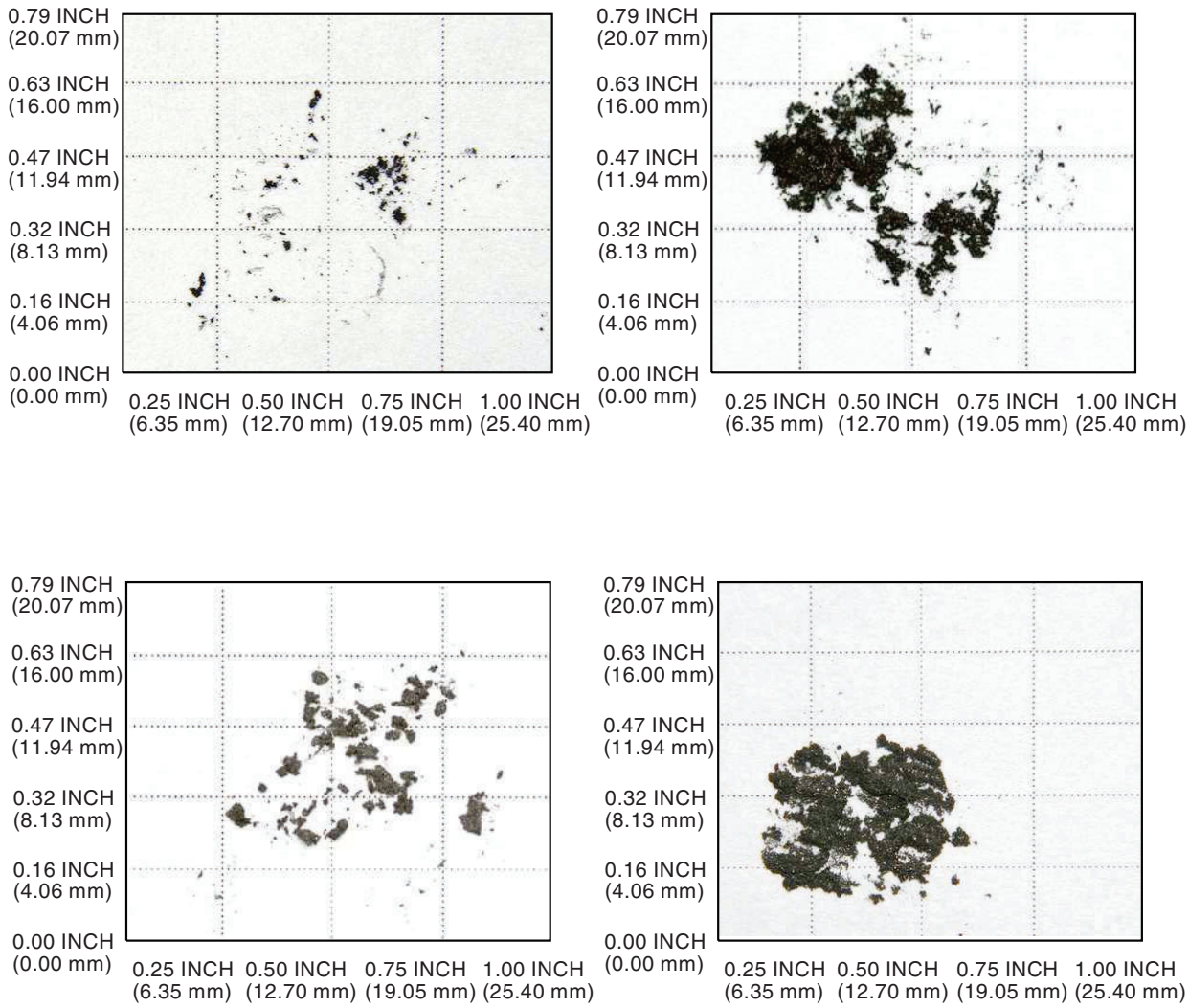
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2457757 S0000571934_V3

Engine Oil System (Oil Debris Monitoring System Sensor (ODMS), Scavenge Screen Plugs, Starter Magnetic Plug and Oil Filter) Detailed Dimensional Check
Figure 605/79-00-00-990-807-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

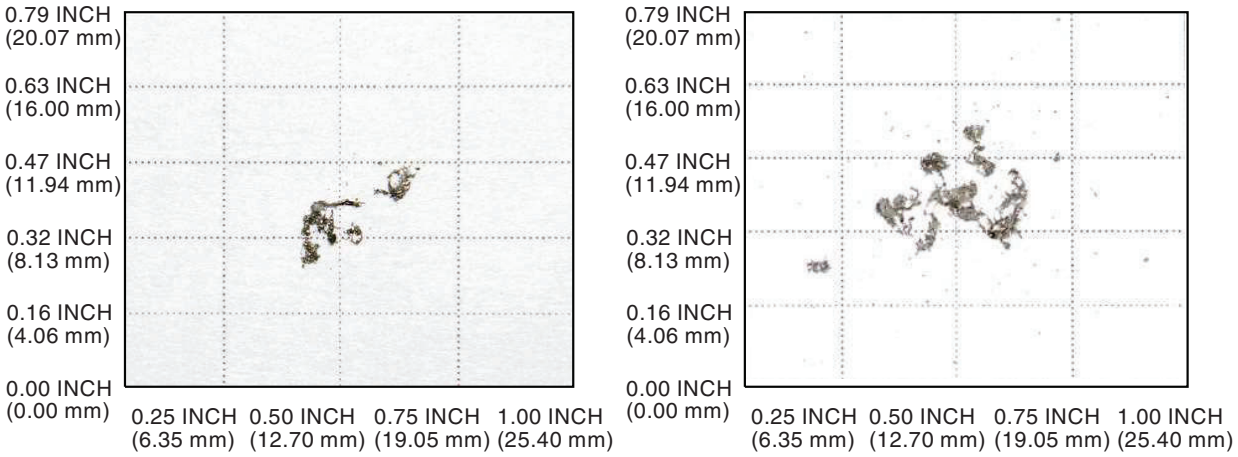


BEARING SHOULDER WEAR DISTRESS (EXAMPLE)

2900623 S0000692893_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 1 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

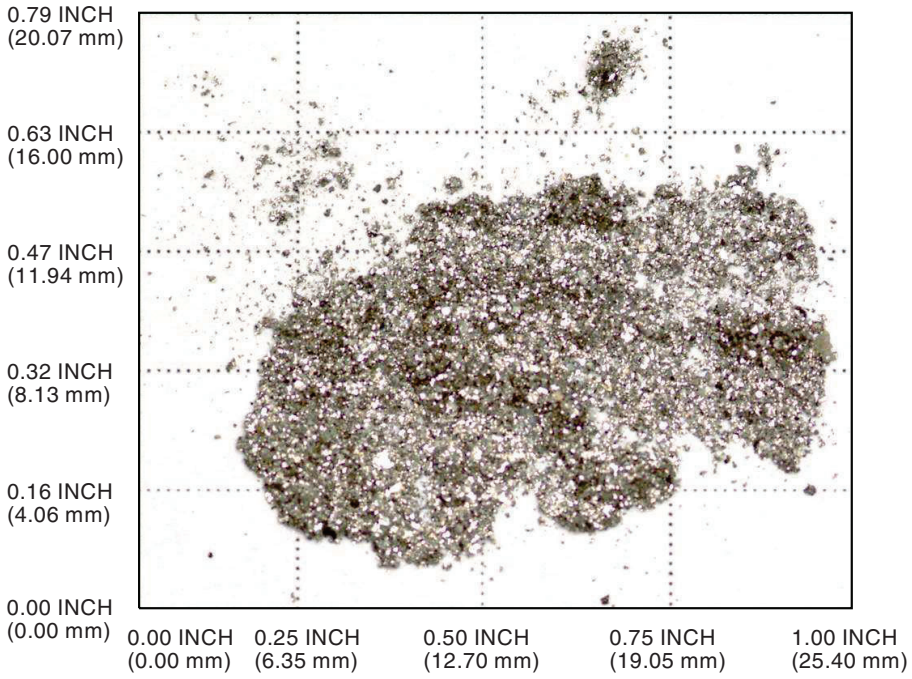


BEARING SPALL DISTRESS
(EXAMPLE)

2900647 S0000692894_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 2 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



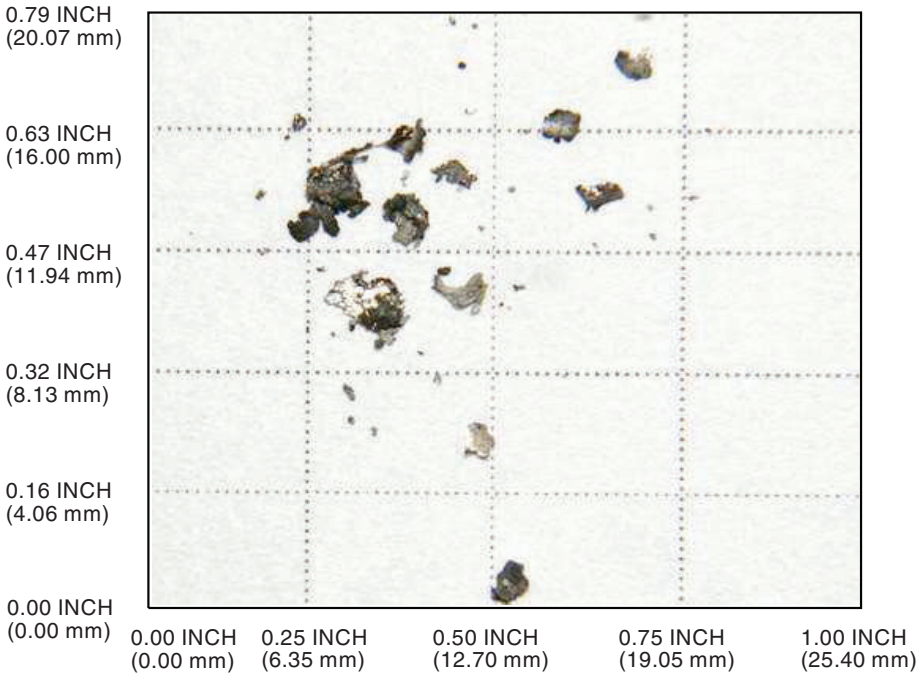
1 GENERAL CHARACTERISTICS

- MAGNETIC
- APPEARANCE
  - METALLIC
  - FLAKES
  - REFLECTIVE
  - POWDER

2900648 S0000692896_V2

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 3 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



M50 2

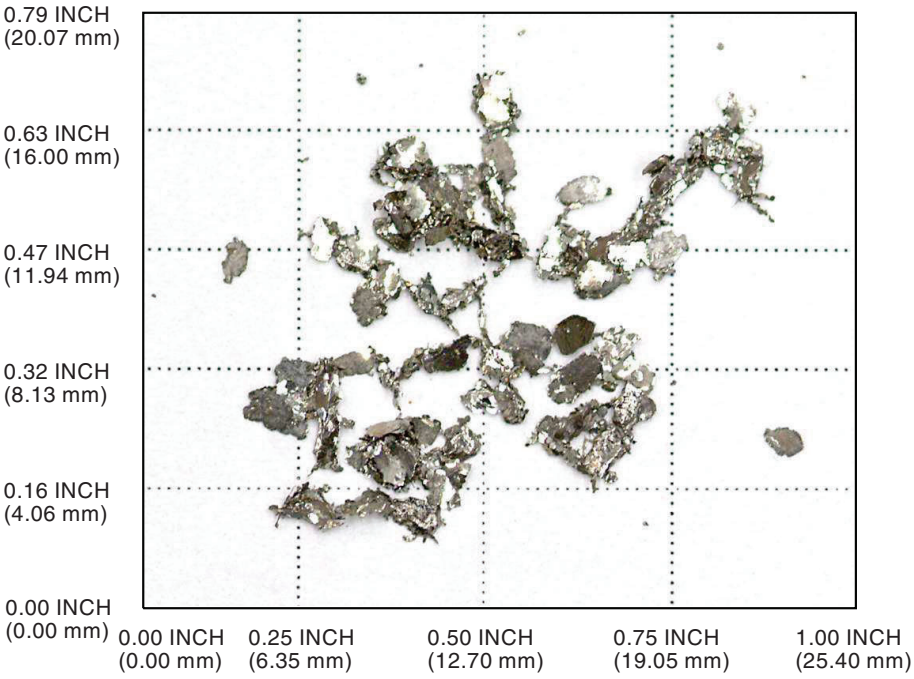
2 GENERAL CHARACTERISTICS

- MAGNETIC
- APPEARANCE
  - METALLIC
  - DULL
  - SMALL FLAKES AND / OR POWDER
  - DARK COLOR

2900658 S0000692899_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 4 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BEARING SPELL CHIP 3

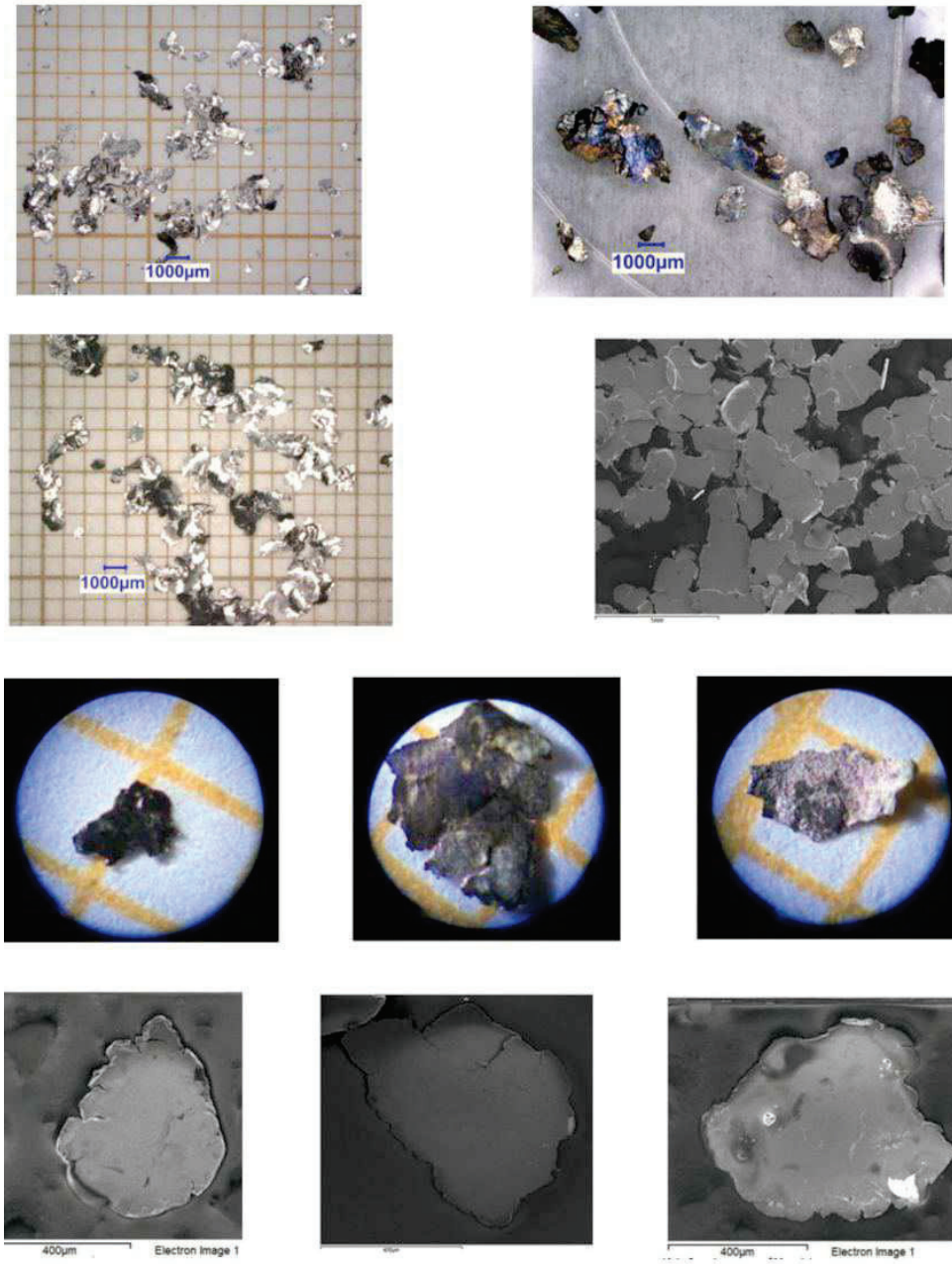
3 GENERAL CHARACTERISTICS

- MAGNETIC
- APPEARANCE
  - METALLIC
  - FLATTENED
  - RAGGED EDGES
  - FLAKES
  - OVAL
  - COINED
  - SMOOTH / REFLECTIVE ON ONE SIDE, DULL ON THE OTHER

2900678 S0000692900_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 5 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

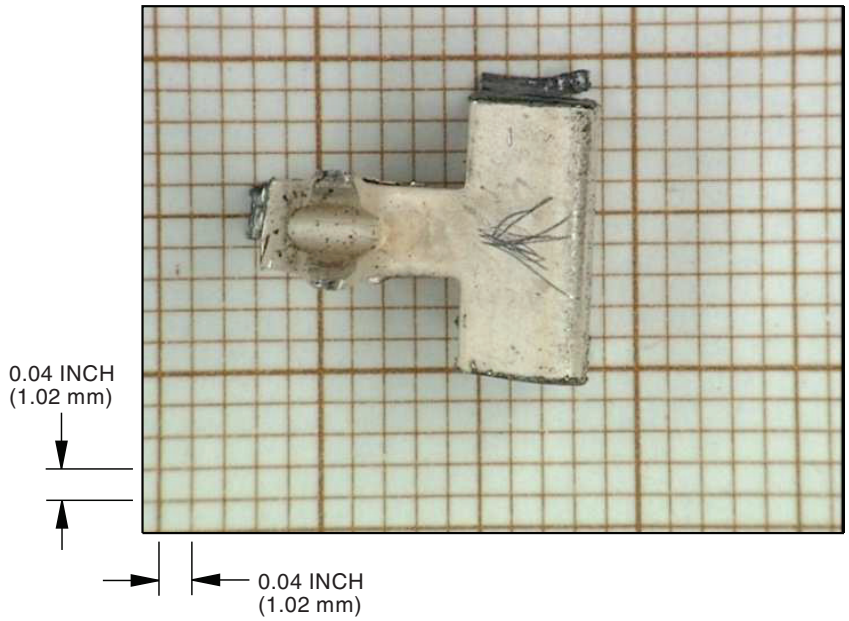


EVIDENCE OF BEARING DEGRADATION:
THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES

ICNLEAP1B790000D5882800079A00101
3101131 S0000846186_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 6 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BEARING CAGE 4

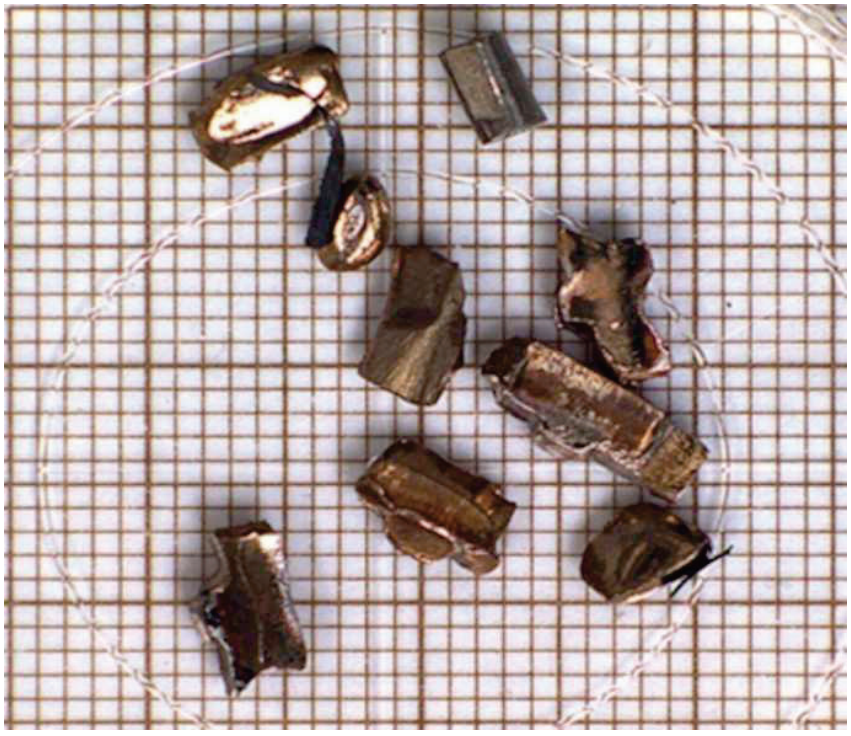
4 GENERAL CHARACTERISTICS

- MAGNETIC OR NON-MAGNETIC
- APPEARANCE
  - METALLIC
  - SHARP EDGES
  - SMOOTH
  - STRUCTURAL IN SHAPE
  - REFLECTIVE
  - SHINY OR DISCOLORED COATING

2900684 S0000692902_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 7 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BEARING CAGE FRAGMENTS

GENERAL CHARACTERISTICS :

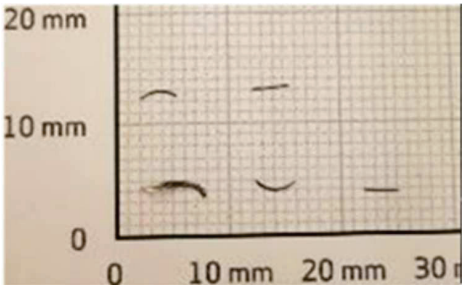
- MAGNETIC
- APPEARANCE :
    - METALLIC
    - SHARP EDGES
    - SMOOTH
    - STRUCTURAL IN SHAPE
    - REFLECTIVE
    - SHINY OR DISCOLORED COATING

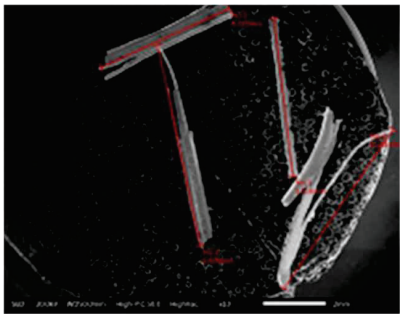
ICNLEAP1B790000D5882800082A00101
3101349 S0000846187_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 8 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

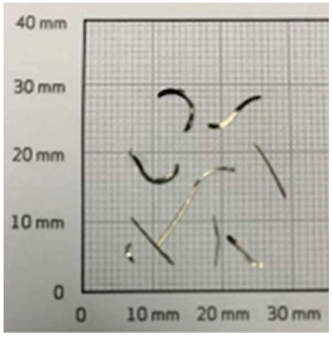


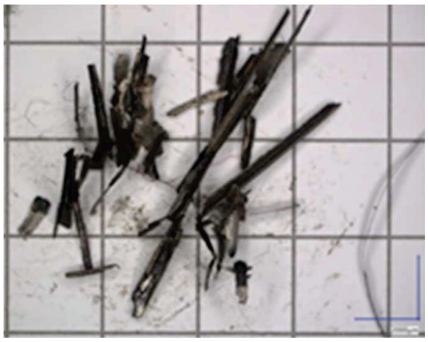










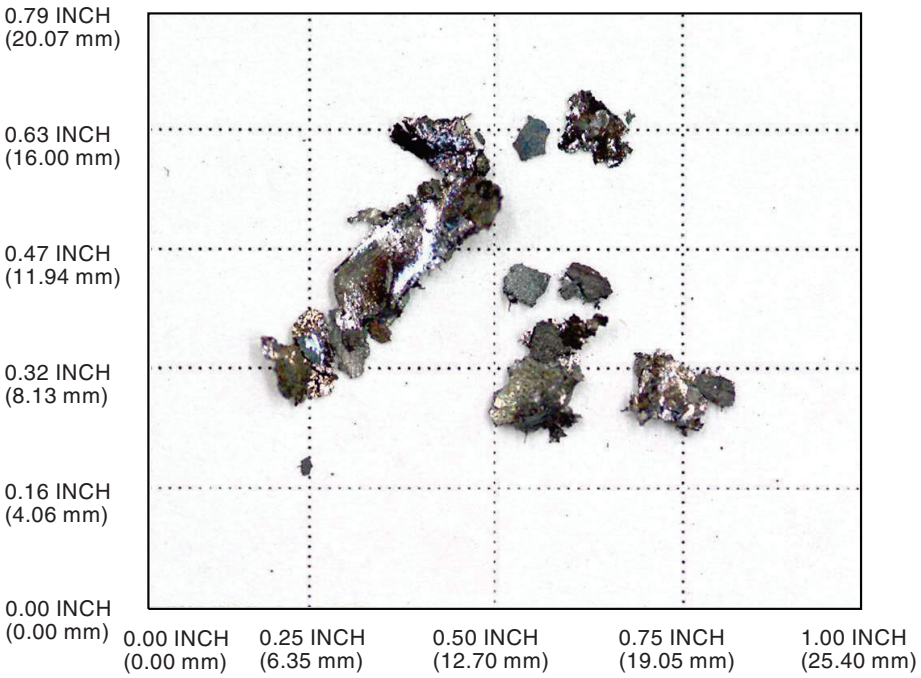


INNER RADIAL DRIVE SHAFT (RDS) CAGE WEAR - PRE SB 72-0258

ICNLEAP1B790000D5882800084A00101
3101350 S0000846188_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 9 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GEARBOX DEBRIS 5

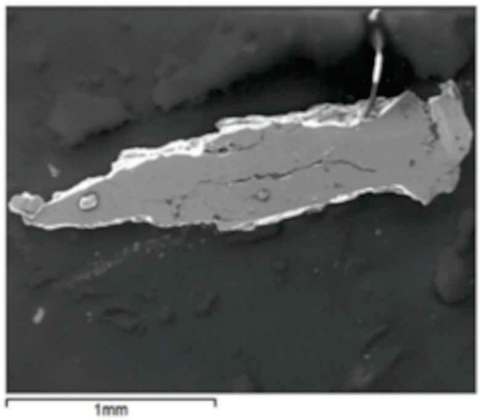
5 GENERAL CHARACTERISTICS

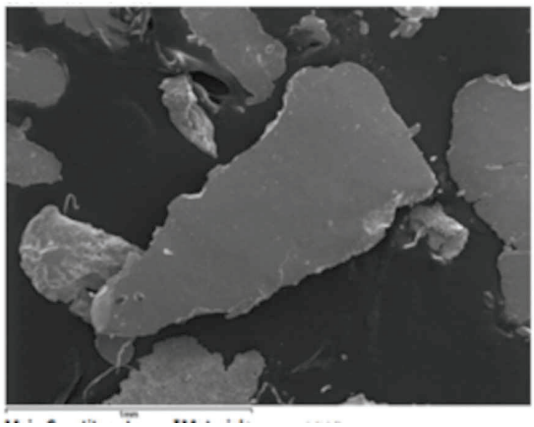
- MAGNETIC
- APPEARANCE
  - METALLIC
  - FLAKES
  - SHARP
  - REFLECTIVE
  - IRREGULAR

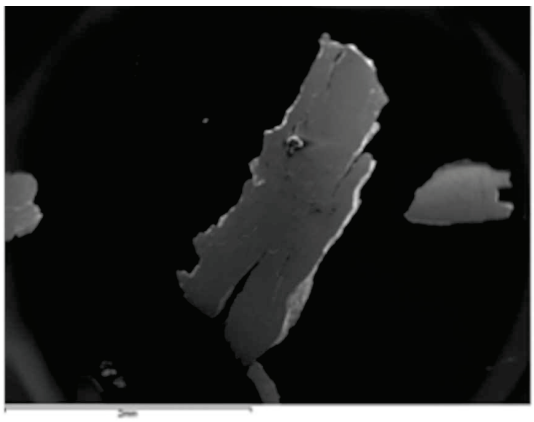
2900695 S0000692903_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 10 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details







GENERAL CHARACTERISTICS :

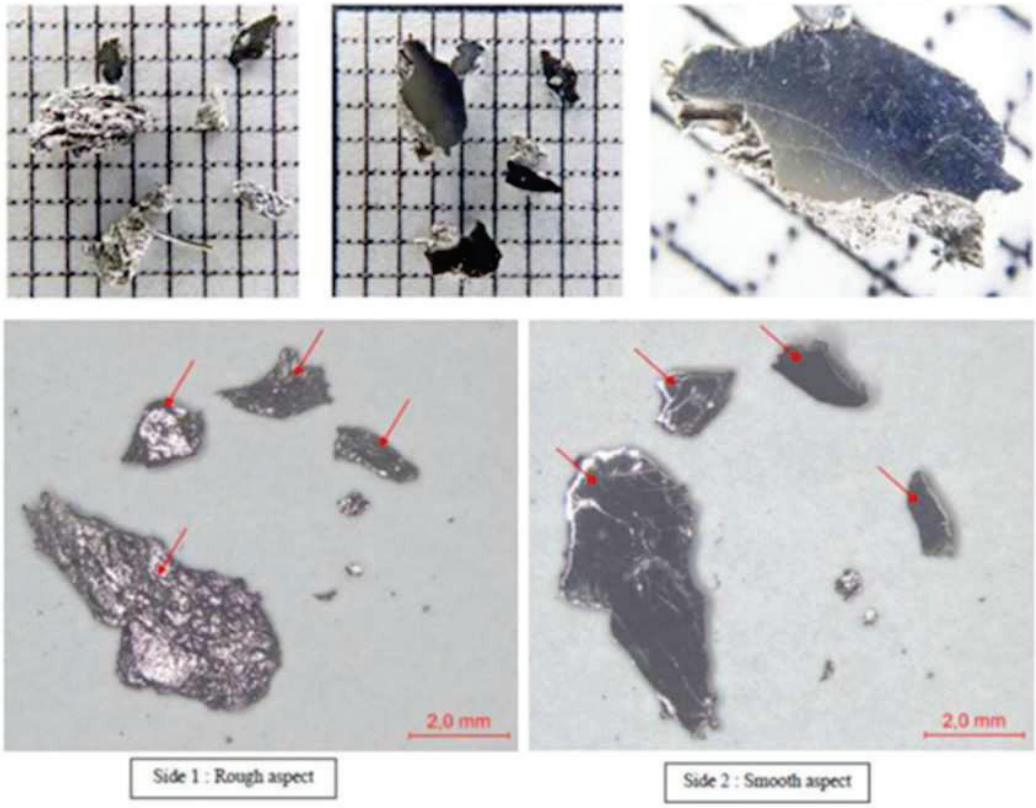
- MAGNETIC
- APPEARANCE :
  - METALLIC
  - FLAKES
  - SHARP
  - IRREGULAR
  - REFLECTIVE

NON LAMINATED FLAKES FROM BEARING DEGRADATION

ICNLEAP1B790000D5882800051A00101
3101351 S0000846189_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 11 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

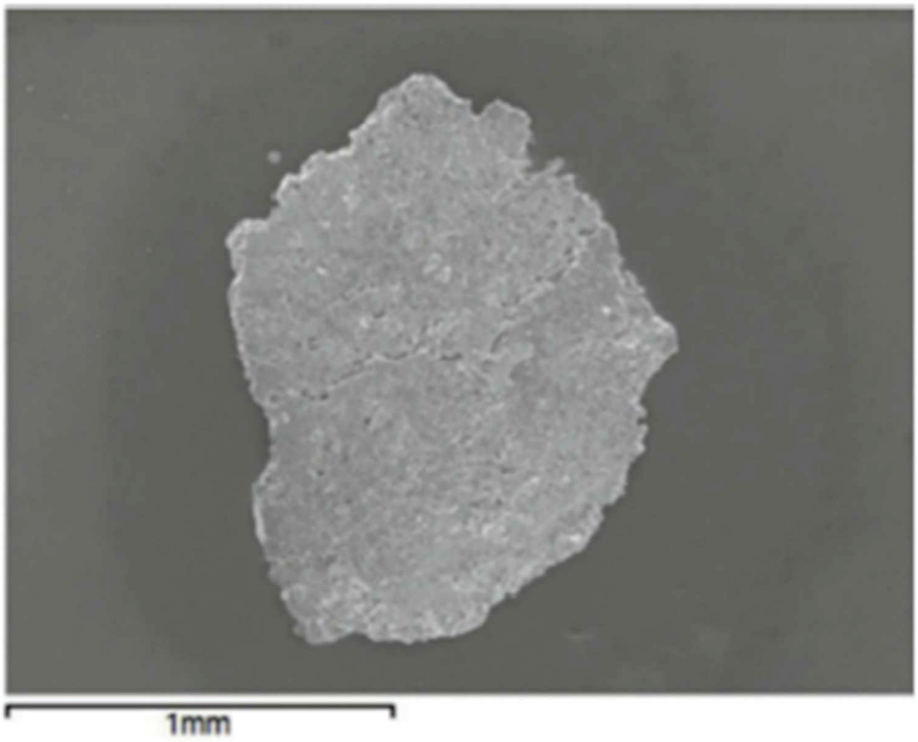


CHUNKS WITH ONE SMOOTH FACE FROM BEARING DEGRADATION

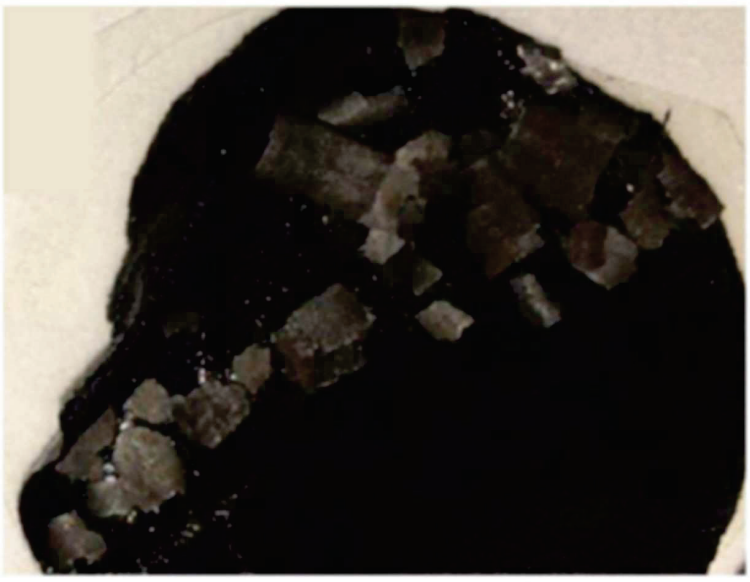
ICNLEAP1B790000D5882800050A00101
3101354 S0000846190_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 12 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details







BEARING HEAT TREAT SCALE

ICNLEAP1B790000D5882800049A00101
3101356 S0000846191_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 13 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SHOTPEENING MEDIA 6

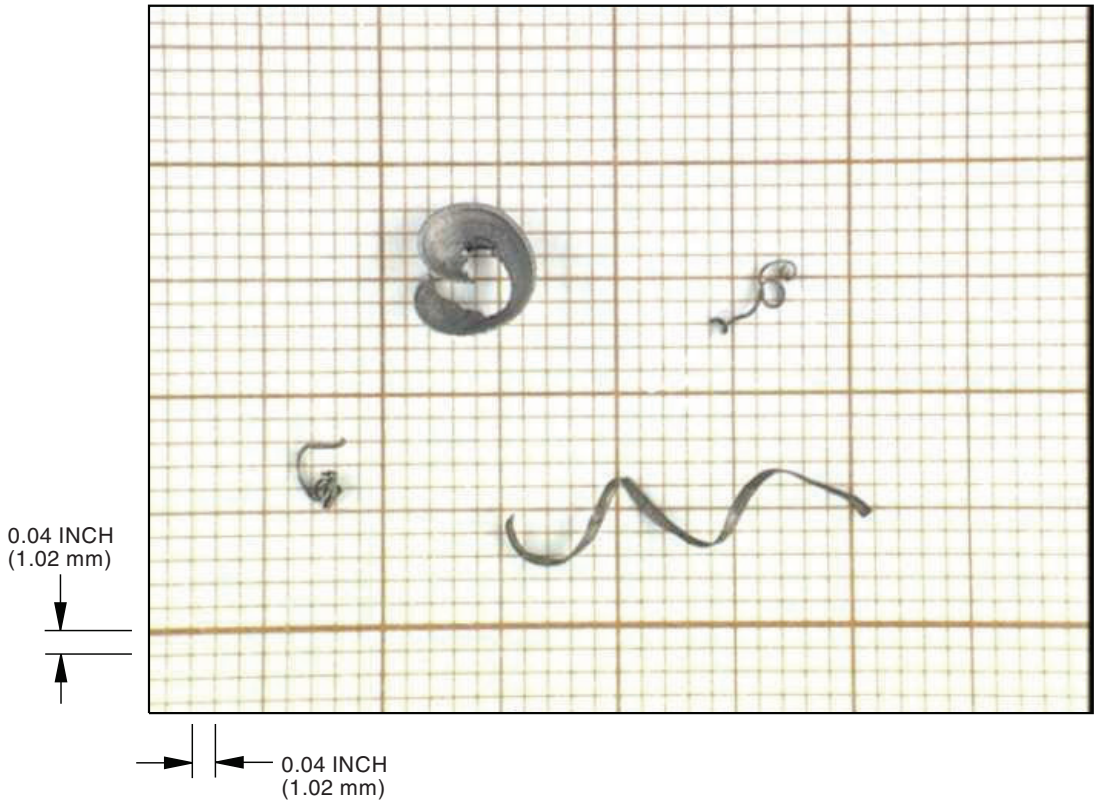
6 GENERAL CHARACTERISTICS

- MAGNETIC
- APPEARANCE
  - SPHERICAL STEEL SHOTPEEN BALLS

2900697 S0000692908_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 14 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MACHINING DEBRIS 7

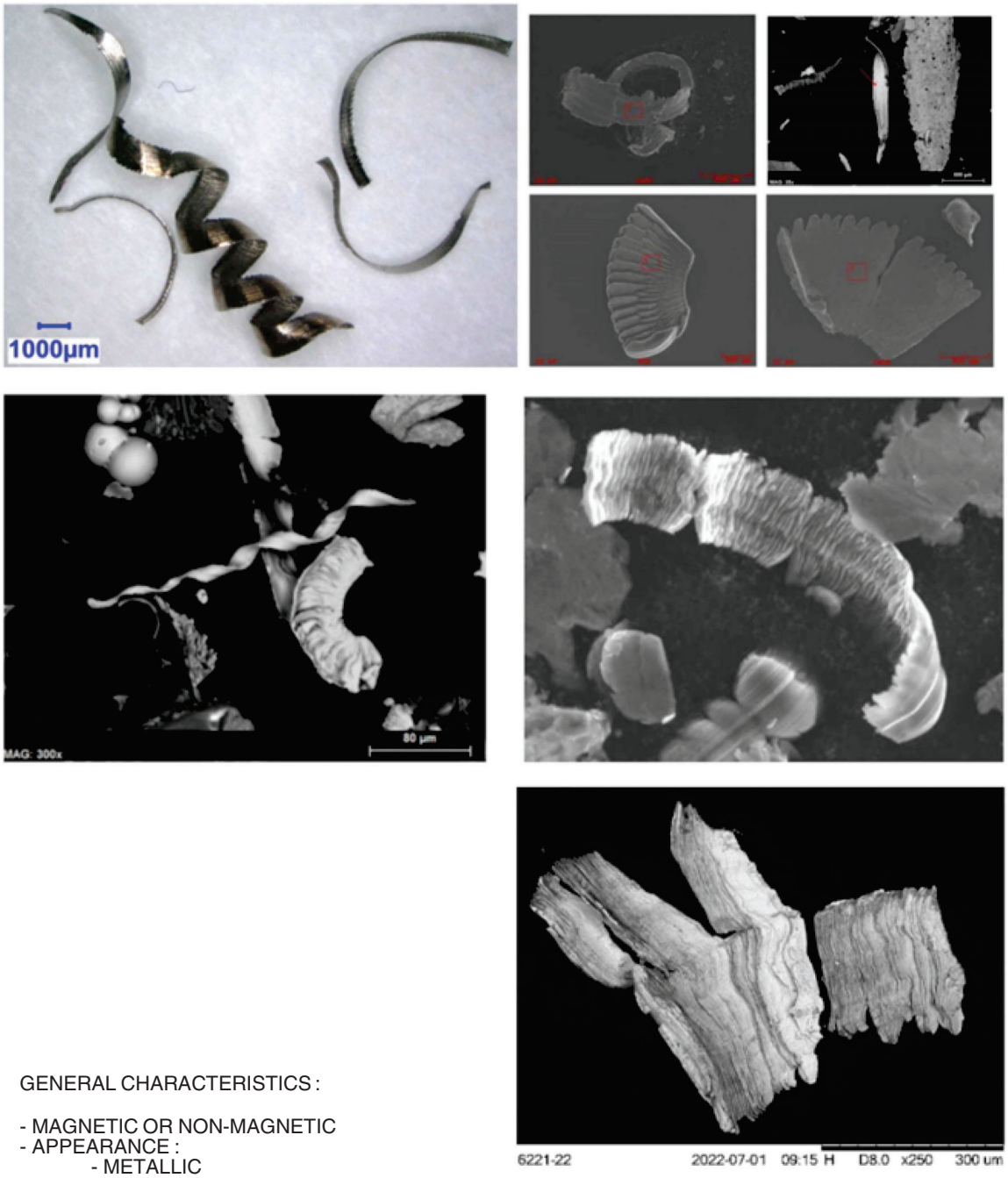
7 GENERAL CHARACTERISTICS

- MAGNETIC OR NON-MAGNETIC
- APPEARANCE
  - METALLIC
  - REFLECTIVE SOLID
  - IRREGULAR

2900699 S0000692905_V2

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 15 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GENERAL CHARACTERISTICS :

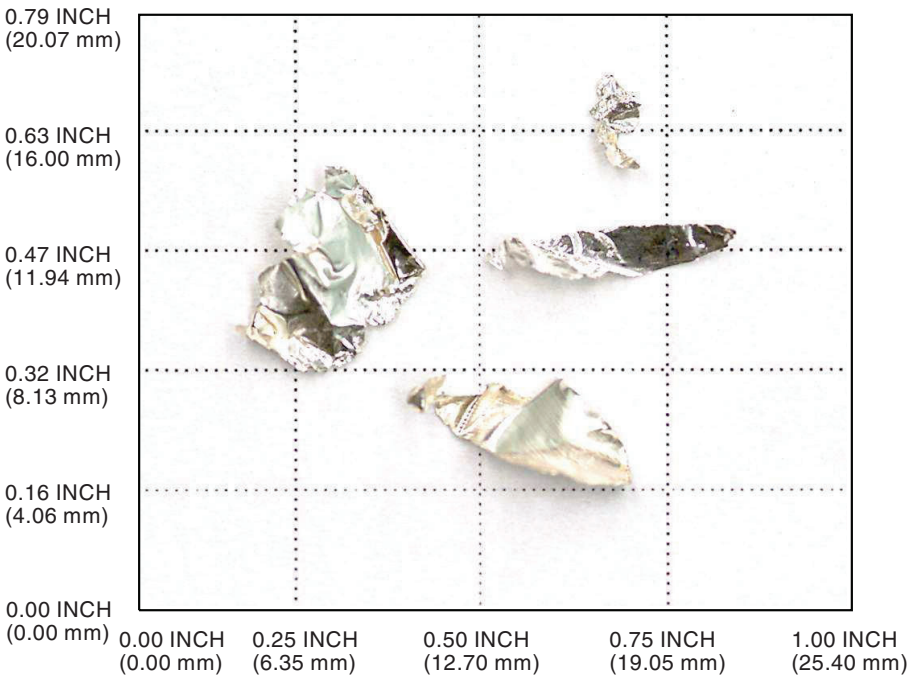
- MAGNETIC OR NON-MAGNETIC
- APPEARANCE :
  - METALLIC
  - REFLECTIVE SOLID
  - IRREGULAR

MACHINING CHIPS

ICNLEAP1B790000D5882800073A00101
3101359 S0000846192_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 16 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SILVER PLATE 8

8 GENERAL CHARACTERISTICS

- NON-MAGNETIC
- APPEARANCE
  - METALLIC – FOIL – IRREGULAR
  - REFLECTIVE – MALLEABLE

2900705 S0000692907_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 17 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





GENERAL CHARACTERISTICS :

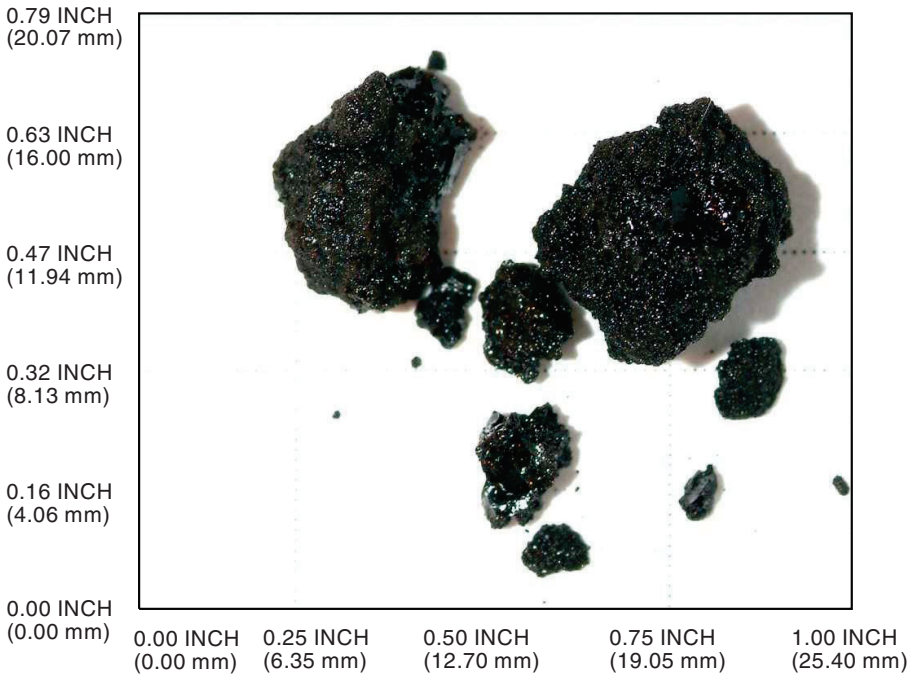
- NON-MAGNETIC
- APPEARANCE :
  - METALLIC
  - REFLECTIVE
  - FOIL
  - MALLEABLE
  - IRREGULAR

SILVER PLATELETS

ICNLEAP1B790000D5882800074A00101
3101366 S0000846193_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 18 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CARBON (COKE) 9

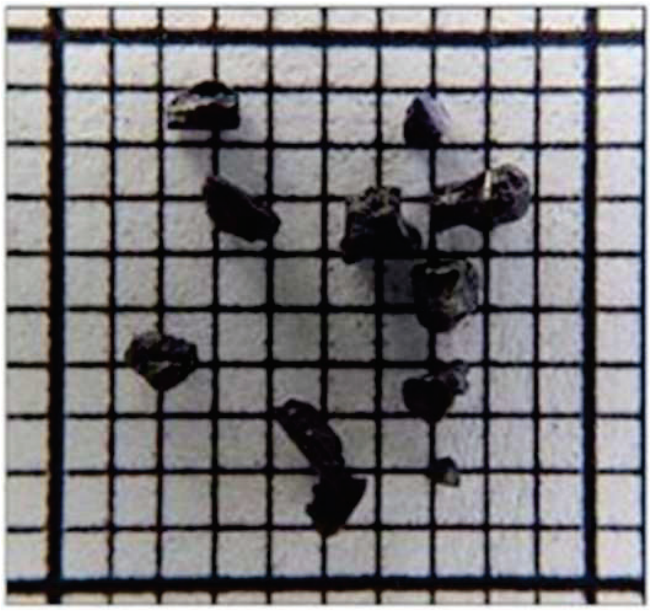
9 GENERAL CHARACTERISTICS

- NON-MAGNETIC
- APPEARANCE
  - NON-METALLIC
  - ROUND
  - SOLID
  - JAGGED EDGES
  - USULLY BLACK
  - BRITTLE TO HARD
  - OILY SMELL

2900708 S0000692904_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 19 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





GENERAL CHARACTERISTICS :

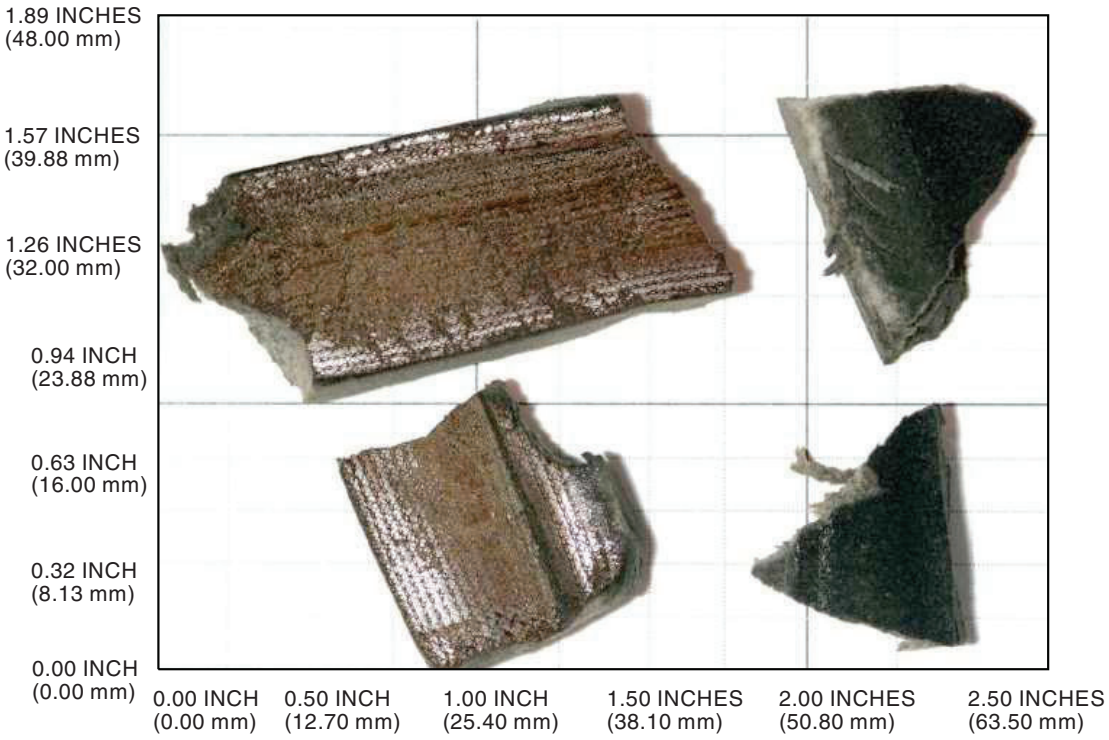
- NON-MAGNETIC
- APPEARANCE :
  - NON-METALLIC
  - ROUND
  - SOLID
  - JAGGED EDGES
  - USUALLY BLACK
  - BRITTLE TO HARD
  - OILY SMELL

CARBON

ICNLEAP1B790000D5882800052A00101
3101367 S0000846194_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 20 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PTFE (TEFLON) SEAL 10

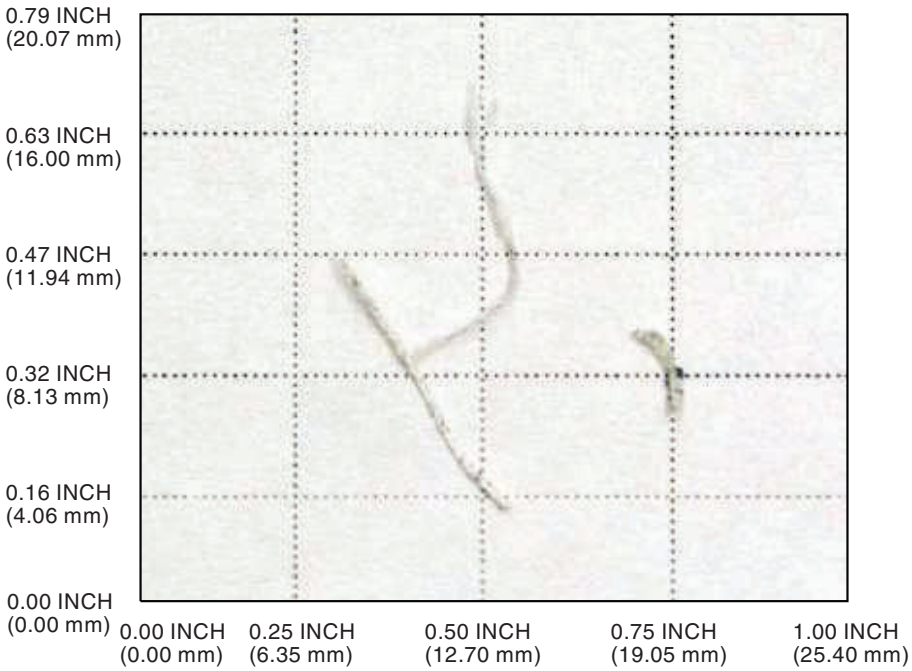
10 GENERAL CHARACTERISTICS

- NON-MAGNETIC
- APPEARANCE
  - NON-METALLIC
  - CHUNKS
  - TAN TO BLACK IN COLOR
  - LIGHTWEIGHT FOR SIZE

2900721 S0000692906_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 21 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PTFE (TEFLON) SEAL 11

11 GENERAL CHARACTERISTICS

- NON-MAGNETIC
- APPEARANCE
  - NON-METALLIC
  - LONG, THIN
  - FLAT
  - LIGHT-WEIGHT
  - STRIP
  - CURLS

2900725 S0000692909_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 22 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



KEY INSERT THREAD MATERIAL 12

12 GENERAL CHARACTERISTICS

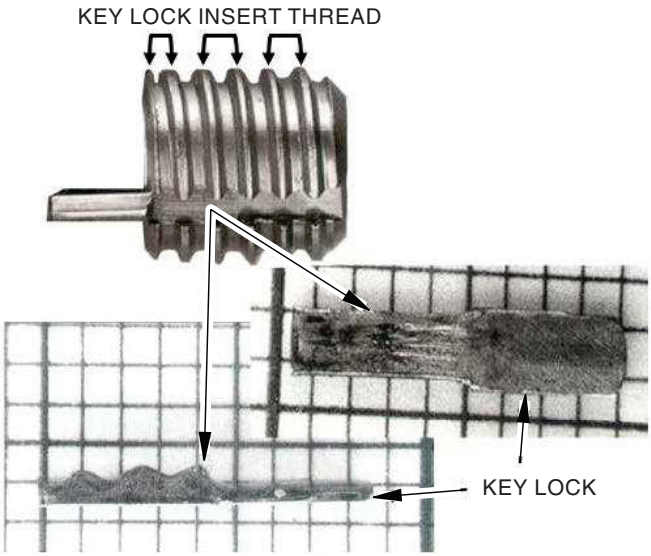
- MAGNETIC
- APPEARANCE
  - METALLIC
  - SHARP EDGES
  - MACHINED CURLS
  - ELONGATED CURLS
  - SLIVER CHUNK

ICNLEAP1B790000D5882800030A00101

2967723 S0000747155_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 23 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



KEY LOCK INSERT MATERIAL 13

13 GENERAL CHARACTERISTICS

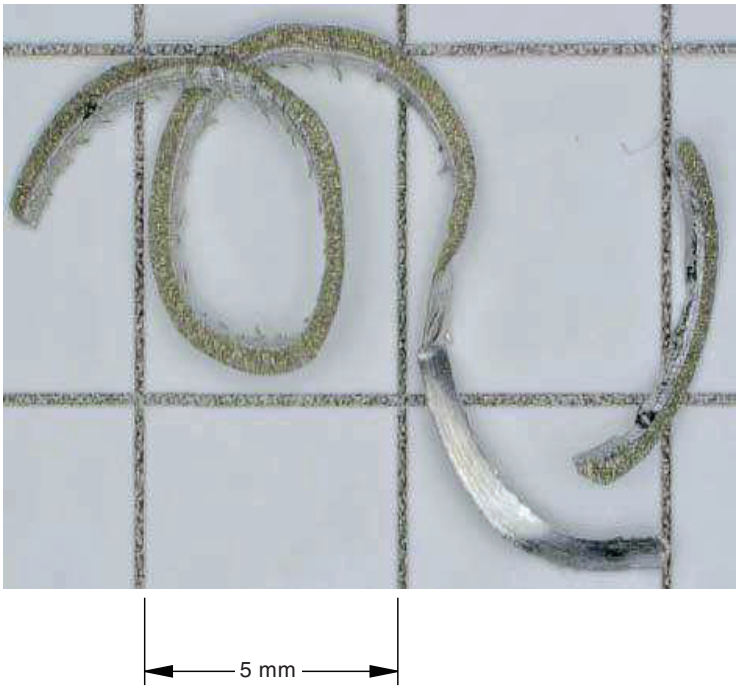
- MAGNETIC
- APPEARANCE
  - METALLIC
  - SHARP EDGES
  - FLAT PIECE WITH CASTELLATED PORTION

ICNLEAP1B790000D5882800031A00101

2967714 S0000747157_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 24 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



AGB OR TGB HOUSING THREAD 14

14 GENERAL CHARACTERISTICS

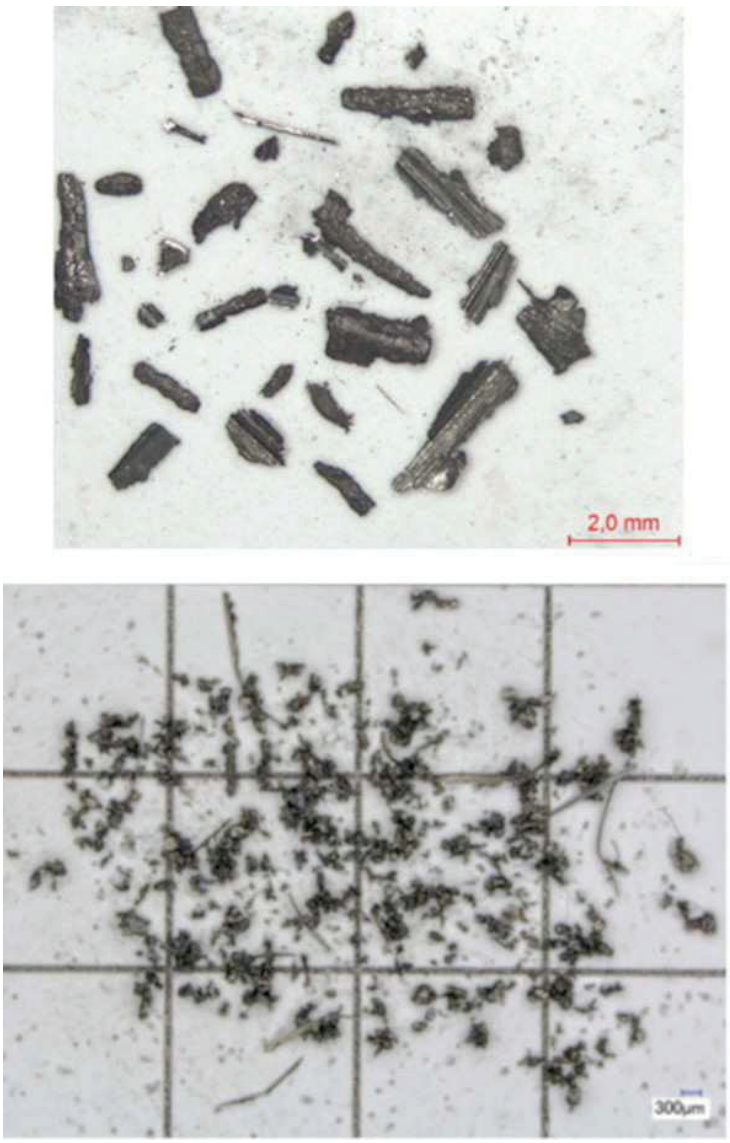
- NON MAGNETIC
- APPEARANCE:
  - METALLIC
  - SHARP EDGES
  - MACHINED CURLS
  - ELONGATED CURLS
  - SLIVER CHUNK

ICNLEAP1B790000D5882800032A00201

2967710 S0000747158_V2

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 25 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



#3 BEARING HOUSING PISTON RING GROOVE WEAR

ICNLEAP1B790000D5882800083A00101
3101368 S0000846195_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 26 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



O-RING 15

15 GENERAL CHARACTERISTICS

- NON-MAGNETIC
- APPEARANCE
  - NON-METALLIC
  - CHUNKS
  - DARK
  - BLACK
  - SOFT

ICNLEAP1B790000D5882800039A00101

2967708 S0000747159_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 27 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



O-RING PIECES

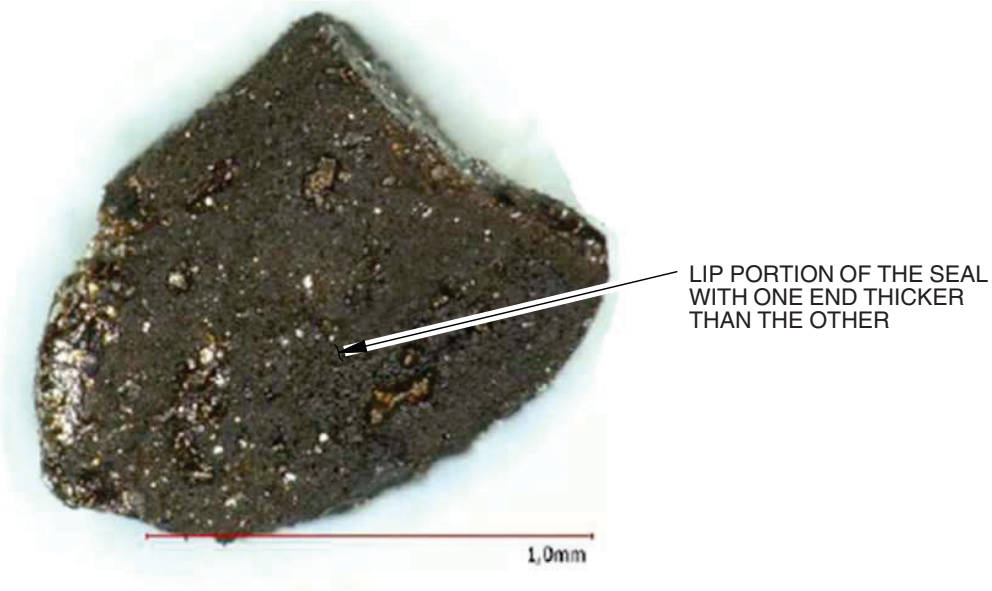
GENERAL CHARACTERISTICS :

- NON-MAGNETIC
- APPEARANCE :
  - NON-METALLIC
  - CHUNKS
  - DARK
  - BLACK
  - SOFT

ICNLEAP1B790000D5882800075A00101
3101371 S0000846196_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 28 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

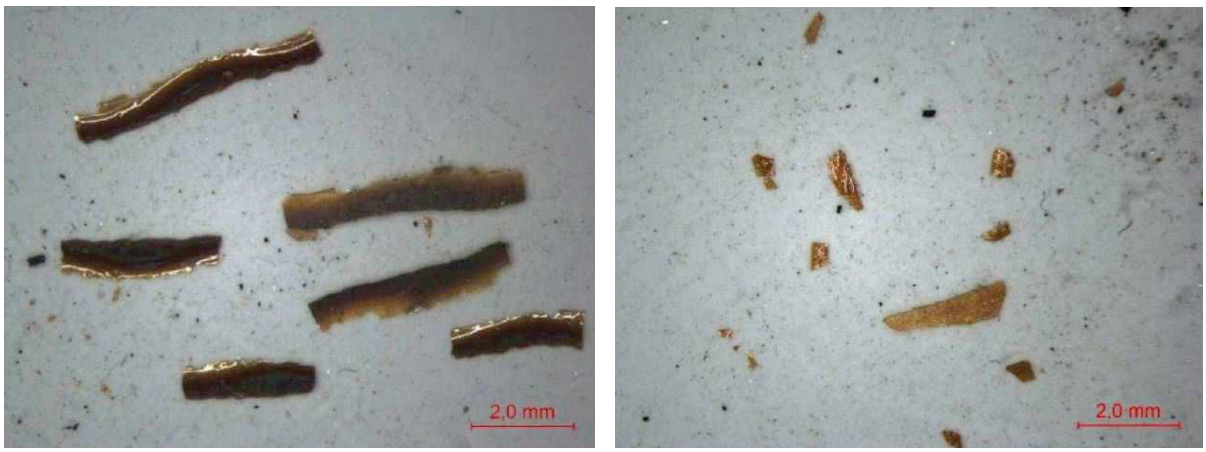


OIL LEVEL RING WITH LIP SEAL SHAPE

ICNLEAP1B790000D5882800076A00101
3101378 S0000846197_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 29 of 31)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GLUE FROM OIL FILTER CARTRIDGE 16

16 GENERAL CHARACTERISTICS

- NON-MAGNETIC
- APPEARANCE
  - NON-METALLIC
  - ELONGATED FRAGMENTS
  - THIN
  - BRITTLE
  - BROWN PARTICLES

ICNLEAP1B790000D5882800040A00101

2967707 S0000747160_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 30 of 31)

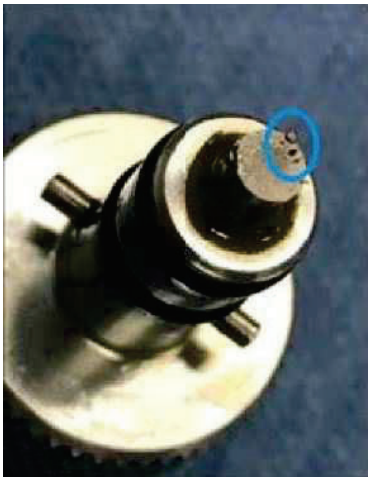
ECCN 9E991 BOEING PROPRIETARY - See title page for details



LIGHT FUZZ



LIGHT-MODERATE FUZZ



MINOR METAL



MAJOR METAL

STARTER MAGNETIC CHIP DETECTOR FINDINGS 17

17 GENERAL CHARACTERISTICS

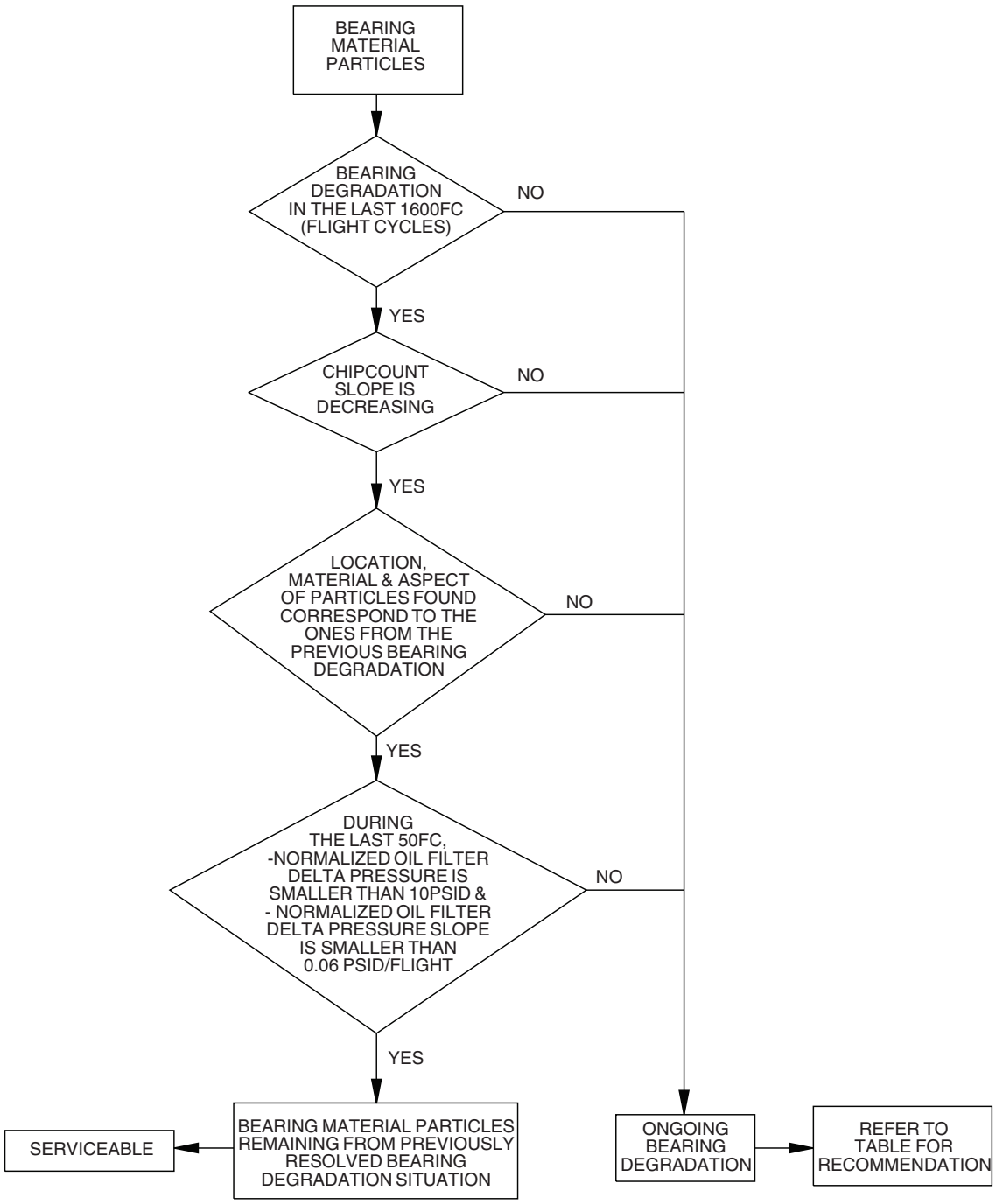
- MAGNETIC
- APPEARANCE
  - METALLIC

ICNLEAP1B790000D5882800041A00101

2967702 S0000747161_V1

Particle Examples of Material Findings
Figure 606/79-00-00-990-813-G00 (Sheet 31 of 31)

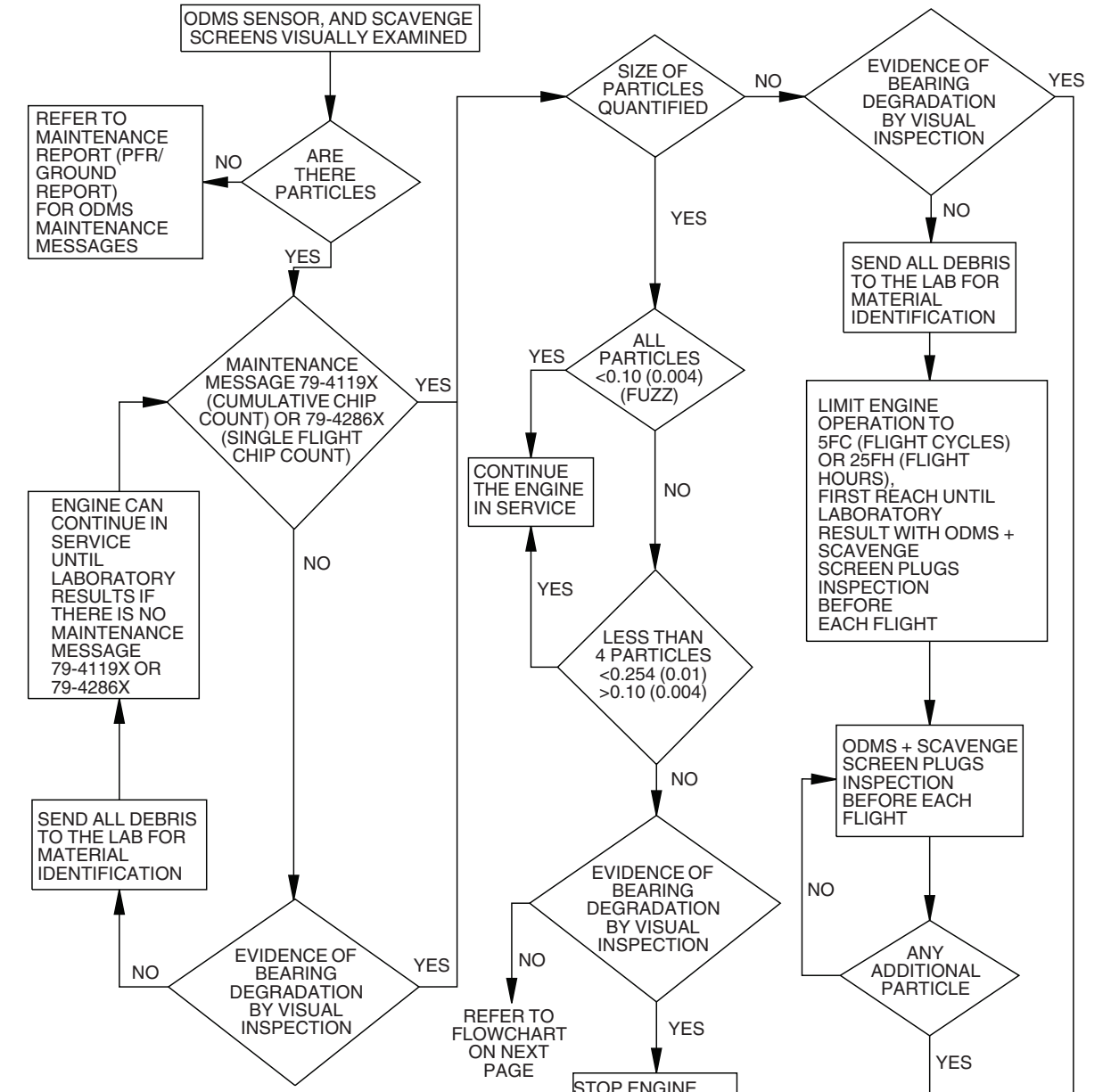
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B790000D5882800044A00101
3101381 S0000846198_V1

Flowchart - Bearing Material Detailed Analysis
Figure 607/79-00-00-990-817-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

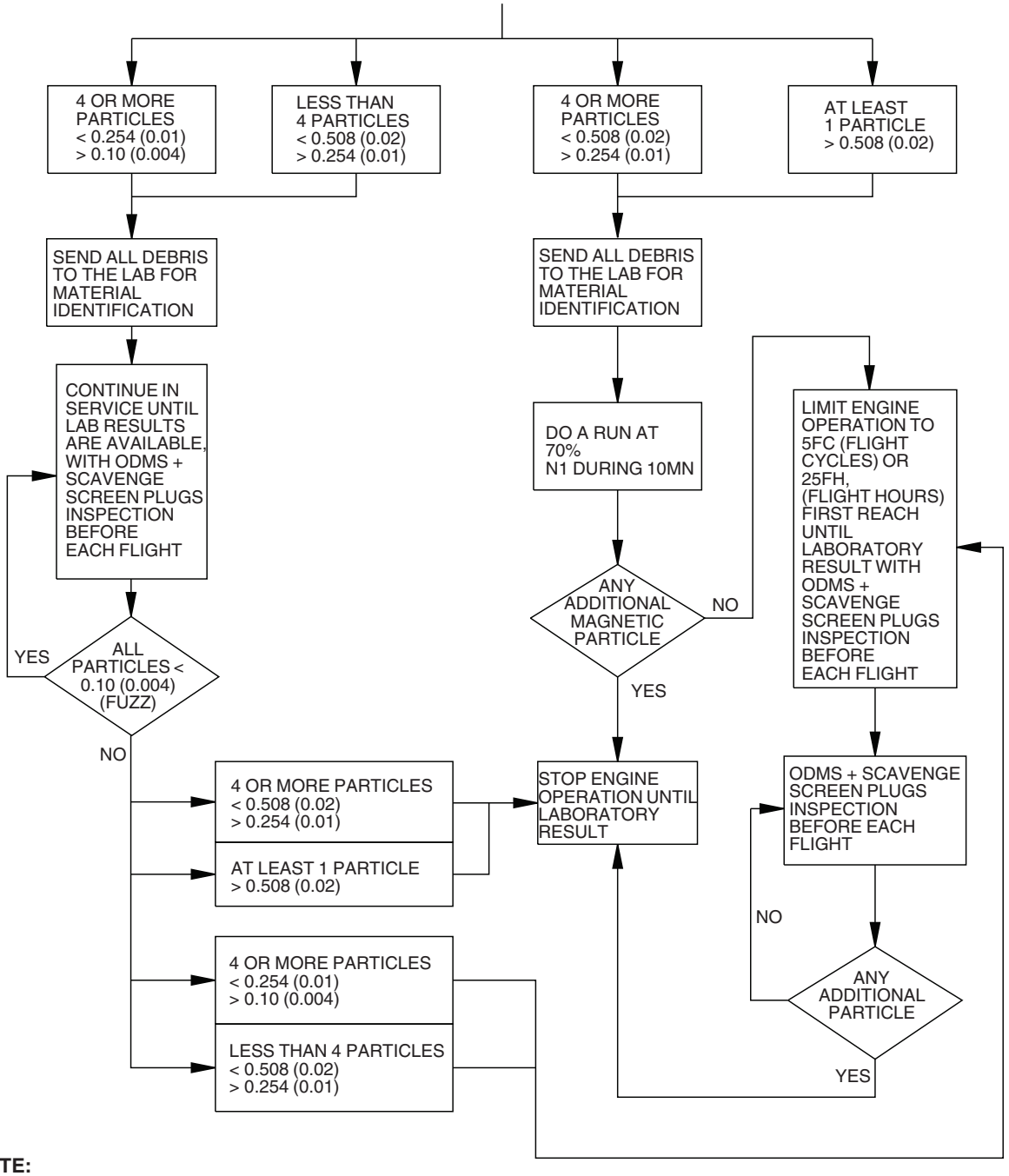
IF A SECOND MAINTENANCE MESSAGE OCCUR, DO NOT OPERATE THE ENGINE UNTIL LAB MATERIAL IDENTIFICATION AND CFM RECOMMENDATION.

DIMENSIONS ARE IN MILLIMETERS WITH INCHES IN PARENTHESES

ICNLEAP1B790000D5882800077A00101
3101383 S0000846199_V1

Flowchart - Decision vs. Magnetic Particle Size/Quantity
Figure 608/79-00-00-990-818-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



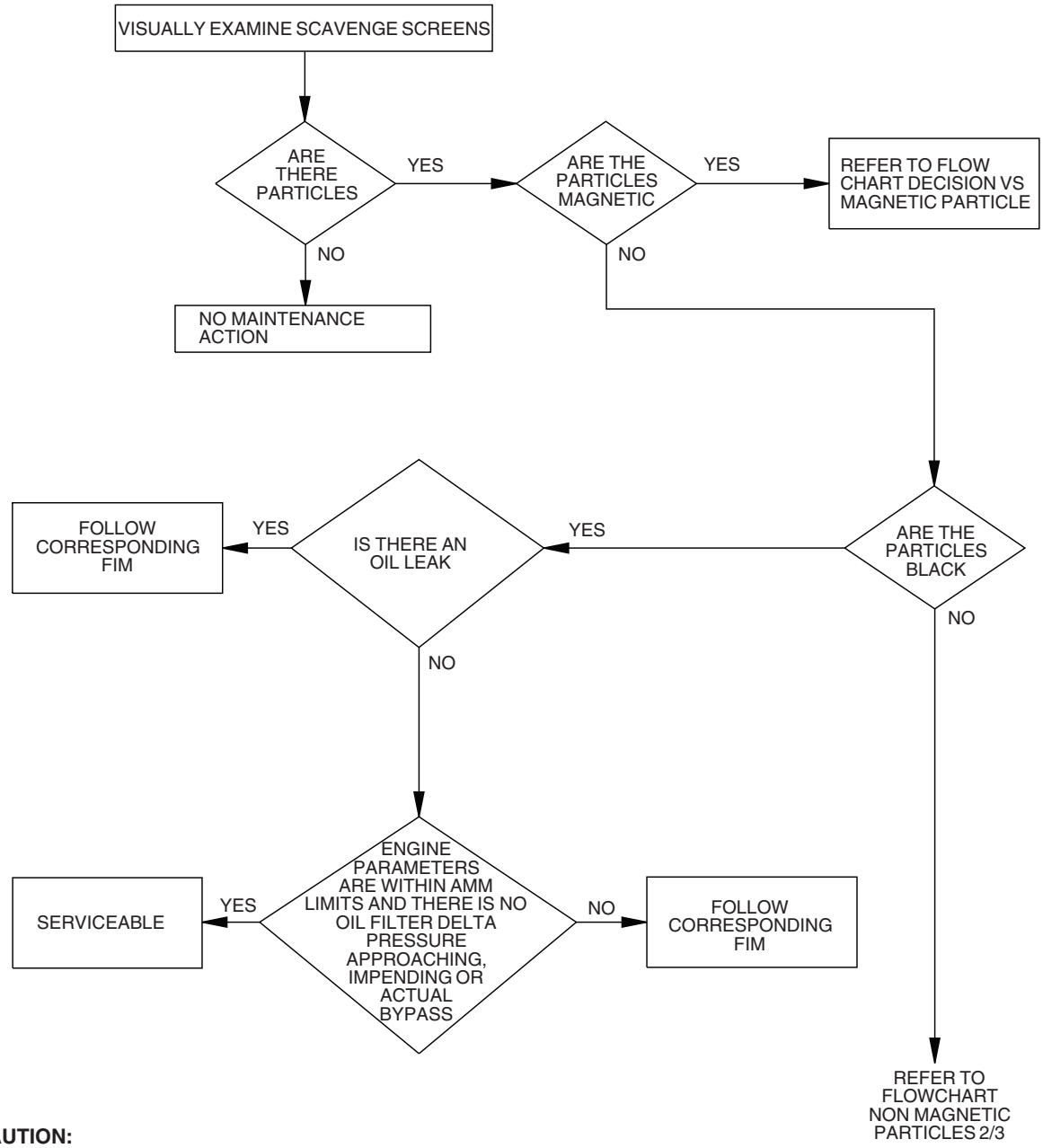
NOTE:

DIMENSIONS ARE IN MILLIMETERS WITH INCHES IN PARENTHESES

ICNLEAP1B790000D5882800078A00101
3101585 S0000846200_V1

Flowchart - Decision vs. Magnetic Particle Size/Quantity
Figure 608/79-00-00-990-818-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



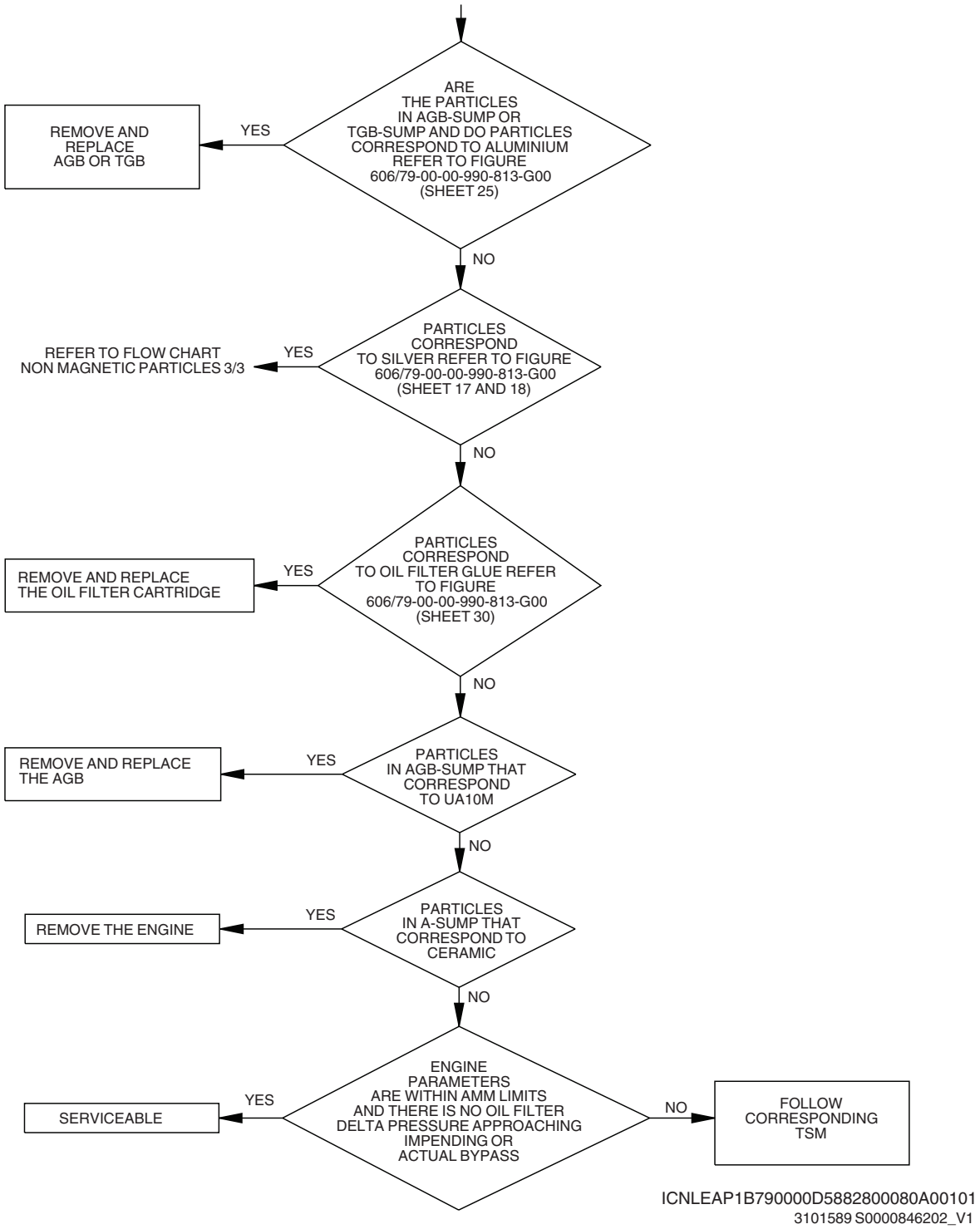
CAUTION:

MAKE SURE THAT THE PARTICLES DO NOT CORRESPOND TO OIL LEVEL RING WITH A LIP SEAL SHAPE.
REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 29).

ICNLEAP1B790000D5882800045A00101
3101587 S0000846201_V1

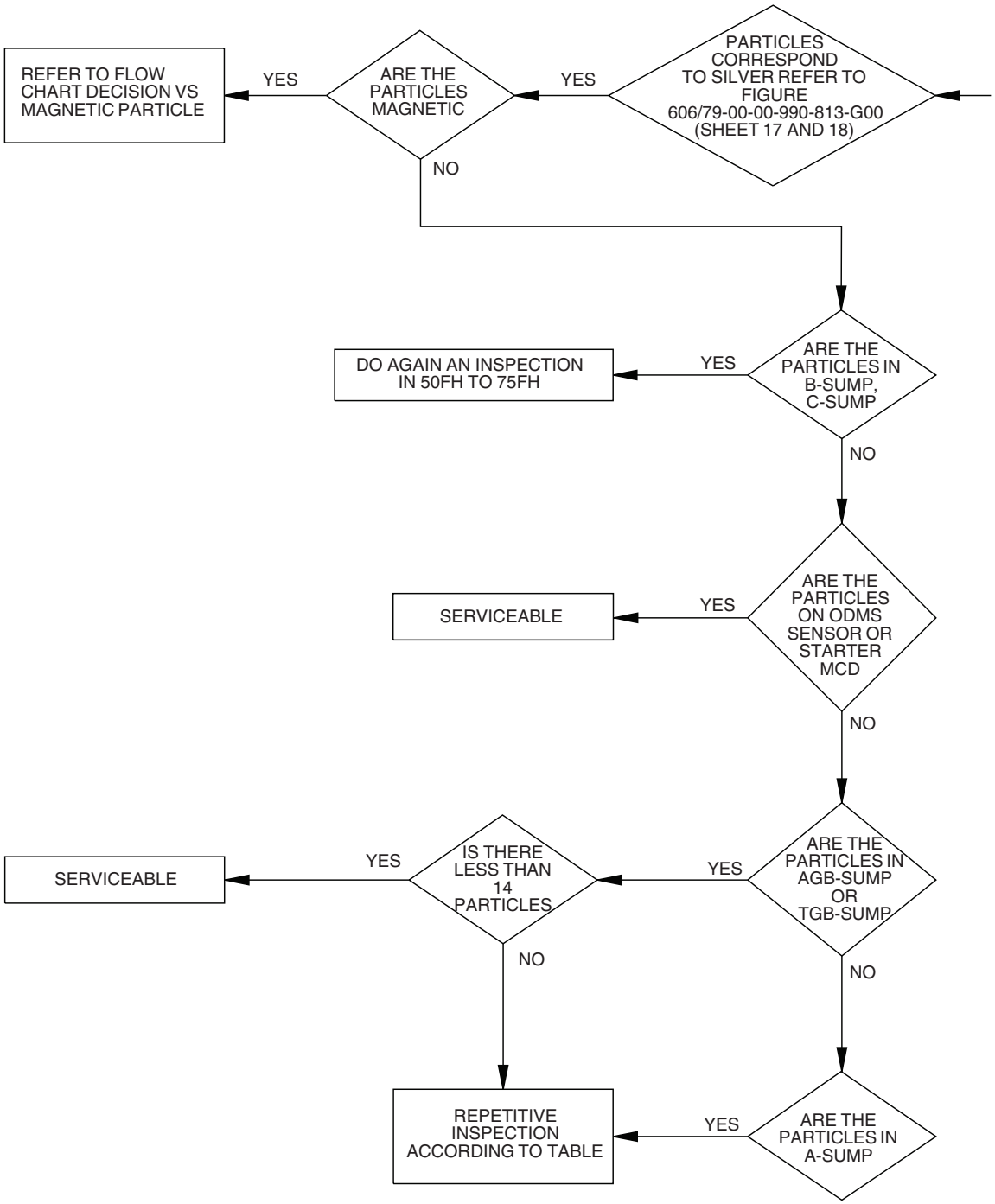
Flowchart - Decision vs. Non-magnetic Particles
Figure 609/79-00-00-990-819-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Flowchart - Decision vs. Non-magnetic Particles
Figure 609/79-00-00-990-819-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B790000D5882800081A00101
3101591 S0000846203_V1

Flowchart - Decision vs. Non-magnetic Particles
Figure 609/79-00-00-990-819-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAGNETIC	ODMS	AGB	TGB	A	B	C	FILTER	STARTER	DESCRIPTION	QUANTITY	MATERIAL COMPOSITION FROM LAB ANALYSIS	POTENTIAL SOURCE	RECOMMENDATION
YES	X	X	X	X	X	X	X	X	<0.004 INCH (0.10 mm) IN LARGEST DIMENSION	ANY	ANY	ANY	SERVICEABLE
YES	X	X	X	X	X	X	X		<0.010 INCH (0.25 mm) IN LARGEST DIMENSION	3 OR FEWER PARTICLES	ANY	ANY	SERVICEABLE, RESET ODMS REF AMM TASK 71-00-00-700-802-G00
YES	X	X*	X*(*)	X	X	X	X		MACHINED CHIPS OR SHINY CURLS OF METAL OR THAT HAVE PARALLEL MARKS FROM MACHINE TOOLS AT MAGNIFYING GLASS POWER 10X. REFER TO FIGURE 606/79- 00-00-990-813-G00 (SHEET 15 AND 16)	ANY	ANY	ANY MACHINED OIL-WETTED SURFACE	SERVICEABLE, RESET ODMS REF AMM TASK 71-00-00-700-802-G00 *: CAUTION IN CASE OF 17-4PH MACHINED SHAPE CHIP, MAKE SURE THAT IT DOES NOT MATCH WITH KEY LOCK INSERT THREAD REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 23). SEE DEDICATED RECOMMENDATION IN CASE OF 17-4PH MATERIAL. (*) : CAUTION FOR RDS BEARING PRE SB 72-0258, IN CASE OF AISI4340 MACHINED SHAPE CHIP, MAKE SURE THAT IT DOES NOT MATCH WITH RDS BEARING CAGE WEAR, REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 9).
YES	X	X	X						FLAT PIECE WITH CASTELLATED PORTION REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 24)	ANY	17-4PH	AGB OR TGB KEY LOCK FROM INSERTS	SERVICEABLE, CONTINUE THE ENGINE IN SERVICE IF THERE IS NO OTHER METALLIC DEBRIS, RESET ODMS REF AMM TASK 71-00-00-700-802-G00
YES	X								SHAPE OTHER THAN THOSE DESCRIBES IN FIGURE 606/79-00-00-990-813-G00 (SHEET 23)	ANY	17-4PH	ALL SUMPS	1) CONTINUE THE ENGINE IN SERVICE IF THE OIL PRESSURE AND CONSUMPTION ARE IN THE LIMITS. 2) DO A CHECK OF OIL PRESSURE AND CONSUMPTION IN 100 TO 200 HOURS. 3) DO A CHECK OF THE ODMS AND SCAVENGE SCREEN PLUGS IN 100-150 HOURS: - IF THERE IS NO ADDITIONAL PARTICLES IN SCAVENGE SCREEN PLUGS RETURN TO NORMAL OPERATION CONDITION, ENGINE IS IN SERVICEABLE CONDITION. RESET ODMS REF AMM TASK 71-00-00-700-802-G00 - IF THERE IS ADDITIONAL PARTICLES, FOLLOW AMM TASK 79-00-00-220-801-G00
YES	X								MACHINED CURLS, ELONGATED CURLS, SLIVER CHUNK REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 23)	ANY	17-4PH	AGB OR TGB KEY INSERTS THREAD	IF THIS IS THE 1ST TIME, PARTICLE IS FOUND IN ODMS, DO A CHECK OF THE ODMS AND SCAVENGE SCREEN PLUGS IN 50-75 HOURS FOR 3 CONSECUTIVE TIMES: - IF THERE IS NO ADDITIONAL PARTICLES IN ODMS AND SCAVENGE SCREEN PLUGS RETURN TO NORMAL OPERATION CONDITION. DO NOT RESET ODMS - IF THERE IS ADDITIONAL PARTICLES, FOLLOW AMM TASK 79-00-00-220-801-G00 IF SAME PARTICLE TYPE IS FOUND ONLY AT ODMS DURING A REPEAT INSPECTION: - REDUCE INSPECTION INTERVAL TO 25/50 HOURS UNTIL PARTICLES ARE FOUND IN AGB OR TGB SCREEN. DO NOT RESET ODMS
YES	X	X	X						MACHINED CURLS, ELONGATED CURLS, SLIVER CHUNK REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 23)	ANY	17-4PH	AGB OR TGB KEY INSERTS THREAD	REMOVE AND REPLACE AGB OR TGB, RESET ODMS REF AMM TASK 71-00-00-700-802-G00 DO OIL SYSTEM FLUSHING AMM TASK 12-13-11-100-801

ICNLEAP1B79000D5882800021A00301 3101597 S0000846204_V1

Disposition and Identification for Magnetic Material Findings
Figure 610/79-00-00-990-820-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAGNETIC	ODMS	AGB	TGB	A	B	C	FILTER	STARTER	DESCRIPTION	QUANTITY	MATERIAL COMPOSITION FROM LAB ANALYSIS	POTENTIAL SOURCE	RECOMMENDATION
YES	X								EVIDENCE OF BEARING DEGRADATION; THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6) OTHER BEARING DEGRADATION SHAPE: - NON-LAMINATED FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 10 AND 11) - CHUNKS WITH ONE SMOOTH FACE. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 12)	4 OR LESS PARTICLES <0.03 INCH (0.76 mm) IN LARGEST DIMENSION	- M50/E-80DCV40 (RACE/BALL/ROLLER MATERIAL), - M50NIL/E-13DCNV40 (RACE MATERIAL), - 32CDV13, NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES	ENGINE BEARINGS	DO NOT RESET THE CHIP COUNT AND PUT THE ENGINE ON WATCH: 1- INSPECT THE ODMS SENSOR AND ALL THE SCAVENGE SCREEN PLUG AND STARTER MCD EVERY DAY OR BEFORE 5 FC MAX FOR 3 TIMES A- IF NO FINDINGS, INSPECT THE ODMS SENSOR AND ALL THE SCAVENGE SCREEN PLUGS AND STARTER MCD EVERY 50-75 FH FOR 3 TIMES I) IF NO FINDINGS, ENGINE IS CLEARED, RESET ODMS REF AMM TASK 71-00-00-700-802-G00 II) IF FINDINGS, REFER TO AMM TASK 79-00-00-220-801-G00 B- IF FINDINGS, REFER TO AMM TASK 79-00-00-220-801-G00
YES	X								EVIDENCE OF BEARING DEGRADATION; THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6) OTHER BEARING DEGRADATION SHAPE: -NON-LAMINATED FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 10 AND 11) - CHUNKS WITH ONE SMOOTH FACE. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 12)	5 OR MORE PARTICLES <0.03 INCH (0.76 mm) IN LARGEST DIMENSION OR ANY SINGLE PARTICLE >0.03 INCH (0.76 mm) IN ANY DIMENSION	- M50/E-80DCV40 (RACE/BALL/ROLLER MATERIAL), - M50NIL/E-13DCNV40 (RACE MATERIAL), - 32CDV13, NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES	ENGINE BEARINGS	REFER TO FIGURE FLOWCHART -BEARING MATERIAL DETAILED ANALYSIS (SHEET 1 OF 1) IN CASE OF ONGOING BEARING DEGRADATION, REMOVE THE ENGINE. IN CASE THE PARTICLES ARE REMNANTS FROM A PREVIOUS RESOLVED BEARING DEGRADATION SITUATION, THOSE ARE ACCEPTABLE. RESET THE ODMS. REFER TO AMM TASK 71-00-00-700-802-G00. BEFORE REMOVING THE ENGINE OR IN CASE OF ENGINE REMOVAL, FOR INVESTIGATION PURPOSE, DOWNLOAD: DIR 1 AND DIR 3 CHANNELS A AND B. REFER TO AMM TASK 73-21-05-970-801-G00 ALL ENGINE VIBRATION DATA (EECxENxVIBE) AVAILABLE CHANNELS A AND B. REFER TO AMM TASK 73-21-05-970-802-G00, TO ENHANCE THE ROBUSTNESS OF THE INVESTIGATION, IT IS RECOMMENDED TO DOWNLOAD AT LEAST 1000 EEC x EN x VIBE FILES.
YES	X								EVIDENCE OF BEARING DEGRADATION; THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6) OTHER BEARING DEGRADATION SHAPE: - PIECES OF SILVER-PLATED STEEL -CAGE FRAGMENTS. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 7 AND 8)	4 OR MORE PARTICLES <0.010 INCH (0.25 mm) IN LARGEST DIMENSION OR ANY SINGLE PARTICLE >0.010 INCH (0.25 mm) IN ANY DIMENSION	- AISI 4340/E-40NCD7 (CAGE MATERIAL), NOTE: THE AISI9310/ AISI9315/16NCD13 (GEAR MATERIAL) CAN BE CONFUSED WITH AISI4340/E-40NCD7 (CAGE MATERIAL)	ENGINE BEARING CAGES	REFER TO FIGURE FLOWCHART- BEARING MATERIAL DETAILED ANALYSIS (SHEET 1 OF 1) IN CASE OF ONGOING BEARING DEGRADATION, REMOVE THE ENGINE. IN CASE THE PARTICLES ARE REMNANTS FROM A PREVIOUS RESOLVED BEARING DEGRADATION SITUATION, THOSE ARE ACCEPTABLE. RESET THE ODMS. REFER TO AMM TASK 71-00-00-700-802-G00 BEFORE REMOVING THE ENGINE OR IN CASE OF ENGINE REMOVAL, FOR INVESTIGATION PURPOSE, DOWNLOAD: DIR 1 AND DIR 3 CHANNELS A AND B. REFER TO AMM TASK 73-21-05-970-801-G00 ALL ENGINE VIBRATION DATA (EECxENxVIBE) AVAILABLE CHANNELS A AND B. REFER TO AMM TASK 73-21-05-970-802-G00, TO ENHANCE THE ROBUSTNESS OF THE INVESTIGATION, IT IS RECOMMENDED TO DOWNLOAD AT LEAST 1000 EEC x EN x VIBE FILES.

ICNLEAP1B790000D5882800022A00301
3101612 S0000846206_V2

Disposition and Identification for Magnetic Material Findings
Figure 610/79-00-00-990-820-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAGNETIC	ODMS	AGB	TGB	A	B	C	FILTER	STARTER	DESCRIPTION	QUANTITY	MATERIAL COMPOSITION FROM LABANALYSIS	POTENTIAL SOURCE	RECOMMENDATION
YES				X	X	X			EVIDENCE OF BEARING DEGRADATION: THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6) OTHER BEARING DEGRADATION SHAPE: - NON-LAMINATED FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 10 AND 11) - CHUNKS WITH ONE SMOOTH FACE. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 12) - PIECES OF SILVER - PLATED STEEL - CAGE FRAGMENTS. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 7 AND 8)	4 OR MORE PARTICLES, <0.010 INCH (0.25 mm) IN LARGEST DIMENSION OR ANY SINGLE PARTICLE >0.010 INCH (0.25 mm) IN ANY DIMENSION.	- M50/E-80DCV40 (RACE/BALL/ROLLER MATERIAL), - M50NIL/E-13DCNV40 (RACE MATERIAL), - AISI 4340 /E-40NDC7 (CAGE MATERIAL), - 32CDV13, NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES. NOTE : THE AISI9310/AISI9315 /16NCD13 (GEAR MATERIAL) CAN BE CONFUSED WITH AISI4340/E-40NCD7 (CAGE MATERIAL)	ENGINE BEARINGS	REFER TO FIGURE FLOWCHART-BEARING MATERIAL DETAILED ANALYSIS (SHEET 1 OF 1). IN CASE OF ONGOING BEARING DEGRADATION, REMOVE THE ENGINE. IN CASE THE PARTICLES ARE REMNANTS FROM A PREVIOUS RESOLVED BEARING DEGRADATION SITUATION, THOSE ARE ACCEPTABLE. RESET THE ODMS. REFER TO AMM TASK 71-00-00-702-802-G00. BEFORE REMOVING THE ENGINE OR IN CASE OF ENGINE REMOVAL, FOR INVESTIGATION PURPOSE, DOWNLOAD: DIR 1 AND DIR 3 CHANNELS A AND B, REFER TO AMM TASK 73-21-05-970-801-G00 ALL ENGINE VIBRATION DATA (EEC x ENG x VIBE) AVAILABLE CHANNELS A AND B, REFER TO AMM TASK 73-21-05-970-802-G00. TO ENHANCE THE ROBUSTNESS OF THE INVESTIGATION, IT IS RECOMMENDED TO DOWNLOAD AT LEAST 1000 EEC x EN x VIBE FILES.
YES		X							EVIDENCE OF BEARING DEGRADATION: THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6). OTHER BEARING DEGRADATION SHAPE: - NON-LAMINATED FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 10 AND 11) - CHUNKS WITH ONE SMOOTH FACE. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 12) - PIECES OF SILVER - PLATED STEEL - CAGE FRAGMENTS. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 7 AND 8)	4 OR MORE PARTICLES, <0.010 INCH (0.25 mm) IN LARGEST DIMENSION OR ANY SINGLE PARTICLE >0.010 INCH (0.25 mm) IN ANY DIMENSION.	- M50/E-80DCV40 (RACE/BALL/ROLLER MATERIAL), - AISI 4340 /E-40NDC7 (CAGE MATERIAL), - 32CDV13, NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES. NOTE : THE AISI9310/AISI9315 /16NCD13 (GEAR MATERIAL) CAN BE CONFUSED WITH AISI4340/E-40NCD7 (CAGE MATERIAL).	AGB BEARINGS	REFER TO FIGURE FLOWCHART-BEARING MATERIAL DETAILED ANALYSIS (SHEET 1 OF 1). IN CASE OF ONGOING BEARING DEGRADATION, REMOVE AND REPLACE THE AGB. RESET THE ODMS. REFER TO AMM TASK 71-00-00-700-801-G00. DO OIL SYSTEM FLUSHING REFER TO AMM TASK 12-13-11-100-801. IN CASE THE PARTICLES ARE REMNANTS FROM A PREVIOUS AND ALREADY RESOLVED BEARING DEGRADATION SITUATION, THOSE ARE ACCEPTABLE. RESET THE ODMS REFER TO AMM TASK 71-00-00-700-802-G00
YES			X						EVIDENCE OF BEARING DEGRADATION: THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6). OTHER BEARING DEGRADATION SHAPE: - NON-LAMINATEDFLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 10 AND 11) - CHUNKS WITH ONE SMOOTH FACE. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 12) - PIECES OF SILVER - PLATED STEEL - CAGE FRAGMENTS. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 7 AND 8)	4 OR MORE PARTICLES, <0.010 INCH (0.25 mm) IN LARGEST DIMENSION OR ANY SINGLE PARTICLE >0.010 INCH (0.25 mm) IN ANY DIMENSION.	- M50/E-80DCV40 (RACE/BALL/ ROLLER MATERIAL), - AISI 4340 /E-40NDC7 (CAGE MATERIAL), - 32CDV13, NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES. NOTE : THE AISI9310/AISI9315 /16NCD13 (GEAR MATERIAL) CAN BE CONFUSED WITH AISI4340/E-40NCD7 (CAGE MATERIAL).	TGB BEARINGS	REFER TO FIGURE FLOWCHART-BEARING MATERIAL DETAILED ANALYSIS (SHEET 1 OF 1). IN CASE OF ONGOING BEARING DEGRADATION, REMOVE AND REPLACE THE TGB. RESET THE ODMS. REFER TO AMM TASK 71-00-00-702-802-G00. DO OIL SYSTEM FLUSHING REFER TO AMM TASK 12-13-11-100-801. IN CASE THE PARTICLES ARE REMNANTS FROM A PREVIOUS AND ALREADY RESOLVED BEARING DEGRADATION SITUATION, THOSE ARE ACCEPTABLE. RESET THE ODMS REFER TO AMM TASK 71-00-00-700-802-G00

ICN LEAP1B 790000 D588280042A00301 3101614 S0000846207_V2

Disposition and Identification for Magnetic Material Findings
Figure 610/79-00-00-990-820-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAGNETIC	ODMS	AGB	TGB	A	B	C	FILTER	STARTER	DESCRIPTION	QUANTITY	MATERIAL COMPOSITION FROM LABANALYSIS	POTENTIAL SOURCE	RECOMMENDATION
YES								X	EVIDENCE OF BEARING DEGRADATION: THIN, SHINY, BRIGHT LAMINATED MAGNETIC FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 1 TO 6) OTHER BEARING DEGRADATION SHAPE: -NON-LAMINATED FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 10 AND 11)	REF AMM TASK 80-11-09-200-801-G00	- M50/E-80DCV40 (BEARING MATERIAL), - ALLOY STEEL 52100/100C6 (BEARING MATERIAL) NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES	STARTER BEARINGS	REFER TO AMM TASK 80-11-09-200-801-G00. IN CASE OF STARTER REPLACEMENT DO OIL SYSTEM FLUSHING AMM TASK 12-13-11-100-801
YES	X								HEAT TREAT SCALES: BRITTTLE, DULL CORRODED FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 13)	ANY	- M50NIL/E-13DCNV40 (RACE MATERIAL)	BEARINGS RACEWAY DRAIN HOLES	IF: - THE PARTICLES ARE FOUND WITHIN 30 DAYS SINCE LAST OPERATION. - NO PRESERVATION OVERDUE BETWEEN ODMS INSPECTION AND LAST ENGINE OPERATION. - ENGINE PARAMETERS ARE WITHIN AMM LIMITS AND THERE IS NO OIL FILTER APPROACHING, IMPENDING OR ACTUAL BYPASS. THEN, THE PARTICLES ARE ACCEPTABLE AND THE ENGINE IS SERVICEABLE. RESET ODMS. REFER TO AMM TASK 71-00-00-700-802-G00.
YES			X						SLIVER REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 9)	ANY	AISI4340	RDS BEARING	FOR RDS BEARING PRE SB 72-0258: DO A BSI OF THE RDS BEARING. REFER TO SB 72-0317.
YES	X								STRANDS, CHUNKS, SPHERES AND FLAKES. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 26)	ANY	15-5PH NOTE : THE 15-5PH CAN BE CONFUSED WITH 17-4PH.	BEARING #3 HOUSING	15-5PH ON THE ODMS MAY BE A SIGN OF WEAR ON THE #3 BEARING HOUSING. REFER TO NSV (NON-SYNCHRONOUS VIBRATION) MONITORING SBS FOR FURTHER DETAILS & MONITORING RECOMMENDATIONS. IF NSV SIGNATURE IS WITHIN ACCEPTABLE LIMITS PER NSV MONITORING SBs: IF ENGINE PARAMETERS ARE WITHIN AMM LIMITS AND THERE IS NO OIL FILTER DELTA PRESSURE APPROACHING, IMPENDING OR ACTUAL BYPASS, SERVICEABLE. RESET ODMS. REFER TO AMM TASK 71-00-00-700-802-G00.
YES	X								ANY	ANY	OTHER THAN THE MATERIALS DESCRIBED IN THE REST OF THE TABLE. NOTE: THE AISI9315/16NCD13 (GEAR MATERIAL) CAN BE CONFUSED WITH AISI4340/E-40NCD7 (CAGE MATERIAL) NOTE: THE MATERIAL 32CDV13 CAN BE CONFUSED WITH M50 OR 40CDV12 WITH SOME LAB ANALYSES.	VARIOUS	IF ENGINE PARAMETERS ARE WITHIN AMM LIMITS AND THERE IS NO OIL FILTER DELTA PRESSURE APPROACHING, IMPENDING OR ACTUAL BYPASS, SERVICEABLE. RESET ODMS. REFER TO AMM TASK 71-00-00-700-802-G00.

ICNLEAP1B79000D5882800046A00101
3101615 S0000846208_V1

Disposition and Identification for Magnetic Material Findings
Figure 610/79-00-00-990-820-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAGNETIC	ODMS	AGB	TGB	A	B	C	FILTER	STARTER	DESCRIPTION	QUANTITY	MATERIAL COMPOSITION FROM LABANALYSIS	POTENTIAL SOURCE	RECOMMENDATION
NO	X								ANY	ANY	ANY	VARIOUS	SERVICEABLE CAUTION: DURING THE CHARACTERIZATION OF PARTICLES AS NON-MAGNETIC, MAKE SURE THAT THERE ARE NO OTHER MAGNETIC PARTICLES ATTACHED/EMBEDDED TO THESE PARTICLES. A MATERIAL-NATURE IDENTIFICATION ERROR CAN CAUSE INCORRECT INTERPRETATION OF THE FINDINGS IN RELATION TO THE SERVICEABLE CRITERIA, AND DAMAGE CAN OCCUR.
NO		X	X						SLIVER, PLATELETS, REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 17 AND 18)	UP TO 14 PARTICLES	SILVER	SILVER COATED PARTS	SERVICEABLE CAUTION: DURING THE CHARACTERIZATION OF PARTICLES AS NON-MAGNETIC, MAKE SURE THAT THERE ARE NO OTHER MAGNETIC PARTICLES ATTACHED/EMBEDDED TO THESE PARTICLES. A MATERIAL-NATURE IDENTIFICATION ERROR CAN CAUSE INCORRECT INTERPRETATION OF THE FINDINGS IN RELATION TO THE SERVICEABLE CRITERIA, AND DAMAGE CAN OCCUR.
NO		X	X						SLIVER, PLATELETS, REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 17 AND 18)	MORE THAN 14 PARTICLES	SILVER	SILVER COATED PARTS	DO TWO INSPECTIONS OF THE AFFECTED SCAVENGE SCREEN ONE AFTER THE OTHER WITHIN 100 TO 115FH EACH. IF LESS THAN 14 SILVER PARTICLES ARE FOUND IN TOTAL AFTER DOING THE TWO INSPECTIONS, STOP THE INSPECTIONS, IT IS SERVICEABLE. IF MORE THAN 14 SILVER PARTICLES ARE FOUND IN TOTAL AFTER DOING THE TWO INSPECTIONS, DO AGAIN THE TWO INSPECTIONS ONE AFTER THE OTHER WITHIN 100 TO 115FH EACH. IF OTHER MATERIAL THAN SILVER IS FOUND DURING THESE INSPECTIONS, FOLLOW AMM TASK 79-00-00-220-801-G00. CAUTION: DURING THE CHARACTERIZATION OF PARTICLES AS NON-MAGNETIC, MAKE SURE THAT THERE ARE NO OTHER MAGNETIC PARTICLES ATTACHED/ EMBEDDED TO THESE PARTICLES. A MATERIAL-NATURE IDENTIFICATION ERROR CAN CAUSE INCORRECT INTERPRETATION OF THE FINDINGS IN RELATION TO THE SERVICEABLE CRITERIA, AND DAMAGE CAN OCCUR.
NO				X					SLIVER, PLATELETS, REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 17 AND 18)	ANY	SILVER	SILVER COATED PARTS	1. WITHIN 10 CYCLES OR 30 FLIGHT HOURS, WHICHEVER COMES FIRST, DO THE FOLLOWING 2. REVIEW MAINTENANCE RECORDS FOR RECENT MAINTENANCE ACTIONS OR OPERATIONAL EVENTS THAT COULD ACCOUNT FOR THE SHIFTS NOTED 3. DO A GENERAL VISUAL INSPECTION OF THE ENGINE OIL LEVEL SENSOR AND/OR HARNESS 4. INSPECT THE A-SUMP SCAVENGE SCREEN (79-21-04-211-801-A) A) IF YOU FOUND SILVER PARTICLES, CLEAN THE A-SUMP SCAVENGE SCREEN FROM ALL SILVER PARTICLES AND THEN GO TO 5. B) IF THERE IS NO SILVER PARTICLES IN THE A-SUMP SCAVENGE SCREEN, GO TO 12 5. DO AN ENGINE GROUND RUN DURING 15 MIN. (71-00-00-790-804-A) 6. INSPECT THE A-SUMP SCAVENGE SCREEN (79-21-04-211-801-A) AND CLEAN THE A-SUMP SCAVENGE SCREEN FROM ALL SILVER PARTICLES IF ANY 7. DRAIN THE OIL TANK (12-13-11-600-803-A) 8. REPLACE THE OIL FILTER (79-21-07-000400-801-A) 9. REFILL THE OIL TANK (12-13-11-600-801-A) 10. DO A SECOND ENGINE GROUND RUN DURING 15 MIN (71-00-00-790-804-A) 11. INSPECT THE A-SUMP SCAVENGE SCREEN TO DETERMINE IF THERE IS LESS PARTICLES THAN DURING THE PREVIOUS INSPECTION (79-21-04-211-801-A) A) CLEAN THE A-SUMP SCAVENGE SCREEN FROM ALL SILVER PARTICLES IF ANY B) IF THERE IS MORE PARTICLES THAN DURING THE PREVIOUS INSPECTION, REPEAT STEPS 5 TO 11 C) IF THERE IS LESS PARTICLES THAN DURING THE PREVIOUS INSPECTION, RETURN INTO SERVICE AND DO A REPETITIVE INSPECTION OF A-SUMP SCAVENGE SCREEN DETAILED BELOW: (1) RE-INSPECT A-SUMP SCAVENGE SCREEN WITHIN 25FH+/-10FH OR 50FH+/-20FH(*) (2) REMOVE THE PARTICLES FROM THE SCREEN AND RETURN INTO SERVICE (3) RE-INSPECT A-SUMP SCAVENGE SCREEN WITHIN 25FH+/-10FH OR 50FH+/-20FH(*) AND STOP AFTER 3 TIMES WITHOUT FINDING (*):25FH+/-10FH FOR ENGINES WHICH EXPERIENCED A LOW OIL LEVEL INDICATION ON THE COCKPIT DURING LAST FLIGHT.50FH+/-20FH FOR ENGINES WHICH DID NOT EXPERIENCED A LOW OIL LEVEL INDICATION ON THE COCKPIT DURING LAST FLIGHT. 12. DO THE TESTS THAT ARE GIVEN IN THE POWER PLANT TEST REFERENCE TABLE (71-00-00-800-804) CAUTION: DURING THE CHARACTERIZATION OF PARTICLES AS NON-MAGNETIC, MAKE SURE THAT THERE ARE NO OTHER MAGNETIC PARTICLES ATTACHED/EMBEDDED TO THESE PARTICLES. A MATERIAL-NATURE IDENTIFICATION ERROR CAN CAUSE INCORRECT INTERPRETATION OF THE FINDINGS IN RELATION TO THE SERVICEABLE CRITERIA, AND DAMAGE CAN OCCUR.

ICN-LEAP1B-790000-D588280004-3A00301
3101621 S0000846209_V1

Disposition and Identification for Non-magnetic Material Findings
Figure 611/79-00-00-990-821-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAGNETIC	ODMS	AGB	TGB	A	B	C	FILTER	STARTER	DESCRIPTION	QUANTITY	MATERIAL COMPOSITION FROM LAB ANALYSIS	POTENTIAL SOURCE	RECOMMENDATION
NO					X	X			SLIVER, PLATELETS, REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 17 AND 18)	ANY	SILVER	BEARING CAGE SILVER COATING	INSPECT THE AFFECTED SCAVENGE SCREEN IN 50-75 FH. IF OTHER MATERIAL THAN SILVER IS FOUND DURING THE INSPECTION, FOLLOW AMM TASK 79-00-00-220-801-G00 CAUTION: DURING THE CHARACTERIZATION OF PARTICLES AS NON-MAGNETIC, MAKE SURE THAT THERE ARE NO OTHER MAGNETIC PARTICLES ATTACHED/EMBEDDED TO THESE PARTICLES. A MATERIAL-NATURE IDENTIFICATION ERROR CAN CAUSE INCORRECT INTERPRETATION OF THE FINDINGS IN RELATION TO THE SERVICEABLE CRITERIA, AND DAMAGE CAN OCCUR.
NO		X	X						MACHINED CURLS, ELONGATED CURLS SLIVER CHUNK REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 25)	ANY	ALUMINIUM	AGB OR TGB HOUSING THREAD	REMOVE AND REPLACE AGB OR TGB DO OIL SYSTEM FLUSHING AMM TASK 12-13-11-100-801
NO		X							THIN LAMINATED FLAKES, CAGE FRAGMENTS	4 OR MORE PARTICLES, <0.010 INCH (0.25mm) IN LARGEST DIMENSION OR ANY SINGLE PARTICLE >0.010 INCH (0.25 mm) IN ANY DIMENSION	UA10M (BRONZE) (CAGE MATERIAL)	AGB BEARINGS CAGES	REMOVE AND REPLACE THE AGB DO OIL SYSTEM FLUSHING AMM TASK 12-13-11-100-801
NO				X					CHUNKS/SHARDS (BLACK)	ANY	CERAMIC	BEARING #3B BALLS	REMOVE AND REPLACE THE ENGINE.
NO		X	X	X	X	X			PIECES, DARK, BLACK, SOFT. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 27 AND 28)	ANY	FLUOROCARBON	O-RING	SERVICEABLE IF : -ENGINE PARAMETERS ARE WITHIN AMM LIMITS AND THERE IS NO OIL FILTER DELTA PRESSURE APPROACHING, IMPENDING OR ACTUAL BYPASS. -THERE IS NO EXTERNAL OIL LEAK. CAUTION: MAKE SURE THAT THE PARTICLES DO NOT CORRESPOND TO OIL LEVEL RING WITH A LIP SEAL SHAPE. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 29)
NO		X	X	X	X	X			PIECES, DARK, BLACK, HARD. REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 19 AND 20)	ANY	CARBON	COKED OIL, SEGMENTED RADIAL SEAL, ROTATING SEAL	SERVICEABLE IF : -ENGINE PARAMETERS ARE WITHIN AMM LIMITS AND THERE IS NO OIL FILTER DELTA PRESSURE APPROACHING, IMPENDING OR ACTUAL BYPASS. -THERE IS NO EXTERNAL OIL LEAK.
NO		X	X	X	X	X			ANY	ANY	OTHER THAN THE MATERIALS DESCRIBED IN THE REST OF THE TABLE	VARIOUS	IF ENGINE PARAMETERS ARE WITHIN AMM LIMITS AND THERE IS NO OIL FILTER DELTA PRESSURE APPROACHING, IMPENDING OR ACTUAL BYPASS, SERVICEABLE.
NO							X		ELONGATED FRAGMENT, THIN, BRITTLE, BROWN PARTICLES REFER TO FIGURE 606/79-00-00-990-813-G00 (SHEET 30)	ANY	NON-METALLIC (ORGANIC) FRAGMENTS (WITH OR WITHOUT MINERAL GLASS FIBERS), RICH IN CARBON, OXYGEN, NITROGEN, SILICIUM AND CHLORINE	GLUE FROM OIL FILTER CARTRIDGE	REMOVE AND REPLACE THE OIL FILTER CARTRIDGE

ICNLEAP1B79000D5882800048A00101
3101625 S0000846210_V1

Disposition and Identification for Non-magnetic Material Findings
Figure 611/79-00-00-990-821-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-211-801-G00

7. Engine Oil System (Fluids (Fuel or Skydrol) and Solids Contamination) Detailed Inspection

(Figure 612)

A. General

(1) This task gives the instructions to examine the engine oil system for fluid (fuel or Skydrol) and solid contamination.

B. References

Reference	Title
12-13-11-170-801	Oil Tank Flushing (P/B 301)
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-11-000-801-G00	Servo Fuel Heater Removal (P/B 401)
73-11-11-400-801-G00	Servo Fuel Heater Installation (P/B 401)
79-21-01-000-801-G00	Main Fuel/Oil Heat Exchanger Removal (P/B 401)
79-21-01-400-801-G00	Main Fuel/Oil Heat Exchanger Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-193	Container
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Prepare for the Inspection

SUBTASK 79-00-00-860-023-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-00-00-010-003-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

H. Engine Oil System (Fluids (Fuel or Skydrol) and Solids Contamination) Detailed Inspection

SUBTASK 79-00-00-010-004-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Drain a small quantity of oil from the oil tank [1]:

(a) Put a 1.32 gal (5.00 l) oil resistant container, STD-193, below the oil tank [1].

(b) Cut the safety wire [5] from the drain plug [3].

(c) Use a clean cotton wiper, G00034, to remove all contamination around the drain plug [3].

(d) Remove the drain plug [3] from the oil tube [2].

(e) Discard the preformed packing [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Drain a small quantity of oil into a clean sample bottle.

(g) Put a protective cap to stop the drain.

(h) Install the drain plug [3] on the oil tube [2]:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [3] with oil, D00599 [CP2442].

3) Lubricate the new preformed packing [4] with pure mineral vaseline, D00672 [CP5070].

4) Install the preformed packing [4] in the groove on the drain plug [3].

5) Remove the protective cap from the oil tube [2].

6) Install the drain plug [3] on the oil tube [2].

7) Tighten the drain plug [3] to 177 in-lb (20 N·m) to 204 in-lb (23 N·m).

8) Install safety wire, G02495 [CP8002], on the drain plug [3].

SUBTASK 79-00-00-211-003-G00

(2) Examine the oil sample for contamination.

SUBTASK 79-00-00-170-001-G00

(3) If you find fluid (Skydrol) or solid contamination, do these steps:

(a) Flush the oil tank (TASK 12-13-11-170-801).

(b) Do the steps again to get the oil sample.

(c) If the sample is clear of contamination, return the engine to service.

(d) If the sample shows contamination, send the sample for contamination analysis.

SUBTASK 79-00-00-960-002-G00

(4) If you find fuel in the engine oil, do these steps:

(a) Replace the main fuel/oil heat exchanger (TASK 79-21-01-000-801-G00 and TASK 79-21-01-400-801-G00).

(b) Replace the servo fuel heater (TASK 73-11-11-000-801-G00 and TASK 73-11-11-400-801-G00).

(c) Flush the oil tank (TASK 12-13-11-170-801).

(d) Do the steps again to get the oil sample.

(e) If the sample shows contamination, flush the engine oil system again until the engine oil does not show contamination (TASK 12-13-11-170-801).

SUBTASK 79-00-00-610-002-G00

(5) If you did not find any contamination, do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 79-00-00-100-001-G00

(6) Clean the area and clear all tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-410-003-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 79-00-00-860-024-G00

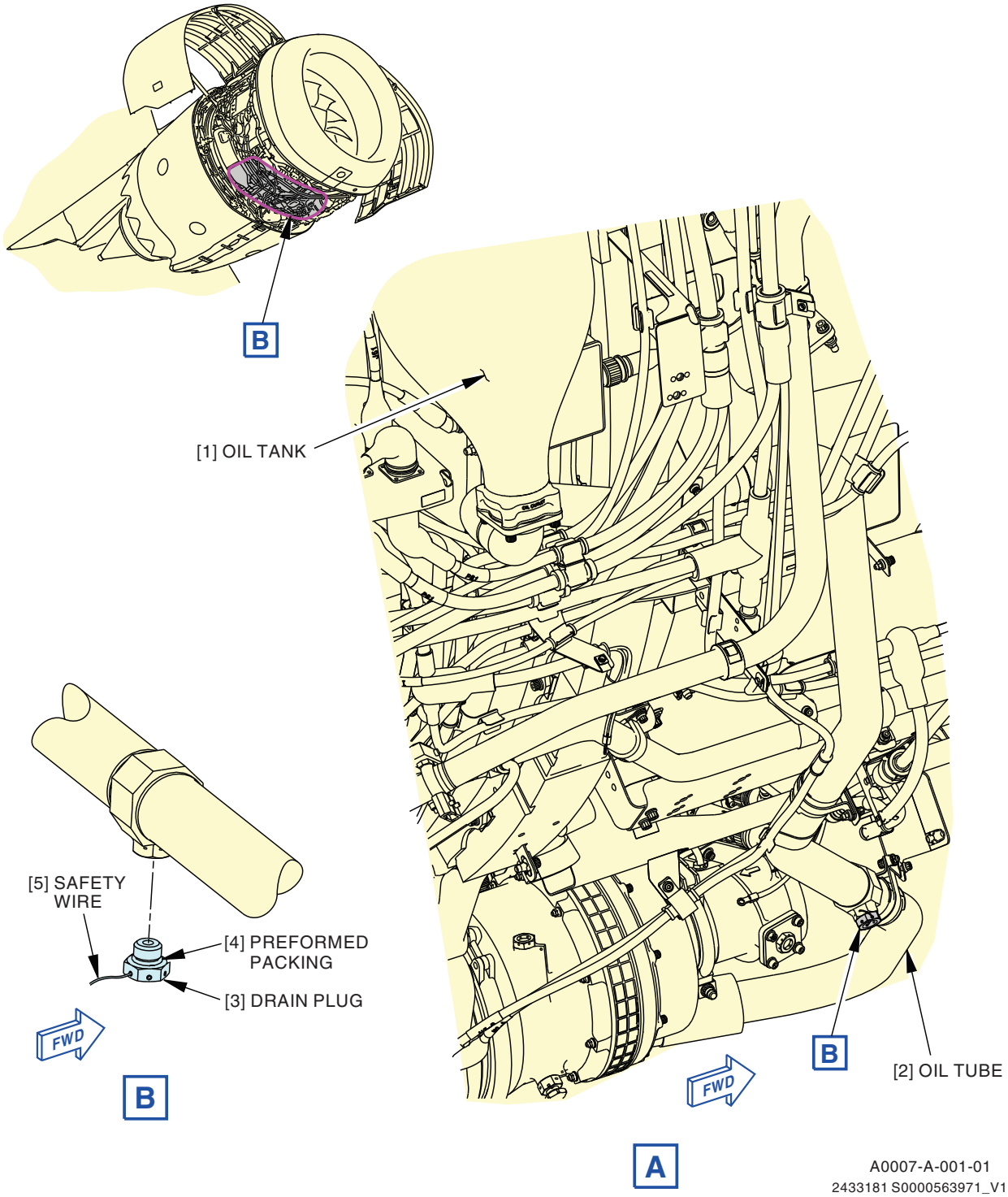
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Oil System Fluids (Fuel or Skydrol) and Solids Contamination Detailed Inspection
Figure 612/79-00-00-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-200-802-G00

8. Engine Oil System (Wear Metal Contamination) Inspection

(Figure 613)

A. General

(1) This task provides the instructions on how to remove engine oil for the oil system wear metal contamination inspection.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

G. Prepare for the Inspection

SUBTASK 79-00-00-860-019-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-00-00-010-005-G00

(2) Open the applicable access panels:

Number	Name/Location
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Engine Oil System (Wear Metal Contamination) Inspection

SUBTASK 79-00-00-010-006-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Open the oil tank filler cap [2] as follows:

(a) Lift the oil tank filler cap handle [3] to the open position.

(b) Open the oil tank filler cap [2].

SUBTASK 79-00-00-280-001-G00

(2) Use an oil sample kit to get an oil sample as follows:

(a) Push the siphon tube [5] down into the oil tank [1] until it touches the bottom of the oil strainer.

NOTE: Do not let the siphon tube stay against the bottom of the oil strainer. The siphon tube can become full with unwanted materials that are on the bottom of the oil strainer.

(b) Move the siphon tube [5] up slightly from the bottom of the strainer.

(c) Use a siphon procedure to remove a sample of the oil from the oil tank [1].

(d) Do these steps to get the oil sample with a siphon tube [5] and an attached plastic bottle [6] (Preferred Method):

NOTE: Oil sample kits come with a plastic bottle attached to a siphon tube or a siphon tube without the attached plastic bottle.

1) Slowly compress the plastic bottle [6] with your hand to move the air out.

2) Slowly release the hand pressure on the plastic bottle [6] to move oil up inside the siphon tube [5] and into plastic bottle [6].

3) Remove the plastic bottle [6] and the siphon tube [5] from the oil tank [1].

4) Put the end of the siphon tube [5] in the oil sample bottle.

5) Slowly move the plastic bottle [6] until all the oil in the siphon tube [5] is in the oil sample bottle.

6) Put a cap on the oil sample bottle and tighten the cap to seal the bottle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

7) Discard the plastic bottle [6] and the siphon tube [5].

NOTE: Use a new plastic bottle and siphon tube with each oil sample. This will make sure that the oil sample shows the present condition of the oil.

(e) Do these steps to get the oil sample with a siphon tube [5] without an attached plastic bottle [6] (Alternate Method):

NOTE: Oil sample kits come with a plastic bottle attached to a siphon tube or a siphon tube without the attached plastic bottle.

1) Put your thumb over the end of the siphon tube [5].
2) Remove the siphon tube [5] from the oil tank [1].
3) Put the end of the siphon tube [5] in the oil sample bottle.
4) Slowly remove your thumb from the end of the siphon tube [5] until all oil in the siphon tube [5] is in the oil sample bottle.
5) Put a cap on the oil sample bottle and tighten it to seal the bottle.
6) Discard the siphon tube [5].

NOTE: Use a new siphon tube with each oil sample. This will make sure that the oil sample shows the present condition of the oil.

SUBTASK 79-00-00-970-001-G00

(3) Write the quantity of oil added for the oil consumption check on a oil consumption sheet.

SUBTASK 79-00-00-610-001-G00

(4) If it is necessary, replenish engine oil (Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801).

SUBTASK 79-00-00-970-002-G00

(5) Make sure that you write the data that follows on the label for oil sample bottle:

NOTE: Make sure that you keep a copy of the label information before you send the sample to the laboratory.

(a) The airplane registration number
(b) The engine S/N
(c) The engine position
(d) The date of the sample
(e) The (type and quantity) of oil that is added.

SUBTASK 79-00-00-910-002-G00

(6) Send the sample to the laboratory as soon as possible to get good results.

SUBTASK 79-00-00-410-004-G00

(7) Close the oil tank filler cap [2] as follows:

(a) Examine the preformed packing [4] for damage.
(b) If you find damage, replace the preformed packing [4] as follows:
1) Remove the preformed packing [4] from the groove of the oil tank filler cap [2].
2) Lubricate a new preformed packing [4] with engine oil, D00599 [CP2442].
3) Install the new preformed packing [4] in the groove of the oil tank filler cap [2].
(c) Push the oil tank filler cap handle [3] to the CLOSED position.
(d) Make sure that the oil tank filler cap [2] is tightly closed and locked.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-00-00-160-002-G00

(8) Clean the work area and remove the tools you use for the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-410-005-G00

(1) Close the applicable access panels:

Number	Name/Location
414AR	Oil Tank Access / Pressure Relief Door, Engine 1
424AR	Oil Tank Access / Pressure Relief Door, Engine 2

(a) Make sure the latches are fully closed.

1) Press the aft “PUSH CLOSE” button to fully close the latch.

2) If the latch appears to be jammed in the fully open position and does not close, press the forward “PUSH OPEN” button to reset the latch.

NOTE: Do not use a prying tool to force the latch closed.

3) Repeat the steps as necessary to fully close the latch.

NOTE: The latch may not close if the aft “PUSH CLOSE” button is released before the latch is fully closed.

SUBTASK 79-00-00-860-020-G00

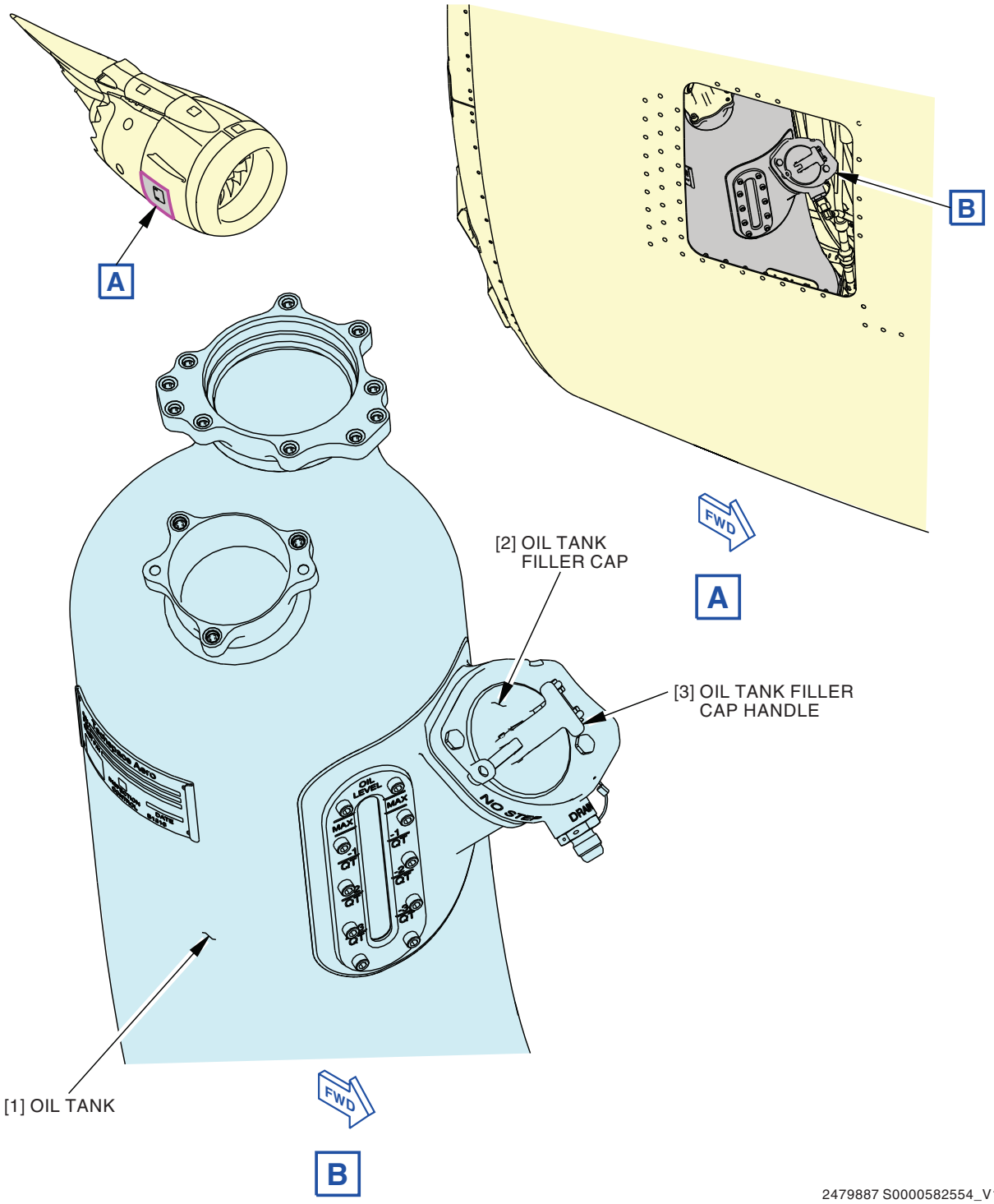
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

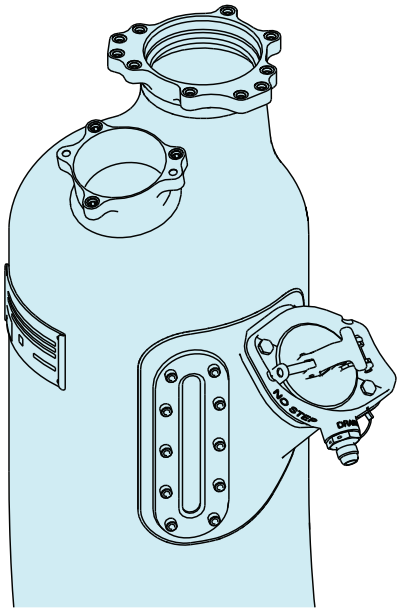
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

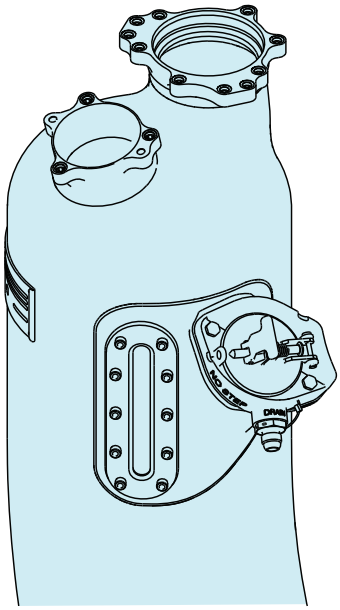


Engine Oil System (Wear Metal Contamination) Inspection
Figure 613/79-00-00-990-810-G00 (Sheet 1 of 2)

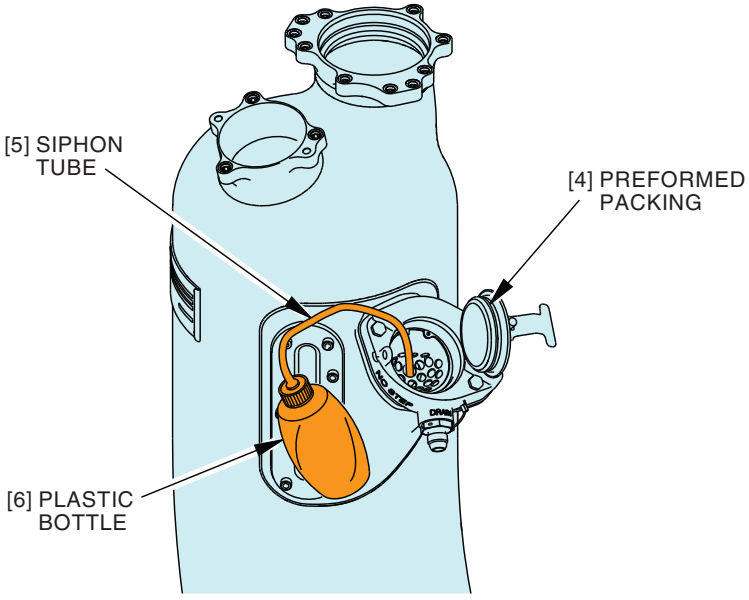
ECCN 9E991 BOEING PROPRIETARY - See title page for details



FILLER CAP HANDLE
(IN LOCKED POSITION)



FILLER CAP HANDLE
(IN UNLOCKED POSITION)



FILLER CAP
(OPEN)

2479903 S0000582555_V1

Engine Oil System (Wear Metal Contamination) Inspection
Figure 613/79-00-00-990-810-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-200-803-G00

9. ODMS Fault Check

A. General

(1) This task provides the instructions on how to do a message fault check of the Oil Debris Monitoring System (ODMS).

B. References

Reference	Title
46-13-00-480-801	Maintenance Laptop Connection (P/B 201)
46-13-02-700-806	Fault History Check (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-11321	Device - Universal Maintenance (UMD) Part #: G45004-44 Supplier: 81205 Part #: G45004-45 Supplier: 81205

D. Location Zones

Zone	Area
211	Flight Compartment - Left

E. Prepare for the Check

SUBTASK 79-00-00-860-025-G00

(1) If the Multi Function Display (MFD) is used to do the fault test, do these steps:

(a) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

(b) Select the L or C, on the P61 maintenance/BITE panel.

SUBTASK 79-00-00-480-001-G00

(2) If the Portable Maintenance Device (PMD) is used to do the fault test, do this task: Maintenance Laptop Connection, TASK 46-13-00-480-801.

F. ODMS Fault Check

SUBTASK 79-00-00-280-002-G00

(1) If the MAX Display System (MDS) display is used, do the fault check as follows:

(a) Use the tabber device, on the P61 maintenance/BITE panel, and select the ONBD MAINT on the MAX display.

(b) Select the FAULT HISTORY from the EXTENDED MAINT menu (TASK 46-13-02-700-806).

(c) Do a check for one or more maintenance messages in chapter 71-80 left or right engine in the last three flight legs (LEG 0, LEG-1, and LEG-2).

1) If a fault message is shown, refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(d) After the check, exit the maintenance menu on the MDS.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-00-00-280-003-G00

(2) If a portable universal maintenance device, SPL-11321, is used, do the fault check as follows (TASK 46-13-02-700-806):

(a) Select the Onboard Maintenance Function (OMF).

(b) From the onboard maintenance menu, select EXTENDED MAINTENANCE.

(c) From the extended maintenance menu select FAULT HISTORY for chapter 71-80 left or right engine.

(d) Do a check for one or more maintenance messages in the last three flight legs (LEG 0, LEG-1, and LEG-2).

1) If a fault message is shown, refer to the IFIM and do the applicable procedure(s).

(e) After the check, exit the maintenance menu on the laptop.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-080-002-G00

(1) If it is necessary, disconnect the ethernet connection from the Onboard Network System (ONS).

SUBTASK 79-00-00-080-003-G00

(2) If it is necessary, disconnect the portable universal maintenance device, SPL-11321.

——— END OF TASK ———

TASK 79-00-00-200-804-G00

10. Drain Tubes Inspection and Cleaning

(Figure 614, Figure 615, Figure 616)

A. General

(1) This task gives instructions to examine the drain tubes for the coking condition.

B. References

Reference	Title
73-11-05-000-801-G00	Fuel Nozzle Removal (P/B 401)
73-11-05-400-801-G00	Fuel Nozzle Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Inspection

SUBTASK 79-00-00-210-014-G00

(1) Do these steps to make sure that the ENGINE START switch and START LEVER switch are not operated:

(a) On the P5 overhead panel, make sure the applicable ENGINE START switch is in the OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, make sure that the applicable START LEVER switch is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

F. Drain Tubes Inspection

SUBTASK 79-00-00-200-007-G00

(1) Criteria for oil coke build-up:

(a) The oil coke build-up inside the drain tubes is not acceptable and must be cleaned if at least one section of any length of one tube:

1) is covered 360° by oil coke build-up (brown surface varnish is accepted), or

2) is not circular anymore because of one-sided heavy oil coke build-up.

(b) Refer to Figure 615 and Figure 616 for acceptable and non-acceptable conditions.

SUBTASK 79-00-00-211-011-G00

(2) B-sump drain

(a) Find the B-sump drain port in the core drain mast [1] (View A, Figure 614).

(b) Put the BSI (Borescope Inspection) equipment into the engine through the B-Sump drain mast port.

NOTE: A 4-mm borescope tip is the recommended size to do the BSI procedure. A 6-mm borescope tip is the maximum size that can be used to do the BSI procedure.

(c) Do a BSI up the B-sump drain through the B-sump external main drain oil tube [2] and B-sump internal main drain oil tube [3] until reaching the B-sump cavity to examine for oil coke build-up, do the following steps (View C, Figure 614):

1) If oil coke build-up is acceptable (defined in F.(1)) in the sections of the B-sump external main drain oil tube [2] and B-sump internal main drain oil tube [3] to the B-sump cavity, the inspection is complete and continued operation is permitted.

NOTE: Small oil traces on the Inside Diameter (ID) of the B-sump external main drain oil tube [2] and B-sump internal main drain oil tube [3] are permitted.

a) Do the steps in the Put the Airplane Back to Its Usual Condition paragraph.

2) If oil coke build-up is not acceptable (defined in F.(1)) in the sections of the B-sump external main drain oil tube [2] and/or of the B-sump internal main drain oil tube [3] to the B-sump cavity, do the following steps:

a) Remove the B-sump external main drain oil tube [2] (TASK 73-11-05-000-801-G00).

b) Clean oil coke build-up in the ID of the removed B-sump external main drain oil tube [2] with a nylon tube brush.

c) Encase the steel wire elements of the nylon tube brush with rubber or plastic coating to prevent scratching.

d) Clean oil coke build-up in the ID of the B-sump internal main drain oil tube [3] with a nylon tube brush. If it is possible, try to remove oil coke build-up out of the B-sump internal main drain oil tube [3] and do not push it in the engine.

e) Do a BSI or a visual inspection to make sure that oil coke build-up in the B-sump internal main drain oil tube [3] is acceptable up to the B-sump cavity.

NOTE: Small oil traces on the ID of the B-sump internal main drain oil tube [3] are permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

f) If there is unwanted material found or if coke build-up is still not acceptable, do steps F(2).2).d and F(2).2).e again.

g) Install the B-sump external main drain oil tube [2] (TASK 73-11-05-400-801-G00).

SUBTASK 79-00-00-211-012-G00

(3) C-sump drain

(a) Find the C-Sump drain port in the core drain mast [1] (View A, Figure 614).

(b) Put the BSI equipment into the engine through the C-sump drain mast port.

NOTE: A 4-mm borescope tip is the recommended size to do the BSI procedure. A 6-mm borescope tip is the maximum size that can be used to do the BSI procedure.

(c) Do a BSI up the C-sump drain through the C-sump external main drain oil tube [4] and C-sump internal main drain oil tube [5] until reaching the C-sump cavity to examine for oil coke build-up (View C, Figure 614).

1) If oil coke build-up is acceptable (defined in F.(1)) in the sections of the C-sump external main drain oil tube [4] and C-sump internal main drain oil tube [5] to the C-sump cavity, the inspection is complete and continued operation is permitted.

NOTE: Small oil traces on the ID of the C-sump external main drain oil tube [4] and C-sump internal main drain oil tube [5] are permitted.

2) If oil coke build-up is not acceptable (defined in F.(1)) in the sections of the C-sump external main drain oil tube [4] and/or of the C-sump internal main drain oil tube [5] to the C-sump cavity, do the following steps:

a) Remove the C-sump external main drain oil tube [4] (TASK 73-11-05-000-801-G00).

b) Clean oil coke build-up in the ID of the removed C-sump external main drain oil tube [4] with a nylon tube brush.

<1> Encase the steel wire elements of the nylon tube brush with rubber or plastic coating to prevent scratching.

c) Clean oil coke build-up in the ID of the C-sump internal main drain oil tube [5] with a nylon tube brush. If it is possible, try to remove oil coke build-up out of the C-sump internal main drain oil tube [5] and do not push it in the engine.

d) Do a BSI or a visual inspection to make sure that oil coke build-up in the C-sump internal main drain oil tube [5] is acceptable up to the C-sump cavity.

NOTE: Small oil traces on the ID of the C-sump internal main drain oil tube [5] are permitted.

e) If there is unwanted material found or if coke build-up is still not acceptable, do steps F(3).2).d and F(3).2).e again.

f) Install the C-sump external main drain oil tube [4] (TASK 73-11-05-400-801-G00).

SUBTASK 79-00-00-211-013-G00

(4) Auxiliary drain

(a) Find the Auxiliary drain port in the core drain mast [1] (View A, Figure 614).

(b) Put the BSI equipment into the engine through the Auxiliary drain mast port.

NOTE: A 4-mm borescope tip is the recommended size to do the BSI procedure. A 6-mm borescope tip is the maximum size that can be used to do the BSI procedure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do a BSI up the Auxiliary drain through the auxiliary external main drain oil tube [6] and auxiliary internal main drain oil tube [7] until reaching the Turbine Center Frame (TCF) cavity to examine for oil coke build-up, do the following steps (View C, Figure 614):

1) If oil coke build-up is acceptable (defined in F.(1)) in the sections of the auxiliary external main drain oil tube [6] and the auxiliary internal main drain oil tube [7] to the TCF cavity, the inspection is complete and continued operation is permitted.

NOTE: Small oil traces on the ID of the auxiliary external main drain oil tube [6] and auxiliary internal main drain oil tube [7] are permitted.

2) If oil coke build-up is not acceptable (defined in F.(1)) in the sections of the auxiliary external main drain oil tube [6] and/or of the auxiliary internal main drain oil tube [7] to the TCF cavity, do the following steps:

a) Remove the auxiliary external main drain oil tube [6] (TASK 73-11-05-000-801-G00).

b) Clean oil coke build-up in the ID of the removed auxiliary external main drain oil tube [6] with a nylon tube brush.

<1> Encase the steel wire elements of the nylon tube brush with rubber or plastic coating to prevent scratching.

c) Clean oil coke build-up in the ID of the auxiliary internal main drain oil tube [7] with a nylon tube brush. If it is possible, try to remove oil coke build-up out of the auxiliary internal main drain oil tube [7] and do not push it in the engine.

d) Do a BSI or a visual inspection to make sure that oil coke build-up in the auxiliary internal main drain oil tube [7] is acceptable up to the TCF cavity.

NOTE: Small oil traces on the ID of the auxiliary internal main drain oil tube [7] are permitted.

e) If there is unwanted material found or if coke build-up is still not acceptable, do steps F(4).2).d and F(4).2).e again.

f) Install the auxiliary external main drain oil tube [6] (TASK 73-11-05-400-801-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 79-00-00-840-001-G00

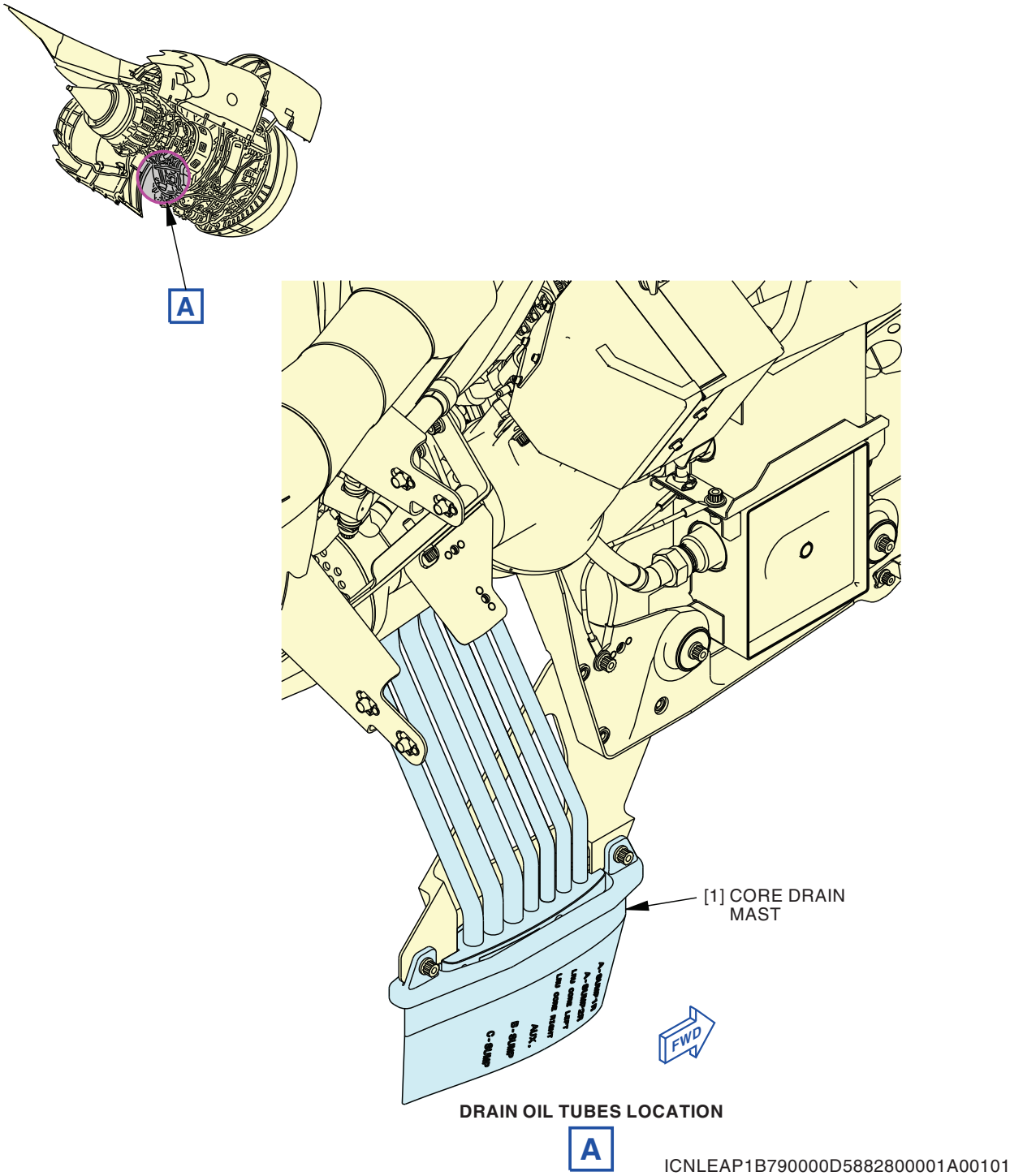
(1) Do these steps to remove the DO NOT OPERATE tag, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, from the applicable START LEVER switch.

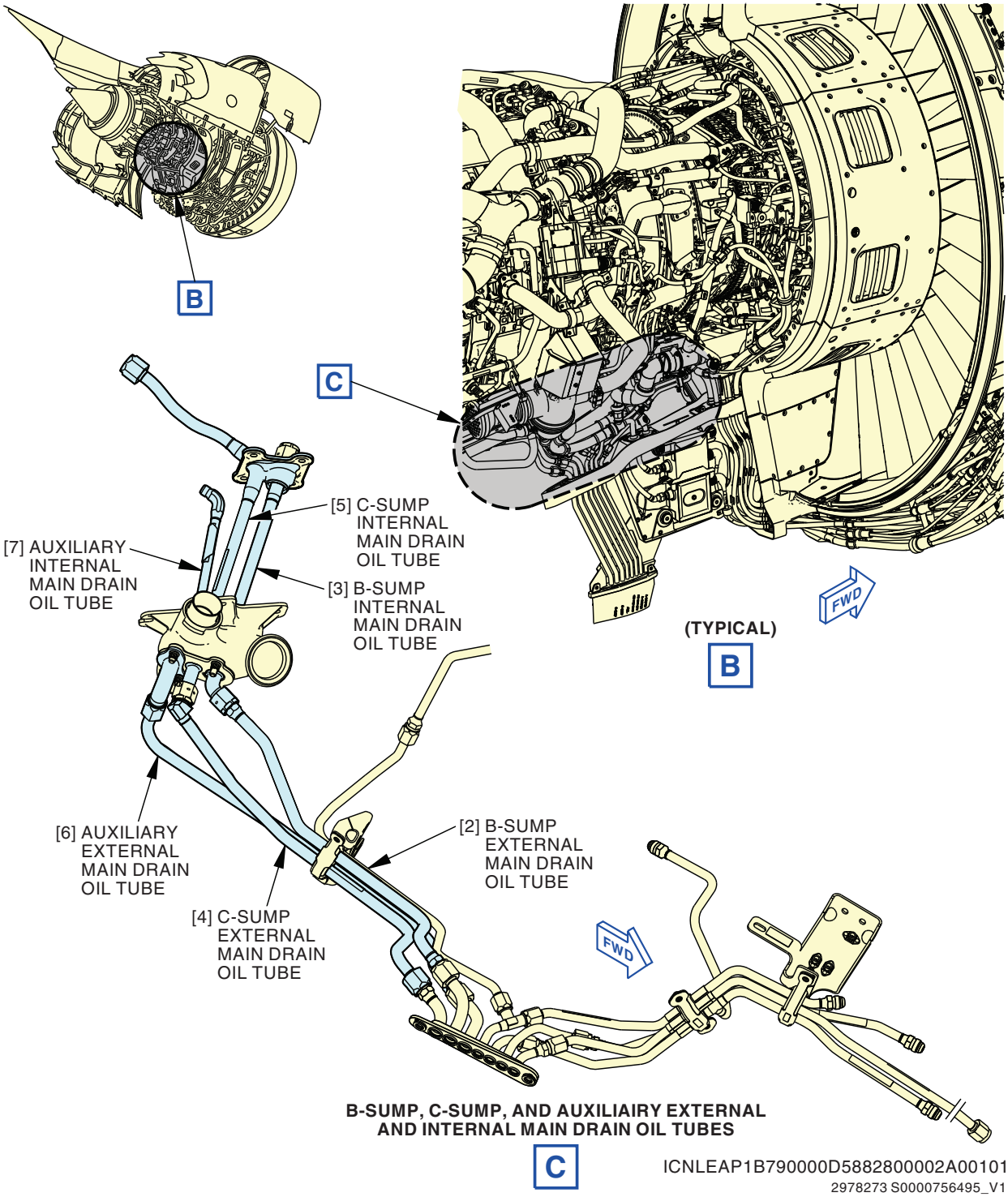
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Drain Oil Tubes Inspection
Figure 614/79-00-00-990-814-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Drain Oil Tubes Inspection
Figure 614/79-00-00-990-814-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





**BROWN SURFACE VARNISH EXAMPLE
(ACCEPTABLE CONDITION)**

2978275 S0000756503_V1

**Brown Surface Varnish Inspection
Figure 615/79-00-00-990-815-G00**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**ONE-SIDED COKE BUILD-UP EXAMPLE
(ACCEPTABLE CONDITION)**



**ONE-SIDED HEAVY COKE BUILD-UP EXAMPLES
(NON-ACCEPTABLE CONDITION)**

2978276 S0000756515_V1

**One-sided Build-up Inspection
Figure 616/79-00-00-990-816-G00**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE OIL - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has the maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks prepare the airplane for flight with components that are inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. This procedure has these tasks:

(1) DDG 79-21-01 Engine Oil Debris Monitoring System Inoperative - Preparation

(2) DDG 79-31-01 Oil Quantity Indicating Systems Inoperative - Preparation.

TASK 79-00-00-040-802-G00

2. DDG 79-31-01 Oil Quantity Indicating Systems Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the engine oil quantity indication systems inoperative.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Oil Quantity Indicating Systems Inoperative

SUBTASK 79-00-00-612-001-G00

(1) At each refueling, do the steps that follow:

(a) Examine the oil level (TASK 71-00-00-800-802-G00).

(b) If necessary, fill the engine oil tank to the recommended capacity (TASK 12-13-11-600-801).

1) Within 30 minutes of the engine shutdown, replenish the engine oil tank again if necessary (TASK 12-13-11-600-801).

(c) Visually examine the applicable engine area for the sign of oil leakage.

(d) Review engine records and verify that the oil usage is within the engine manufacturer's limits or operator's approved limits (if more restrictive).

SUBTASK 79-00-00-800-001-G00

(2) Do a check of the aircraft flight log to make sure that no faults were recorded which relate to the oil pressure and oil temperature indications and that the indications were within limits shown in Engine Operation Limits, TASK 71-00-00-800-802-G00.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-00-00-040-804-G00

3. DDG 79-21-01 Engine Oil Debris Monitoring System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the engine Oil Debris Monitoring System (ODMS) inoperative.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
79-00-00-210-804-G00	Oil Debris Monitoring System Sensor Inspection (P/B 601)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Engine Oil Debris Monitoring System Inoperative

SUBTASK 79-00-00-200-002-G00

(1) Do the following task to make sure that the malfunction is in the applicable engine ODMS:

(a) Oil Debris Monitoring System Sensor Inspection, TASK 79-00-00-210-804-G00.

(2) If the engine oil system inspections show no contaminants, then the malfunction is in the ODMS.

(3) If it is necessary, extinguish the MAINT light on the P5-1 panel. Do this task: Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL TANK - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil tank removal

(2) Oil tank installation.

TASK 79-11-01-000-801-G00

2. Oil Tank Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the oil tank.

B. References

Reference	Title
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
79-21-21-000-801-G00	Air Oil Separator Removal (P/B 401)
79-31-01-000-801-G00	Oil Level Sensor Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-11-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-11-01-010-001-G00

(2) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Oil Tank Removal

SUBTASK 79-11-01-680-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Drain the oil tank [1] as follows:

(a) Put the oil resistant container (5 gal)(19 Liter), STD-1055 below the oil tank [1].

(b) Drain the oil tank [1] (TASK 12-13-11-600-803).

SUBTASK 79-11-01-020-001-G00

(2) Do this task: Air Oil Separator Removal, TASK 79-21-21-000-801-G00.

SUBTASK 79-11-01-010-002-G00

(3) Do this task: Oil Level Sensor Removal, TASK 79-31-01-000-801-G00.

SUBTASK 79-11-01-020-002-G00

(4) If necessary, disconnect the scupper drain tube [4] from the oil tank [1] as follows:

(a) Cut and remove the safety wire [6] from the coupling nut [5] of the scupper drain tube [4].

(b) Loosen the coupling nut [5] of the scupper drain tube [4].

(c) Disconnect the scupper drain tube [4] from the oil tank [1].

(d) Install the protective caps on the openings.

SUBTASK 79-11-01-020-003-G00

(5) Disconnect the oil tube [8] from the oil tank [1] as follows:

NOTE: Be careful. There is approximately 0.13 gal (0.5 liter) of oil remain in the oil tank [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the four bolts [9] from the oil tube [8].

(b) Disconnect the oil tube [8] from the oil tank [1].

(c) Remove and discard the gasket seal [10] from the oil tube [8].

(d) Install the protective caps on the openings.

SUBTASK 79-11-01-020-004-G00

(6) Disconnect the drain tube [11] from the oil tank [1] as follows:

(a) Loosen the coupling nut [12] of the drain tube [11].

(b) Remove the bolt [13] from the clamp [14] that attaches the drain tube [11] to the bracket [15].

(c) Remove the bolt [16] from the clamp [17] that attaches the drain tube [11] to the bracket [18].

(d) Remove the bolt [19] from the clamp [20] that attaches the drain tube [11] to the bracket [21].

(e) Install the protective caps on the openings.

SUBTASK 79-11-01-020-005-G00

(7) Remove the oil tank [1] from the mounting bracket [23], the mounting bracket [25], and the mounting bracket [27] as follows:

NOTE: It is necessary for two people to do this procedure.

(a) Remove the two bolts [22] that attach the oil tank [1] to the mounting bracket [23].

(b) Remove the two bolts [24] that attach the oil tank [1] to the mounting bracket [25].

(c) Loosen, but do not remove the four bolts [28] that attach the mounting bracket [25] to the engine.

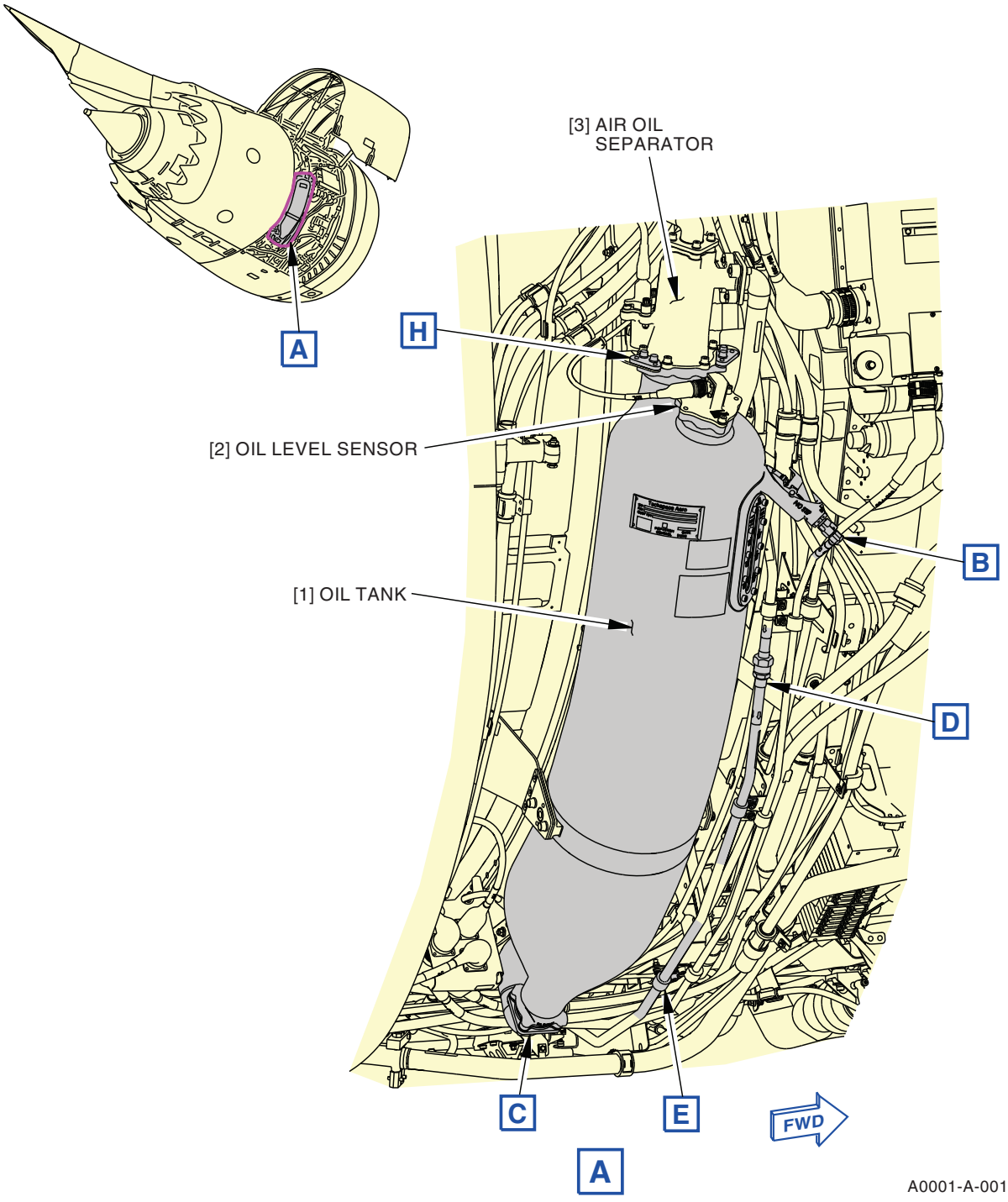
(d) Remove the four bolts [26] that attach the oil tank [1] to the mounting bracket [27].

(e) Remove the oil tank [1] from the engine.

(f) Install the protective covers on the openings.

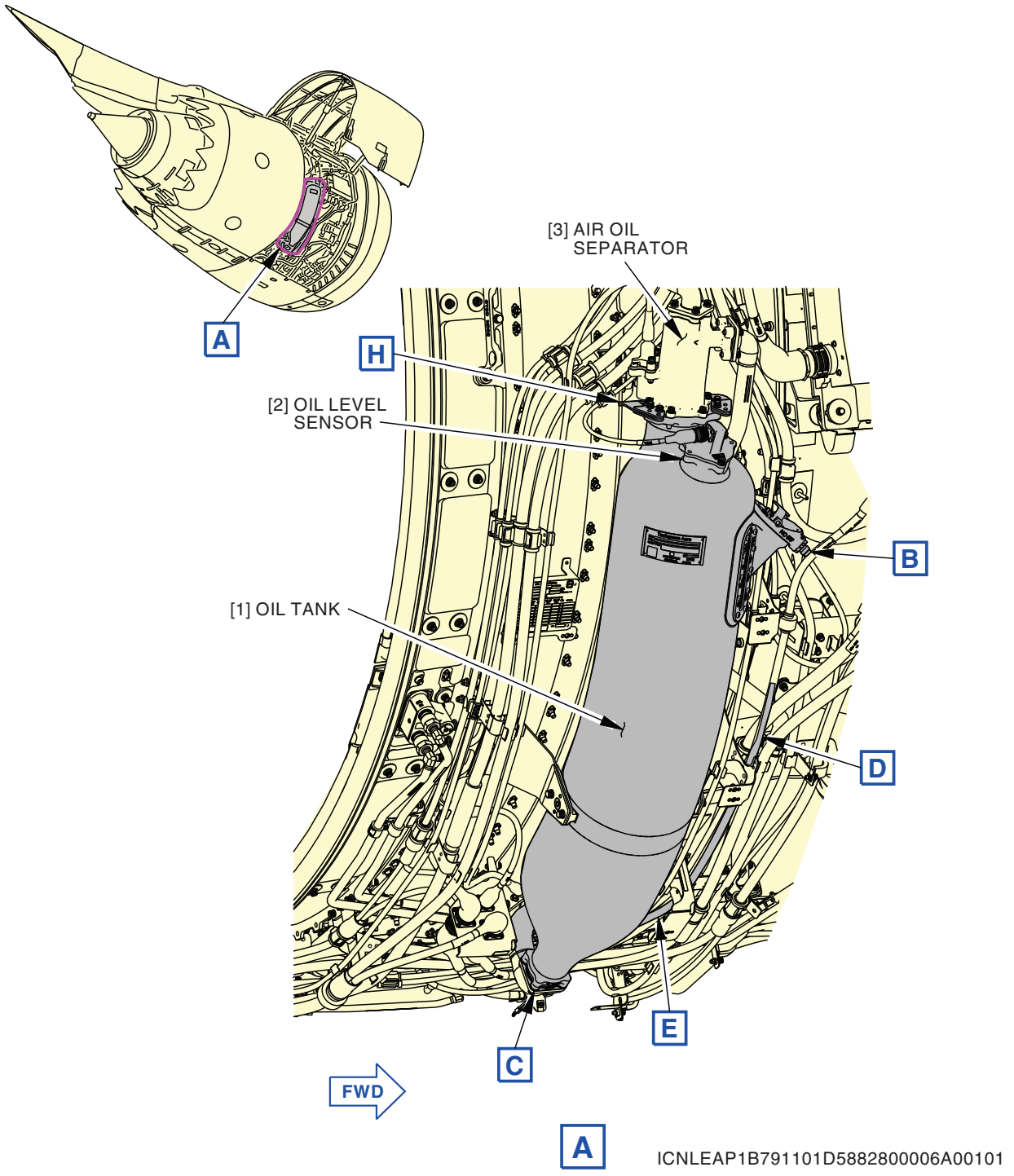
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



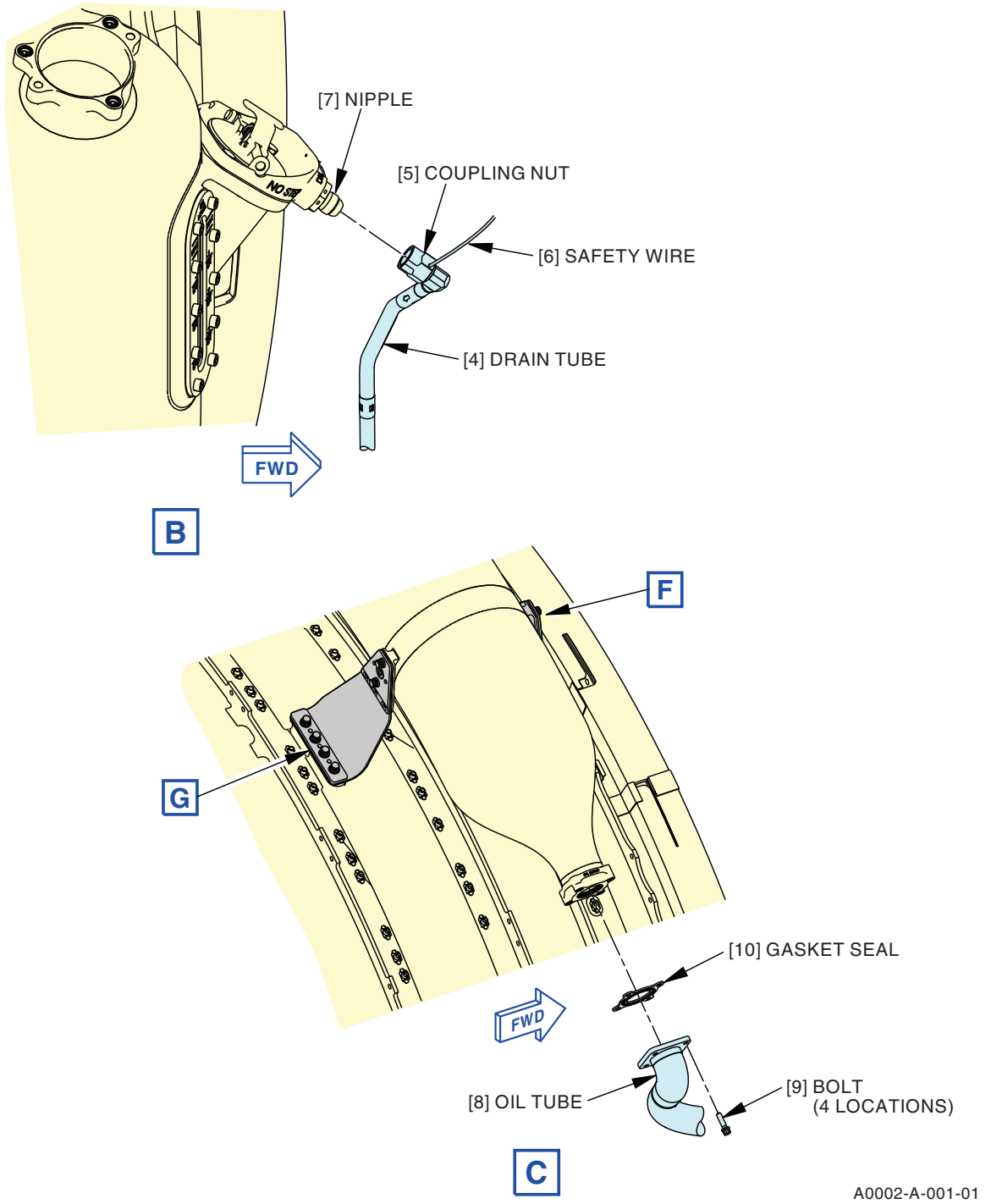
Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



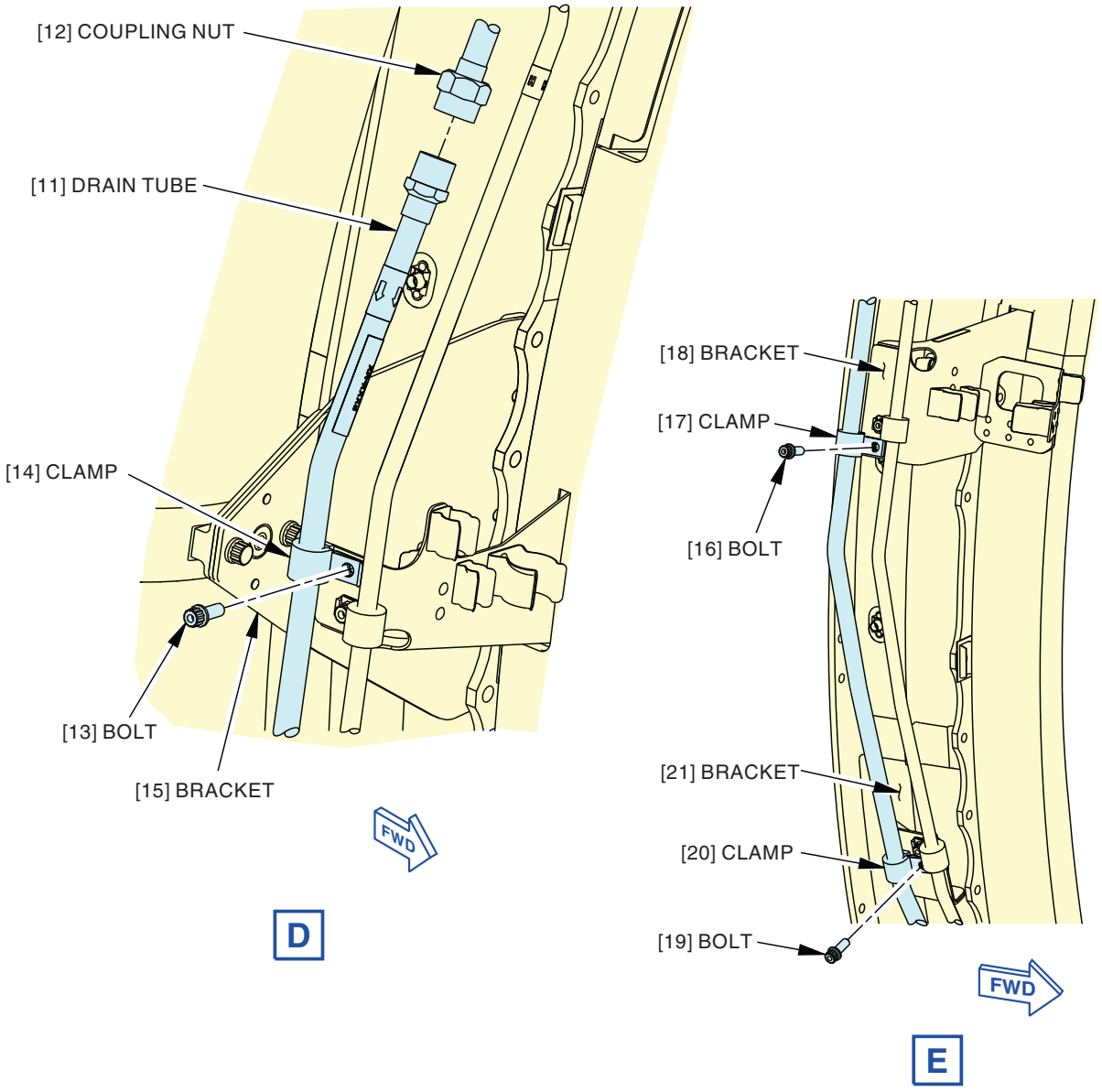
Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 3 of 7)

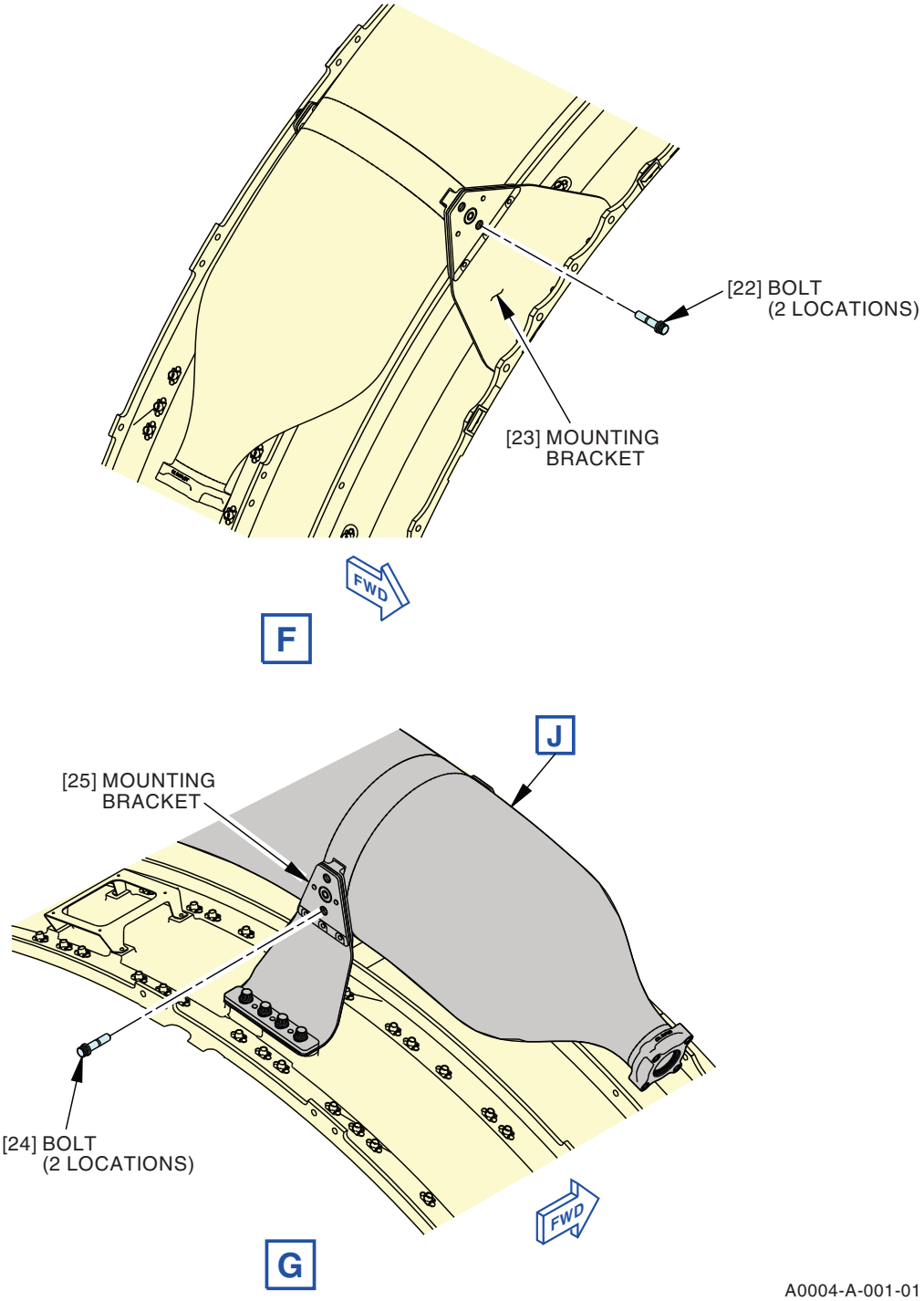
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0003-A-001-01
2355321 S00061517007_V1

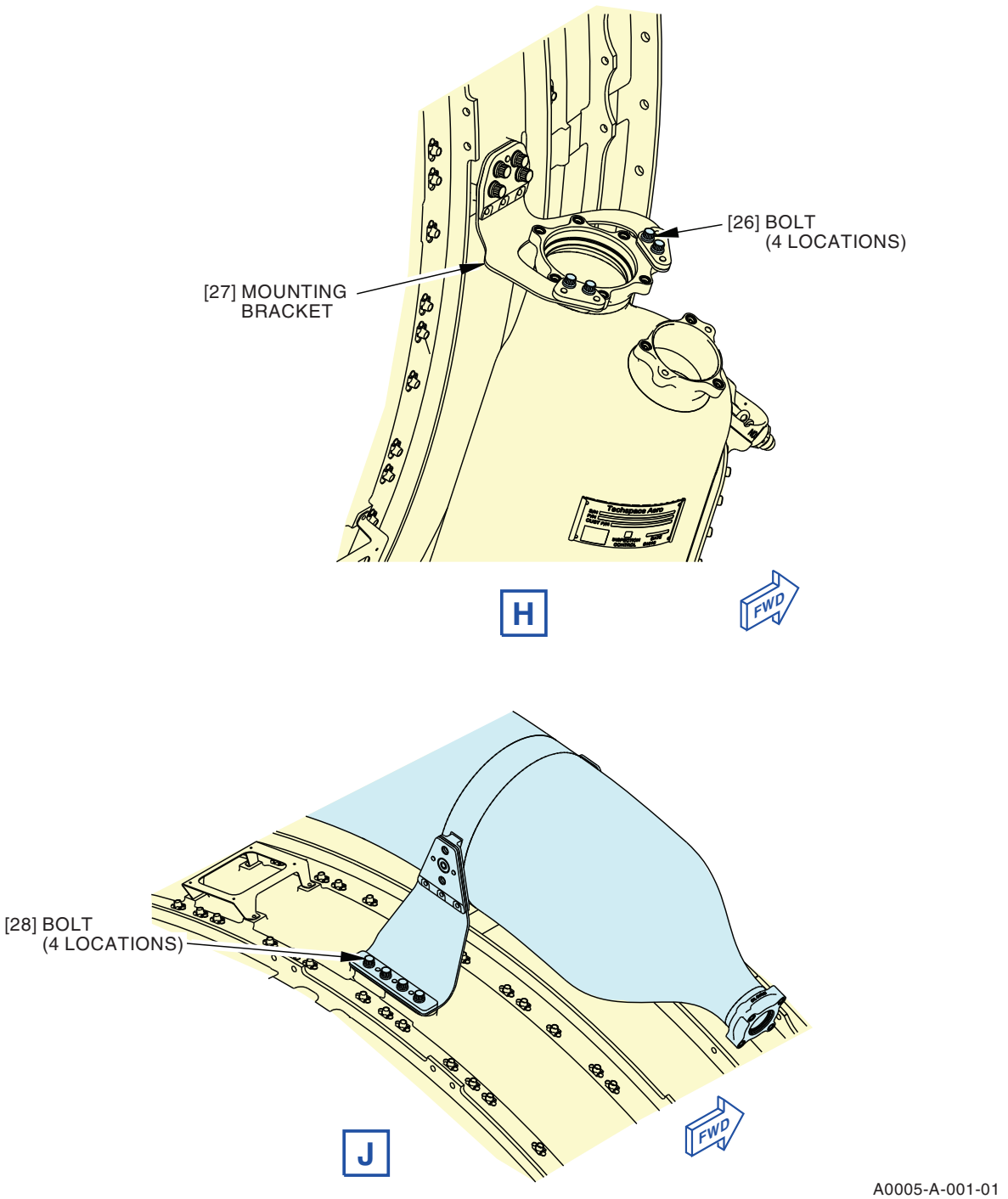
Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



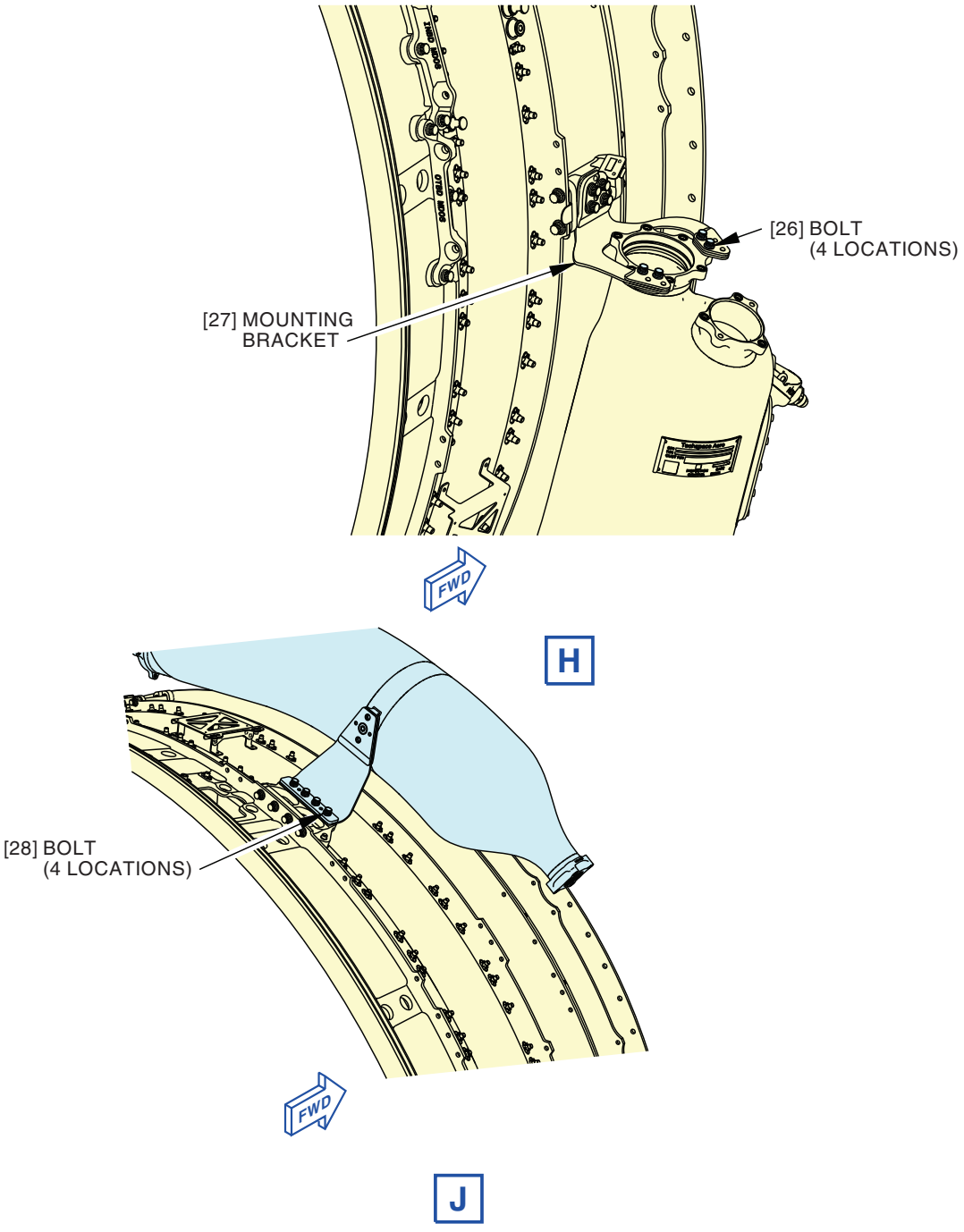
Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B791101D5882800007A00101

2872563 S0000678923_V1

Oil Tank Installation
Figure 401/79-11-01-990-801-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-11-01-400-801-G00

3. Oil Tank Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the oil tank.

(2) You must do the tests that are listed in the power plant test reference table after you install the oil tank.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
79-21-21-400-801-G00	Air Oil Separator Installation (P/B 401)
79-31-01-400-801-G00	Oil Level Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
10	Gasket seal	79-11-01-01-020	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

H. Oil Tank Installation

SUBTASK 79-11-01-420-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Install the oil tank [1] on the mounting bracket [23], mounting bracket [25], and mounting bracket [27] as follows:

NOTE: It is necessary for two people to do this procedure.

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and the pressure faces of the two bolts [22], the two bolts [24], the four bolts [26], and the four bolts [28] with graphite mineral vaseline, D00601 [CP2101].

(d) Install the oil tank [1] in a correct position on the engine.

(e) Install the four bolts [26] that attach the oil tank [1] to the mounting bracket [27].

(f) Secure the four bolts [28] that attach the mounting bracket [25] to the engine.

(g) Install the two bolts [24] that attach the oil tank [1] to the mounting bracket [25].

(h) Install the two bolts [22] that attach the oil tank [1] to the mounting bracket [23].

SUBTASK 79-11-01-420-003-G00

(2) Connect the oil tube [8] to the oil tank [1] as follows:

(a) Remove the protective caps from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads and the pressure faces of the four bolts [9] with graphite mineral vaseline, D00601 [CP2101].

(d) Lubricate each side of the elastomer part of a new gasket seal [10] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

(e) Install the gasket seal [10] and oil tube [8] on the oil tank [1].

(f) Install the four bolts [9] that attach the oil tube [8] to the oil tank [1].

1) Tighten the four bolts [9] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-11-01-010-003-G00

(3) Do this task: Air Oil Separator Installation, TASK 79-21-21-400-801-G00.

SUBTASK 79-11-01-420-005-G00

(4) Tighten the bolts [28] to 227 in-lb (25.65 N·m) - 251 in-lb (28.35 N·m).

SUBTASK 79-11-01-420-006-G00

(5) Tighten the bolts [26] in a criss cross pattern to 133 in-lb (15 N·m) - 186 in-lb (21 N·m).

SUBTASK 79-11-01-420-007-G00

(6) Tighten the bolts [24] to 266 in-lb (30 N·m) - 327 in-lb (37 N·m).

SUBTASK 79-11-01-420-008-G00

(7) Tighten the bolts [22] to 227 in-lb (25.65 N·m) - 251 in-lb (28.35 N·m).

SUBTASK 79-11-01-420-004-G00

(8) Connect the scupper drain tube [4] to the oil tank [1] as follows:

(a) Remove the protective caps from the openings.

(b) Lubricate the threads of the nipple [7] with oil, D00599 [CP2442].

(c) Engage and tighten the coupling nut [5] to the nipple [7].

(d) Tighten the coupling nut [5] to 257.0 in-lb (29.0 N·m) - 283.0 in-lb (32.0 N·m).

(e) Install MS20995C32 lockwire, G01048, on the coupling nut [5].

SUBTASK 79-11-01-010-004-G00

(9) Do this task: Oil Level Sensor Installation, TASK 79-31-01-400-801-G00.

SUBTASK 79-11-01-420-002-G00

(10) Connect the drain tube [11] as follows:

(a) Remove the protective caps from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the threads of the coupling nut [12], bolt [13], bolt [16], and bolt [19] with graphite mineral vaseline, D00601 [CP2101].

(d) Engage and tighten the coupling nut [12].

(e) Install the bolt [19] and clamp [20] that attach the drain tube [11] to the bracket [21].

1) Tighten the bolt [19] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(f) Install the bolt [16] and clamp [17] that attach the drain tube [11] to the bracket [18].

1) Tighten the bolt [16] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

(g) Install the bolt [13] and clamp [14] that attach the drain tube [11] to the bracket [15].

1) Tighten the bolt [13] to 78.0 in-lb (8.8 N·m) - 82.0 in-lb (9.3 N·m).

SUBTASK 79-11-01-020-006-G00

(11) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

SUBTASK 79-11-01-610-001-G00

(12) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 79-11-01-280-001-G00

(13) Clean the work area and clear the tools and other items.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-11-01-410-001-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 79-11-01-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Oil Tank Installation Test

SUBTASK 79-11-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL TANK - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) The visual inspection of the oil tank.

TASK 79-11-01-200-801-G00

2. Oil Tank Inspection

(Figure 601 and Figure 602)

A. General

(1) This task provides the instructions on how to do a visual inspection of the oil tank.

(2) Surface Types

(a) The four primary surface types are:

1) Rough material (casting). Casting surfaces that are not machined.

2) Tank body. The tank body is the global shell of the oil tank, that is made from sheet metal and is not machined.

3) Machined surface. All machined surfaces.

4) Weld bead. It includes the weld bead, weld root and maximum area of 0.12 in. (3 mm) from root.

(b) The secondary surface types are:

1) Low stress. Areas that are not much affected (by pressure and vibrations) during component functioning. These zones have high design margins.

2) High stress. Areas that are more affected (by pressure and vibrations) during component functioning. These zones have lower design margins.

3) Functional areas. Areas that have a specified function that includes sealing areas, contact areas and zones for inserts or bushings.

(3) Indication Types

(a) Type 0 – Coloring

1) Spot of color. Occurrence of a different color on a surface caused by the repair of an indication before a surface treatment, tests or part manufacturing process.

2) Stain. Surface discoloration caused by liquid drying.

3) Surface finish. Appearance of the oil tank body, because of the type of surface finish used in the production or repair process. The four different surface finishes are:

- Polished surface finish
- Raw forming surface finish
- Rubbed with iron mesh surface finish
- Sand blasting surface finish.

(b) Type 1 - Smoothed loss of material

1) Blending. Removal of an irregularity from a surface, that results in a shallow, smooth depression.

2) Cut. Indication on an edge that makes a chamfer.

(c) Type 2 - Smoothed distortion of surface without loss of material

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Bump. A distortion of the oil tank body that is caused by contact with another blunt object. It is a smooth indication without an angular step and no thinning effect.

2) Dent. Indication on the surface that has smooth edges and rounded bottom. Impact of a non-cutting object on the part can cause this type of indications. The material is twisted but not removed.

(d) Type 3 - Loss of material on a large surface without measurable depth

1) Burnish marks. Burnish marks are shiny surfaces that result from rubbing against another surface. It can contain scratches with no measurable depth.

2) Tool marks. Tool marks are recurring patterns in a machined surface caused by machining tools.

3) Tool lines. Tools lines are light marks caused by the work of a cutting tool.

4) Chatter marks. Vibrations between the cutting tool and machined surface can cause chatter marks.

5) Wear. Typical wear is because of usual usage of the oil tank. For example, wear due to friction or relative motion between different parts.

(e) Type 4 - Sharp loss of material

1) Nicks. Negative indications with raised sharp edges, corners or V formed bottom. Impact of an object with sharp edges can cause these indications.

2) Scratches. Linear depressions in the surface with sharp bottoms and edges. Sharp object moving along the surface can cause scratches. Usually the material is not removed.

3) Scorings/grooves. They are the same as scratches, but more marked and multiple.

(f) Type 5 - Superficial imperfections

1) Sulfuric Anodizing indication. Surface on an aluminum part where the necessary anodizing is not fully or totally removed.

2) Passivation indication. Surface on a steel part where the necessary passivation is not fully or totally removed.

3) Corrosion. It is the destruction of materials gradually by chemical reaction with its environment.

4) Pitting. A type of localized corrosion that causes the creation of small holes in the metal.

5) Fretting. Corrosion indication that occurs at the contact area between two materials under load and subject to relative motion by vibration or some other force.

(g) Type 6 - Raised imperfections

1) Burrs. Burrs are small metal chips that stay attached to the part after a machining operation.

2) Roll overs. Deformations seen at the edges of a part, usually after blasting operation. This type of indication can also be because of a marking operation or by a contact with another piece.

3) Protruding materials. It is raised materials because of displacement of material. Usually seen at the edges of type 4 indications.

(h) Type 7 - Material indications

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Step. Step is a type of indication seen on a machined surface or a surface on an adjacent radius. This type of finding can appear on a surface made by casting, at a mismatched connection between cores (sand) or molds.

2) Surface porosity. Circular or elongated, smooth-edged voids that occurs individually, in clusters or distributed during the casting. Usually they are caused by air or gas bubbles during the solidification.

3) Inclusions. Particles of unwanted material that are embedded in the material.

(4) To do the check of the indication with the inspection scribe, the inspection scribe must be slightly held between the thumb and forefinger, bent at 60° to the imperfection and pushed forward along the surface, without applying pressure other than its own weight, into a direction perpendicular to the lay of the imperfections (Figure 601).

B. References

Reference	Title
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
79-11-01-000-801-G00	Oil Tank Removal (P/B 401)
79-11-01-400-801-G00	Oil Tank Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-26194	Scribe - Inspection, 0.010" ball tip Part #: SDS-0100 Supplier: 8D486
COM-26334	Scribe - Inspection, 0.060" ball tip Part #: SDS-0600 Supplier: 8D486

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Prepare for the Oil Tank Inspection

SUBTASK 79-11-01-010-005-G00

(1) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

F. Oil Tank Inspection

SUBTASK 79-11-01-900-001-G00

(1) If you find damage that is more than the limits, replace the oil tank, unless you are given other instructions, these are the tasks:

- Oil Tank Removal, TASK 79-11-01-000-801-G00
- Oil Tank Installation, TASK 79-11-01-400-801-G00.

SUBTASK 79-11-01-210-001-G00

(2) Examine the low stress areas of the rough surfaces.

(a) Type 0 finding: Coloring

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Spot of color and stain

a) Serviceable if the surface roughness is not modified between impacted and not impacted area.

b) If the surface roughness is modified, change the type of indication and follow the guidelines associated to this type.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending and cut

a) Serviceable if indication depth is less than 0.0196 in. (0.5 mm).

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Dent

a) Check indication with the 0.060" inspection scribe, COM-26334.

<1> If, and only if, the 0.060" inspection scribe, COM-26334 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.0078 in. (0.20 mm).

<3> If indication depth is more than 0.0078 in. (0.20 mm), hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

1) Nick

a) Check indication with the 0.060" inspection scribe, COM-26334.

<1> If, and only if, the 0.060" inspection scribe, COM-26334 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.0039 in. (0.10 mm).

<3> If indication depth is more than 0.0039 in. (0.10 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

2) Scratch and scoring/groove

a) Serviceable if indication depth is less than 0.0039 in. (0.10 mm).

b) If indication depth is more than 0.0039 in. (0.10 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization indication

a) Serviceable if the affected area is internal and oil wetted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Serviceable if no impact on functionality and height is less than 0.0311 in. (0.79 mm).

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

SUBTASK 79-11-01-210-002-G00

(3) Examine the high stress areas of the rough surfaces.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Serviceable if the surface roughness is not modified between impacted and not impacted area.

b) If the surface roughness is modified, change the type of indication and follow the guidelines associated to this type.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending and cut

a) Not serviceable.

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump and dent

a) Not serviceable.

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

1) Nick, scratch, and scoring/groove

a) Not serviceable.

(f) Type 5 finding: superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

2) Surface pits - Raised metal (investment casting)

a) Not serviceable.

SUBTASK 79-11-01-210-003-G00

(4) Examine the low stress areas of the tank body.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Serviceable if the surface roughness is not modified between impacted and not impacted area.

b) If the surface roughness is modified, change the type of indication and follow the guidelines associated to this type.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending and cut

a) Serviceable if indication depth is less than 0.0118 in. (0.30 mm).

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump

a) Not serviceable.

2) Dent

a) Check indication with the 0.060" inspection scribe, COM-26334.

<1> If, and only if, the 0.060" inspection scribe, COM-26334 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.0078 in. (0.20 mm).

<3> If indication depth is more than 0.0078 in. (0.20 mm), hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Nick

a) Check indication with the 0.060" inspection scribe, COM-26334.

<1> If, and only if, the 0.060" inspection scribe, COM-26334 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.0039 in. (0.10 mm).

<3> If indication depth is more than 0.0039 in. (0.10 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

2) Scratch and scoring/groove

a) Serviceable if indication depth is less than 0.0039 in. (0.10 mm).

b) If indication depth is more than 0.0039 in. (0.10 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

SUBTASK 79-11-01-210-004-G00

(5) Examine the high stress areas of the tank body.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Serviceable if the surface roughness is not modified between impacted and not impacted area.

b) If the surface roughness is modified, change the type of indication and follow the guidelines associated to this type.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending

a) Serviceable if indication depth less than 0.0039 in. (0.10 mm) and indication minimum diameter above 0.1181 in. (3.00 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Cut

a) Not serviceable.

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump

a) Not serviceable.

2) Dent

a) Check indication with the 0.010" inspection scribe, COM-26194.

<1> If, and only if, the 0.010" inspection scribe, COM-26194 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.00098 in. (0.025 mm).

<3> If indication depth is more than 0.00098 in. (0.025 mm), hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

1) Nick

a) Check indication with the 0.010" inspection scribe, COM-26194.

<1> If, and only if, the 0.010" inspection scribe, COM-26194 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.00098 in. (0.025 mm).

<3> If indication depth is more than 0.00098 in. (0.025 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

2) Scratch and scoring/groove

a) Not serviceable.

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-11-01-210-009-G00

(6) Examine the very high stress areas of the tank body.

(a) Type 1 finding: Smoothed loss of material

1) Blending

a) Not serviceable.

(b) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump and dent

a) Not serviceable.

(c) Type 4 finding: Sharp loss of material

1) Nick

a) Not serviceable.

SUBTASK 79-11-01-210-005-G00

(7) Examine the low stress areas of the machined surfaces.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Serviceable if the surface roughness is not modified between impacted and not impacted area.

b) If the surface roughness is modified, change the type of indication and follow the guidelines associated to this type.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending, cut

a) Serviceable if indication depth is less than 0.0157 in. (0.40 mm).

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump

a) Not serviceable.

2) Dent

a) Check indication with the 0.060" inspection scribe, COM-26334.

<1> If, and only if, the 0.060" inspection scribe, COM-26334 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.0078 in. (0.20 mm).

<3> If indication depth is more than 0.0078 in. (0.20 mm), hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Type 4 finding: Sharp loss of material

1) Nick

a) Check indication with the 0.060" inspection scribe, COM-26334.

<1> If, and only if, the 0.060" inspection scribe, COM-26334 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.0039 in. (0.10 mm).

<3> If indication depth is more than 0.0039 in. (0.10 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

2) Scratch and scoring/groove

a) Serviceable if indication depth is less than 0.0039 in. (0.10 mm).

b) If indication depth is more than 0.0039 in. (0.10 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

SUBTASK 79-11-01-210-006-G00

(8) Examine the high stress areas of the machined surfaces.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Serviceable if the surface roughness is not modified between impacted and not impacted area.

b) If the surface roughness is modified, change the type of indication and follow the guidelines associated to this type.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Serviceable if indication depth less than 0.0039 in. (0.10 mm) and indication minimum diameter above 0.1181 in. (3.00 mm).

2) Cut

a) Not serviceable.

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump

a) Not serviceable.

2) Dent

a) Check indication with the 0.010" inspection scribe, COM-26194.

<1> If, and only if, the 0.010" inspection scribe, COM-26194 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.00098 in. (0.025 mm).

<3> If indication depth is more than 0.00098 in. (0.025 mm), hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

1) Nick

a) Check indication with the 0.010" inspection scribe, COM-26194.

<1> If, and only if, the 0.010" inspection scribe, COM-26194 hooks or bumps, the indication must be measured.

<2> Serviceable if indication depth is less than 0.00098 in. (0.025 mm).

<3> If indication depth is more than 0.00098 in. (0.025 mm), hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

2) Scratch and scoring/groove

a) Not serviceable.

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Step, surface porosity, and inclusion

a) Not serviceable.

SUBTASK 79-11-01-210-010-G00

(9) Examine the very high stress areas of the machined surfaces

(a) Type 1 finding: Smoothed loss of material

1) Blending

a) Not serviceable.

(b) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump and dent

a) Not serviceable.

(c) Type 4 finding: Sharp loss of material

1) Nick

a) Not serviceable.

SUBTASK 79-11-01-210-007-G00

(10) Examine the low stress areas of the weld bead surfaces.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Not serviceable.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending, cut

a) Serviceable if transitions between impacted area and non impacted area are smooth, without sharp edges.

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump

a) Not serviceable.

2) Dent

a) Hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

1) Nick

a) Hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

2) Scratch and scoring/groove

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Not serviceable.

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

SUBTASK 79-11-01-210-008-G00

(11) Examine the high stress areas of the weld bead surfaces.

(a) Type 0 finding: Coloring

1) Spot of color and stain

a) Not serviceable.

2) Surface finish

a) Serviceable if the surface roughness is not modified and the surface finish is equal along the entire surface or on a simple geometric shape if there is a rework.

(b) Type 1 finding: Smoothed loss of material

1) Blending, cut

a) Serviceable if transitions between impacted area and non impacted area are smooth, without sharp edges.

(c) Type 2 finding: Smoothed distortion of surface without loss of material

1) Bump

a) Not serviceable.

2) Dent

a) Hand blend to remove stress concentration and to smooth the Type 2 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

(d) Type 3 finding: Loss of material on a large surface without measurable depth

1) Burnish mark, tool mark, tool lines, chatter mark, and wear

a) Serviceable if surface roughness is not modified between impacted and not impacted area.

(e) Type 4 finding: Sharp loss of material

1) Nick

a) Hand blend to remove stress concentration and to smooth the Type 4 finding so it is classified as acceptable Type 1 finding (TASK 70-43-42-300-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Scratch, and scoring/groove

a) Not serviceable.

(f) Type 5 finding: Superficial imperfections

1) Sulfuric anodization

a) Serviceable if the affected area is internal and oil wetted.

2) Passivation indication

a) Serviceable if the affected area is internal and oil wetted.

3) Corrosion, pitting, and fretting

a) Not serviceable.

(g) Type 6 finding: Raised imperfections

1) Burr, roll over, and protruding material

a) Not serviceable.

(h) Type 7 finding: Material indications

1) Step, surface porosity, and inclusion

a) Not serviceable.

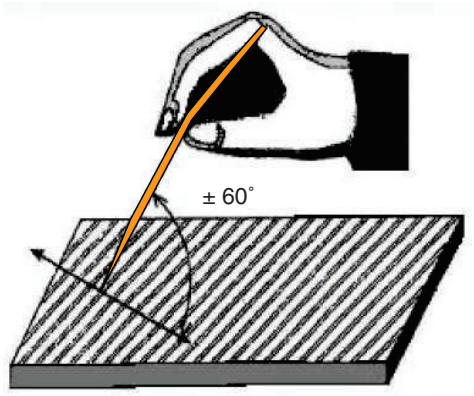
G. Put the Airplane Back to Its Usual Condition

SUBTASK 79-11-01-410-002-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

——— END OF TASK ———

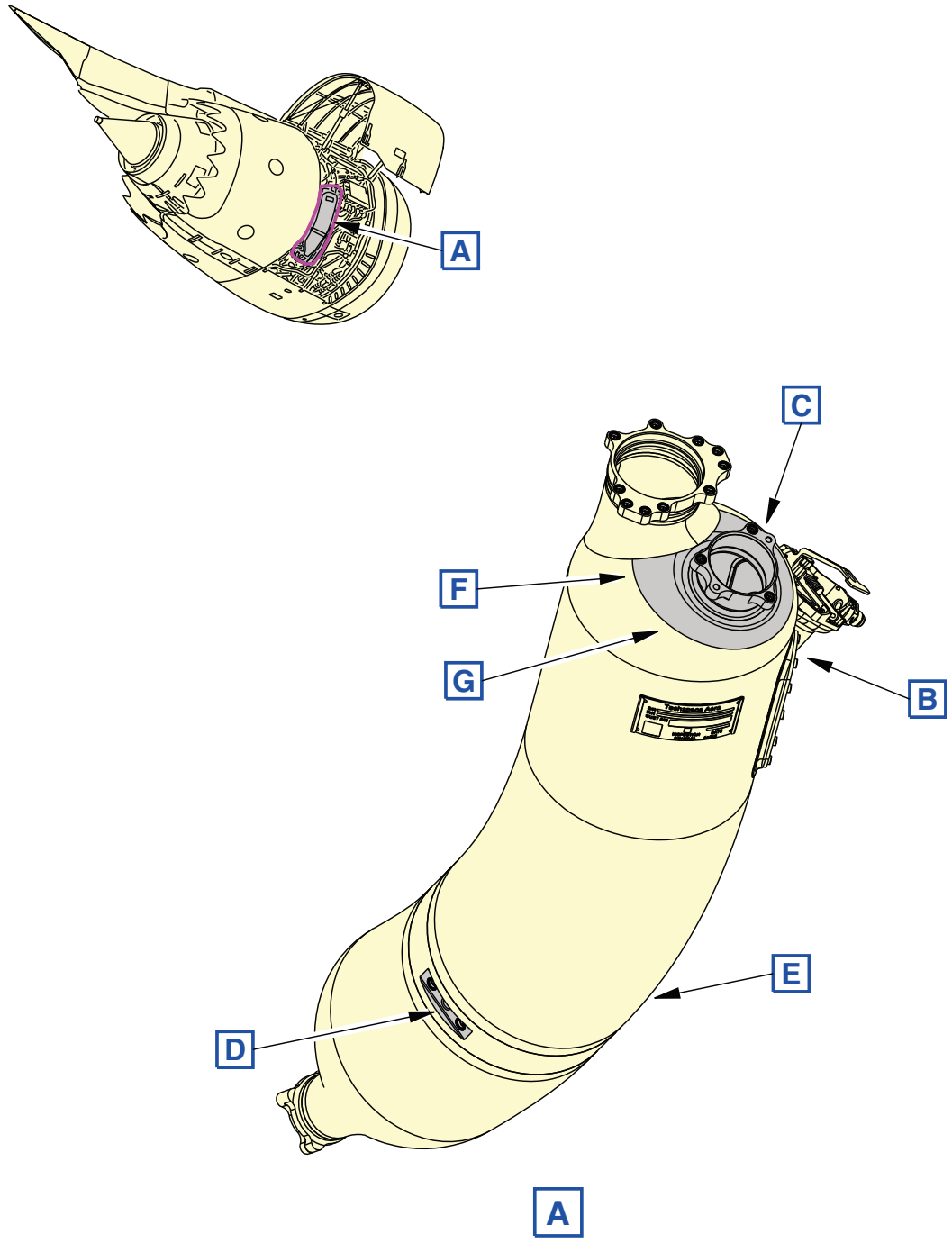
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B791101D5882800001A00101
3122306 S0000856884_V1

Indication Check with Inspection Scribe
Figure 601/79-11-01-990-803-G00

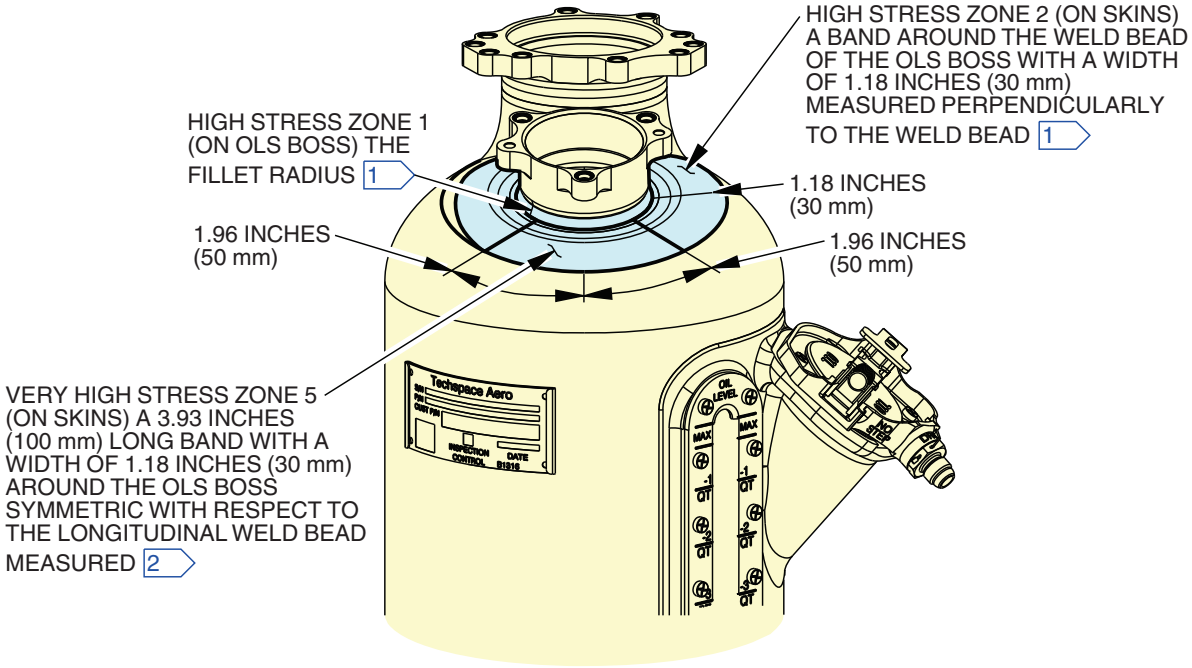
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B791101D5882800002A00101
3122308 S0000856885_V1

Oil Tank Inspection
Figure 602/79-11-01-990-802-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

NON-BLUE AREAS ARE CONSIDERED AS LOW STRESS AREA.

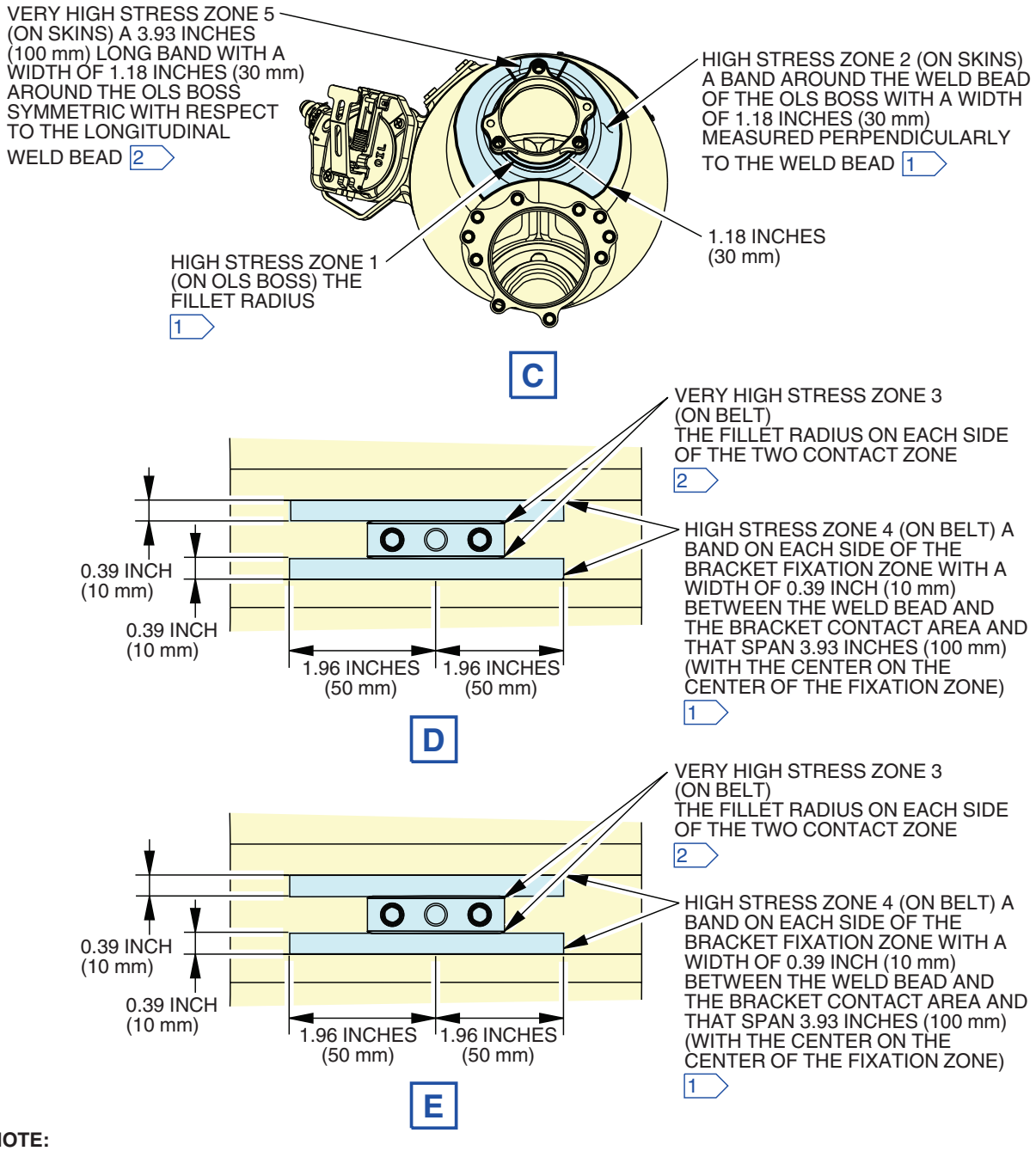
1 HIGH STRESS AREA

2 VERY HIGH STRESS AREA

ICNLEAP1B791101D5882800002A00101
3122313 S0000856886_V1

Oil Tank Inspection
Figure 602/79-11-01-990-802-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

NON-BLUE AREAS ARE CONSIDERED AS LOW STRESS AREA.

1 HIGH STRESS AREA

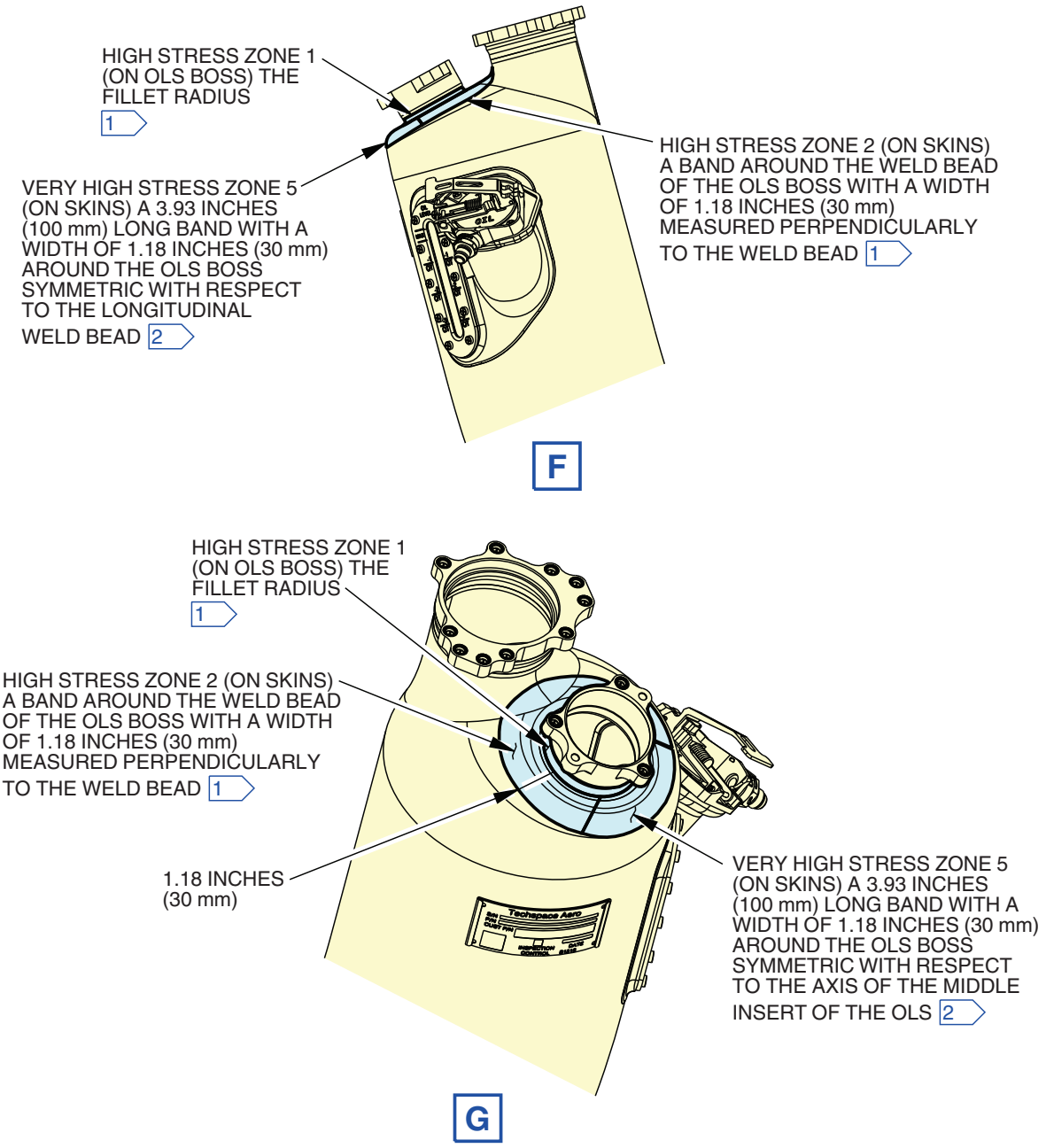
2 VERY HIGH STRESS AREA

ICNLEAP1B791101D5882800003A00101

3122317 S0000856887_V1

Oil Tank Inspection
Figure 602/79-11-01-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

NON-BLUE AREAS ARE CONSIDERED AS LOW STRESS AREA.

1 HIGH STRESS AREA

2 VERY HIGH STRESS AREA

ICNLEAP1B791101D5882800004A00101

3122377 S0000856888_V1

Oil Tank Inspection
Figure 602/79-11-01-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OIL TANK FILLER CAP - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Oil Tank Filler Cap Removal

(2) Oil Tank Filler Cap Installation.

TASK 79-11-03-000-801-G00

2. Oil Tank Filler Cap Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the oil tank filler cap.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-11-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-11-03-860-002-G00

(2) Do this task: Open the Fan Cowl Panels (Selection), TASK 71-11-04-010-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the applicable engine, open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

G. Oil Tank Filler Cap Removal

SUBTASK 79-11-03-020-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the oil tank filler cap [6].

(a) Disconnect the drain tube [7] from the oil tank filler cap [6]:

1) Loosen the coupling nut [8] of the drain tube [7].

2) Disconnect the drain tube [7] from the oil tank filler cap [6].

(b) Remove the two screws [10] that attach the oil tank filler cap [6] to the oil tank service panel [11].

(c) Remove the oil tank filler cap [6] from the oil tank service panel [11].

PTF ALL PRE SB 737-CFM-LEAP-1B-79-0016

NOTE: Do not remove the adapter assembly.

PTF ALL POST SB 737-CFM-LEAP-1B-79-0016

(d) Remove and discard the scupper O-ring [13] from the oil tank filler cap [6].

(e) Remove the adapter assembly [14].

1) Remove the adapter insulator [15] from the top of the adapter assembly [14].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-79-0016 (Continued)

2) Remove the adapter assembly [14] from the oil tank service panel [11].

3) Remove and discard the O-ring [16] from the groove of the adapter assembly [14].

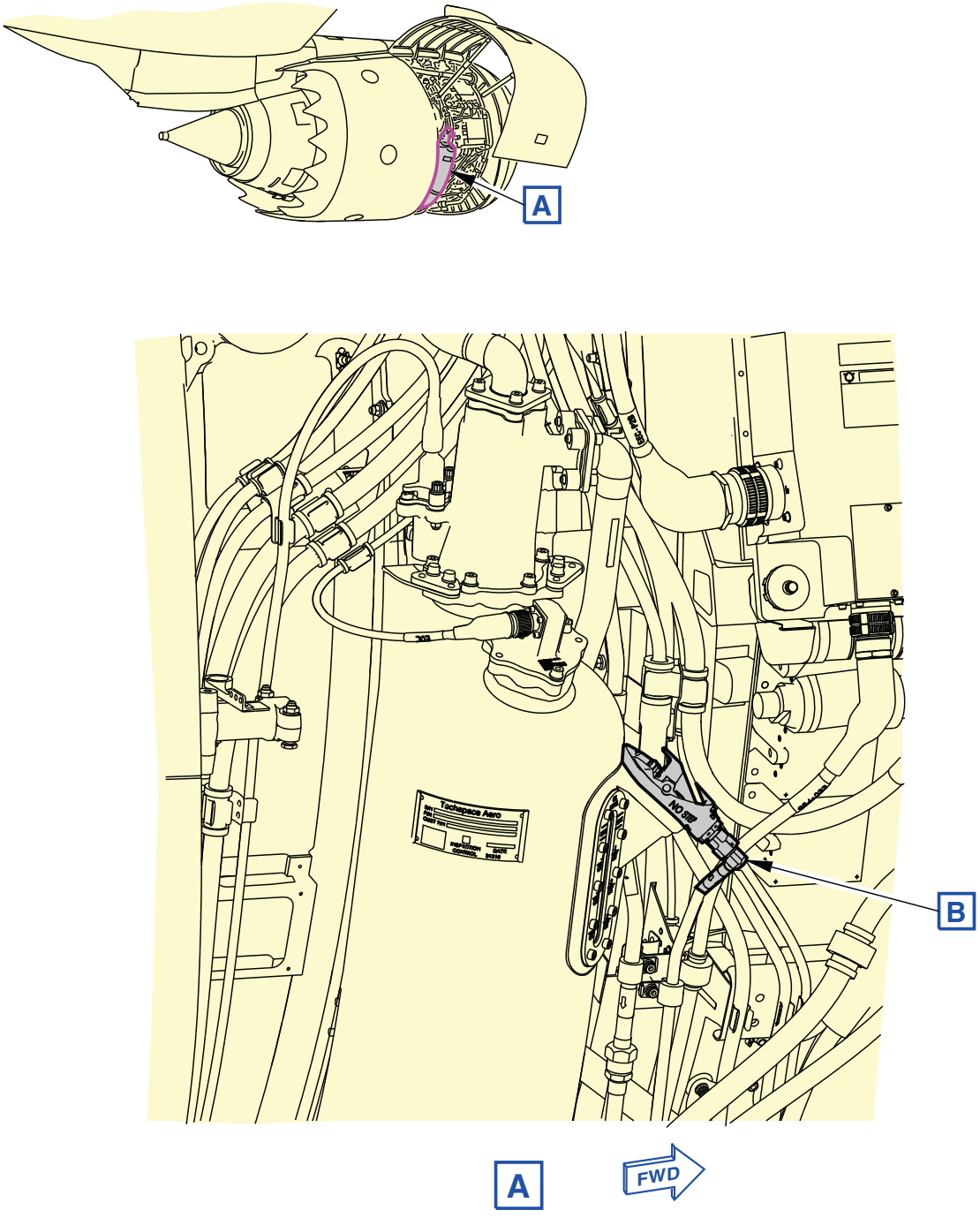
PTF ALL

(f) Remove and discard the O-ring [12] from the groove of the oil tank filler cap [6].

(g) Install the protective covers on the drain tube [7], nipple [9], and oil tank filler cap [6] port on the oil tank service panel [11].

END OF TASK

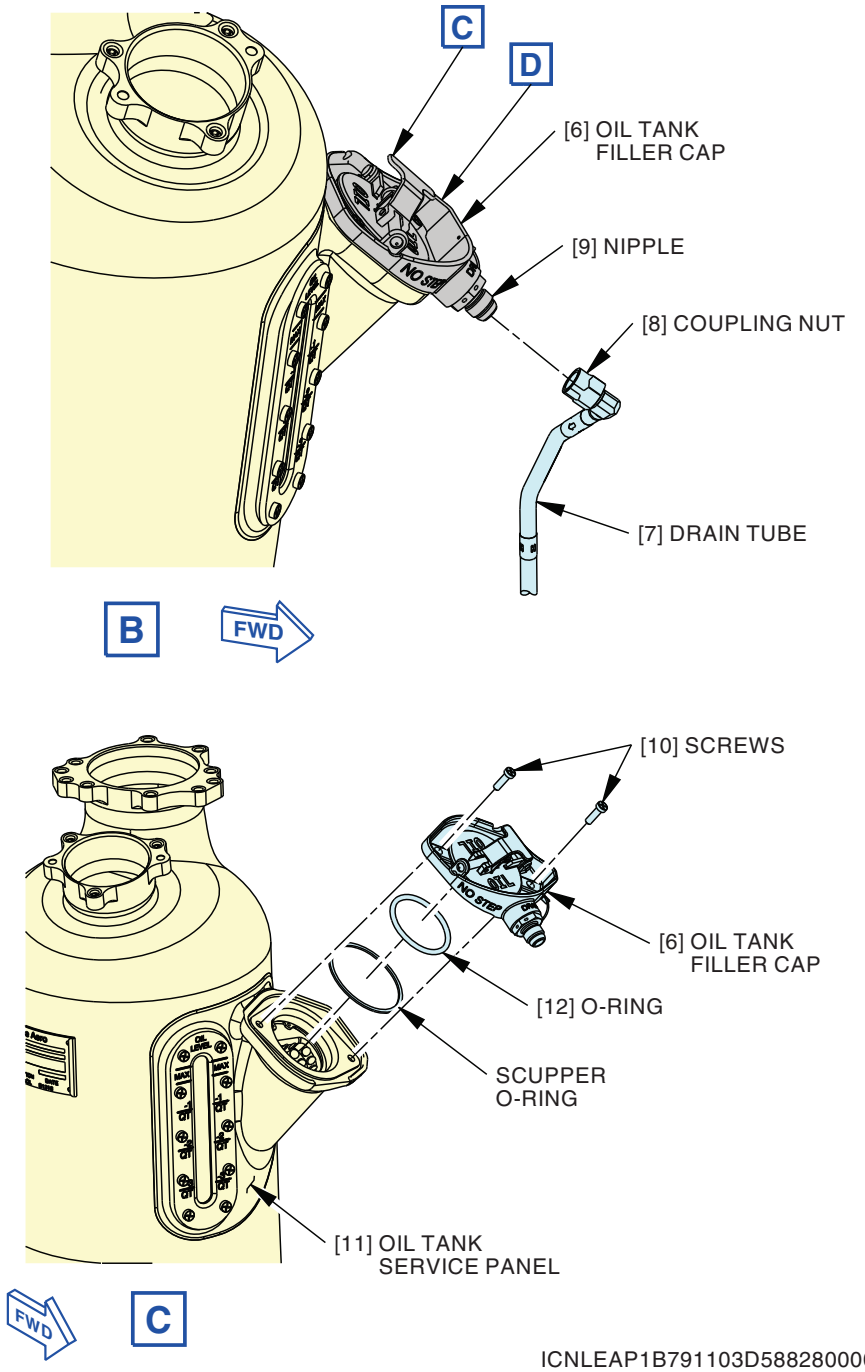
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0002-A-001-01
2428716 S0000561637_V3

Oil Tank Filler Cap Installation
Figure 401/79-11-03-990-801-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

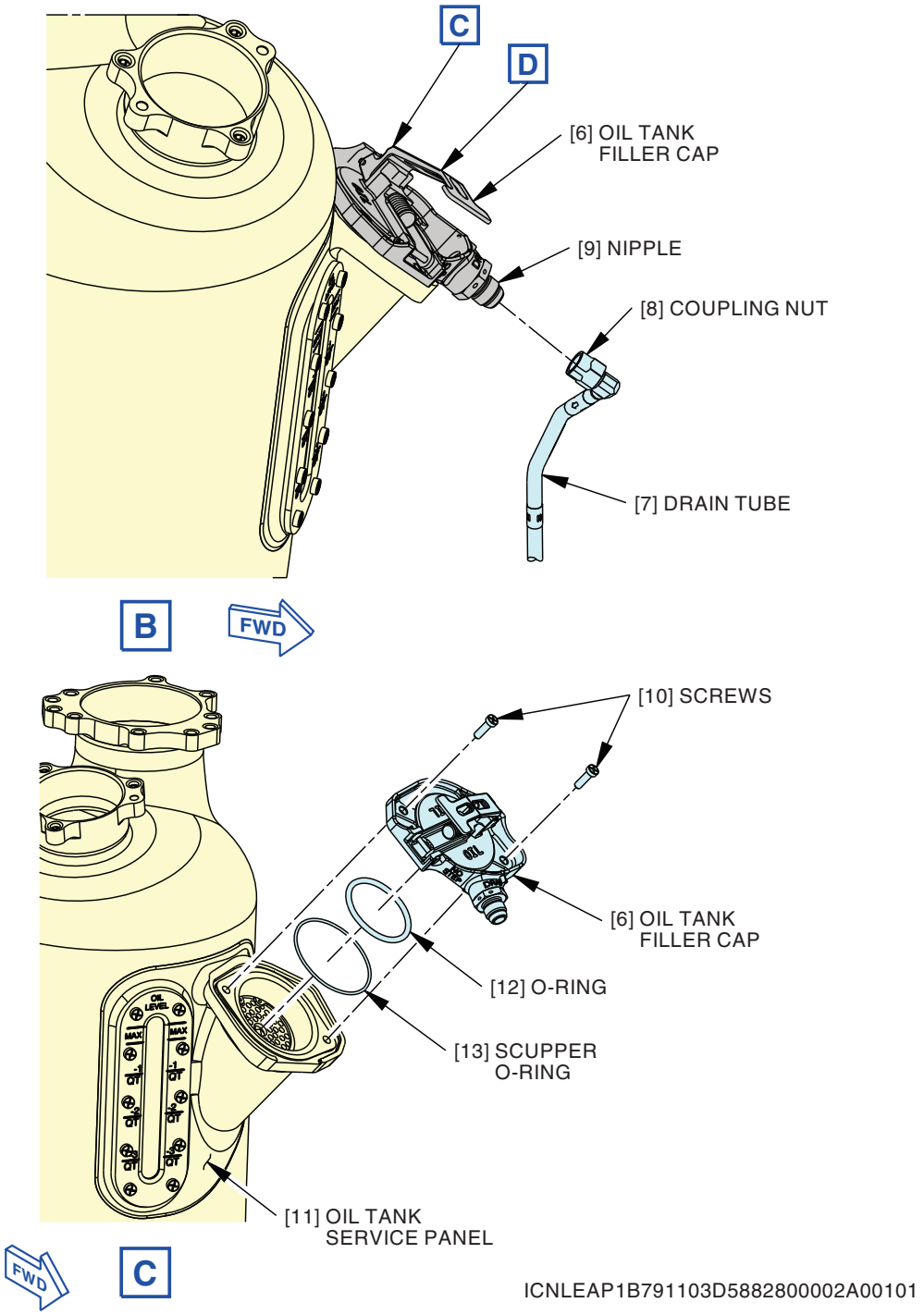


ICNLEAP1B791103D5882800001A00101

2428719 S0000561638_V3

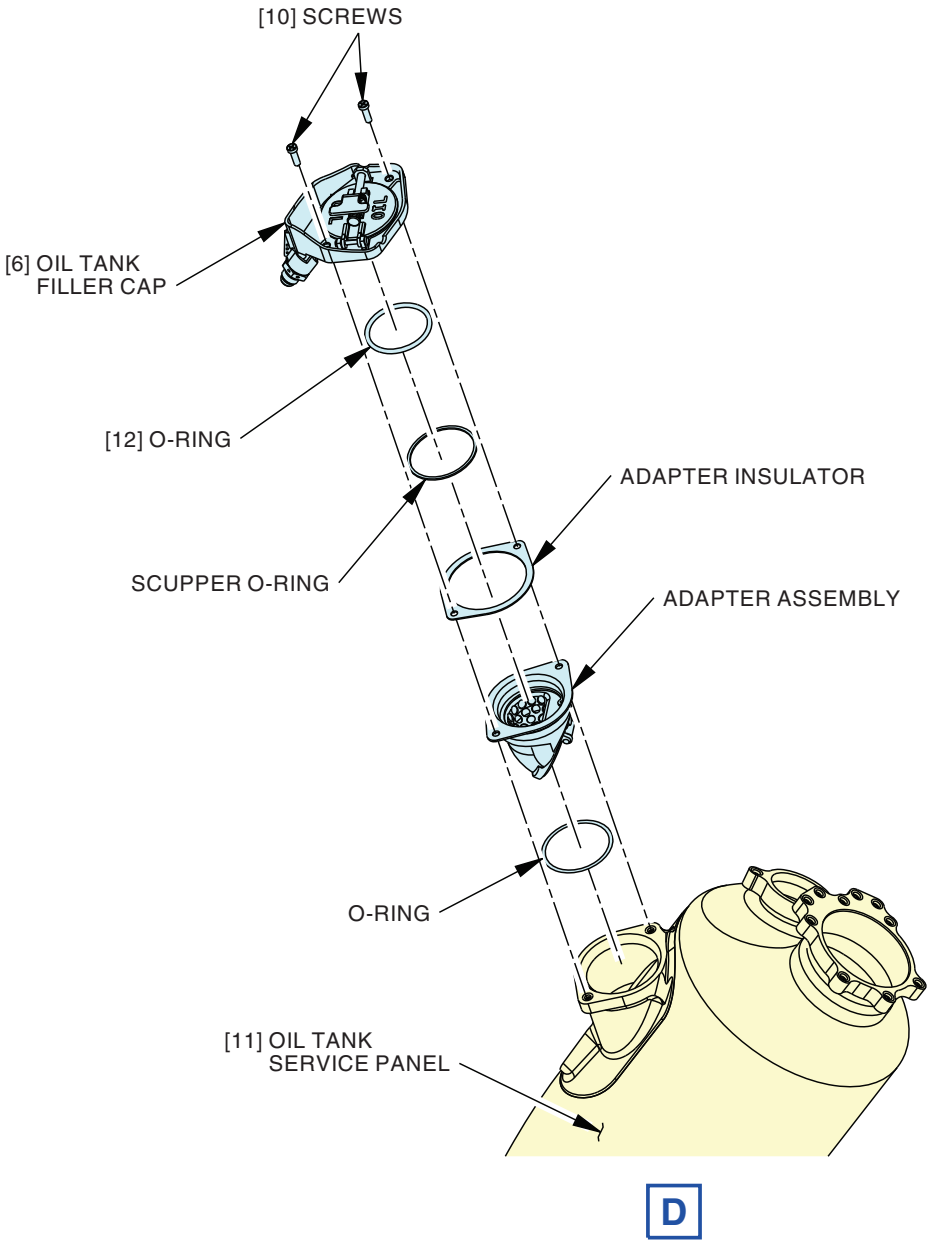
Oil Tank Filler Cap Installation
Figure 401/79-11-03-990-801-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Oil Tank Filler Cap Installation
Figure 401/79-11-03-990-801-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

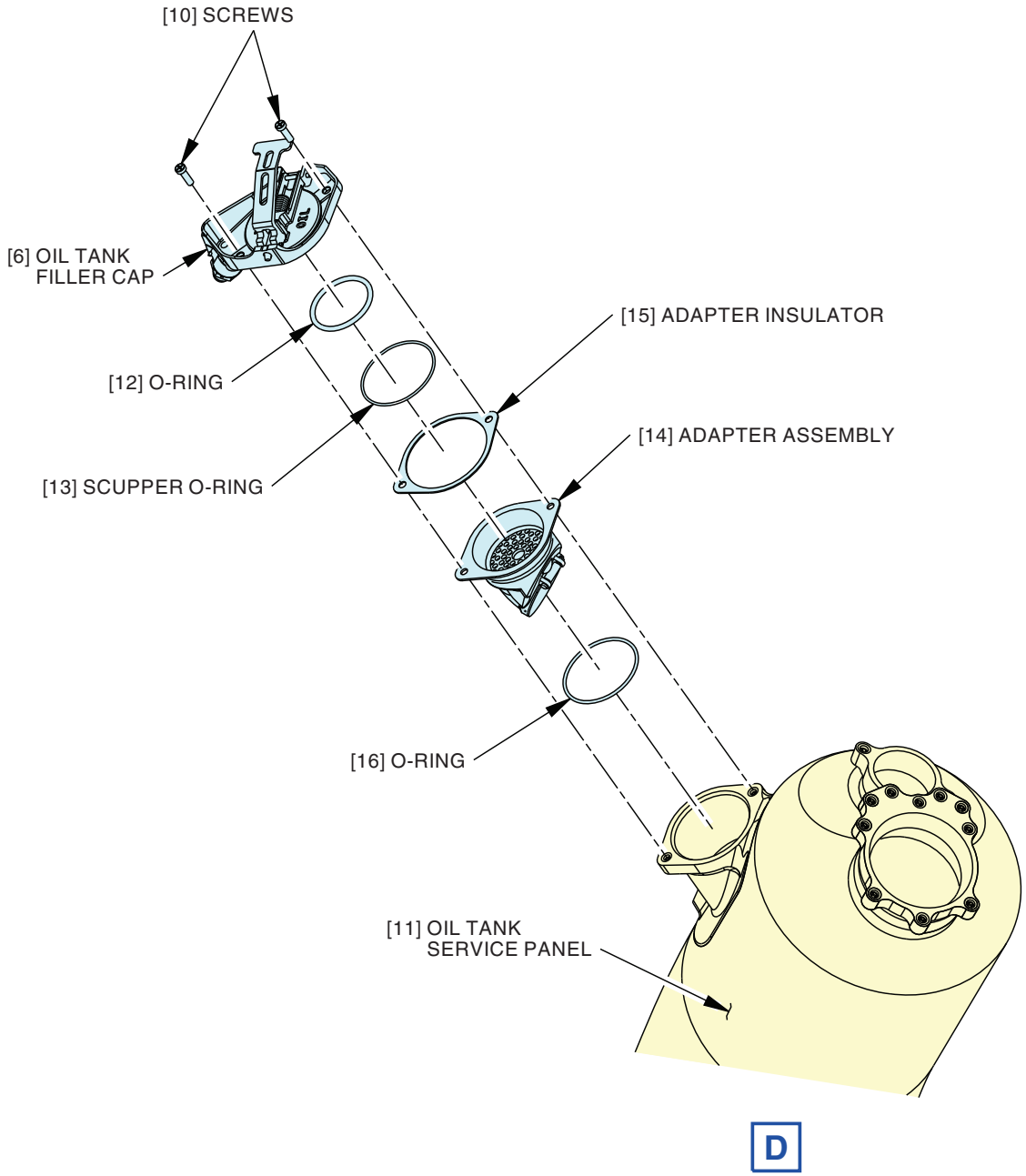


ICNLEAP1B791103D5882800003A00101

3059034 S0000822287_V1

Oil Tank Filler Cap Installation
Figure 401/79-11-03-990-801-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B791103D5882800004A00101

3059036 S0000822289_V1

Oil Tank Filler Cap Installation
Figure 401/79-11-03-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-11-03-400-801-G00

3. Oil Tank Filler Cap Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the oil tank filler cap.

(2) You must do the tests that are listed in the power plant test reference table after you install the oil tank filler cap.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
12	O-ring	79-11-01-01-100	PTF ALL
		79-11-01-01-125	PTF ALL
13	Scupper O-ring	79-11-01-01-130	PTF ALL
16	O-ring	79-11-01-01-145	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

H. Oil Tank Filler Cap Installation

SUBTASK 79-11-03-420-001-G00

(1) Install the oil tank filler cap [6].

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Remove the protective covers on the drain tube [7], nipple [9], and oil tank filler cap [6] port on the oil tank service panel [11].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Lubricate all the inserts and threads of the two screws [10] with graphite mineral vaseline, D00601 [CP2101].

PTF ALL POST SB 737-CFM-LEAP-1B-79-0016

(d) Install the adapter assembly [14].

1) Lubricate the new O-ring [16] with oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

2) Install the new O-ring [16] in the groove of the adapter assembly [14].

3) Install the adapter assembly [14] in its position on the oil tank service panel [11].

4) Install the adapter insulator [15] on the top of the adapter assembly [14].

(e) Lubricate the new scupper O-ring [13] with oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

(f) Install the new scupper O-ring [13] on the oil tank filler cap [6].

PTF ALL

(g) Lubricate the new O-ring [12] with oil, D00599 [CP2442], or vaseline, D00672 [CP5070].

(h) Install the O-ring [12] onto the groove of the oil tank filler cap [6].

(i) Install the oil tank filler cap [6] in a correct position on the oil tank service panel [11].

(j) Install the two screws [10] that attach the oil tank filler cap [6] to the oil tank service panel [11].

(k) Tighten the screws [10] to 32.0 in-lb (3.6 N·m) to 35.0 in-lb (4.0 N·m).

(l) Connect the drain tube [7] to the oil tank filler cap [6].

1) Lubricate the threads of the nipple [9] with oil, D00599 [CP2442].

2) Engage the coupling nut [8] to the nipple [9].

3) Tighten the coupling nut [8] to 257.0 in-lb (29.0 N·m) to 283.0 in-lb (32.0 N·m).

SUBTASK 79-11-03-160-001-G00

(2) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-11-03-860-003-G00

(1) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

(a) On the applicable engine, close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

SUBTASK 79-11-03-860-004-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Oil Tank Filler Cap Installation Test

SUBTASK 79-11-03-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN FUEL/OIL HEAT EXCHANGER - REMOVAL/INSTALLATION

1. General

A. This procedure provides two tasks:

(1) Main fuel/oil heat exchanger removal

(2) Main fuel/oil heat exchanger installation.

TASK 79-21-01-000-801-G00

2. Main Fuel/Oil Heat Exchanger Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the main fuel/oil heat exchanger.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-11-12-000-801-G00	Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-21-01-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 79-21-01-860-003-G00

(3) For the left engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 79-21-01-860-004-G00

(4) For the right engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 79-21-01-010-001-G00

(5) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Main Fuel/Oil Heat Exchanger Removal

SUBTASK 79-21-01-030-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(1) Remove the main fuel/oil heat exchanger [1]:

(a) Put a fuel resistant container (5 gal), STD-201 below the main fuel/oil heat exchanger [1].

(b) Remove the main fuel/oil heat exchanger DP sensor [2] from the main fuel/oil heat exchanger [1] (TASK 73-11-12-000-801-G00).

(c) Disconnect the oil tube [3] from the main fuel/oil heat exchanger [1]:

1) Disconnect the fitting nut [4] from the oil tube [3].

2) Install the protective covers on the openings.

(d) Disconnect the oil tube [6] from the main fuel/oil heat exchanger [1]:

1) Remove the four bolts [7] from the oil tube [6].

2) Disconnect the oil tube [6] from the main fuel/oil heat exchanger [1].

3) Remove and discard the gasket [8].

4) Install the protective covers on the openings.

(e) Disconnect the oil tube [9] from the main fuel/oil heat exchanger [1]:

1) Remove the four bolts [10] from the oil tube [9].

2) Disconnect the oil tube [9] from the main fuel/oil heat exchanger [1].

3) Remove and discard the gasket [11].

4) Install the protective covers on the openings.

(f) Disconnect the fuel tube [12] from the main fuel/oil heat exchanger [1]:

1) Remove the four bolts [13] from the fuel tube [12].

2) Disconnect the fuel tube [12] from the main fuel/oil heat exchanger [1].

3) Remove and discard the gasket [14].

4) Install the protective covers on the openings.

(g) Disconnect the fuel tube [15] from the main fuel/oil heat exchanger [1]:

1) Remove the four bolts [16] from the fuel tube [15].

2) Disconnect the fuel tube [15] from the main fuel/oil heat exchanger [1].

3) Remove and discard the gasket [17].

4) Install the protective covers on the openings.

5) Remove the bolt [18] that attaches the clamp [19] to the fuel tube [15].

(h) Remove the main fuel/oil heat exchanger [1] from the three mounting brackets [24]:

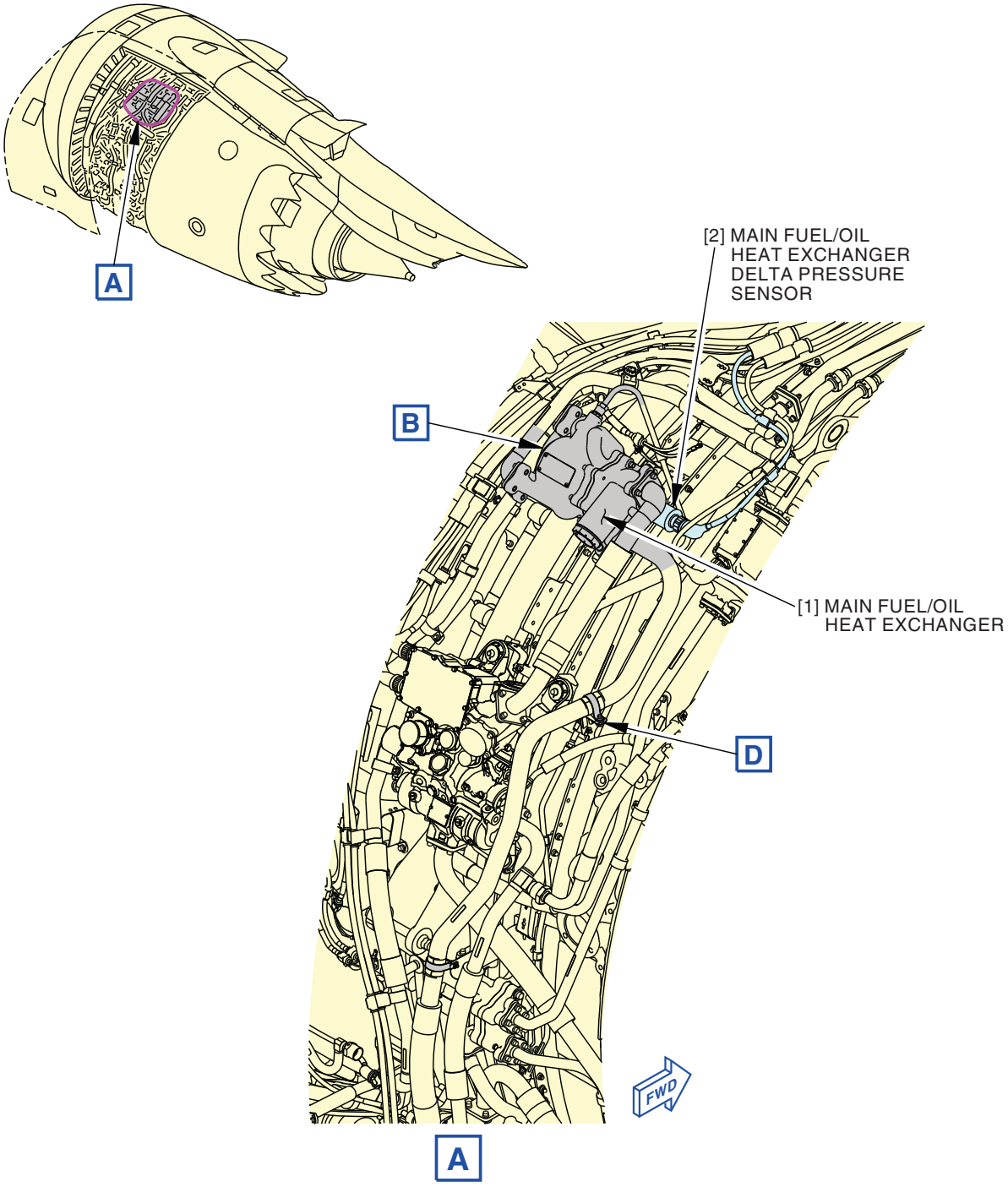
1) Remove the four bolts [22] and the four washers [23] that attach the main fuel/oil heat exchanger [1] to the three mounting brackets [24].

2) Remove the main fuel/oil heat exchanger [1] from the three mounting brackets [24].

3) Install the protective covers on the openings.

END OF TASK

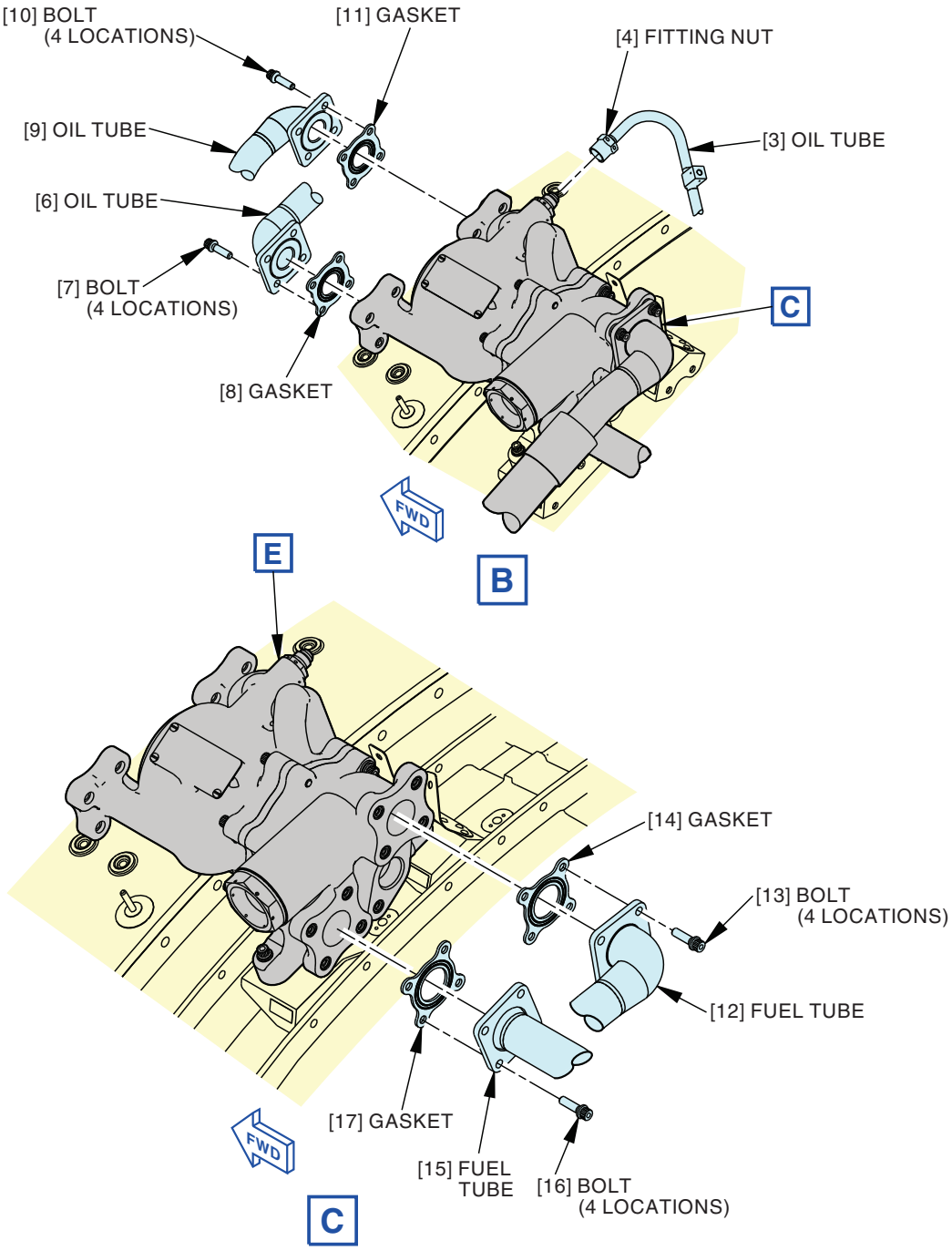
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-792101-B-F0301-A0001-A-001-01
2355334 S00061517045_V3

Main Fuel/Oil Heat Exchanger Installation
Figure 401/79-21-01-990-802-G00 (Sheet 1 of 4)

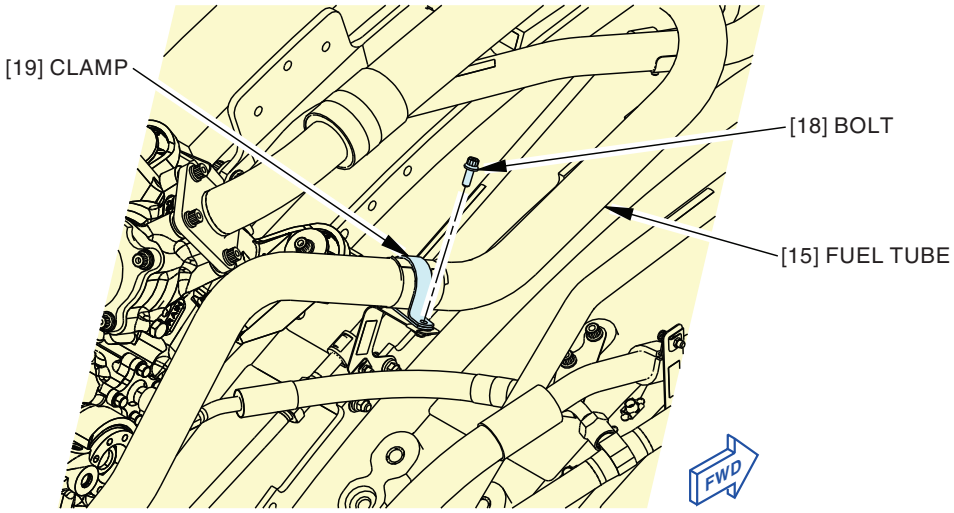
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-792101-B-F0301-A0002-A-001-01
2355335 S00061517046_V4

Main Fuel/Oil Heat Exchanger Installation
Figure 401/79-21-01-990-802-G00 (Sheet 2 of 4)

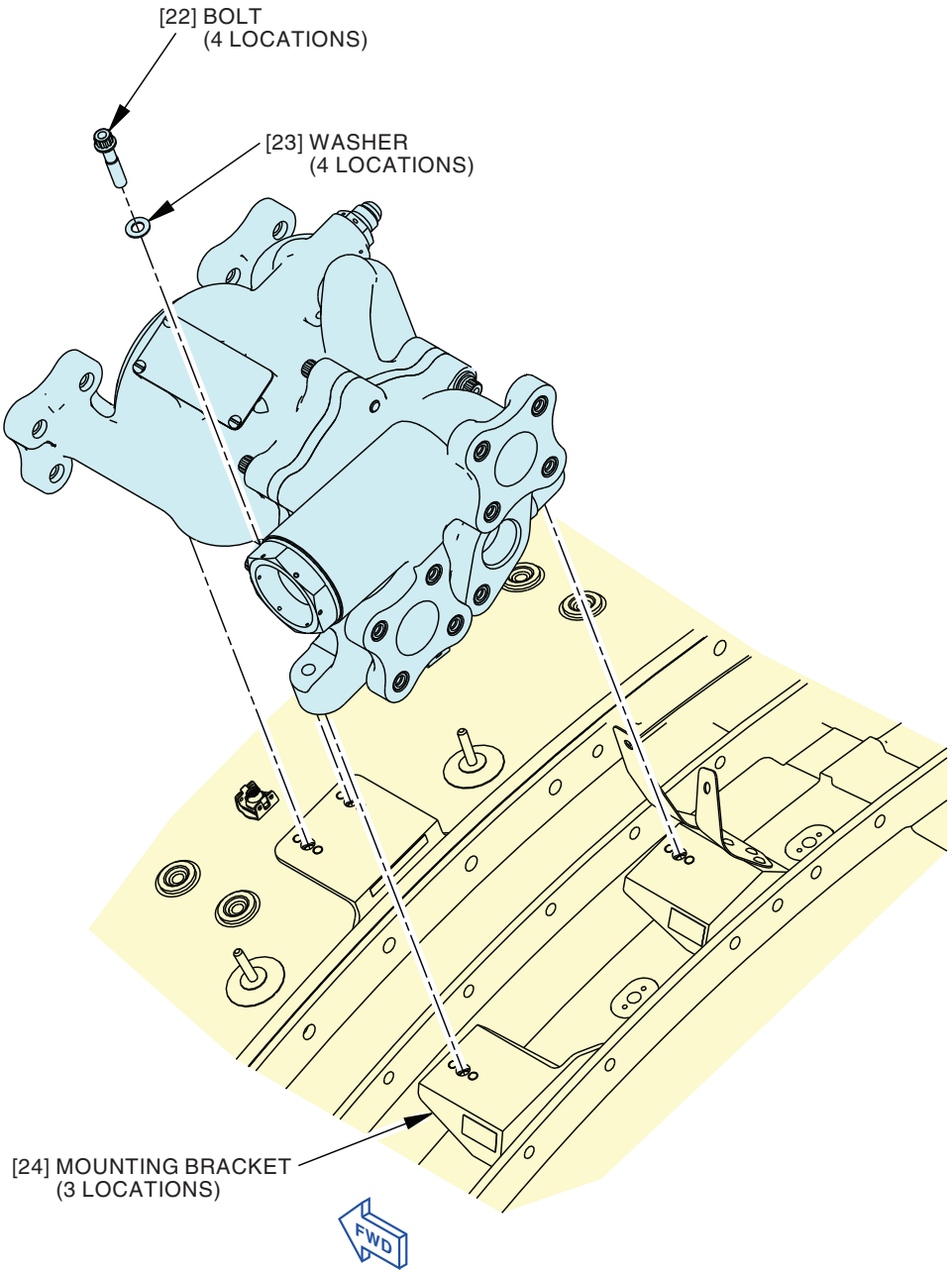
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-792101-B-F0301-A0003-A-001-01
2402886 S0000555519_V3

Main Fuel/Oil Heat Exchanger Installation
Figure 401/79-21-01-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



E

A0004-A-001-01
2403164 S0000555520_V1

Main Fuel/Oil Heat Exchanger Installation
Figure 401/79-21-01-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-01-400-801-G00

3. Main Fuel/Oil Heat Exchanger Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the main fuel/oil heat exchanger.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-12-400-801-G00	Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
8	Gasket	79-21-01-01-020	PTF ALL
14	Gasket	79-21-01-01-060	PTF ALL
17	Gasket	79-21-01-01-080	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Main Fuel/Oil Heat Exchanger Installation

SUBTASK 79-21-01-420-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT CAUSE DAMAGE TO THE PREFORMED PACKINGS WHEN YOU INSTALL THE HEAT EXCHANGER. A DAMAGED PACKING CAN CAUSE A LEAK IN THE ENGINE FUEL OR OIL SYSTEMS. THE LEAK CAN CAUSE AN ENGINE SHUTDOWN IN-FLIGHT.

(1) Install the main fuel/oil heat exchanger [1]:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Install the main fuel/oil heat exchanger [1] on the three mounting brackets [24].

1) Lubricate the threads and pressure faces of the four bolts [22] with engine oil, D00599 [CP2442].

2) Install the four bolts [22] and the four washers [23] on the main fuel/oil heat exchanger [1].

3) Tighten the bolts [22] in a criss cross pattern to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.42 N·m).

4) Remove the unwanted engine oil, D00599 [CP2442].

(d) Connect the fuel tube [15] to the main fuel/oil heat exchanger [1]:

1) Lubricate the threads and pressure faces of the four bolts [16] with graphite mineral vaseline, D00601 [CP2101].

2) Lubricate each side of the elastomer part of a new gasket [17] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

3) Install the gasket [17] and the fuel tube [15] on the main fuel/oil heat exchanger [1].

4) Install the bolts [16] on the fuel tube [15].

5) Tighten the four bolts [16] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

6) Install the bolt [18] and the clamp [19] that attach the fuel tube [15] to the engine bracket.

7) Tighten the bolt [18] to 42 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

8) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(e) Connect the fuel tube [12] to the main fuel/oil heat exchanger [1]:

1) Lubricate the threads and pressure faces of the four bolts [13] with graphite mineral vaseline, D00601 [CP2101].

2) Lubricate each side of the elastomer part of a new gasket [14] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

3) Install the gasket [14] and the fuel tube [12] on the main fuel/oil heat exchanger [1].

4) Install the four bolts [13] on the fuel tube [12].

5) Tighten the four bolts [13] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

6) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(f) Connect the oil tube [9] to the main fuel/oil heat exchanger [1]:

1) Lubricate the threads and pressure faces of the four bolts [10] with graphite mineral vaseline, D00601 [CP2101].

2) Lubricate each side of the elastomer part of a new gasket [11] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

3) Install the gasket [11] and the oil tube [9] on the main fuel/oil heat exchanger [1].

4) Install the four bolts [10] on the oil tube [9].

5) Tighten the four bolts [10] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

6) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101] with a lint-free cloth, G51335 [CP2674].

(g) Connect the oil tube [6] to the main fuel/oil heat exchanger [1]:

1) Lubricate the threads and pressure faces of the four bolts [7] with graphite mineral vaseline, D00601 [CP2101].

2) Lubricate each side of the elastomer part of a new gasket [8] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].

3) Install the gasket [8] and the oil tube [6] on the main fuel/oil heat exchanger [1].

4) Install the four bolts [7] on the oil tube [6].

5) Tighten the four bolts [7] in a criss cross pattern to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

6) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(h) Connect the oil tube [3] to the main fuel/oil heat exchanger [1]:

1) Lubricate the threads of the fitting nut [4] with engine oil, D00599 [CP2442].

2) Install the fitting nut [4] on the bolt [7].

3) Tighten the fitting nut [4] to 266 in-lb (30 N·m) - 310 in-lb (35 N·m).

(i) Do this task: Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation, TASK 73-11-12-400-801-G00.

(j) Clean the work area and clear the tools you used to do the task.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-01-860-005-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 79-21-01-860-006-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 79-21-01-860-007-G00

(3) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 79-21-01-860-008-G00

(4) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 79-21-01-410-001-G00

(5) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Main Fuel/Oil Heat Exchanger Installation Test

SUBTASK 79-21-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE SURFACE AIR COOLING OIL COOLER - REMOVAL/INSTALLATION

1. General

A. This procedure has four tasks:

(1) Engine Surface Air Cooling Oil Cooler 1 Removal

(2) Engine Surface Air Cooling Oil Cooler 1 Installation

(3) Engine Surface Air Cooling Oil Cooler 2 Removal

(4) Engine Surface Air Cooling Oil Cooler 2 Installation.

TASK 79-21-02-000-801-G00

2. Engine Surface Air Cooling Oil Cooler 1 Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the engine Surface Air Cooled Oil Cooler (SACOC) 1.

B. References

Reference	Title
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-02-010-001-G00

(2) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Engine Surface Air Cooling Oil Cooler 1 Removal

SUBTASK 79-21-02-030-001-G00

WARNING DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION

DO NOT LET OIL GET ON THE ENGINE, OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE AREAS THAT OIL FALLS ON. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(1) Drain the engine oil system (TASK 12-13-11-600-803).

SUBTASK 79-21-02-020-001-G00

(2) Disconnect the oil tube [6] from the engine SACOC 1 [1] as follows:

(a) Loosen the coupling nut [5] that attaches the oil tube [6] to the engine SACOC 1 [1].

(b) Disconnect the oil tube [6] from the engine SACOC 1 [1].

SUBTASK 79-21-02-020-002-G00

(3) Disconnect the oil tube [7] from the engine SACOC 1 [1] as follows:

(a) Loosen the coupling nut [8] that attaches the oil tube [7] to the engine SACOC 1 [1].

(b) Disconnect the oil tube [7] from the engine SACOC 1 [1].

SUBTASK 79-21-02-480-002-G00

(4) Install protective covers on the oil tube [6], oil tube [7], and ports on the engine SACOC 1 [1].

SUBTASK 79-21-02-020-003-G00

(5) Remove the engine SACOC 1 [1] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(a) Remove the 14 bolts with plates [9] that attach the engine SACOC 1 [1] to the engine.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(b) Remove the 14 bolts with integrated washers [9] that attach the engine SACOC 1 [1] to the engine.

PTF ALL

(c) Remove the two bolts with integrated washers [4] that attach the engine SACOC 1 [1] to the engine.

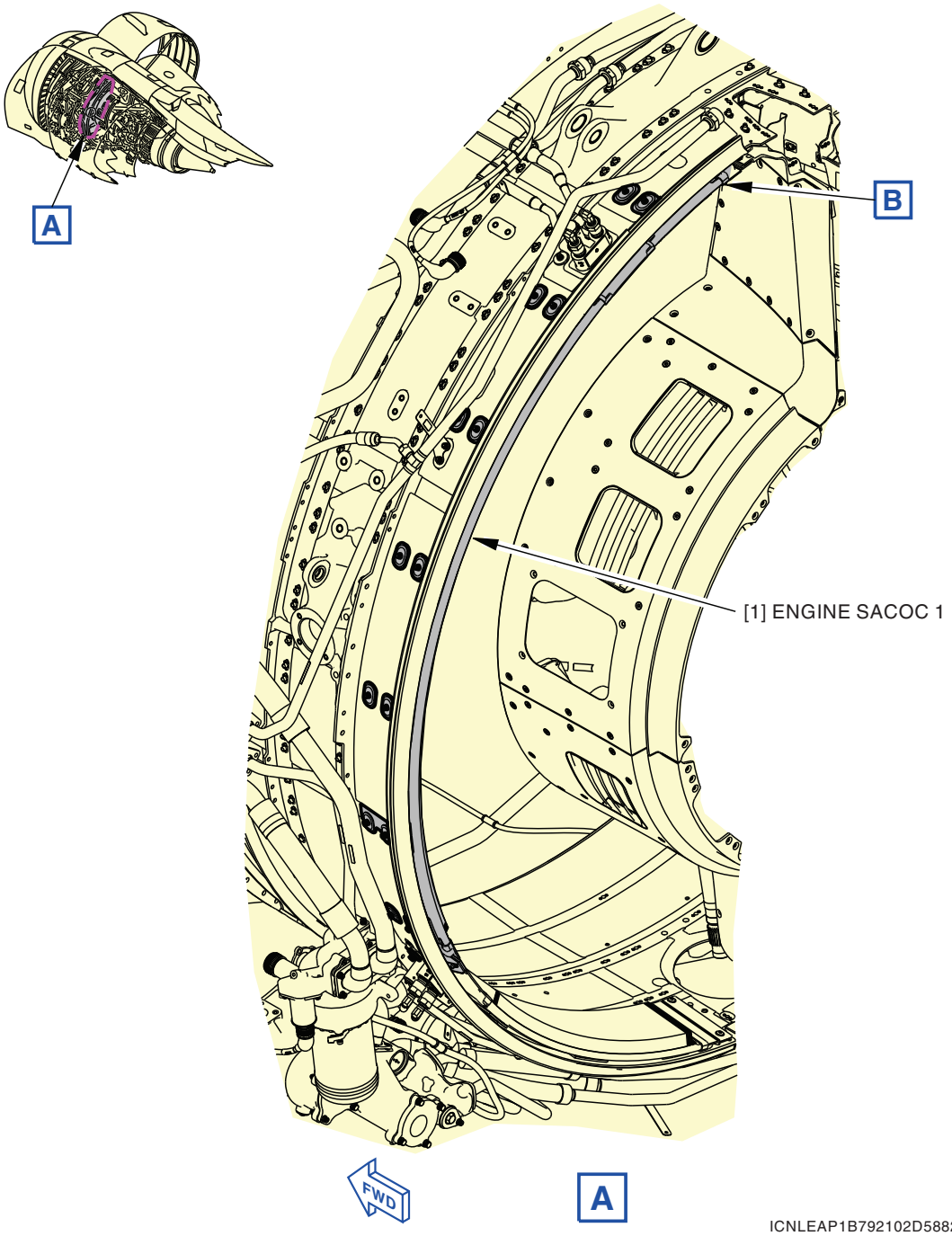
(d) Remove the two screws [13] and attachment clamp [14] that attach the engine SACOC 1 [1] to the bracket [27].

(e) Remove the two screws [12] and attachment clamp [11] that attach the engine SACOC 1 [1] to the bracket [10].

(f) Remove the engine SACOC 1 [1] from the engine.

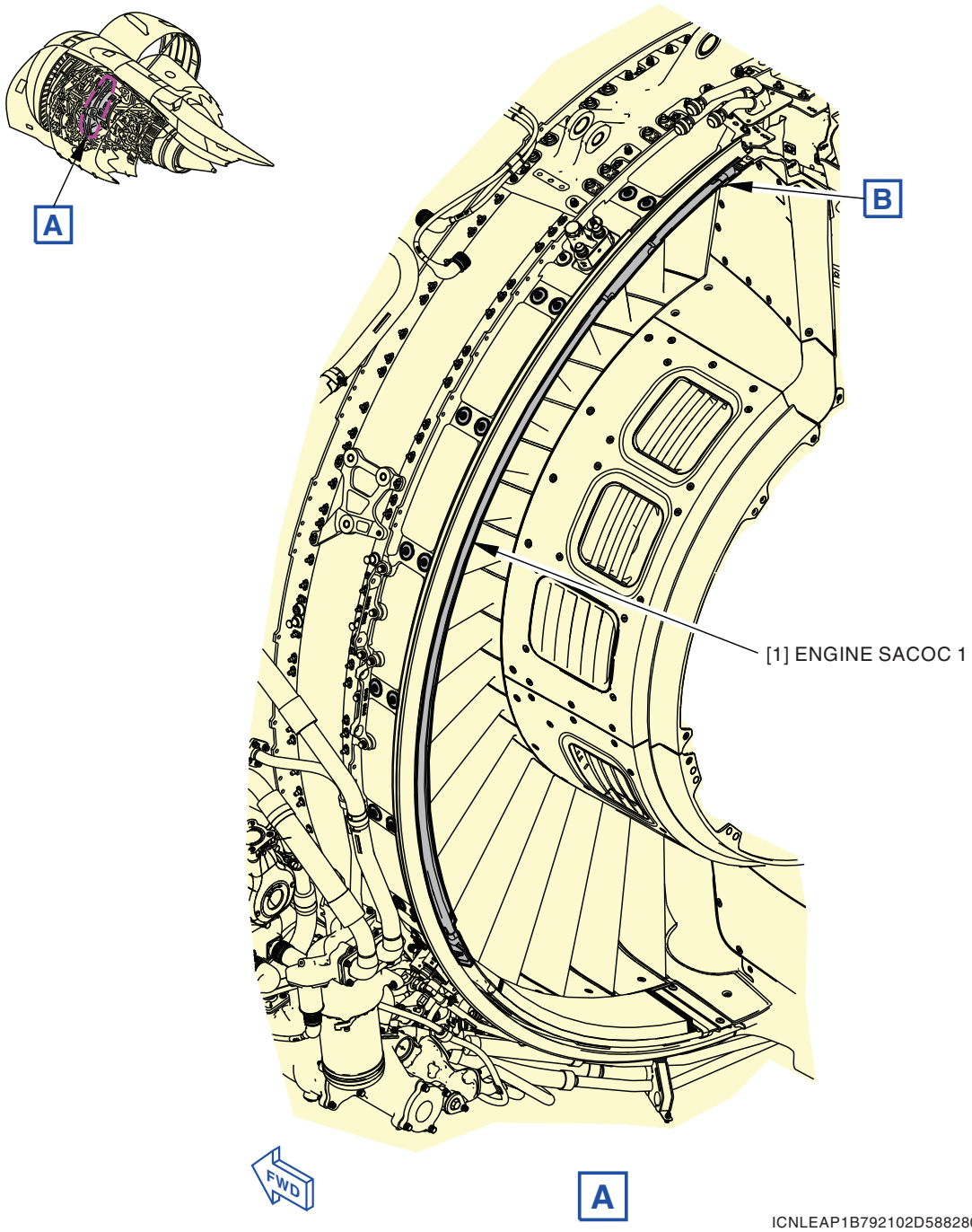
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



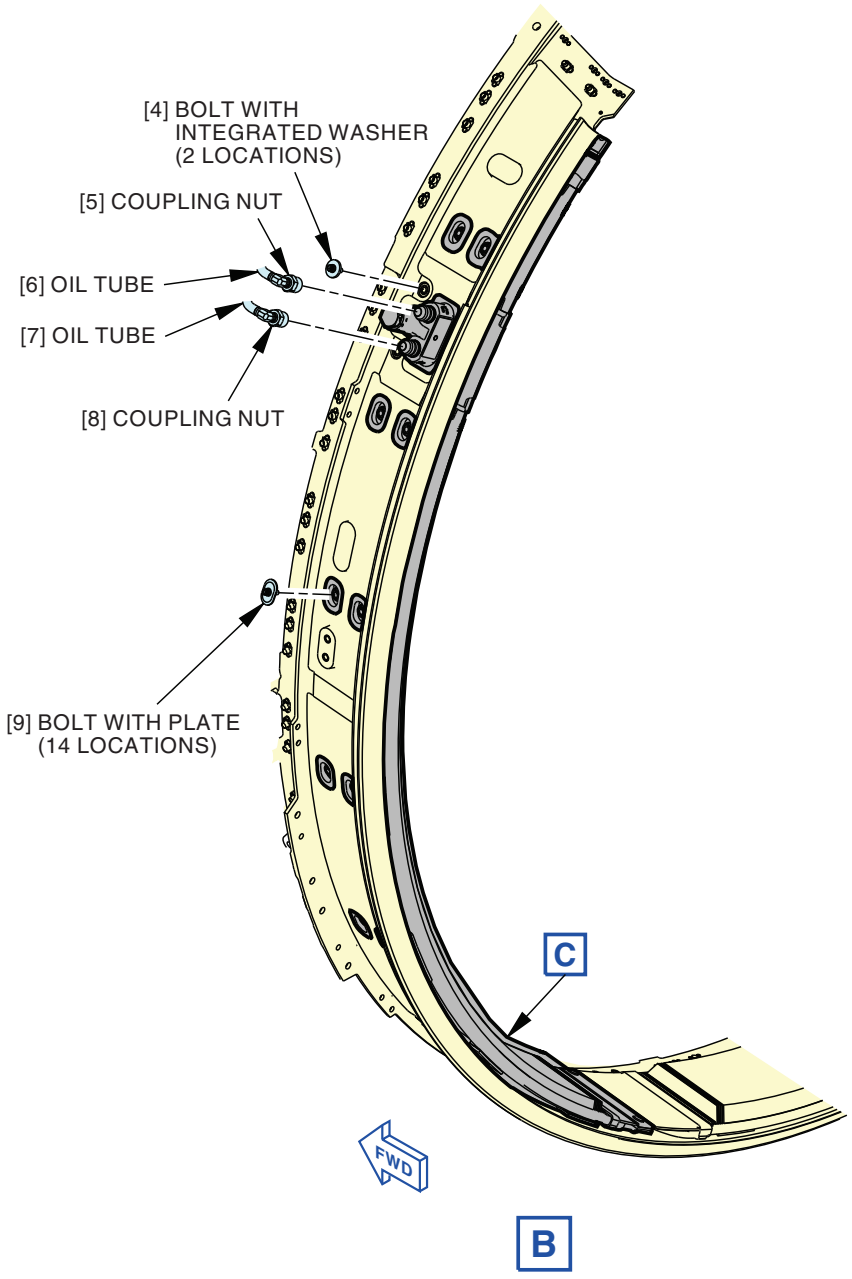
Engine Surface Air Cooling Oil Cooler 1 Installation
Figure 401/79-21-02-990-802-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Surface Air Cooling Oil Cooler 1 Installation
Figure 401/79-21-02-990-802-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

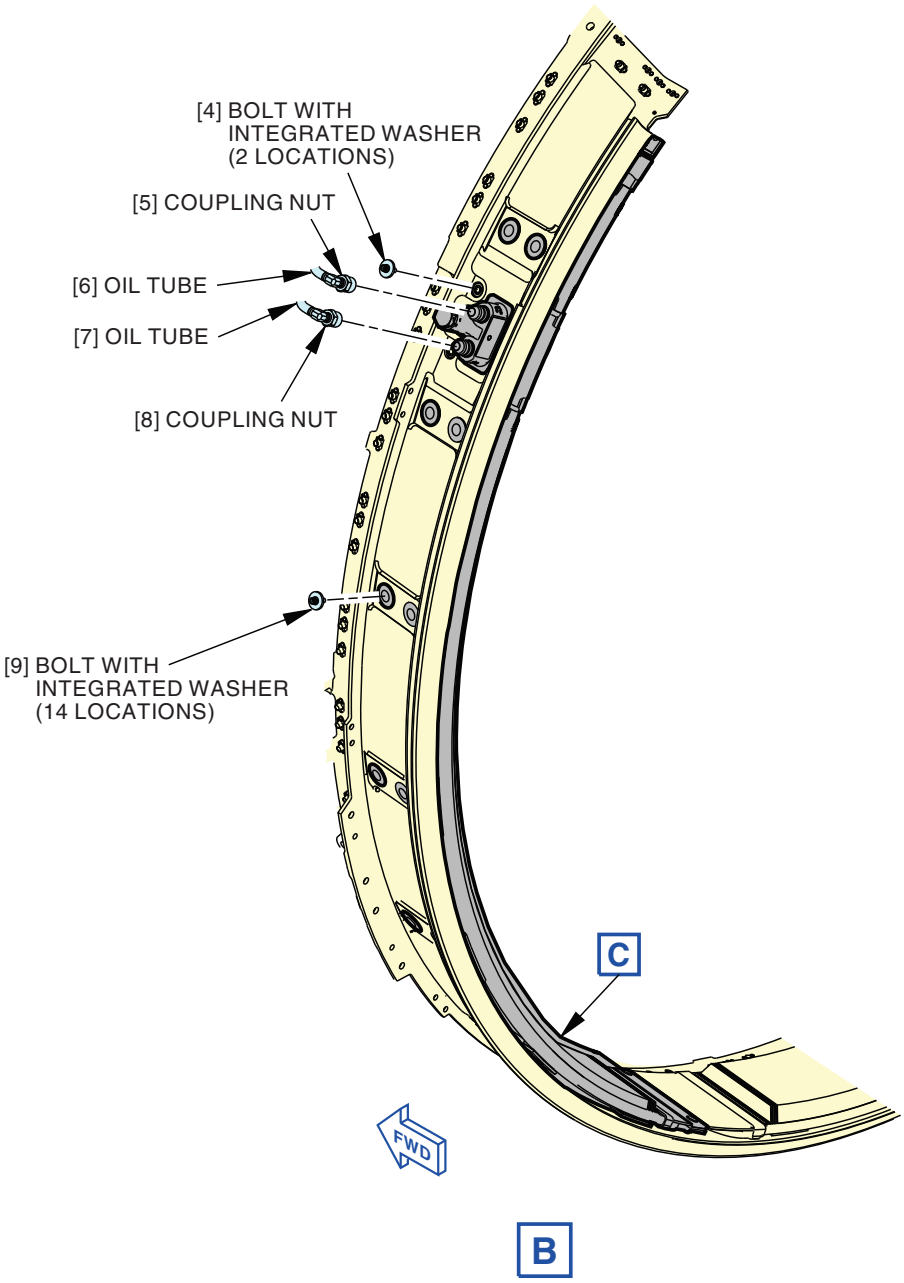


ICNLEAP1B792102D5882800002A00101

2355339 S00061517054_V5

Engine Surface Air Cooling Oil Cooler 1 Installation
Figure 401/79-21-02-990-802-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

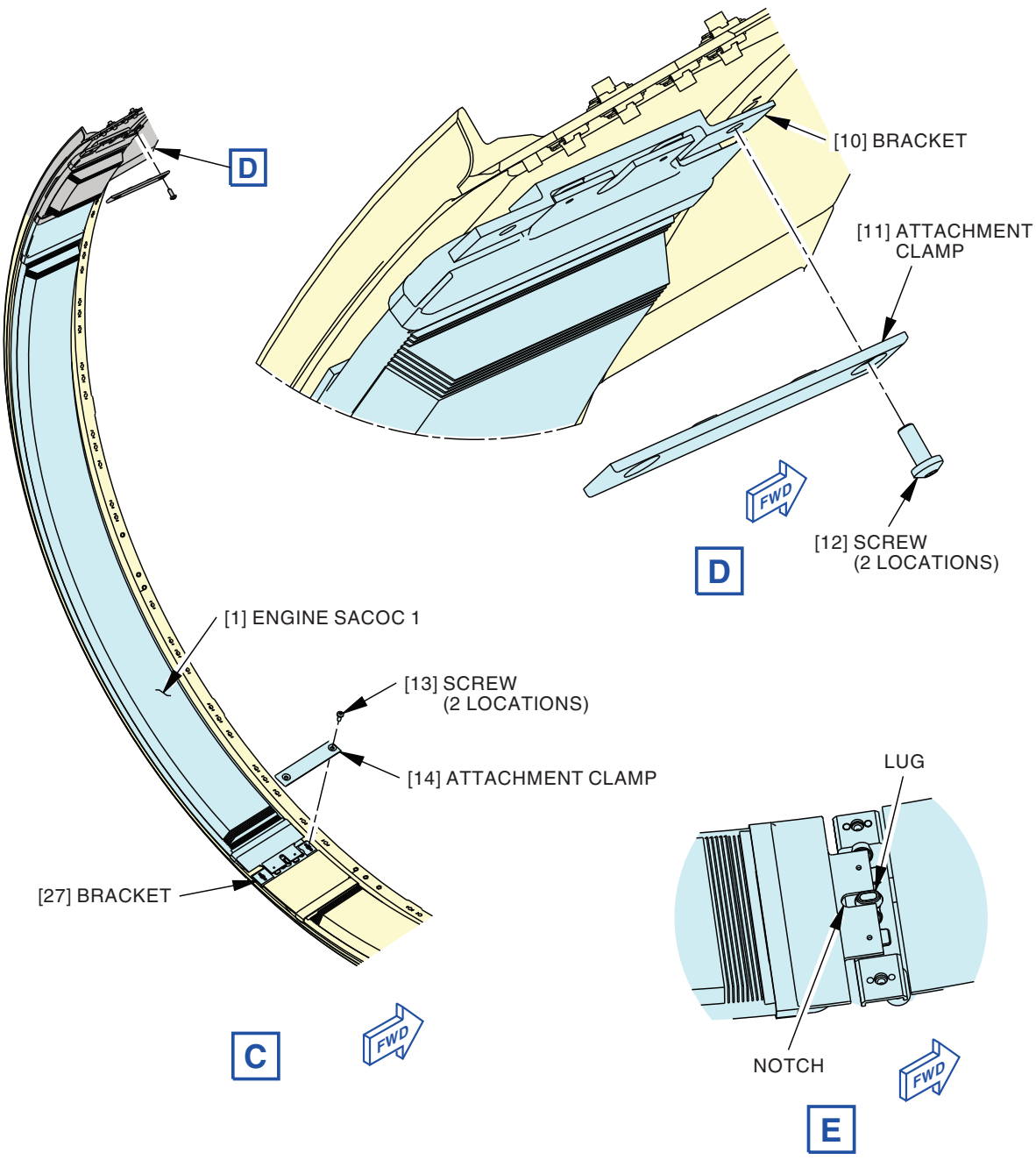


ICNLEAP1B792102D5882800011A00101

2841248 S0000663124_V1

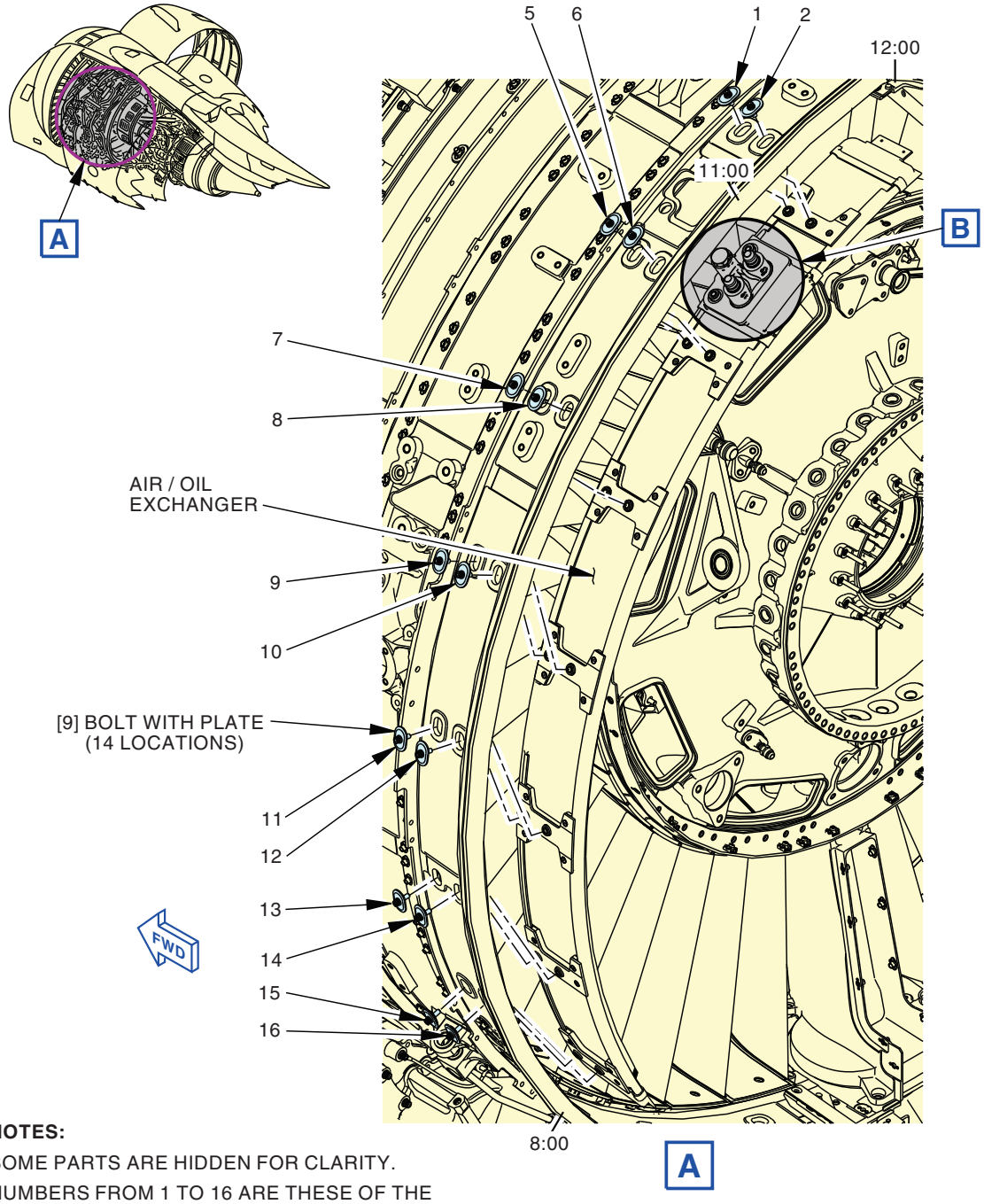
Engine Surface Air Cooling Oil Cooler 1 Installation
Figure 401/79-21-02-990-802-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Engine Surface Air Cooling Oil Cooler 1 Installation
Figure 401/79-21-02-990-802-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTES:

SOME PARTS ARE HIDDEN FOR CLARITY.

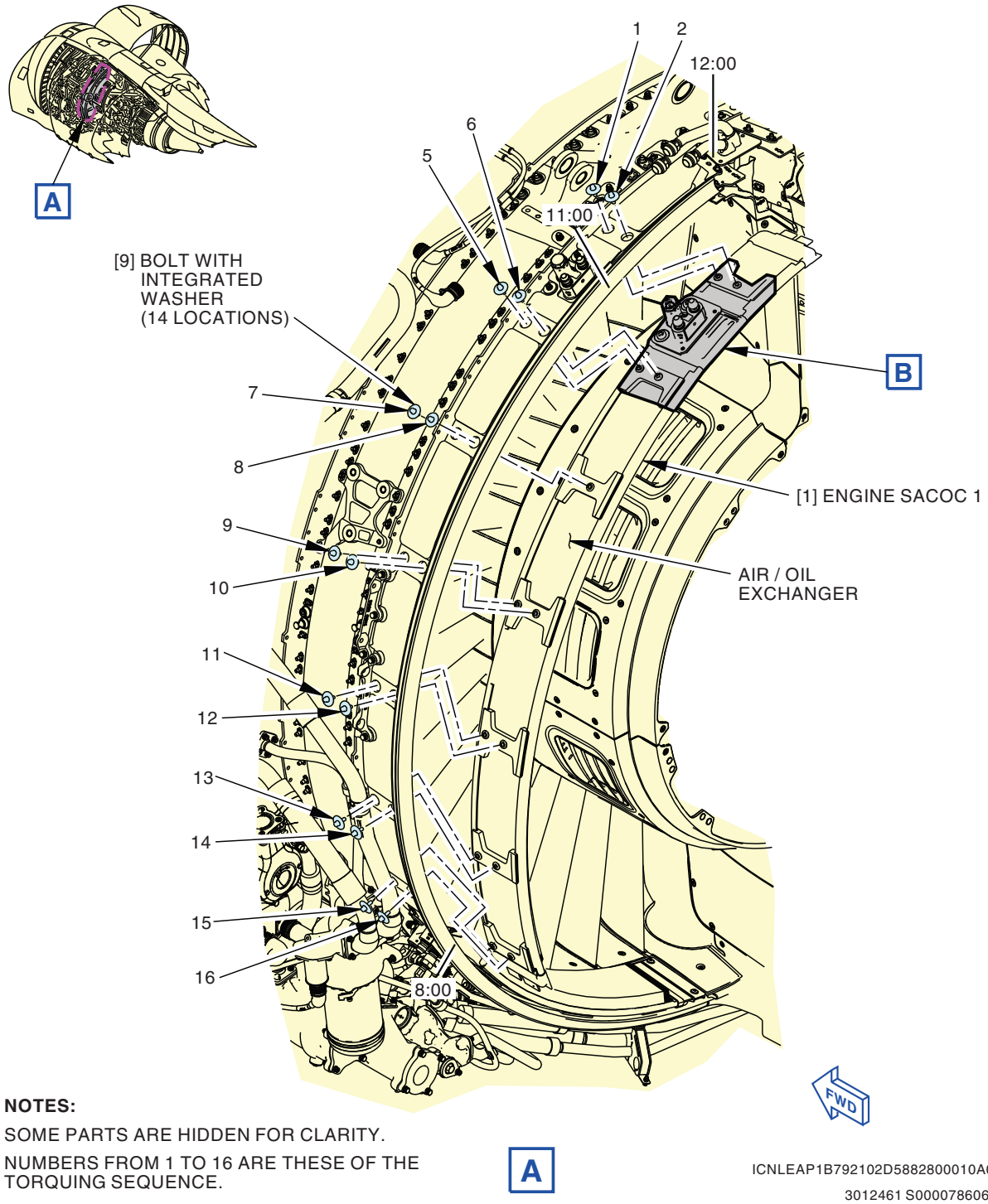
NUMBERS FROM 1 TO 16 ARE THESE OF THE TORQUING SEQUENCE.

ICNLEAP1B792102D5882800014A00101

2946276 S0000723171_V2

Bolt Tightening Sequence
Figure 402/79-21-02-990-806-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTES:

SOME PARTS ARE HIDDEN FOR CLARITY.

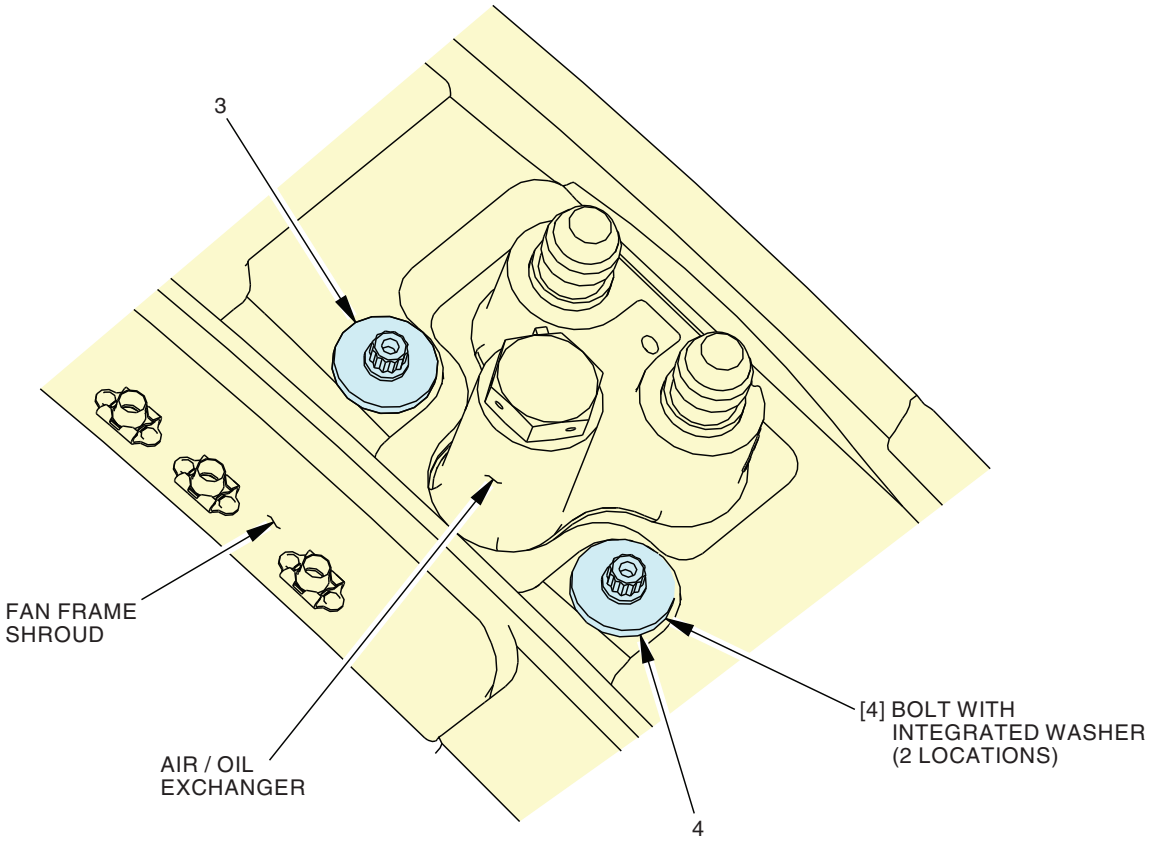
NUMBERS FROM 1 TO 16 ARE THESE OF THE TORQUING SEQUENCE.

ICNLEAP1B792102D5882800010A00101

3012461 S0000786061_V1

Bolt Tightening Sequence
Figure 402/79-21-02-990-806-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

NUMBERS FROM 3 AND 4 ARE THESE OF THE TORQUING SEQUENCE.

ICNLEAP1B792102D5882800015A00101

2946282 S0000723172_V1

Bolt Tightening Sequence
Figure 402/79-21-02-990-806-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 79-21-02-400-801-G00

3. Engine Surface Air Cooling Oil Cooler 1 Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the engine Surface Air Cooled Oil Cooler (SACOC) 1.

B. References

Reference	Title
12-13-11-600-801	Engine Servicing (Oil Replenishing) (P/B 301)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50474 [CP2256]	Locking Agent - Liquid	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

G. Engine Surface Air Cooling Oil Cooler 1 Installation

SUBTASK 79-21-02-420-001-G00

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION

DO NOT LET OIL GET ON THE ENGINE, OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE AREAS THAT OIL FALLS ON. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(1) Install the engine SACOC 1 [1]:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

MAKE SURE THAT YOU CORRECTLY ALIGN THE LUG OF THE BRACKET WITH THE NOTCH OF THE ENGINE SURFACE AIR COOLED OIL COOLER (SACOC) 1. IF YOU DO NOT OBEY, DAMAGE TO THE ENGINE SACOC 1 CAN OCCUR.

(b) Install the engine SACOC 1 [1] on the engine.

(c) Remove the protective covers from the engine SACOC 1 [1], oil tube [6], and oil tube [7].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(d) Lubricate the threads of the 14 bolts with plates [9] and two bolts with integrated washers [4] with graphite mineral vaseline, D00601 [CP2101].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(e) Lubricate the threads of the 14 bolts with integrated washers [9] and two bolts with integrated washers [4] with graphite mineral vaseline, D00601 [CP2101].

PTF ALL

(f) Put the bracket [10] and engine SACOC 1 [1] in position at the same time.

1) Slide the bracket [10] between the engine SACOC 1 [1] and fan case.

2) Make sure that the assembly direction is correct.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

CAUTION

BE CAREFUL WHEN YOU INSTALL THE BOLTS ON THE AIR/OIL EX-CHANGER. IF YOU DO NOT OBEY, DAMAGE TO THE LIP SEAL OF THE AIR/OIL EX-CHANGER CAN OCCUR.

(g) Install the 14 bolts with plates [9] and two bolts with integrated washers [4] that attach the engine SACOC 1 [1] to the engine.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

CAUTION

BE CAREFUL WHEN YOU INSTALL THE BOLTS ON THE AIR/OIL EX-CHANGER. IF YOU DO NOT OBEY, DAMAGE TO THE LIP SEAL OF THE AIR/OIL EX-CHANGER CAN OCCUR.

(h) Install the 14 bolts with integrated washers [9] and two bolts with integrated washers [4] that attach the engine SACOC 1 [1] to the engine.

PTF ALL

(i) Remove the covers.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(j) Tighten the bolts with plates [9] from the top to the bottom, following the sequence 1 to 16 then re-tighten the bolts with plates [9] a second time following the same sequence order as follows (Figure 402):

1) Tighten the two bolts with plates [9], following the specific sequence (start by 1 and finish by 2), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

2) Tighten the two bolts with integrated washers [4], following the specific sequence (start by 3 and finish by 4), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181 (Continued)

3) Tighten the 12 remaining bolts with plates [9], from the top to the bottom following the specific sequence (start by 5 until 16), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

4) Tighten the bolts with plates [9] and bolts with integrated washers [4] again in accordance with the requested torque value, follow the sequence order from the top to the bottom.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(k) Tighten the bolts with integrated washers [9] from the top to the bottom, following the sequence 1 to 16 then re-tighten the bolts with plates [9] a second time following the same sequence order as follows (Figure 402):

1) Tighten the two bolts with integrated washers [9], following the specific sequence (start by 1 and finish by 2), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

2) Tighten the two bolts with integrated washers [4], following the specific sequence (start by 3 and finish by 4), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

3) Tighten the 12 remaining bolts with integrated washers [9], from the top to the bottom following the specific sequence (start by 5 until 16), to 110.6 in-lb (12.5 N·m) - 119.5 in-lb (13.5 N·m).

4) Tighten the bolts with integrated washers [9] and bolts with integrated washers [4] again in accordance with the requested torque value, follow the sequence order from the top to the bottom.

PTF ALL

(l) Install the attachment clamp [14] and attachment clamp [11] to secure the engine SACOC 1 [1] as follows:

1) Apply Loctite 241, A50474 [CP2256], on the threads and under the head of the two screws [12].

2) Lubricate the threads of the two screws [13] with graphite mineral vaseline, D00601 [CP2101].

3) Install the attachment clamp [14] and two screws [13] that attach the engine SACOC 1 [1] to the bracket [27].

4) Install the attachment clamp [11] and two screws [12] that attach the engine SACOC 1 [1] to the bracket [10].

5) Tighten the two screws [12] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

6) Tighten the two screws [13] to 84.1 in-lb (9.5 N·m) - 92.9 in-lb (10.5 N·m).

(m) Connect the oil tube [7] to the engine SACOC 1 [1] as follows:

1) Lubricate the threads of the coupling nut [8] with oil, D00599 [CP2442].

2) Connect the oil tube [7] with the coupling nut [8] to the engine SACOC 1 [1].

3) Tighten the coupling nut [8] to 443 in-lb (50 N·m) - 531 in-lb (60 N·m).

(n) Connect the oil tube [6] to the engine SACOC 1 [1] as follows:

1) Lubricate the threads of the coupling nut [5] with oil, D00599 [CP2442].

2) Connect the oil tube [6] with the coupling nut [5] to the engine SACOC 1 [1].

3) Tighten the coupling nut [5] to 443 in-lb (50 N·m) - 531 in-lb (60 N·m).

(o) Clean the work area and clear the tools used to do the procedure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 79-21-02-600-001-G00

(1) Do this task: Engine Servicing (Oil Replenishing), TASK 12-13-11-600-801.

SUBTASK 79-21-02-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 79-21-02-860-002-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

I. Engine Surface Air Cooling Oil Cooler 1 Installation Test

SUBTASK 79-21-02-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 79-21-02-000-802-G00

4. Engine Surface Air Cooling Oil Cooler 2 Removal

(Figure 403)

A. General

(1) This task gives the instructions to remove the engine Surface Air Cooled Oil Cooler (SACOC) 2.

B. References

Reference	Title
12-13-11-600-803	Engine Oil Change Servicing (P/B 301)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 79-21-02-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 79-21-02-010-002-G00

(2) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

G. Engine Surface Air Cooling Oil Cooler 2 Removal

SUBTASK 79-21-02-680-001-G00

WARNING

DO NOT TOUCH THE COMPONENTS OF THE OIL SYSTEM IF THE ENGINE IS HOT. THESE COMPONENTS STAY HOTTER THAN OTHER COMPONENTS. HOT COMPONENTS CAN BURN YOU.

WARNING

DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE STOPS. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING

DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION DO NOT LET OIL GET ON THE ENGINE, OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE AREAS THAT OIL FALLS ON. OIL CAN CAUSE DAMAGE TO PAINT AND RUBBER.

(1) Drain the engine oil system (TASK 12-13-11-600-803).

SUBTASK 79-21-02-010-003-G00

(2) Disconnect the oil tube [17] from the engine SACOC 2 [2] as follows:

(a) Loosen the coupling nut [16] that attaches the oil tube [17] to the engine SACOC 2 [2].

(b) Disconnect the oil tube [17] from the engine SACOC 2 [2].

SUBTASK 79-21-02-010-004-G00

(3) Disconnect the oil tube [18] from the engine SACOC 2 [2] as follows:

(a) Loosen the coupling nut [19] that attaches the oil tube [18] to the engine SACOC 2 [2].

(b) Disconnect the oil tube [18] from the engine SACOC 2 [2].

SUBTASK 79-21-02-480-001-G00

(4) Install the protective covers on the oil tube [17], oil tube [18], and engine SACOC 2 [2].

SUBTASK 79-21-02-020-004-G00

(5) Remove the engine SACOC 2 [2] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0181

(a) Remove the 14 bolts with plates [20] that attach the engine SACOC 2 [2] to the engine.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0181

(b) Remove the 14 bolts with integrated washers [20] that attach the engine SACOC 2 [2] to the engine.

PTF ALL

(c) Remove the two bolts with integrated washers [15] that attach the engine SACOC 2 [2] to the engine.

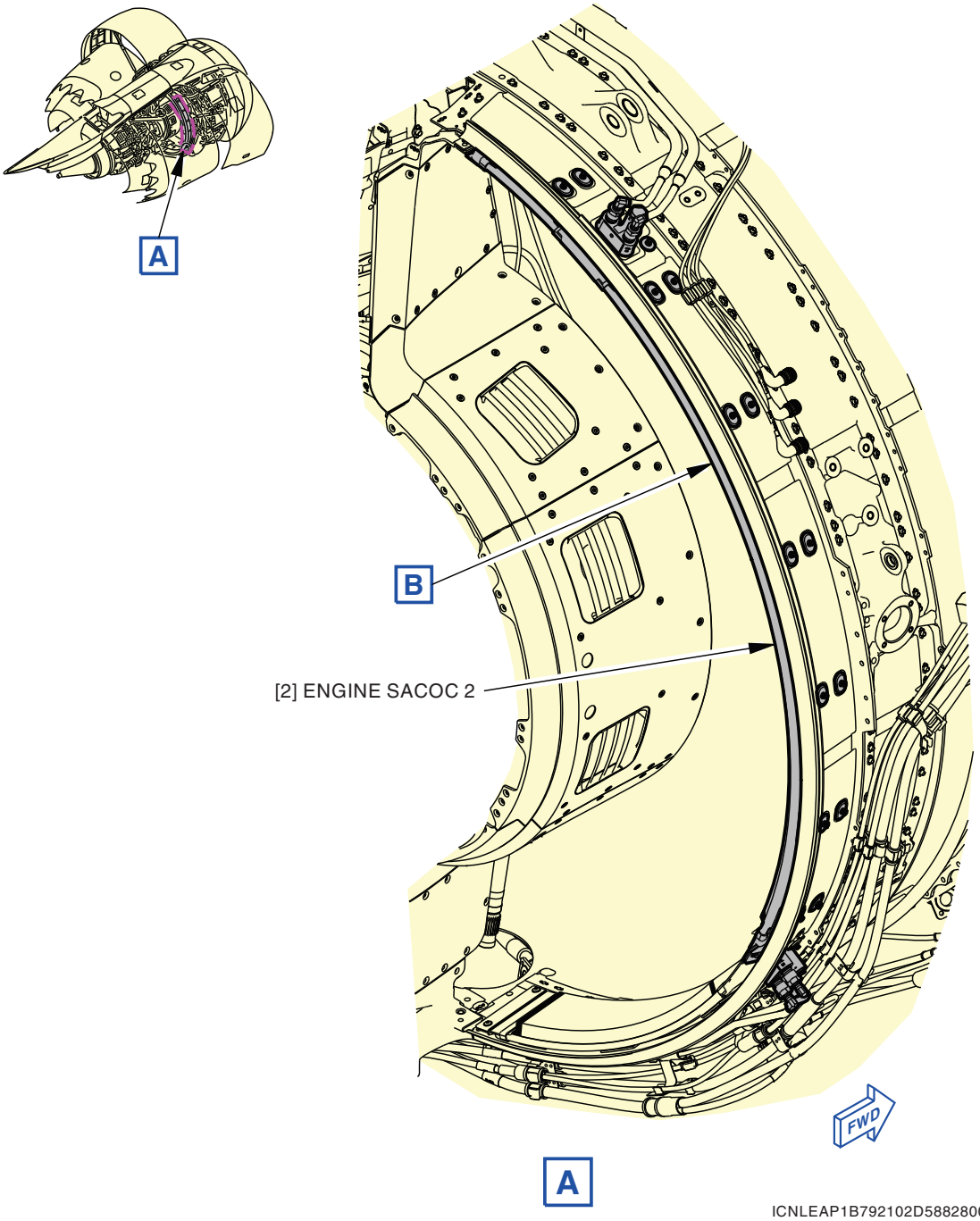
(d) Remove the two screws [23] and attachment clamp [22] that attach the engine SACOC 2 [2] to the bracket [21].

(e) Remove the two screws [24] and attachment clamp [25] that attach the engine SACOC 2 [2] to the bracket [26].

(f) Remove the engine SACOC 2 [2] from the engine.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

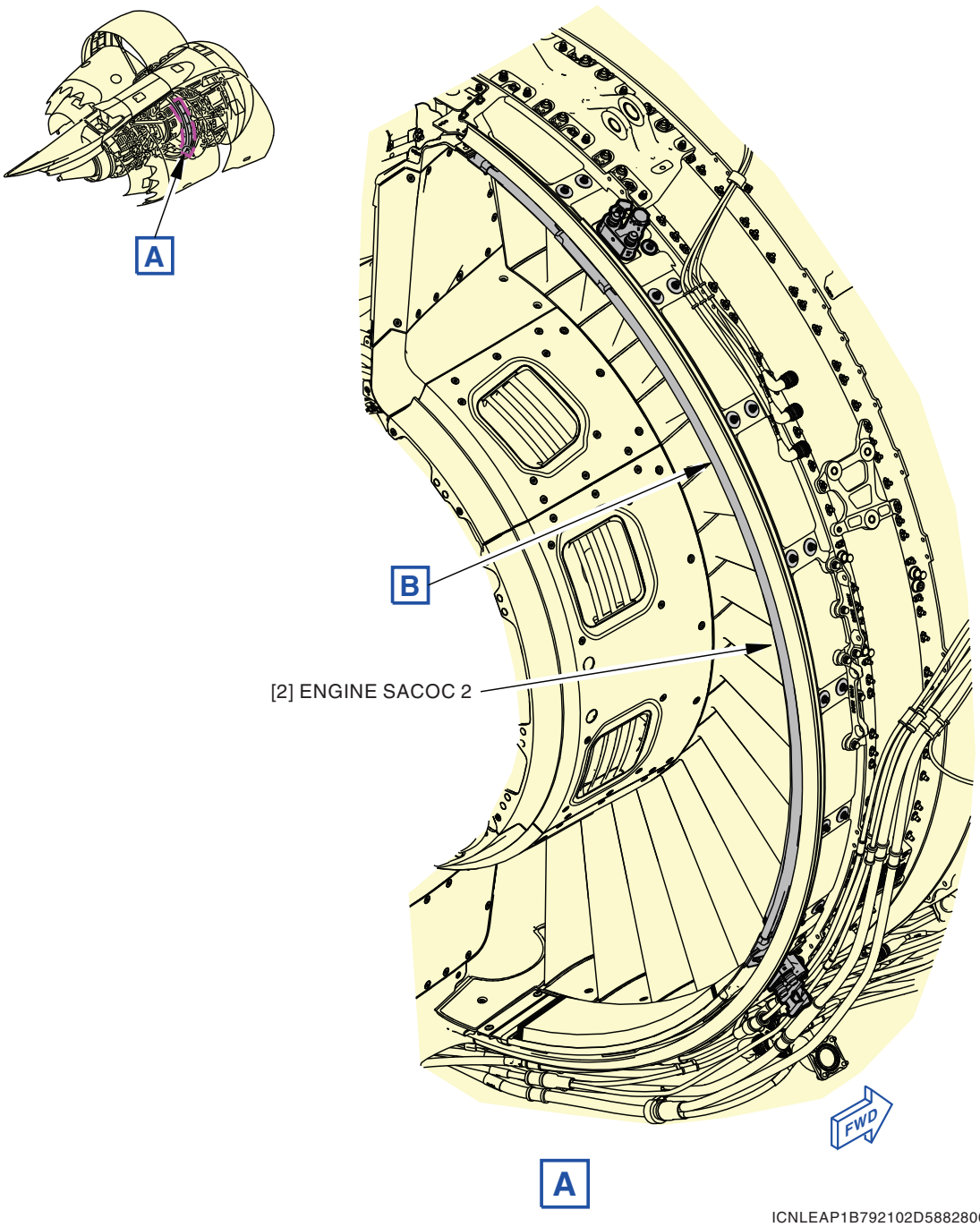


ICNLEAP1B792102D5882800004A00101

2355345 S00061517062_V4

Engine Surface Air Cooling Oil Cooler 2 Installation
Figure 403/79-21-02-990-804-G00 (Sheet 1 of 5)

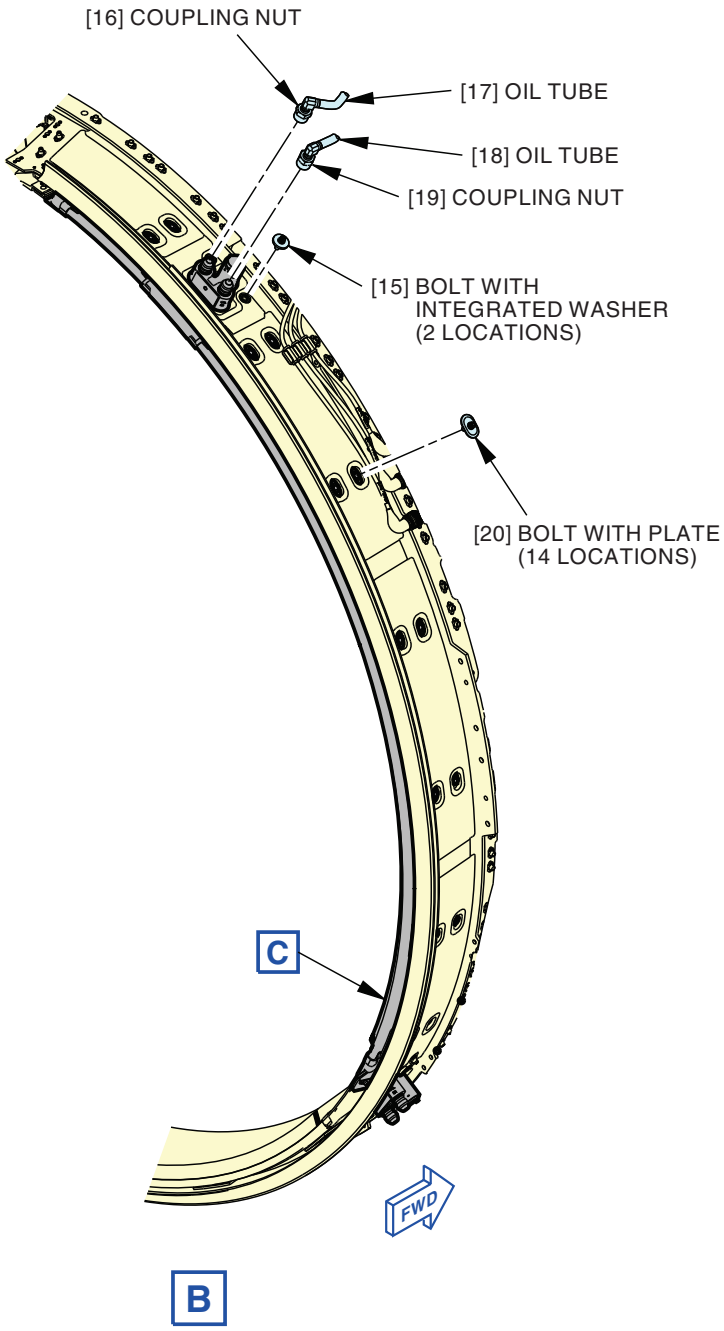
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B792102D5882800012A00101
2841258 S0000663127_V2

Engine Surface Air Cooling Oil Cooler 2 Installation
Figure 403/79-21-02-990-804-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

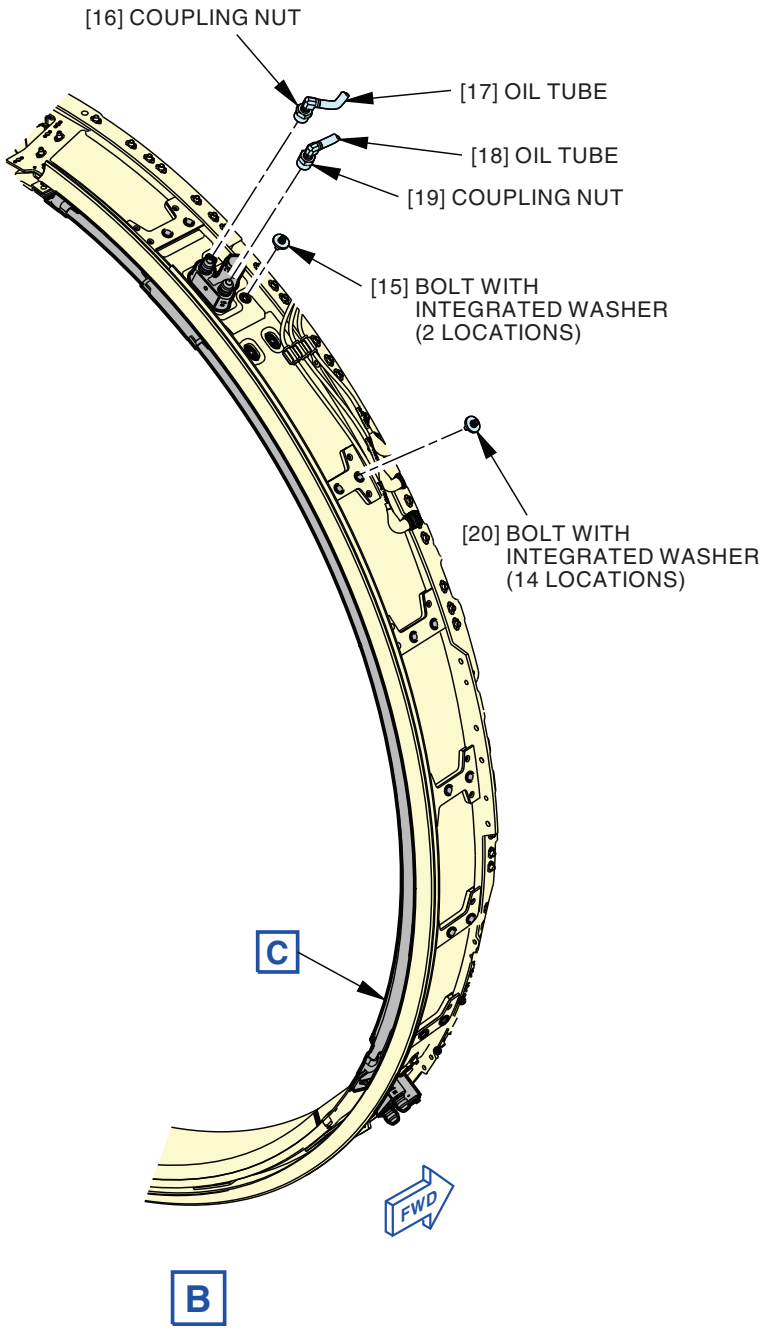


ICNLEAP1B792102D5882800005A00101

2422245 S0000556951_V5

Engine Surface Air Cooling Oil Cooler 2 Installation
Figure 403/79-21-02-990-804-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

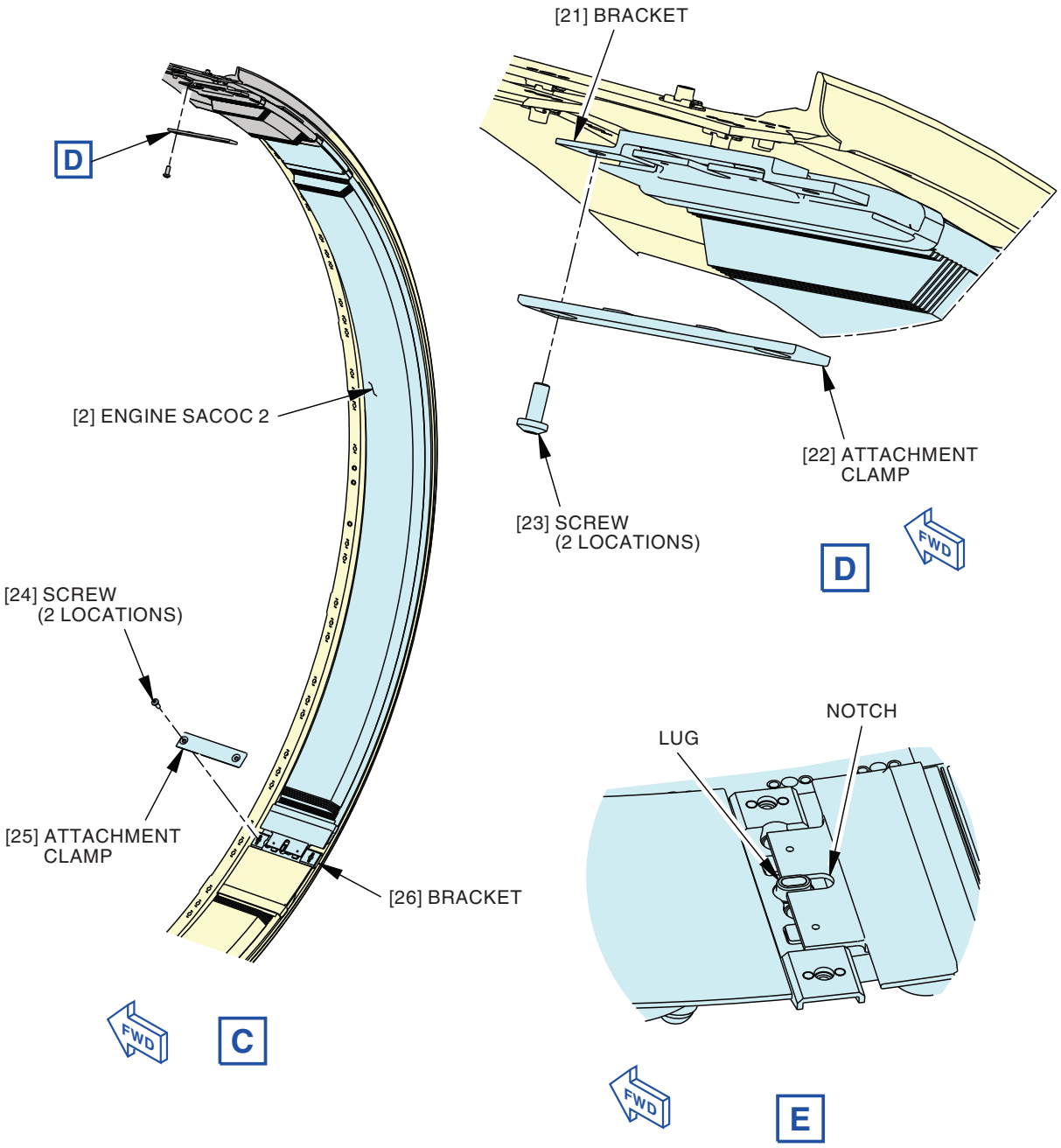


ICNLEAP1B792102D5882800013A00101

2841260 S0000663128_V1

Engine Surface Air Cooling Oil Cooler 2 Installation
Figure 403/79-21-02-990-804-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

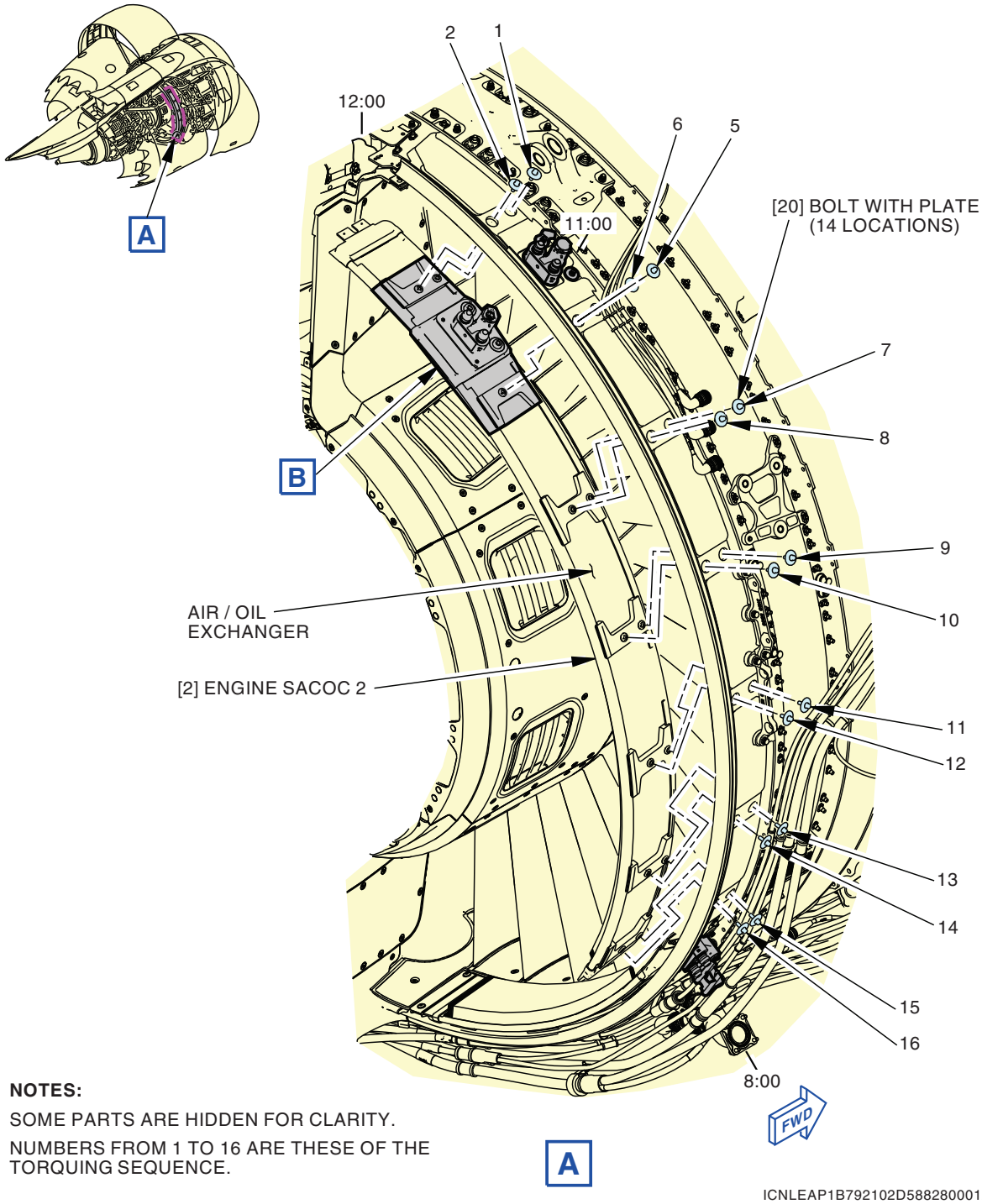


ICNLEAP1B792102D5882800006A00201

2422391 S0000556952_V6

Engine Surface Air Cooling Oil Cooler 2 Installation
Figure 403/79-21-02-990-804-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTES:

SOME PARTS ARE HIDDEN FOR CLARITY.

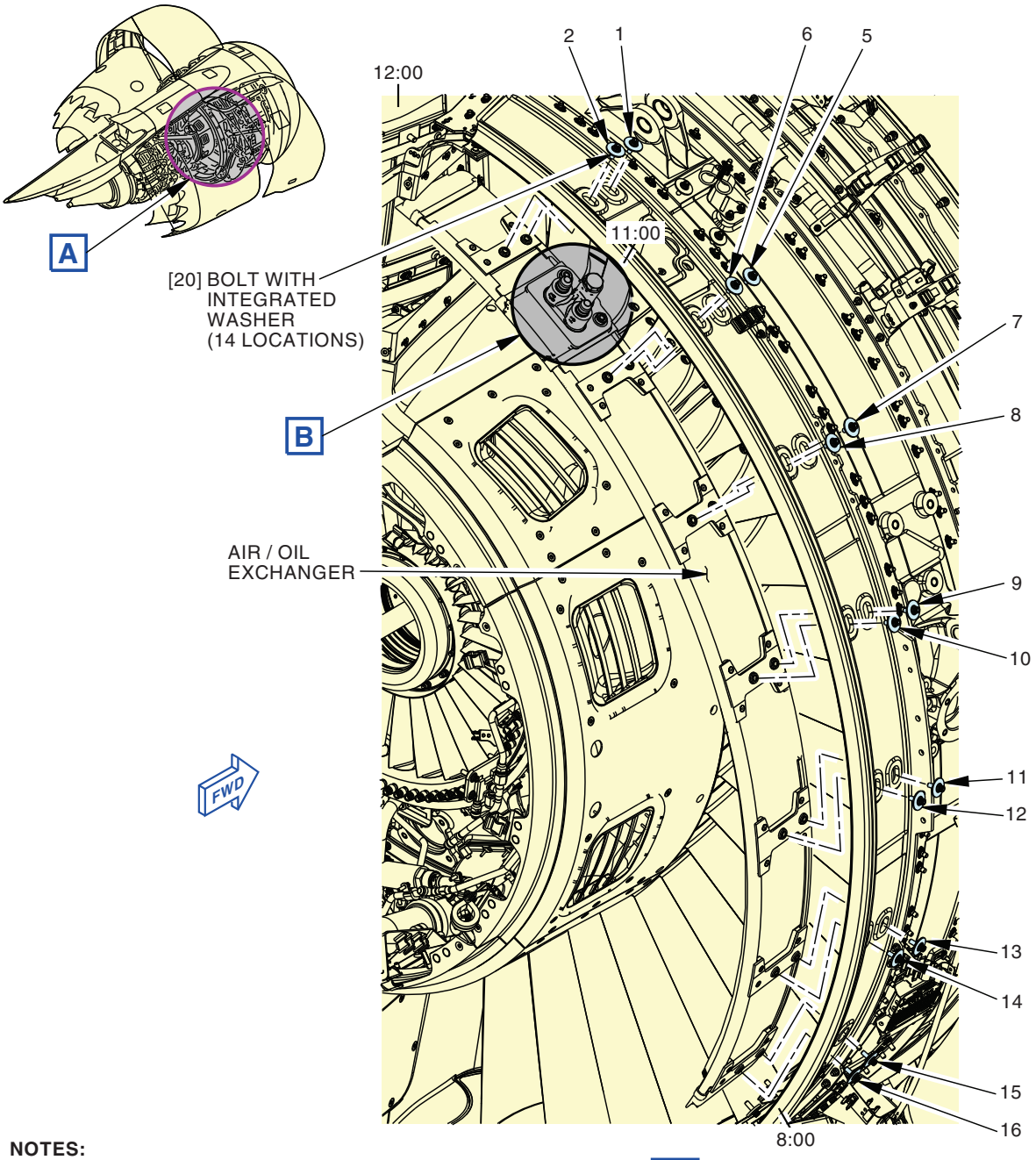
NUMBERS FROM 1 TO 16 ARE THESE OF THE TORQUING SEQUENCE.

ICNLEAP1B792102D5882800012A00101

3012581 S0000786064_V1

Bolt Tightening Sequence
Figure 404/79-21-02-990-807-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTES:

SOME PARTS ARE HIDDEN FOR CLARITY.

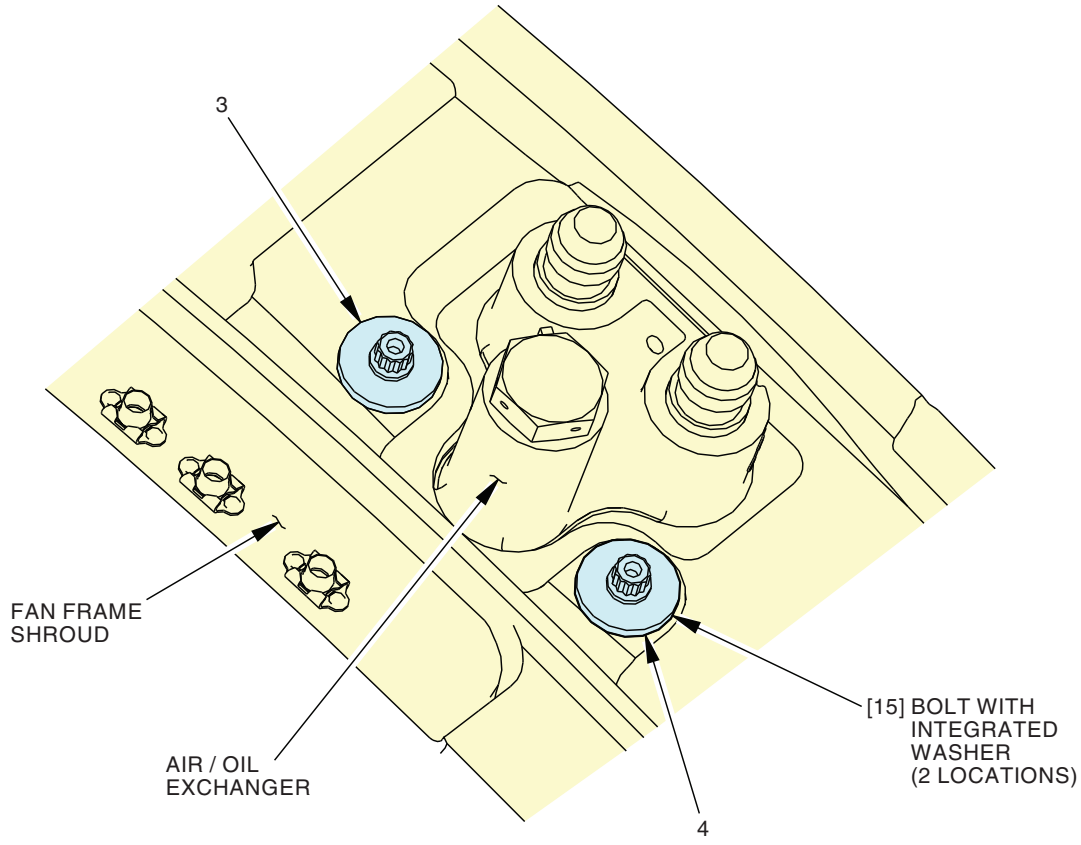
NUMBERS FROM 1 TO 16 ARE THESE OF THE TORQUING SEQUENCE.

ICNLEAP1B792102D5882800016A00101

2946454 S0000723173_V2

Bolt Tightening Sequence
Figure 404/79-21-02-990-807-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

NUMBERS FROM 3 AND 4 ARE THESE OF THE TORQUING SEQUENCE.

ICNLEAP1B792102D5882800017A00101

2946456 S0000723174_V1

Bolt Tightening Sequence
Figure 404/79-21-02-990-807-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details